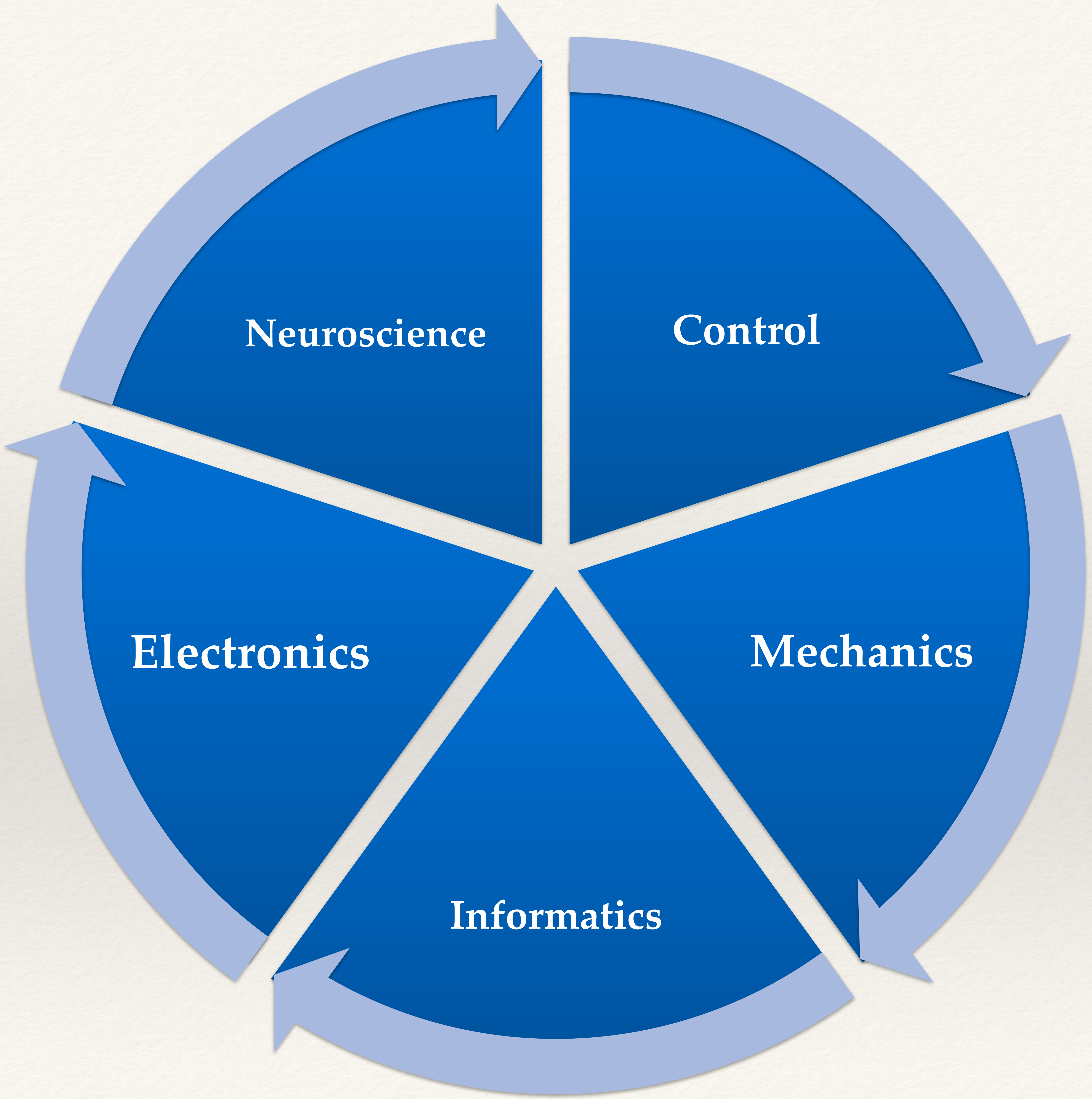
Biomimetic Neuroscience

Deep Learning

Rodrigo Ramele
Lab Neurotrónica
Departamento de Ingeniería Informática


Neuroscience
Control
Electronics
Mechanics
Informatics

The Brain

CNS and PNS

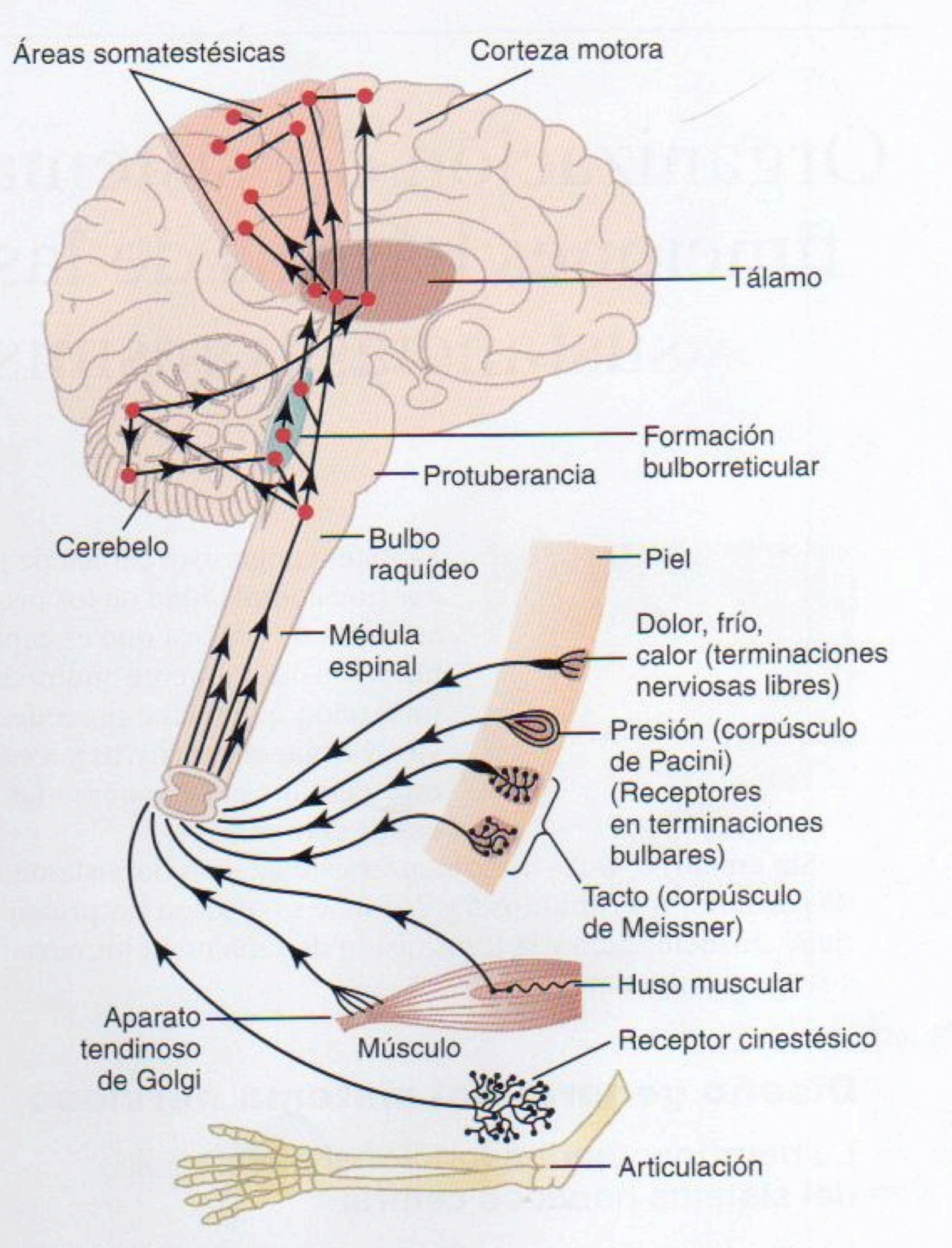
CNS (skull)

PNS

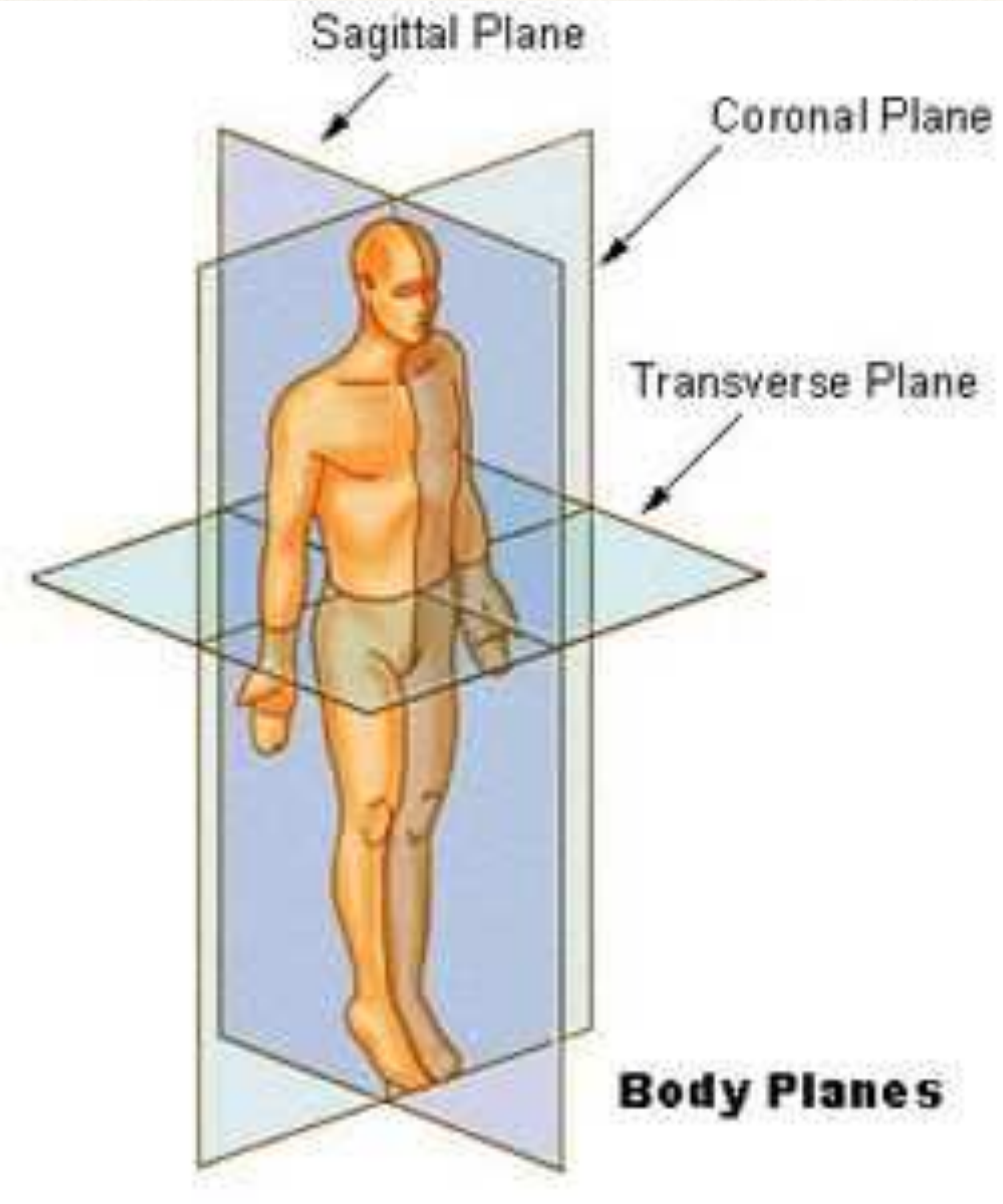
- Spinal Cord
- Subcortical Level
- Cortical Level

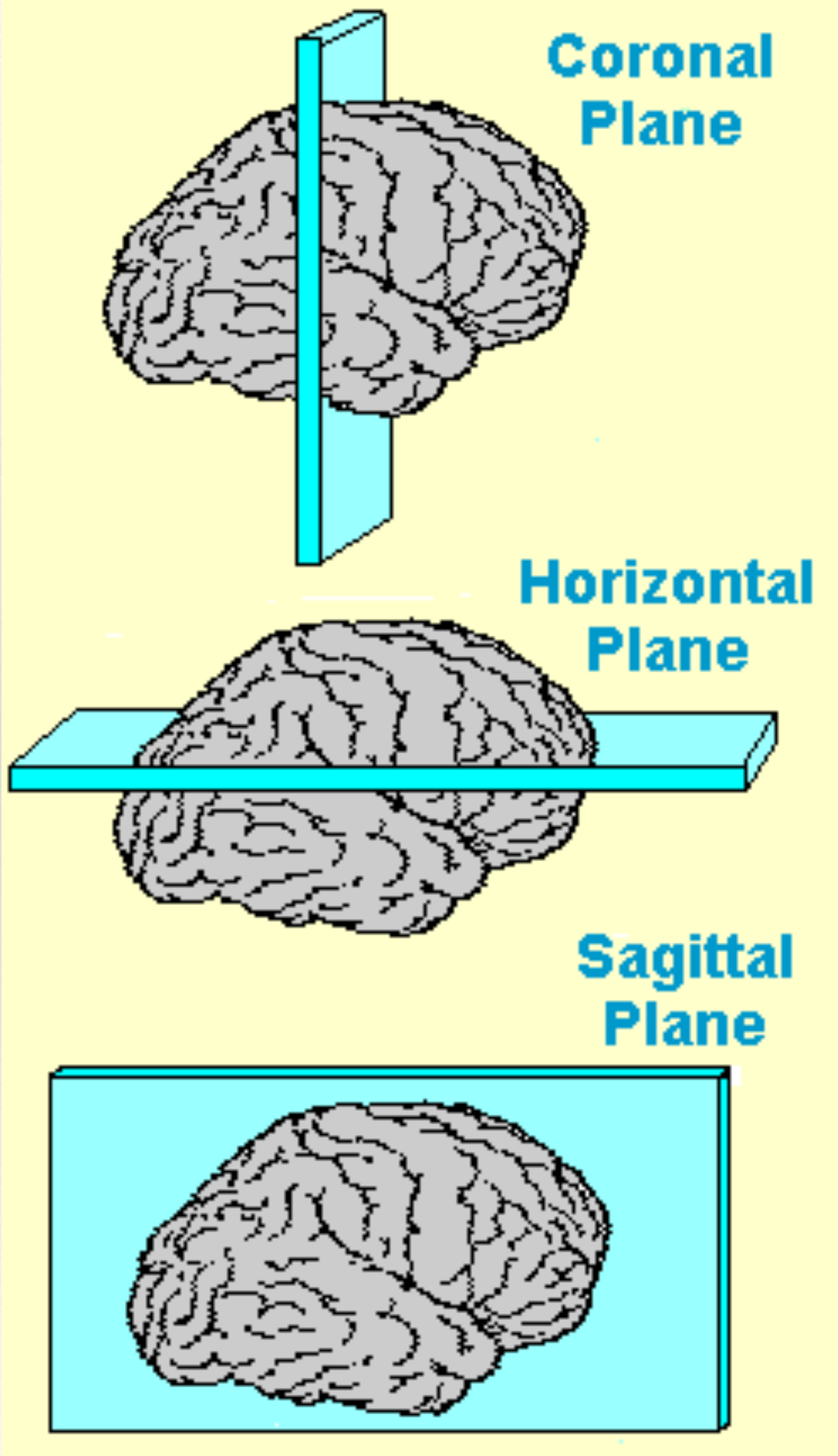
- **Autonomic system**
  - Parasympathetic: rest and digest
  - Sympathetic: fight-or-flight-response

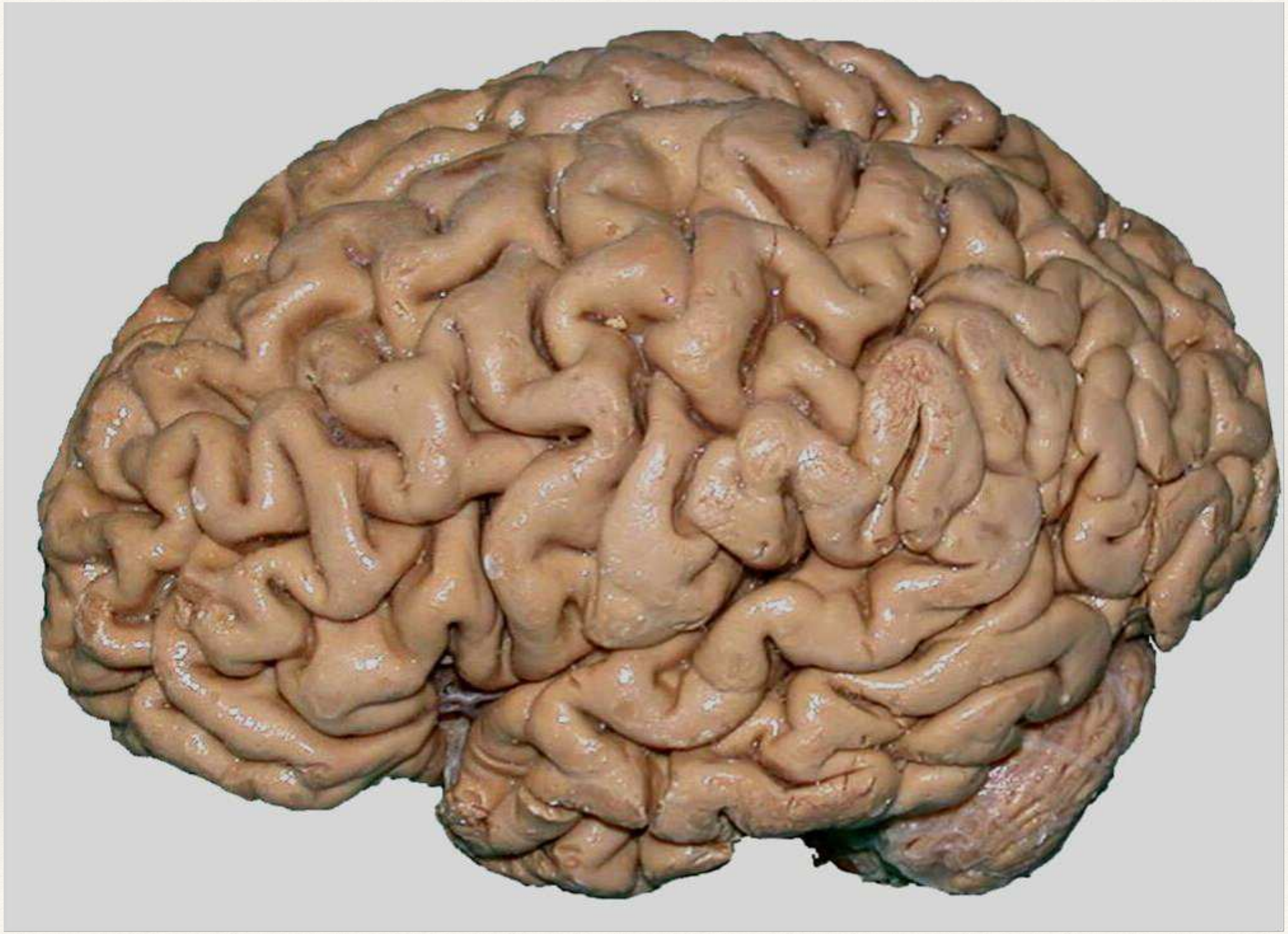
- **Nervous endings**
  - Sense
  - Motor
  - Mixed

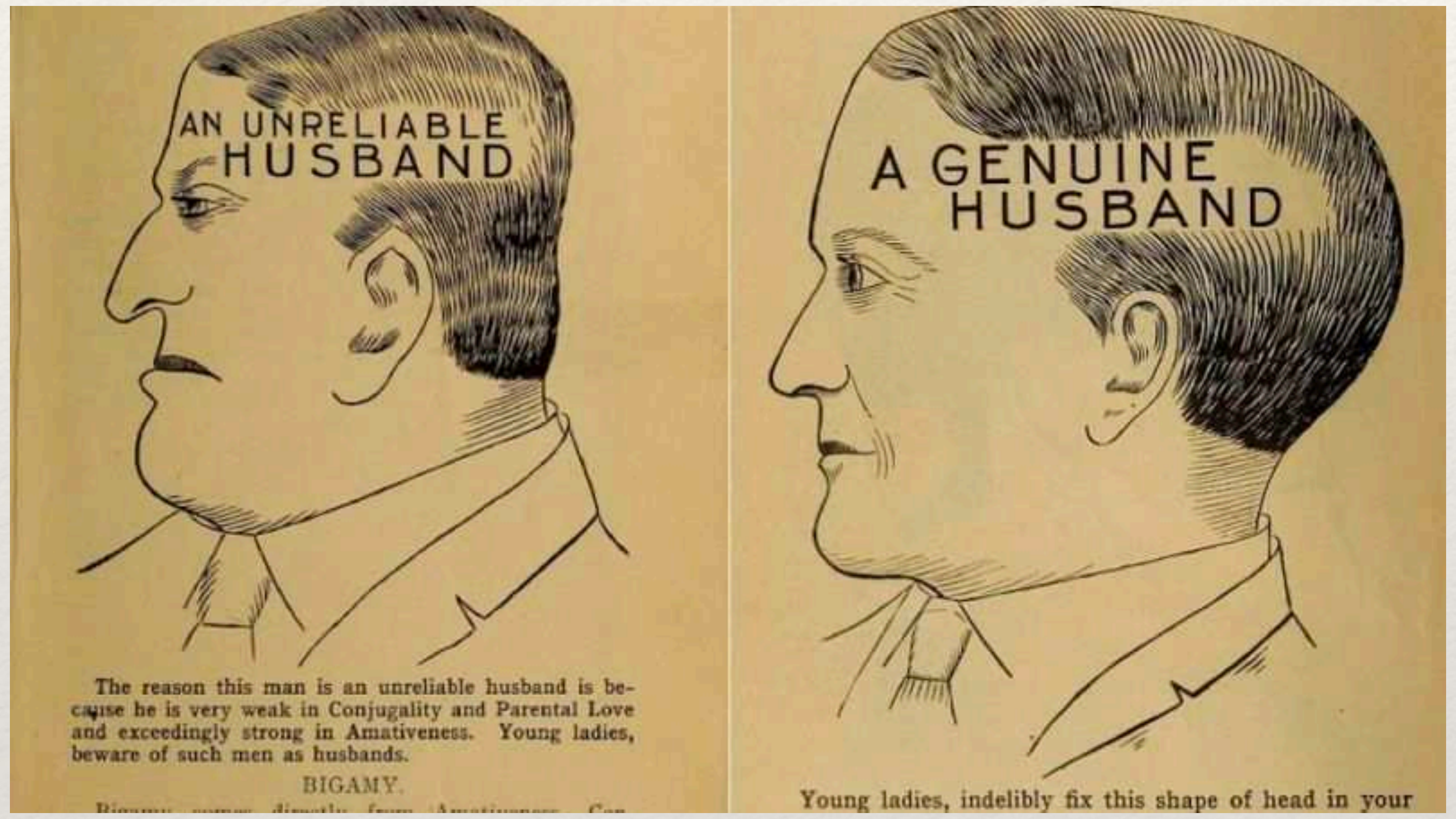


Brain Anatomy

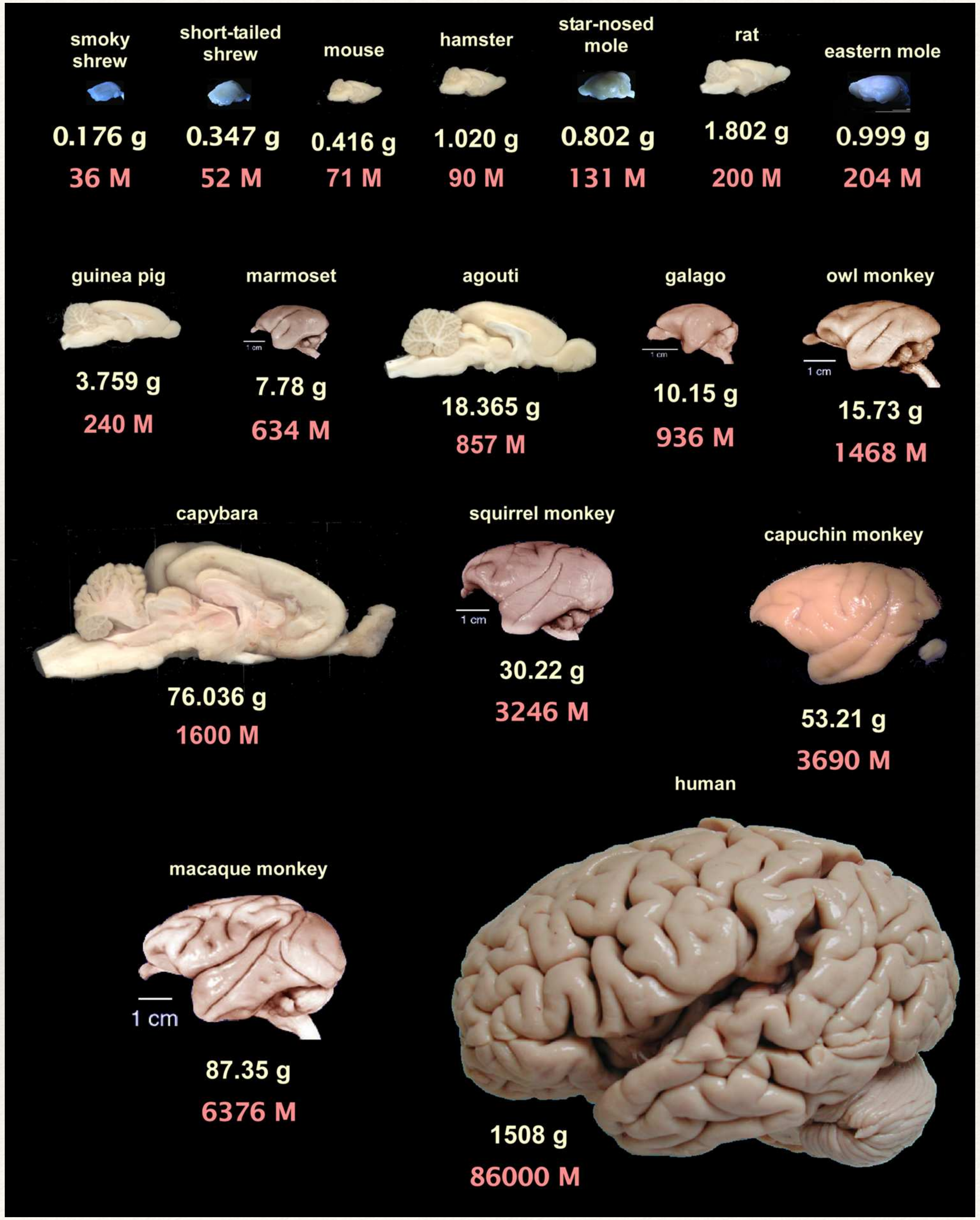








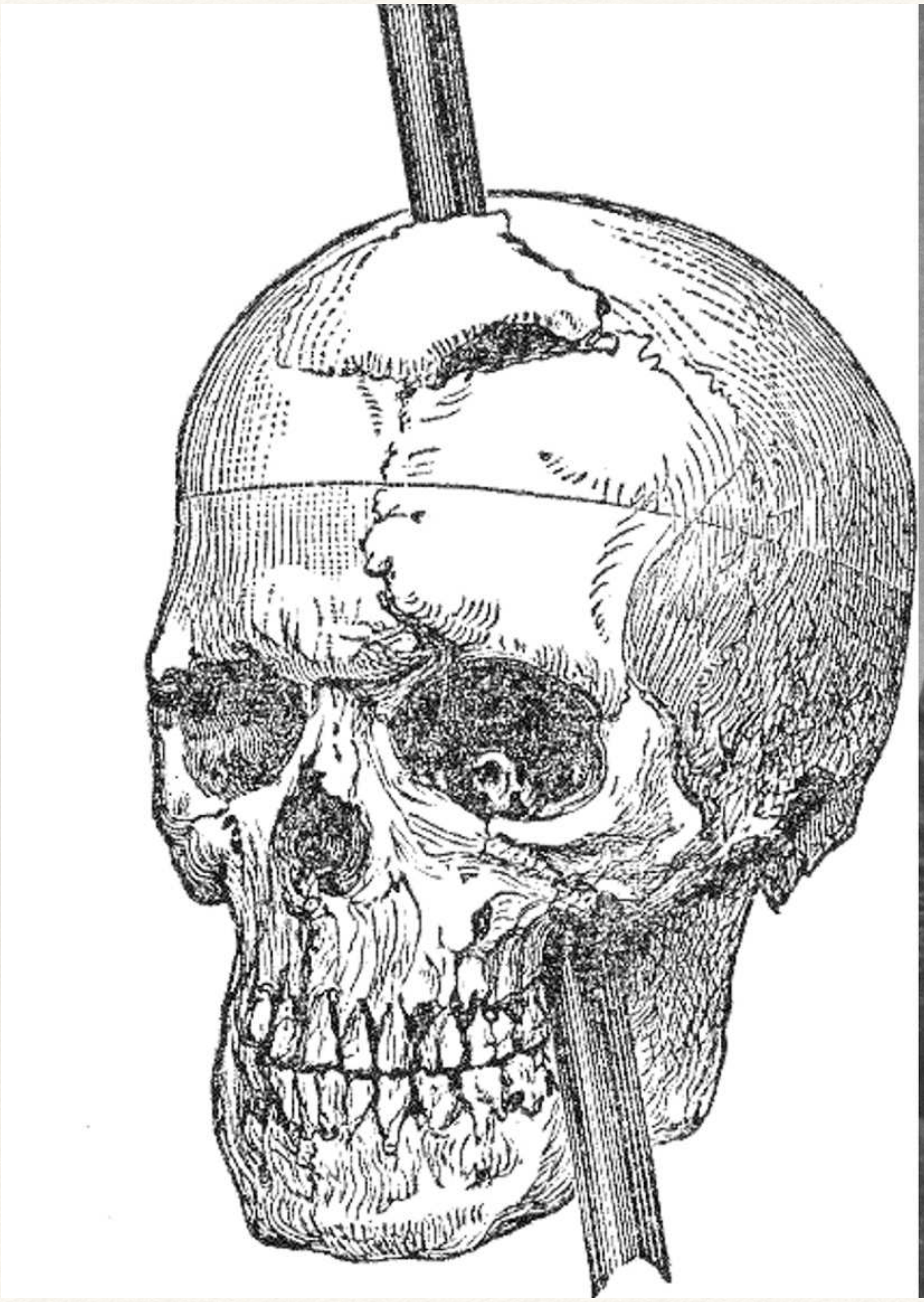
Frenología

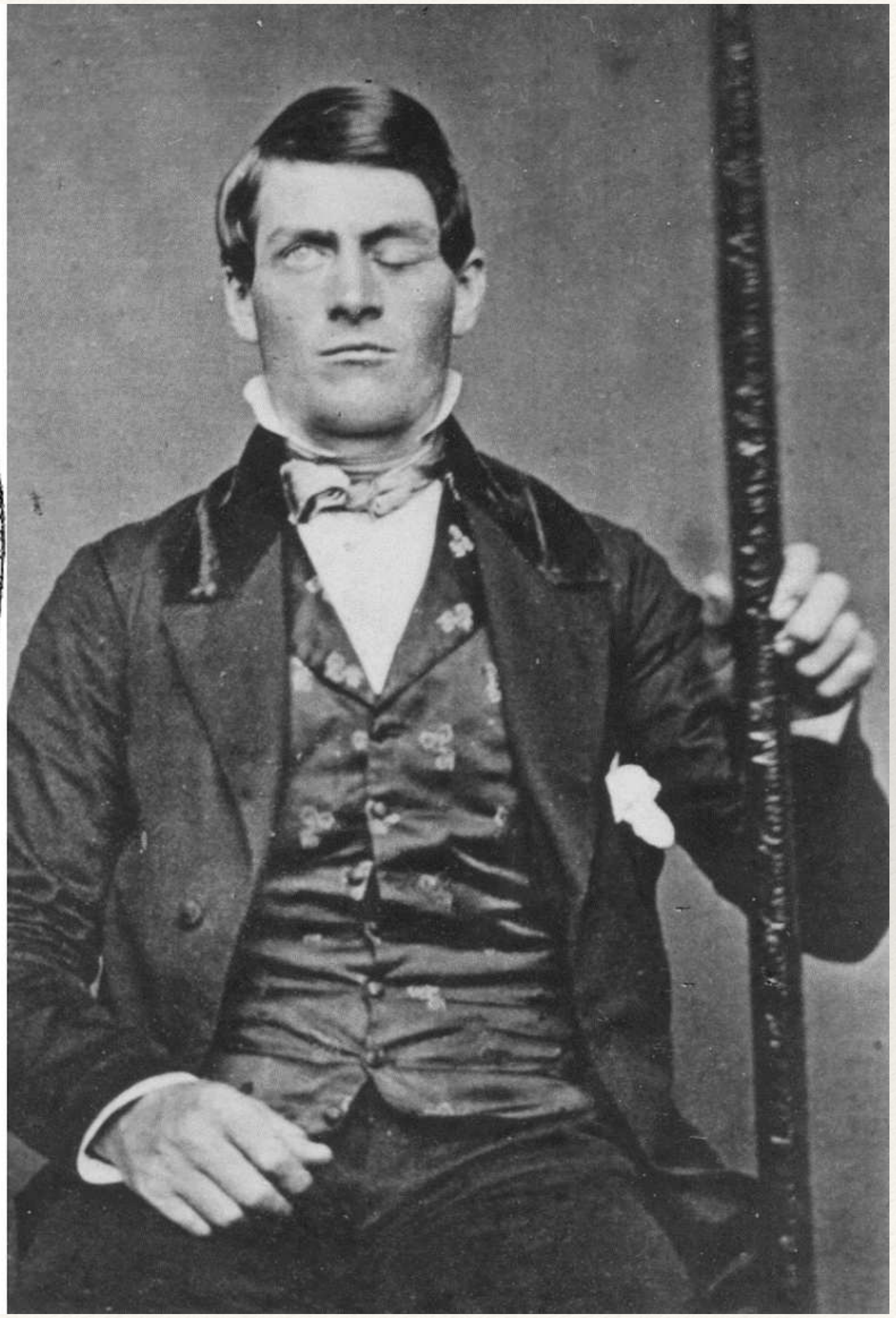


Herculano-Souzel (2009)

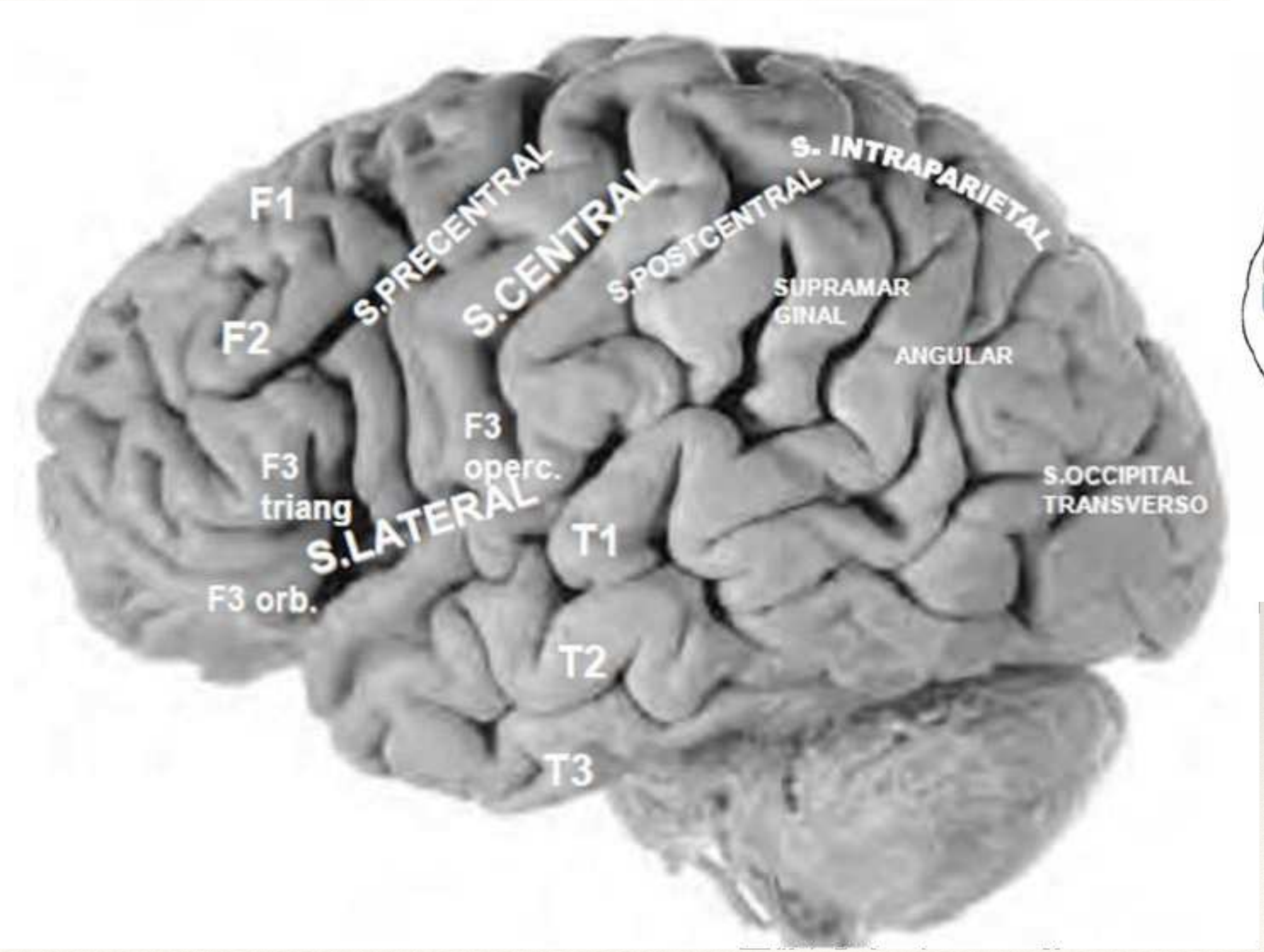

USC Mark and Mary Stevens
roimaging and Informatics Institute
Play
Keck Schoo
Medicine of

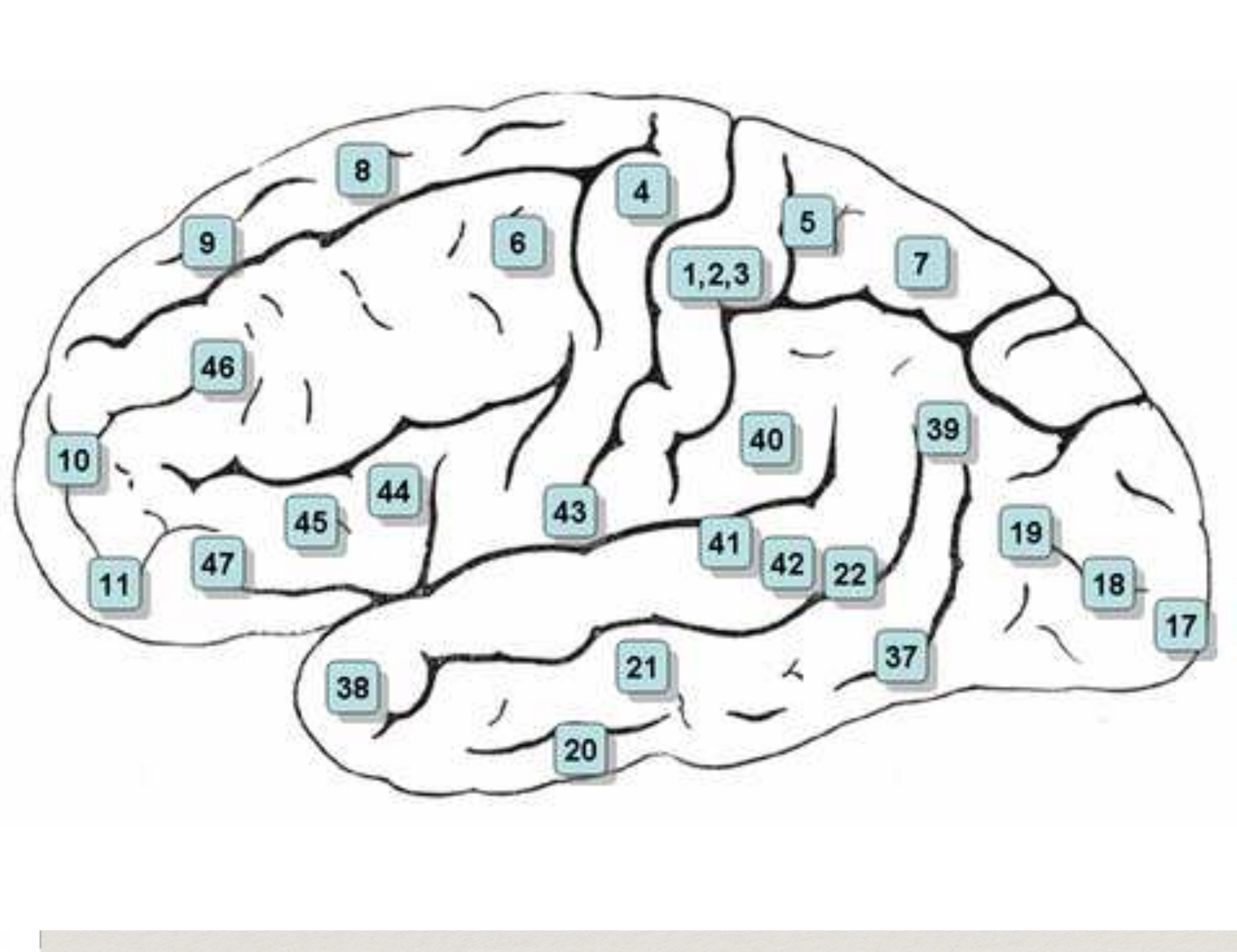
Thanks Phineas!



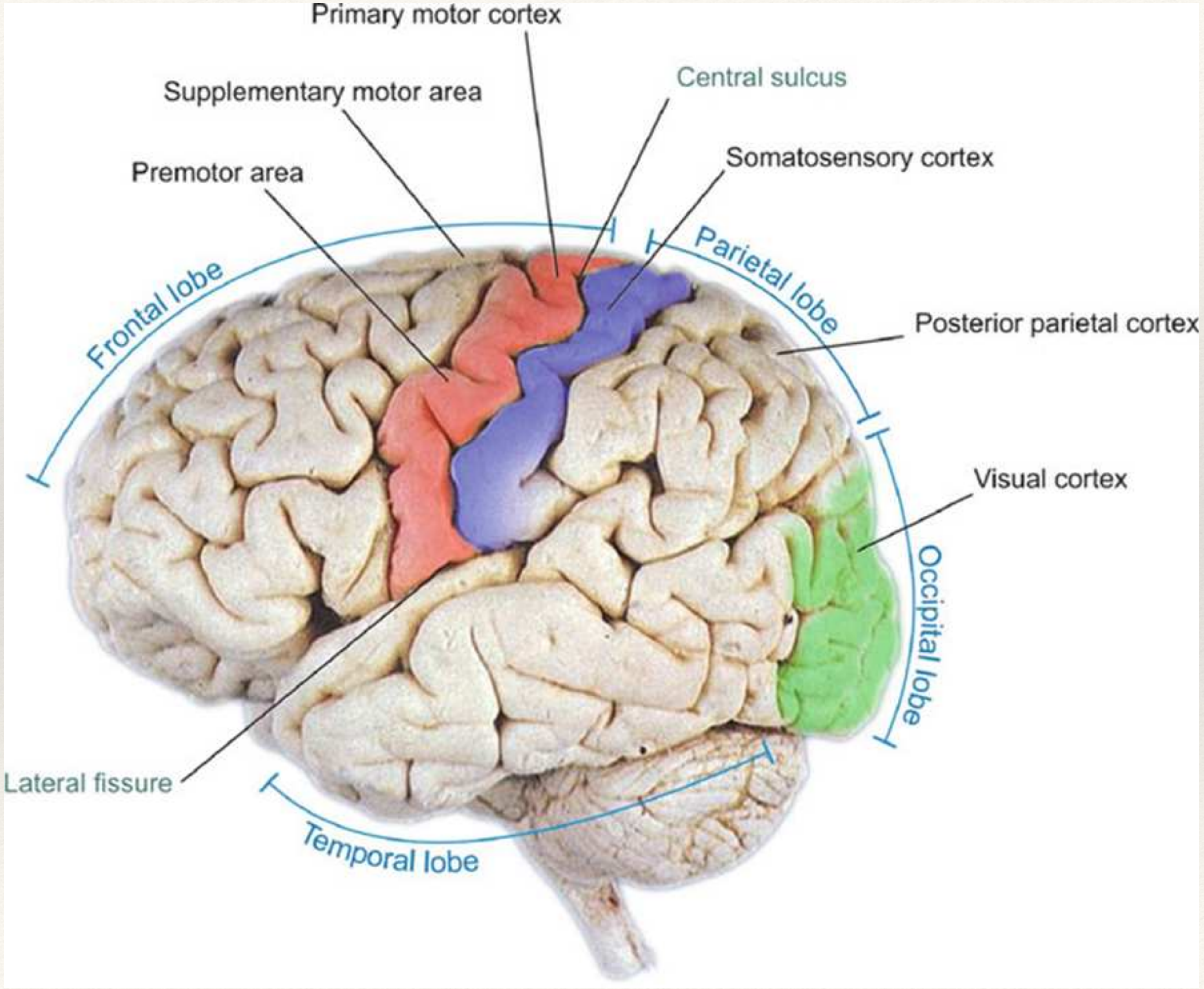


Divisions and Broadmann Mappings

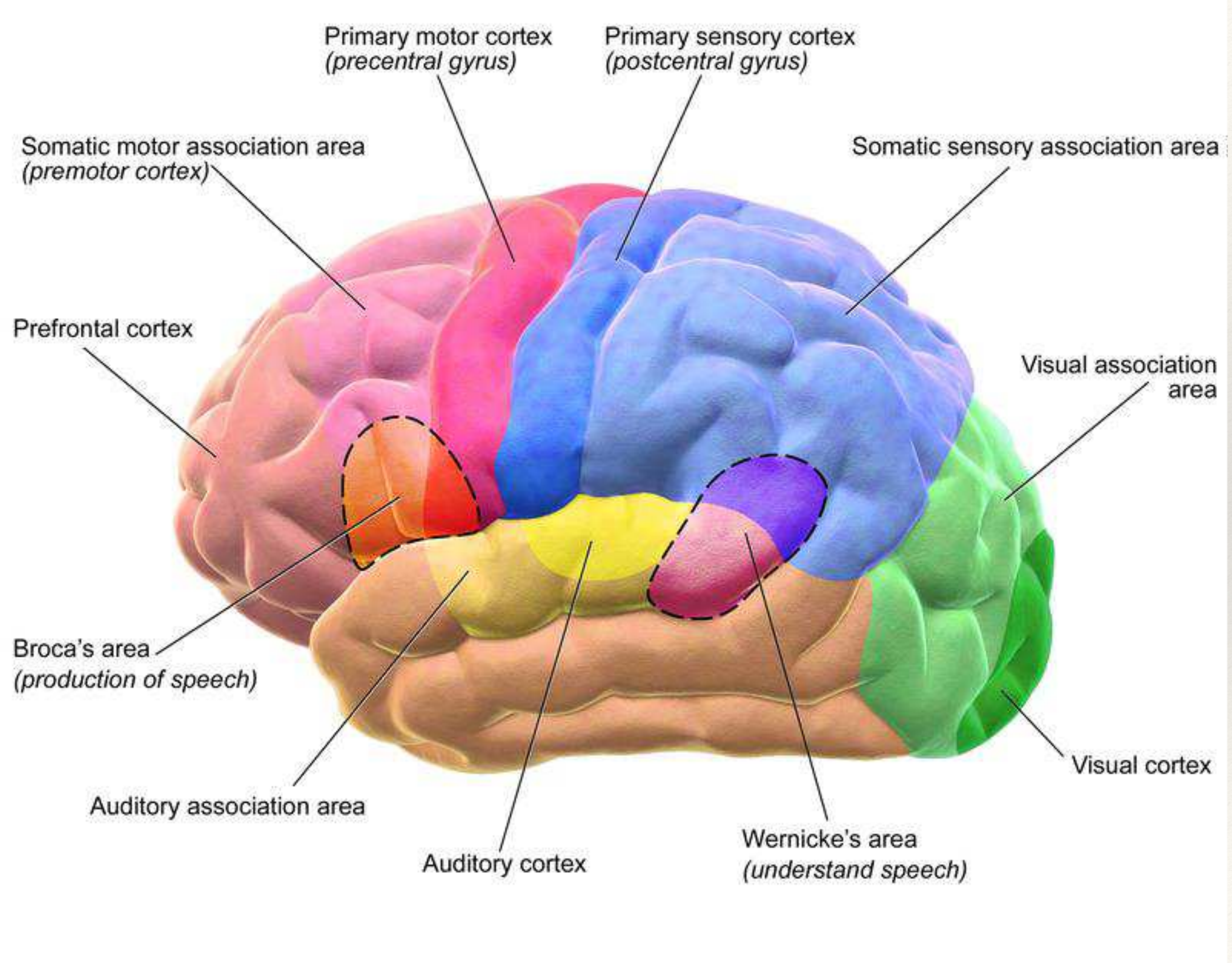




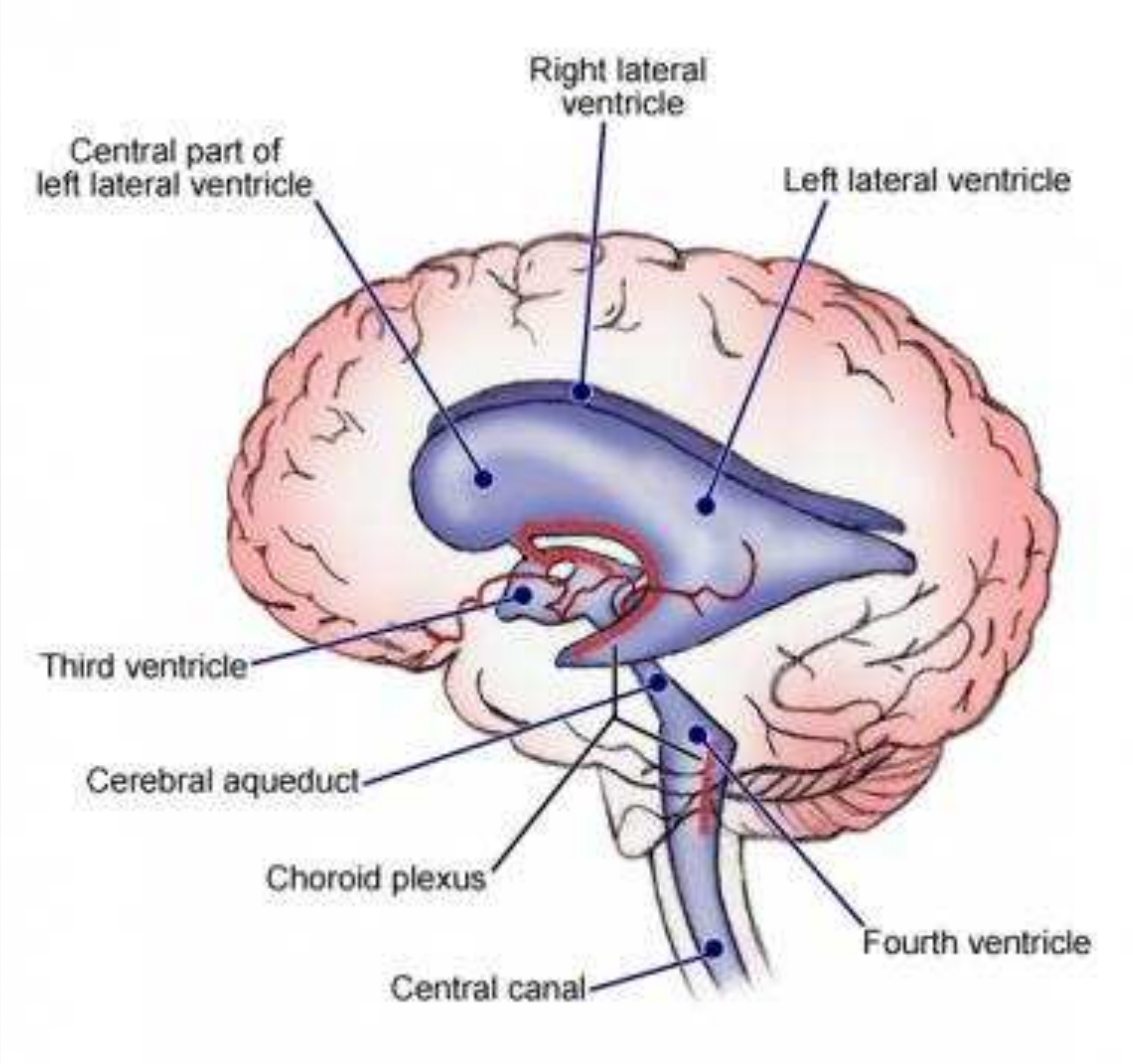
Primary Areas

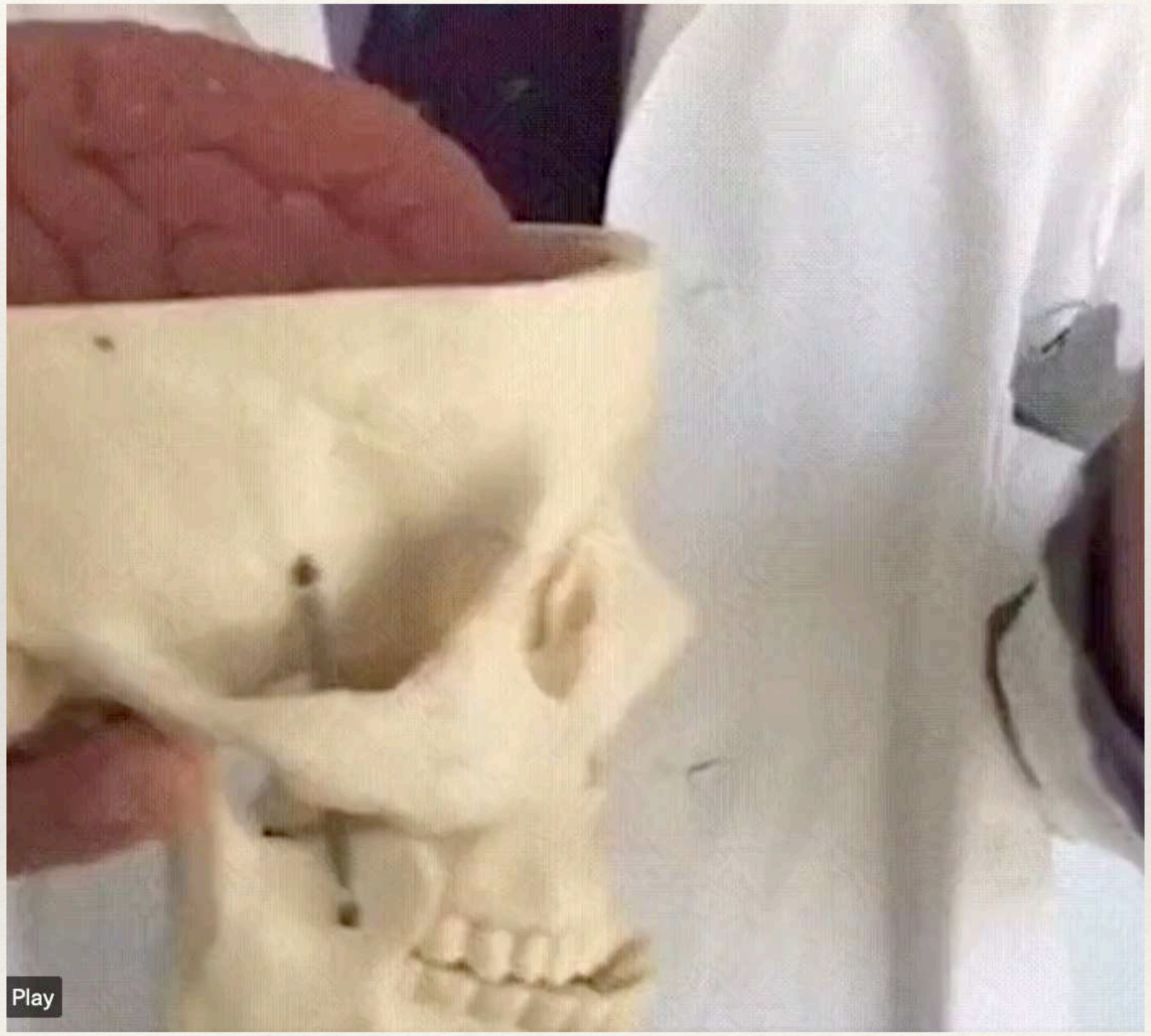


Primary Areas

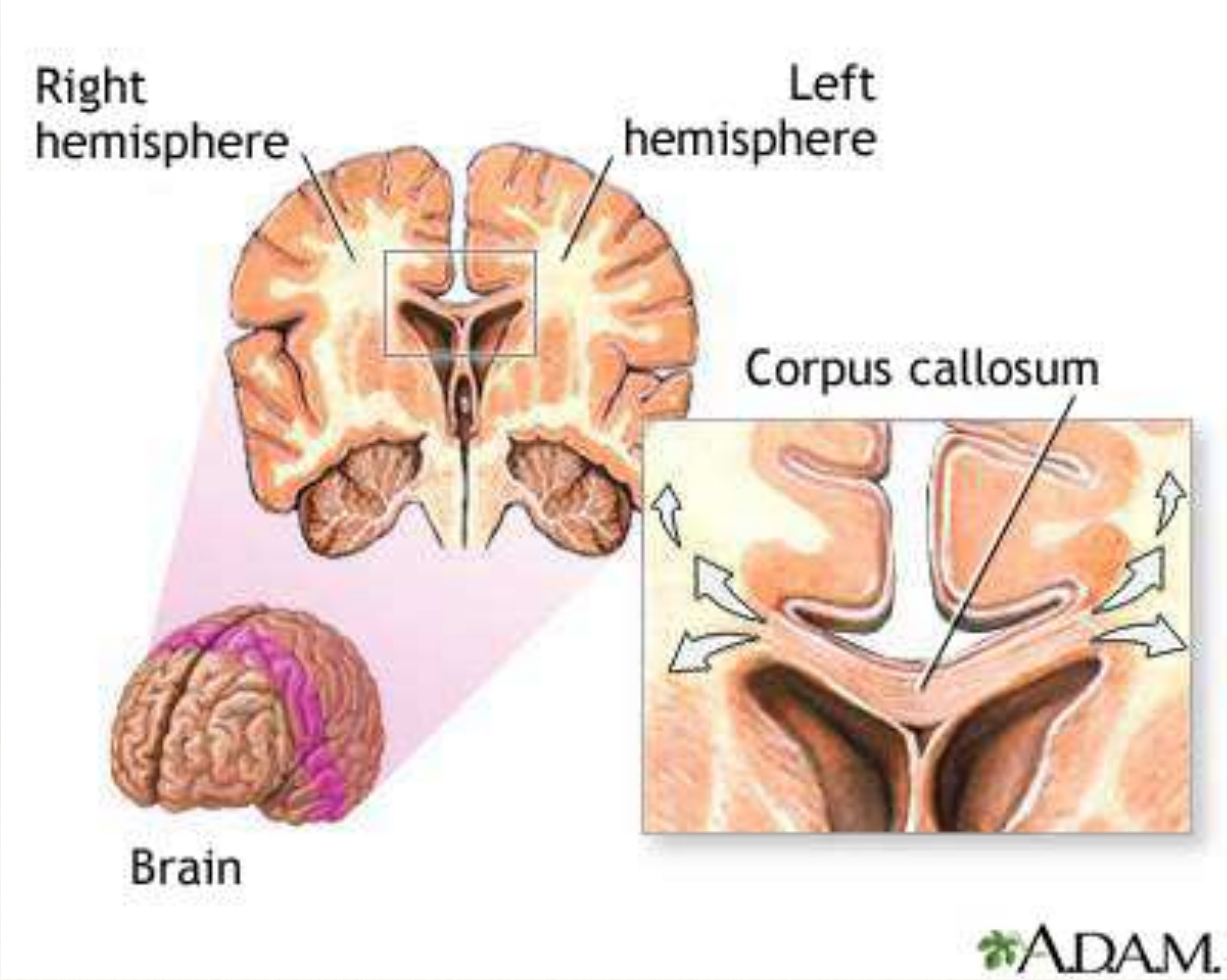


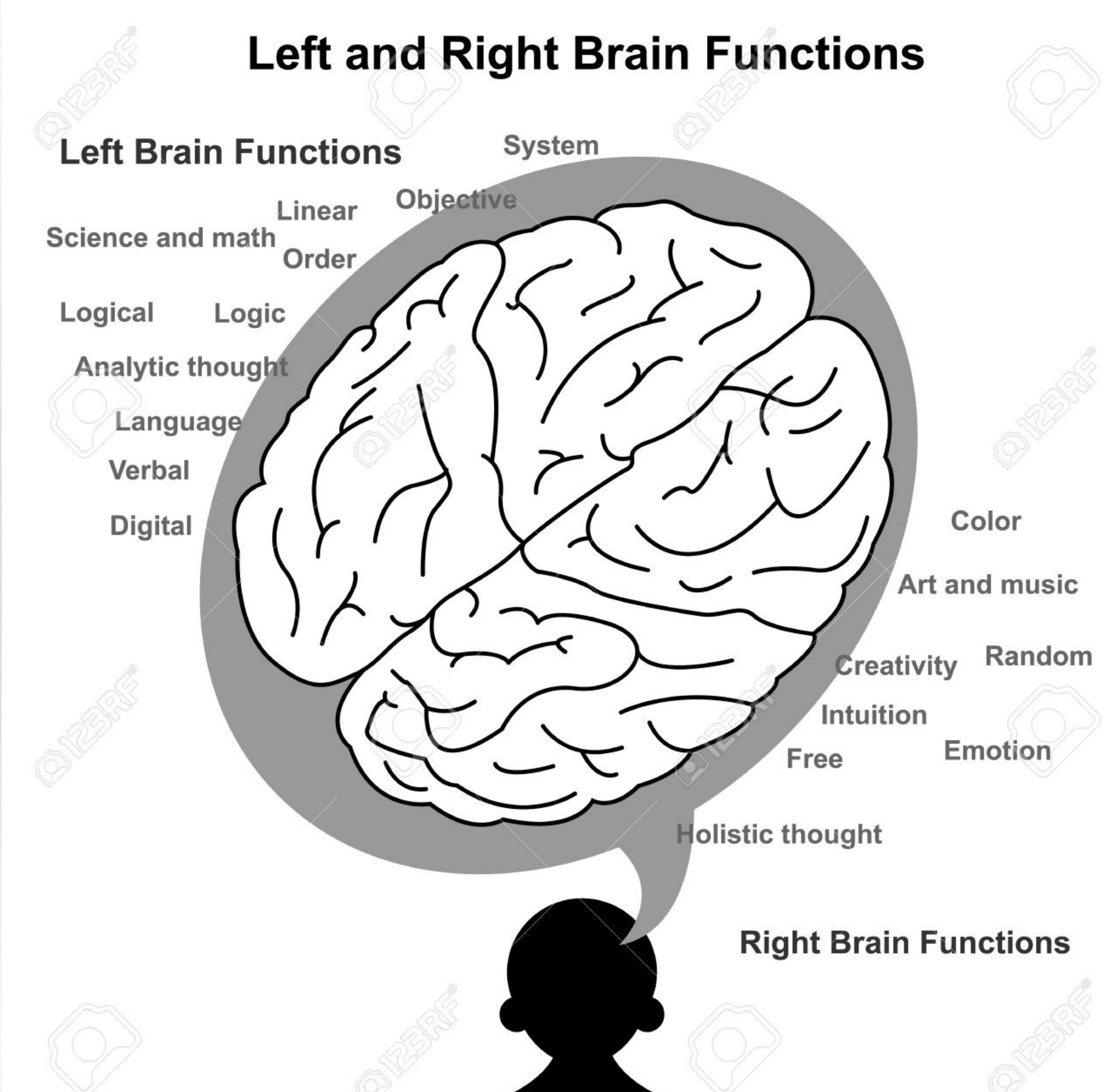
Brain Ventricles and Cerebrospinal Fluid



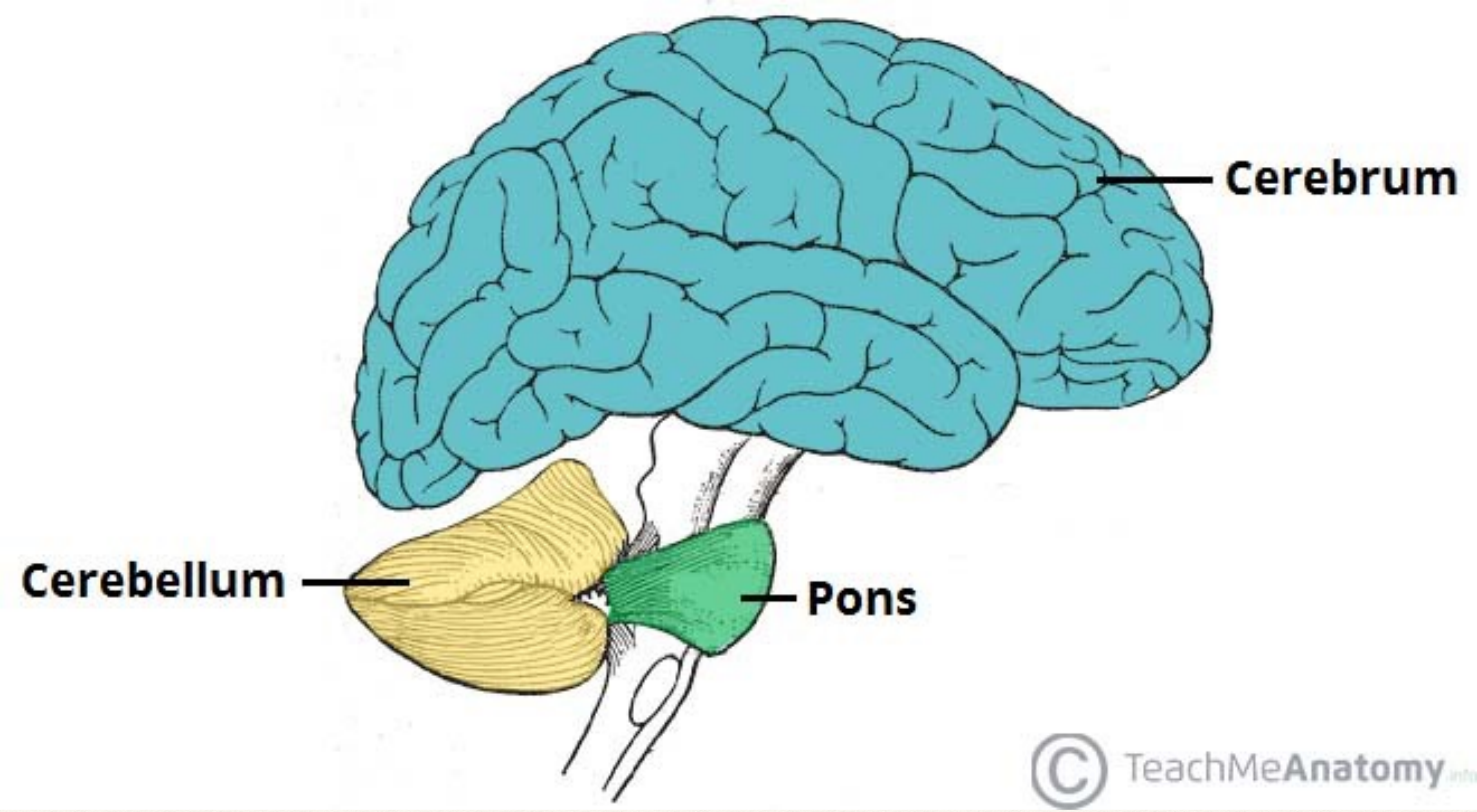


Hemispheres and Corpus Callosum - Network

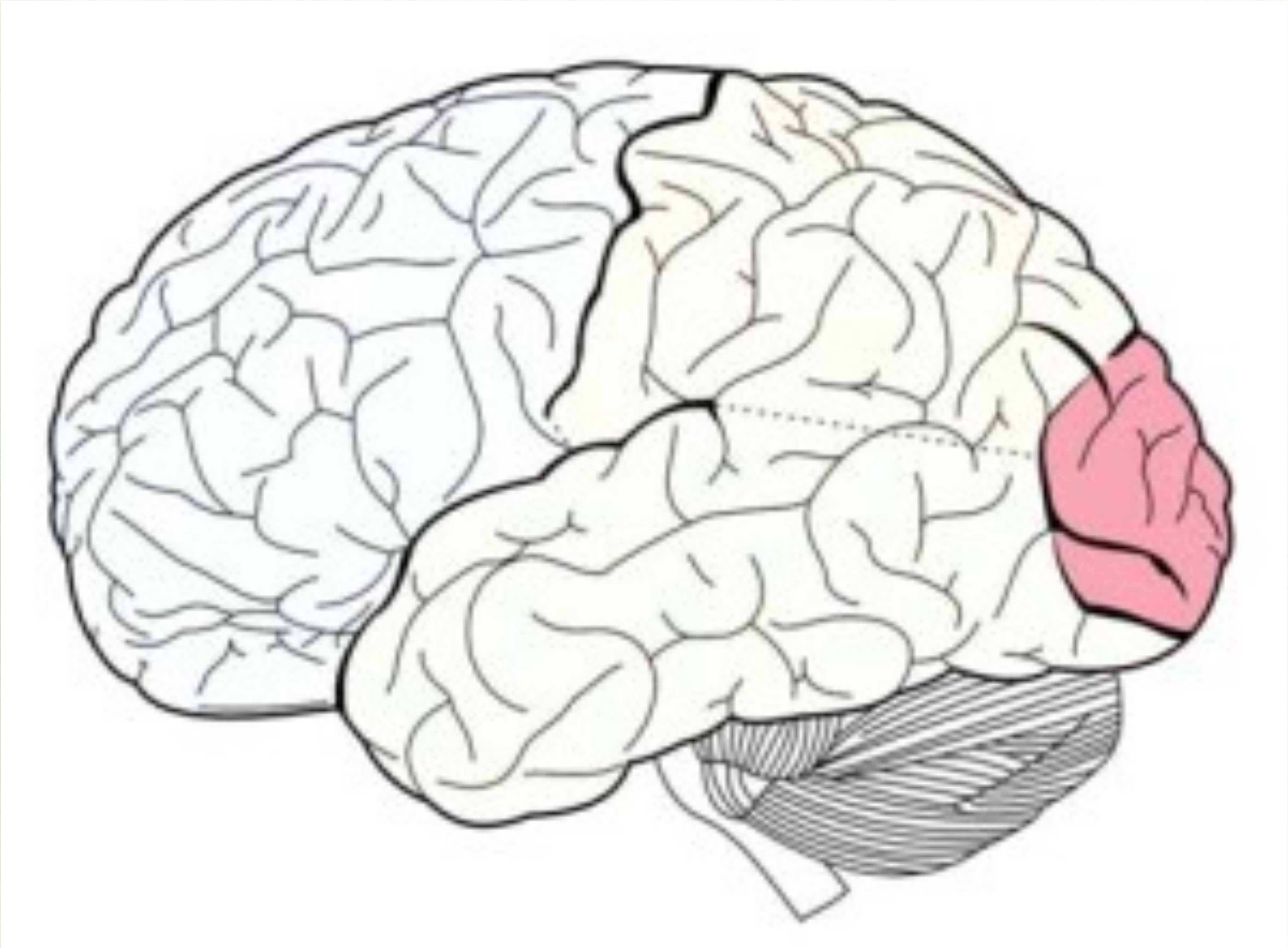


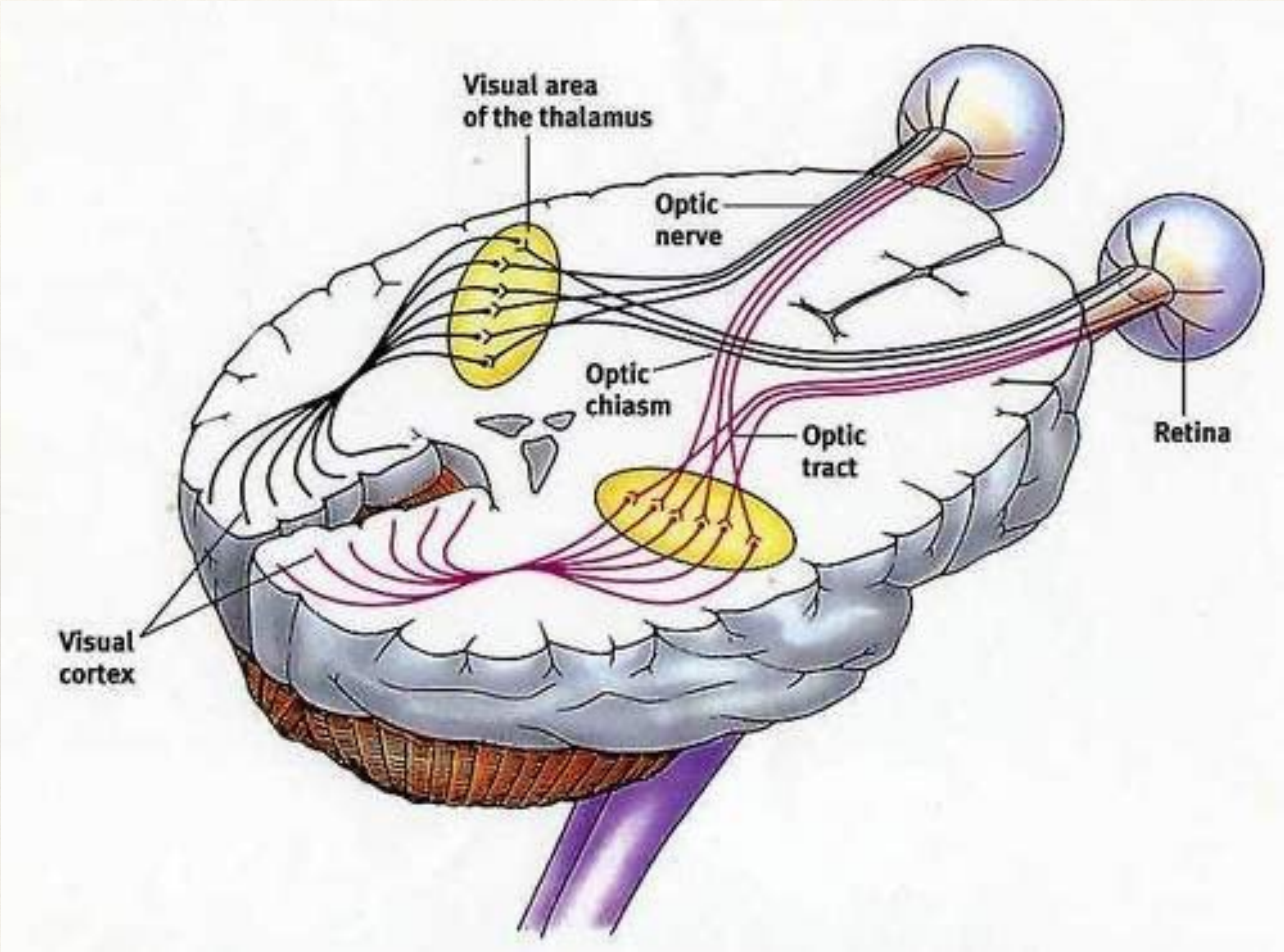


Subcortical - firmware



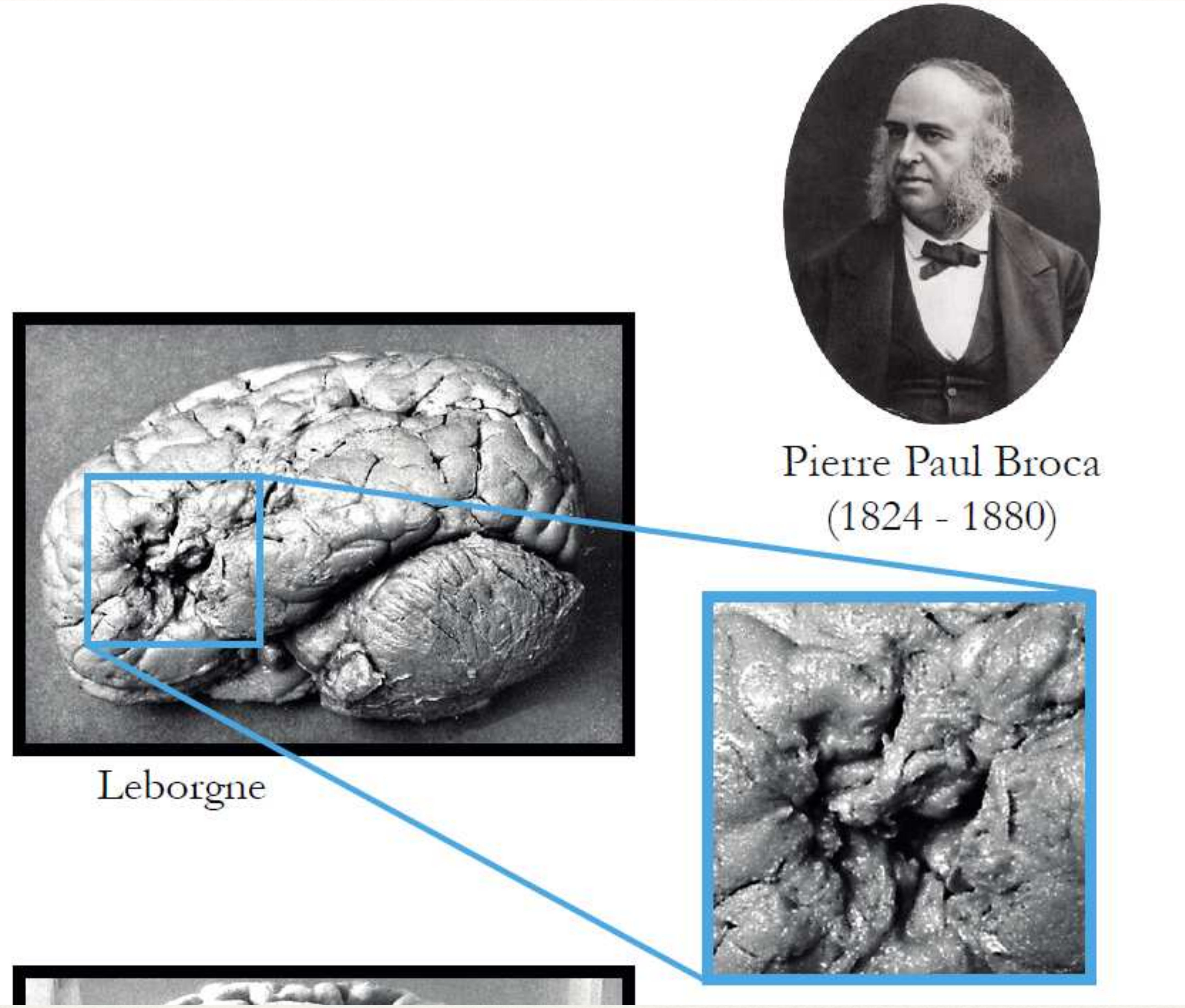
Occipital - Human GPU



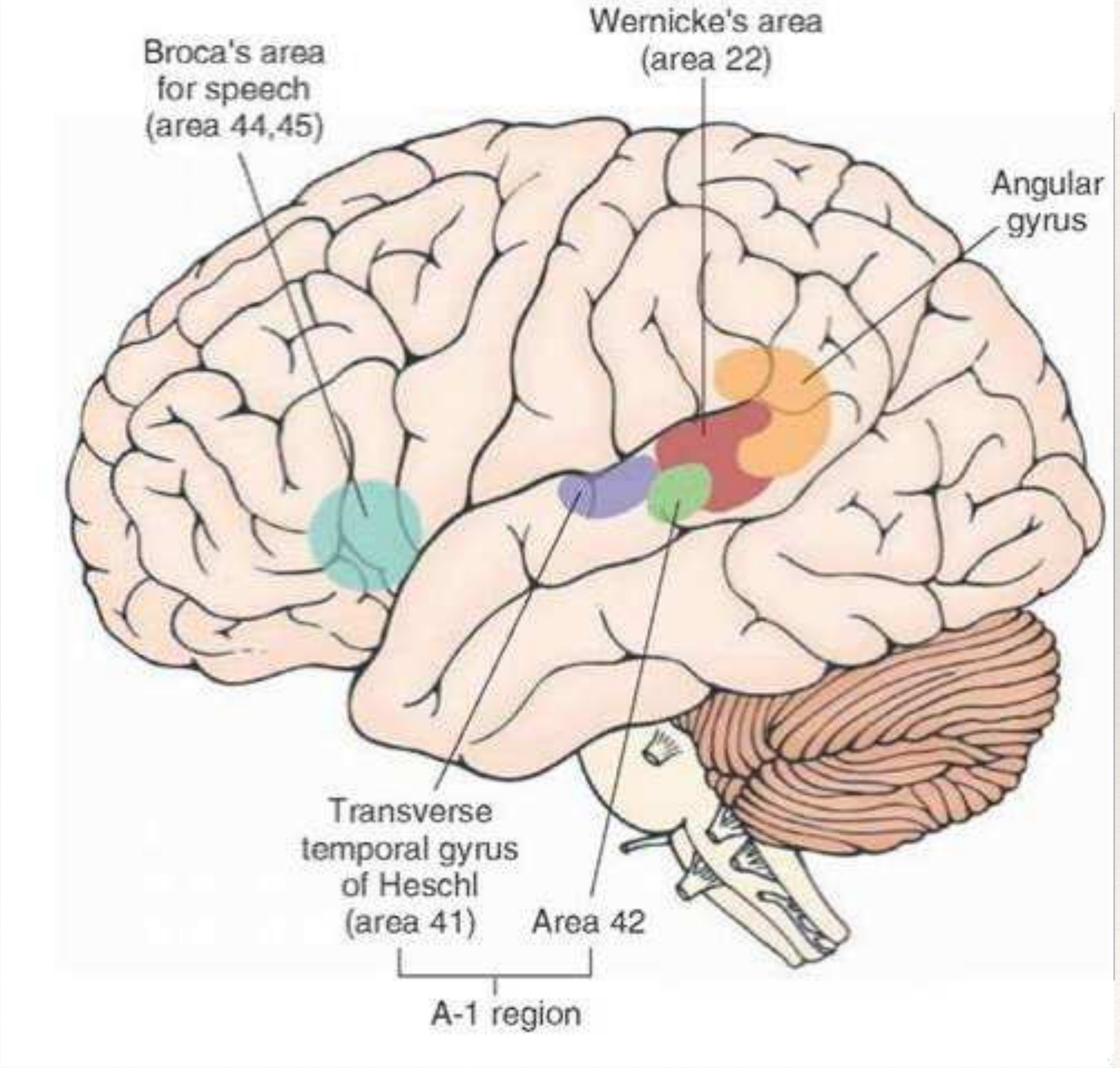


Frontal Area - Brocca

Natural Language Processing



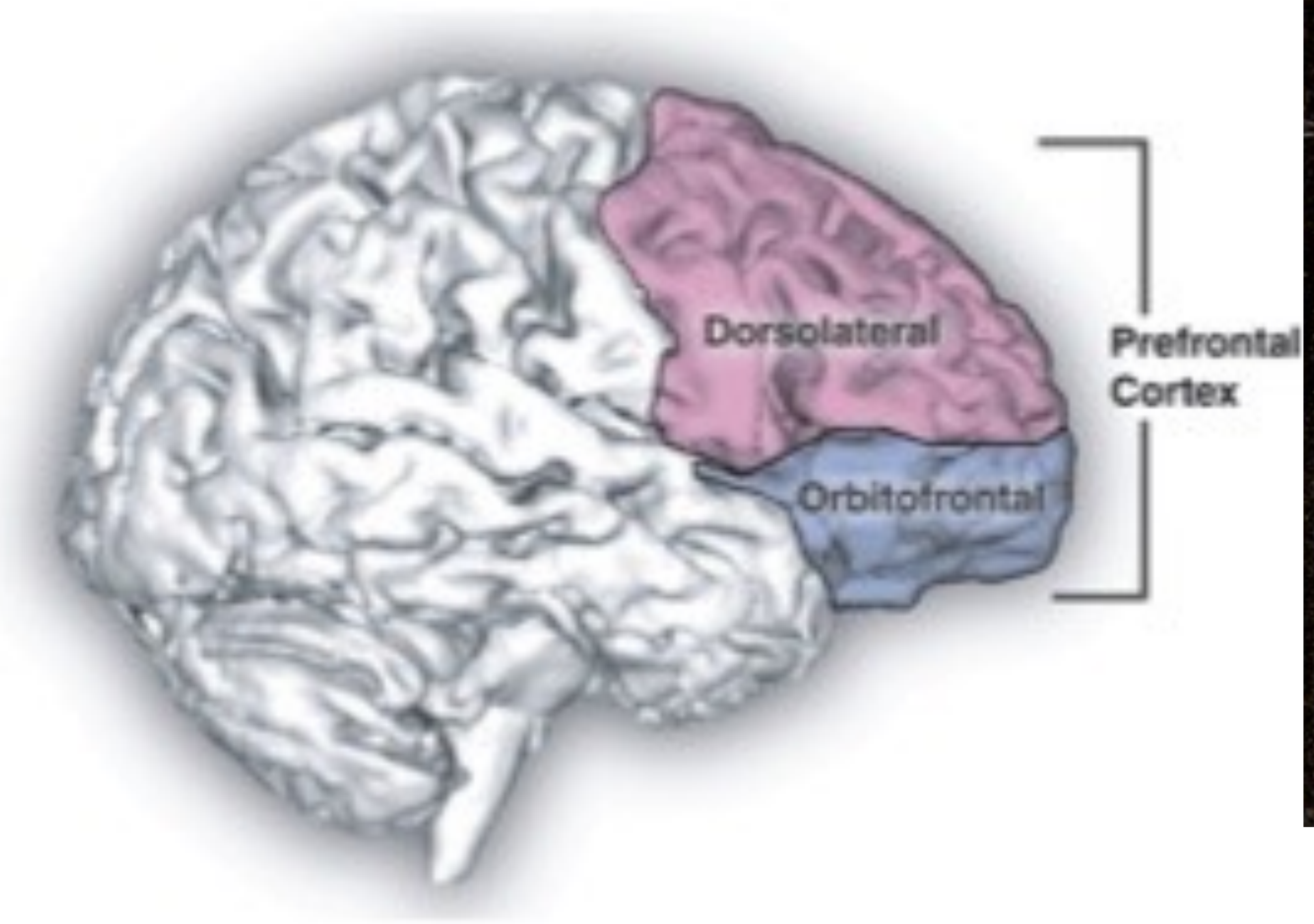
Frontal - Language



Frontal - Neocortex

The prefrontal cortex as a controller

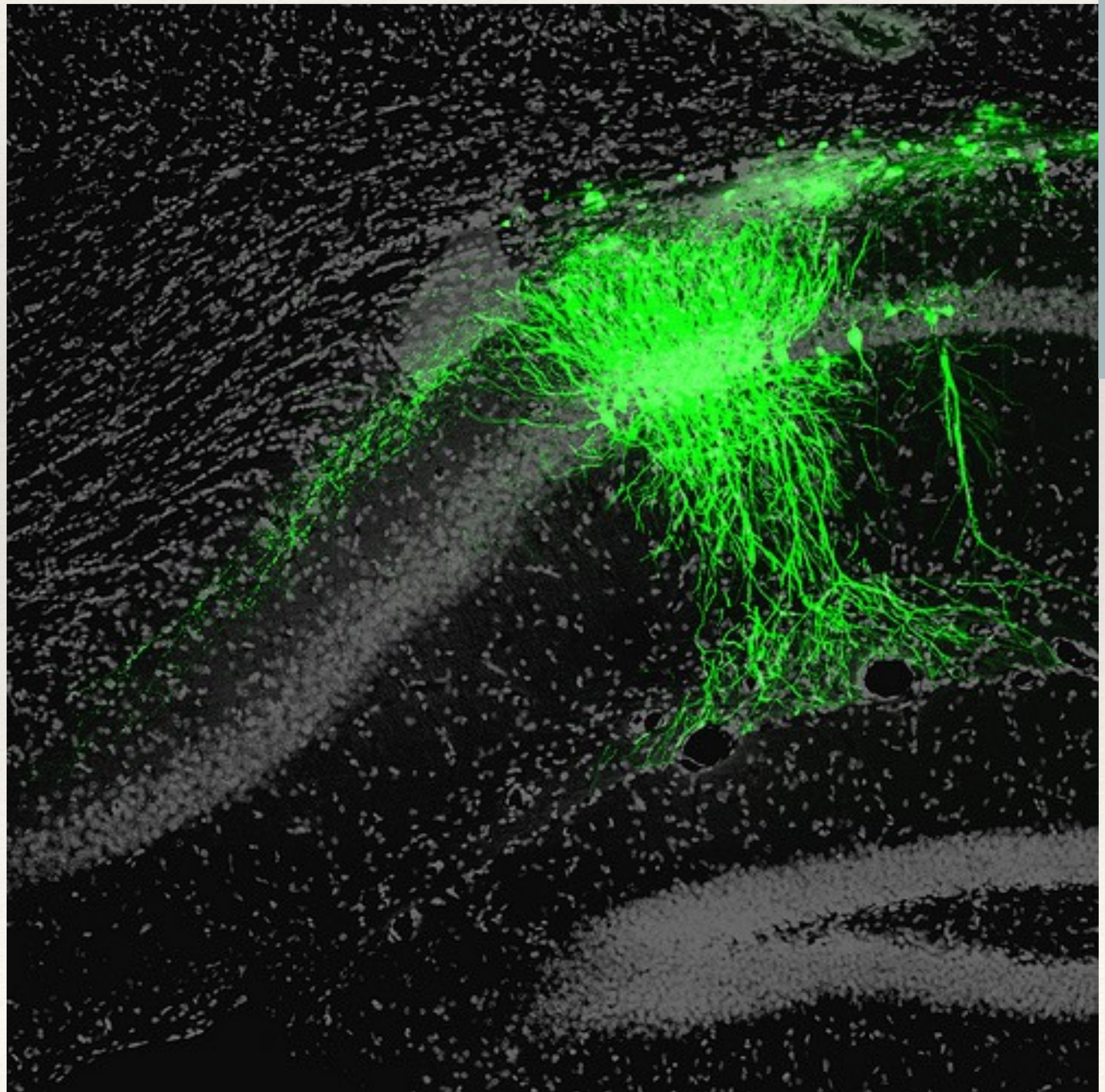
- Collection of interconnected neocortical areas in the frontal lobe
- Performs **executive functions** that are important for the **conscious control of thought and action** in accordance with **internal goals**
- Dorsolateral PFC: abstract reasoning & problem solving
- Orbitofrontal and medial cortex: affective & motivational functions



http://pubs.niaaa.nih.gov/publications/arh313/215-230.htm

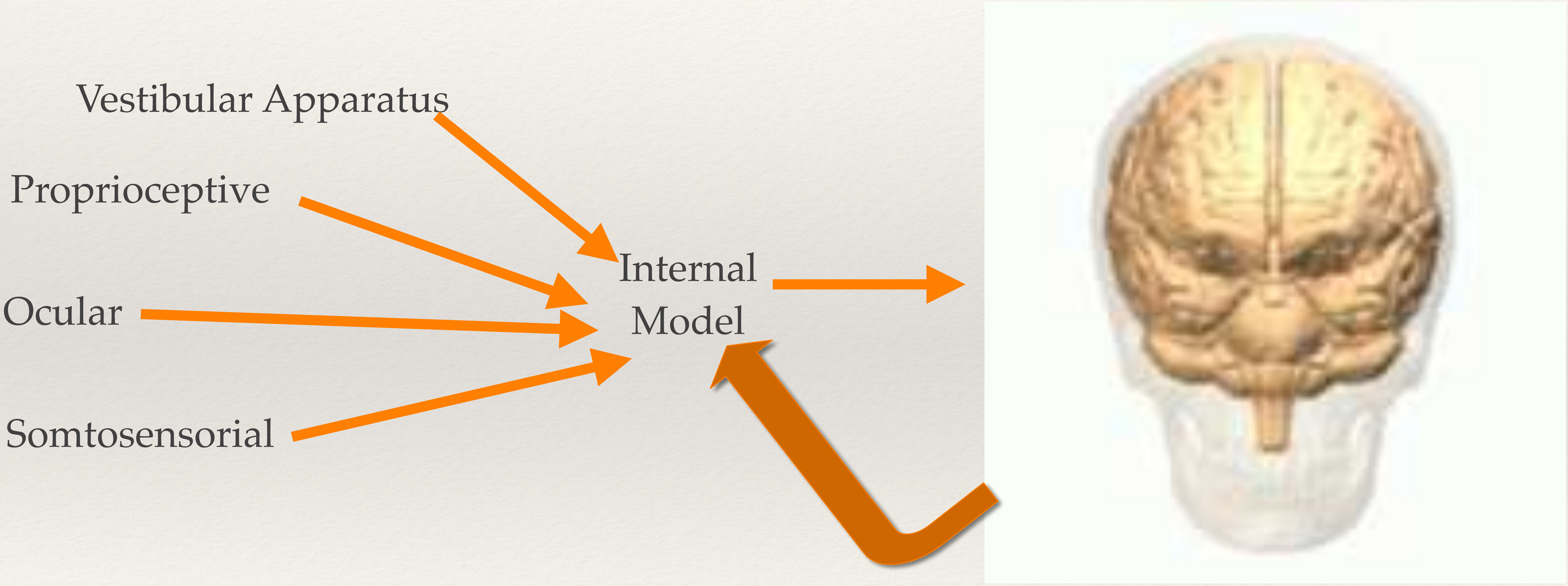
Temporal - Pre, Motor and Somatosensorial

Temporal - Hippocampus and Neurogenesis (RAM)



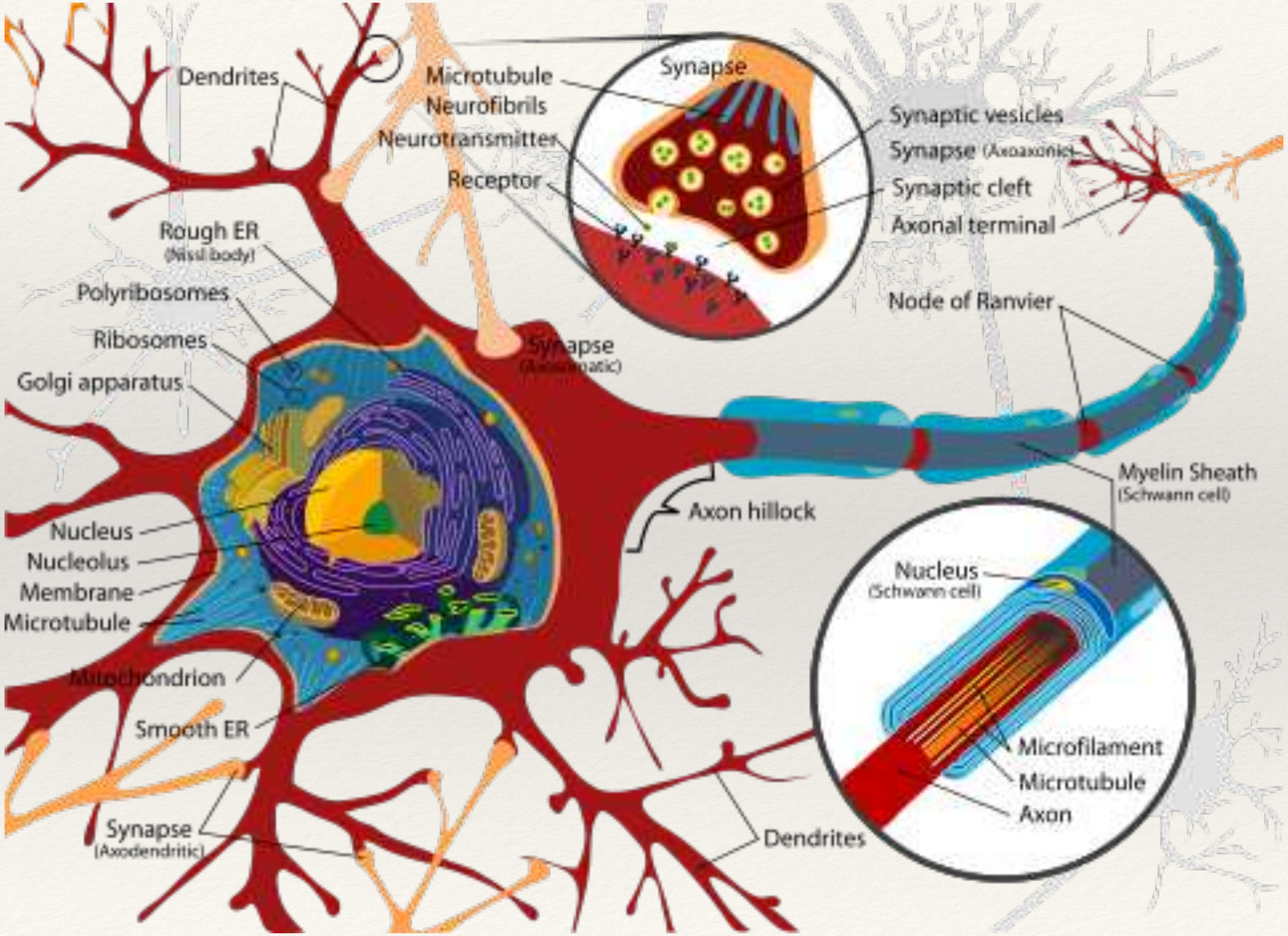
Krzisch, Temprana, Mongiat 2014

Temporal - Tempo parietal junction



Neuron

Neuron

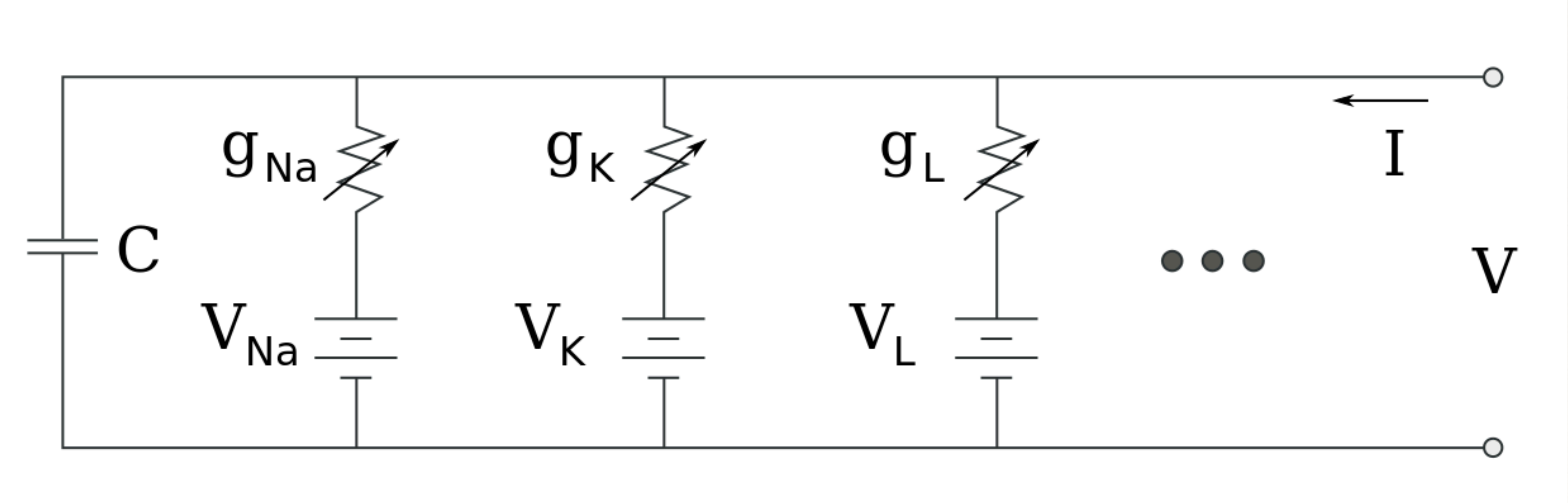


Dendrites
Microtubule
Neurofibrils
Neurotransmitter
Receptor
Synapse
Synaptic vesicles
Synapse (Axoaxonic)
Synaptic cleft
Axonal terminal
Rough ER
(Nissl body)
Polyribosomes
Ribosomes
Golgi apparatus
Synapse
(Axosomatic)
Node of Ranvier
Myelin Sheath
(Schwann cell)
Axon hillock
Nucleus
Nucleolus
Membrane
Microtubule
Mitochondrion
Smooth ER
Nucleus
(Schwann cell)
Microfilament
Microtubule
Axon
Synapse
(Axodendritic)
Dendrites

Hodgkin-Huxley Model

$$I = -(C_m \frac{dV}{dt}) + \frac{Ie}{A}$$

$$I = C_m \frac{dV_m}{dt} + g_K(V_m - V_K) + g_{Na}(V_m - V_{Na}) + g_l(V_m - V_l)$$

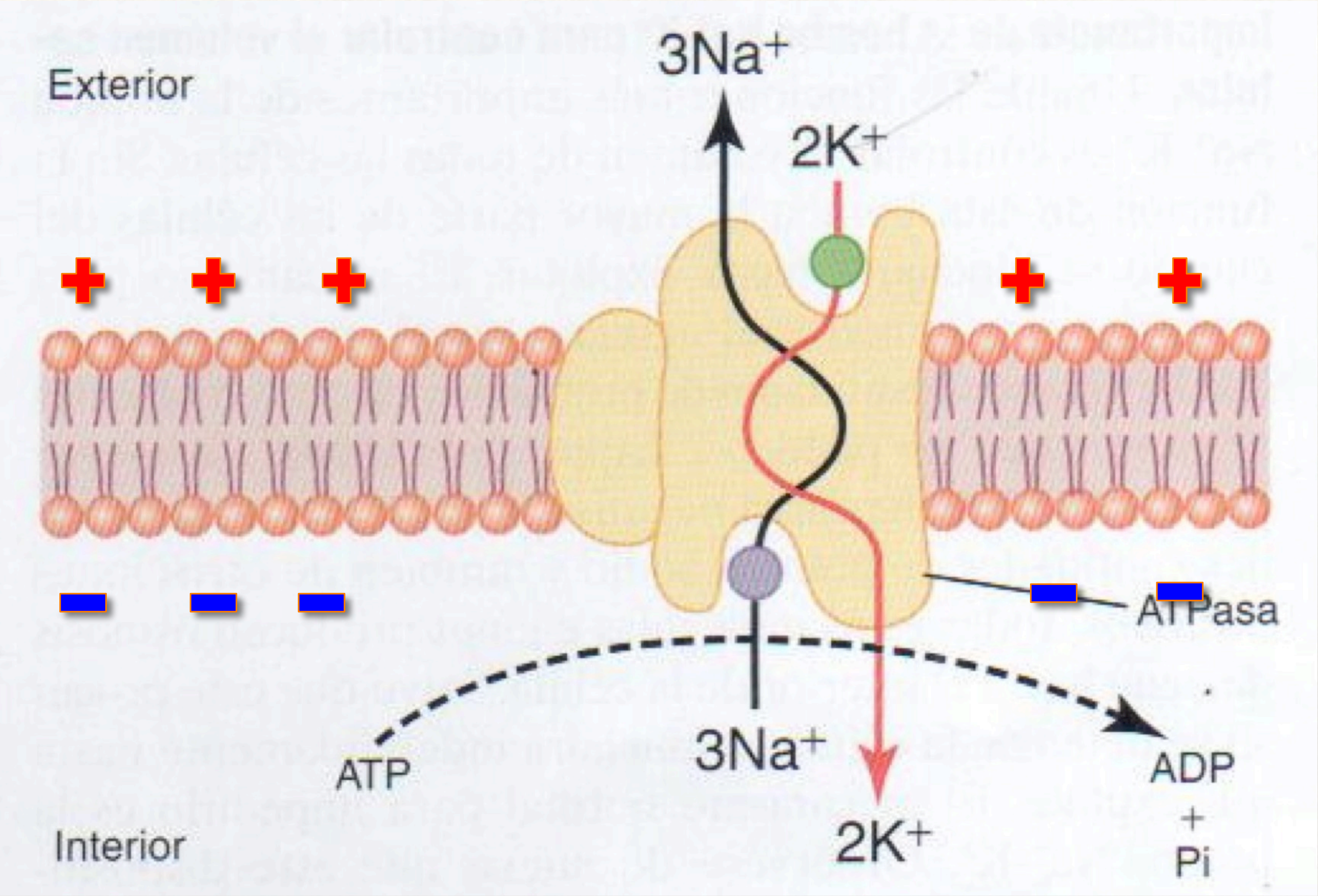


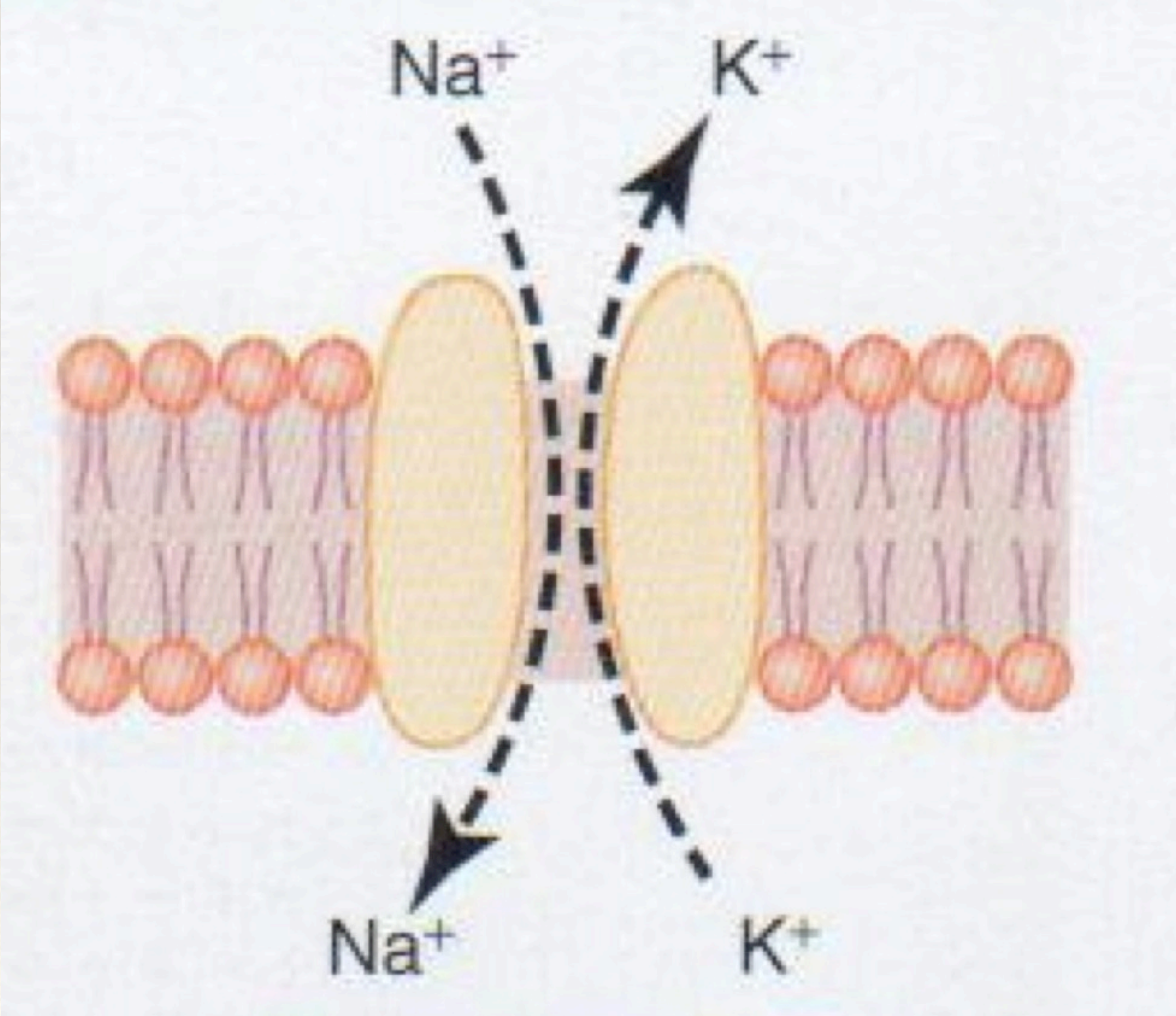
Hodgkin 1952, *Abott* "Theoretical Neuroscience"

Plasmatic Membrane - Na+/K+ Pumps

+Mantiene el volumen de la célula.

+Puede funcionar en reversa.





Canales de «fuga»
K+-Na+

+Canal de K^+ selectivo de cationes mediante afinidad de terminaciones con grupos hidroxilos con el K^+ deshidratado.

+Sensible al voltaje

+Determina la permeabilidad.

Membrane Potential

- El Na^+ por los Canales de fuga **positiviza** el interior al entrar.

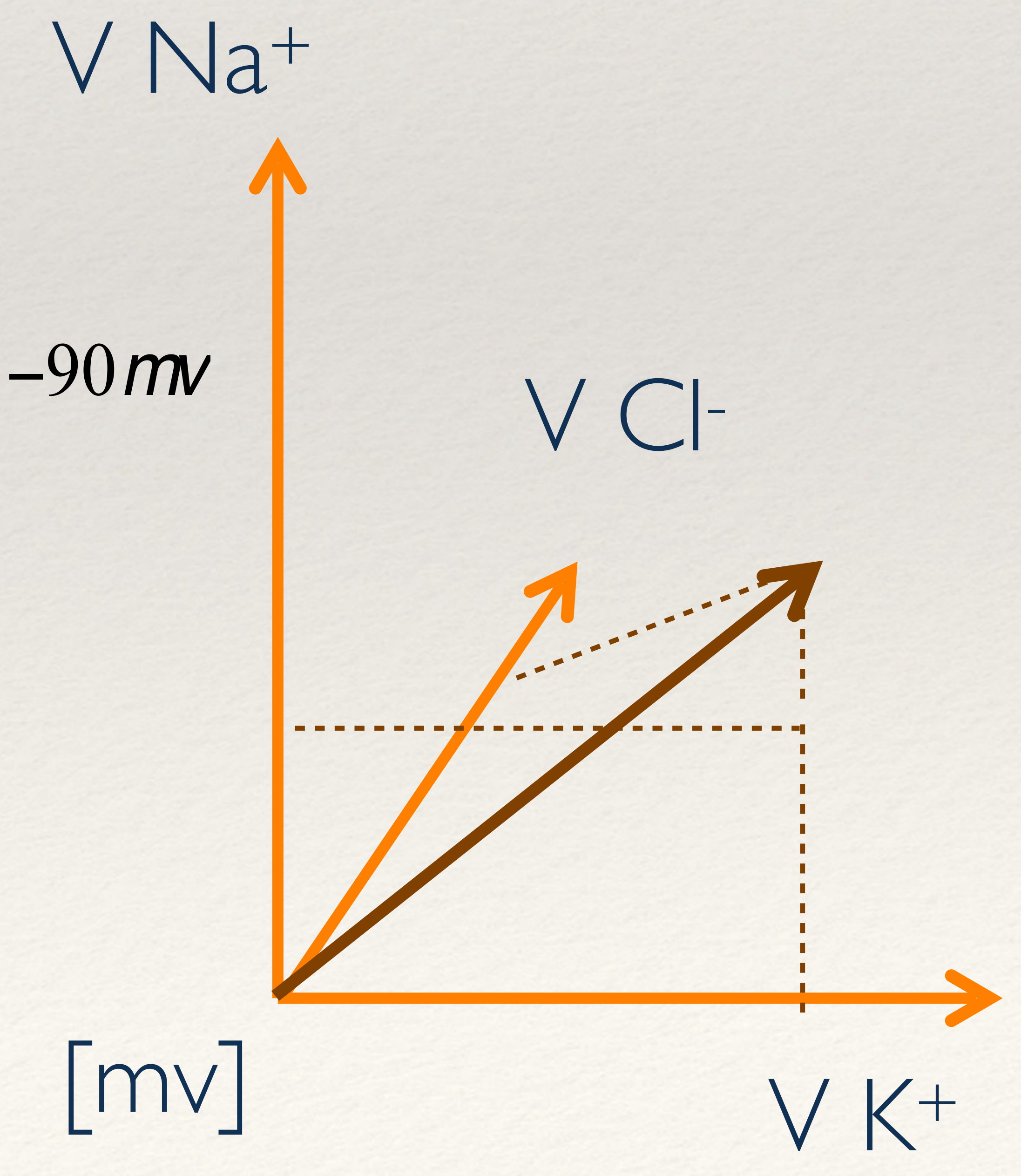
- El K^+ por los Canales de fuga **negativiza** el interior al salir.

- La bomba de Na^+-K^+ saca 3 contra 2 (todos positivos) con lo que **negativiza** el interior.

Ecuación Nerst para cada ion.

$$V = -61 \cdot \log\left(\frac{CNa_i^+ PNa^+ + CK_i^+ PK^+ + CCl_o^- PCl^-}{CNa_o^+ PNa^+ + CK_o^+ PK^+ + CCl_i^- PCl^-}\right)$$

$$+61\,mV - 94\,mV = -86\,mV - 4\,mv = -90\,mv$$



Membrane Potential

Equilibrio para cada ion

$$Na^{+} \quad 62 \ \log \frac{145}{5} = 90mV$$

$$62 \ \log \frac{145}{15} = 61mV$$

$$K^{+} \quad 62 \ \log \frac{5}{140} = -90mV$$

$$Cl^{-} \ -62 \ \log \frac{110}{4} = -89mV$$

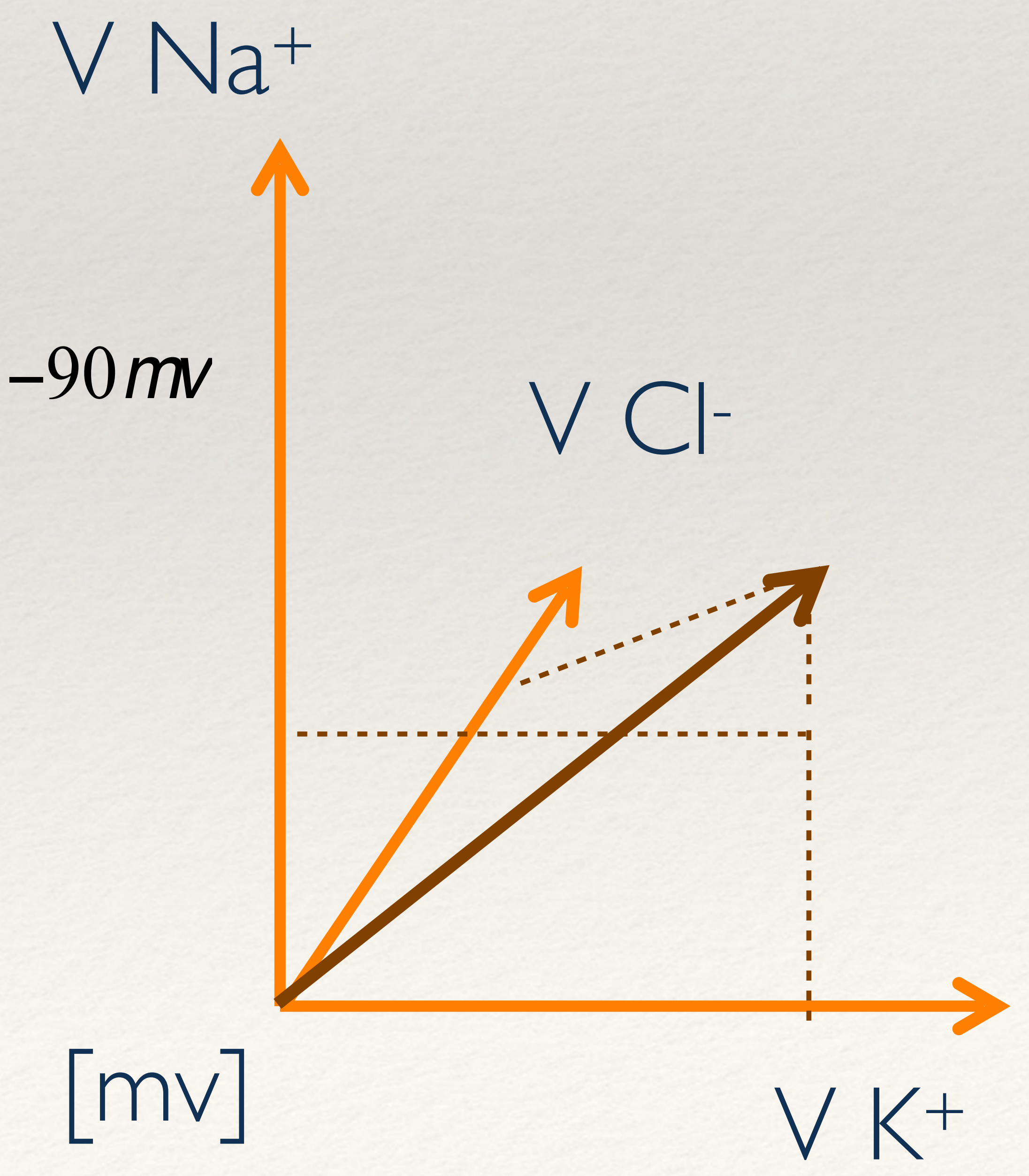
$$Ca^{2+} \quad 31 \ \log \frac{2.5}{10^{-4}} = 136mV$$

$$31 \ \log \frac{5}{10^{-4}} = 146mV$$

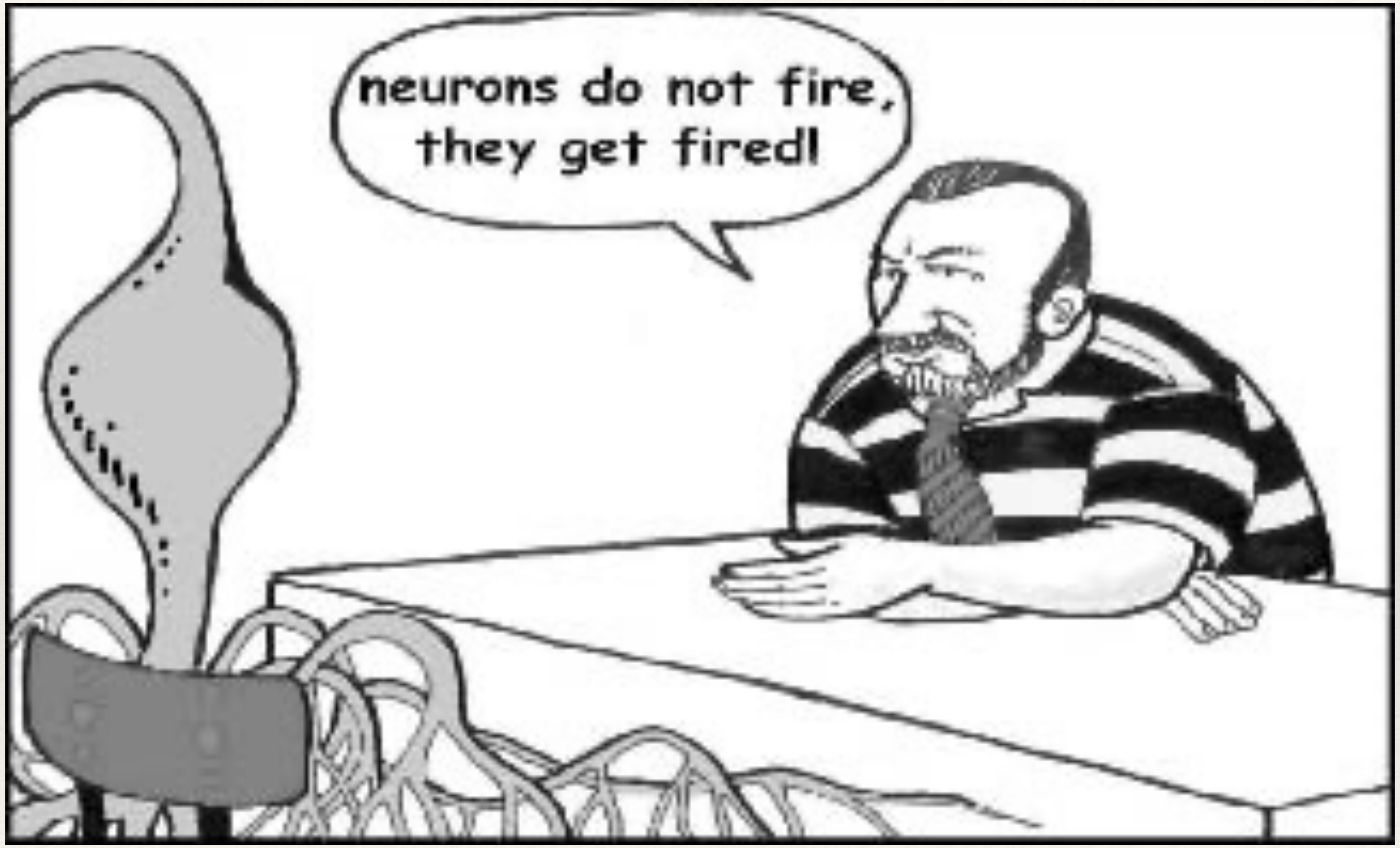
Ecuación Nerst para cada ion.

$$V = -61 \cdot \log\left(\frac{CNa_i^{+} PNa^{+} + CK_i^{+} PK^{+} + CCl_o^{-} PCl^{-}}{CNa_o^{+} PNa^{+} + CK_o^{+} PK^{+} + CCl_i^{-} PCl^{-}} \right)$$

$$+61mV - 94mV = -86mV - 4mv = -90mv$$

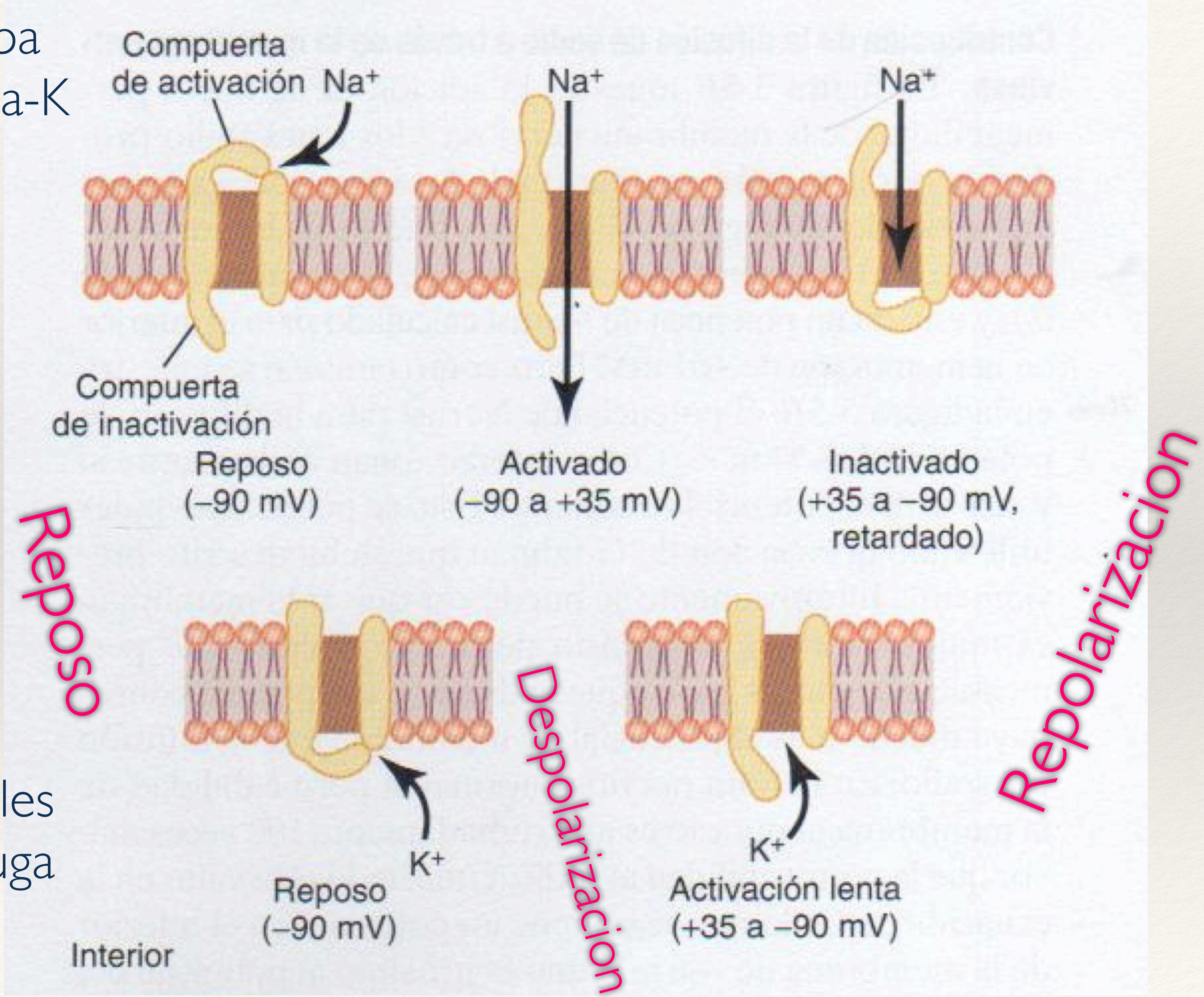


Action Potential

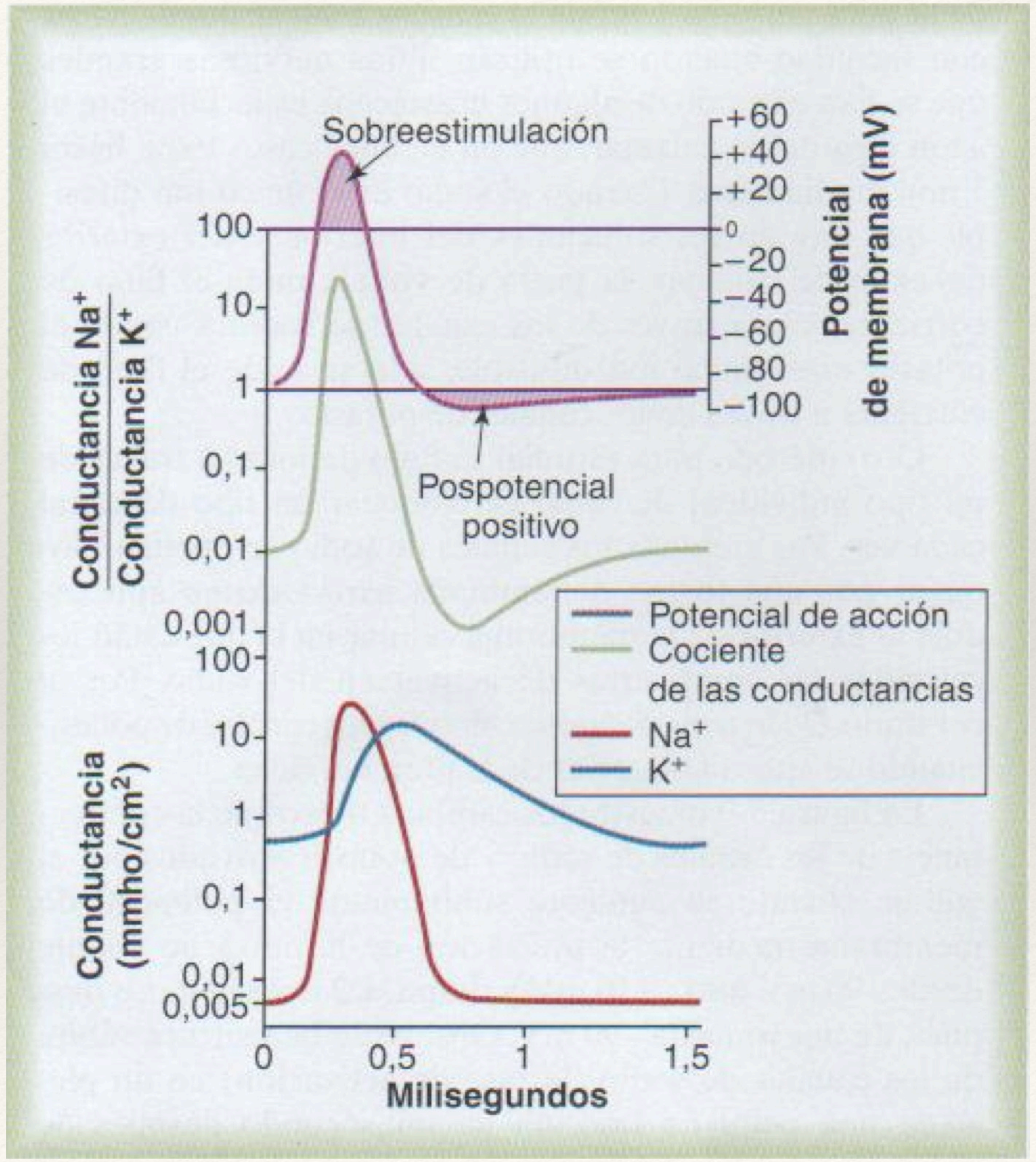


Action Potential

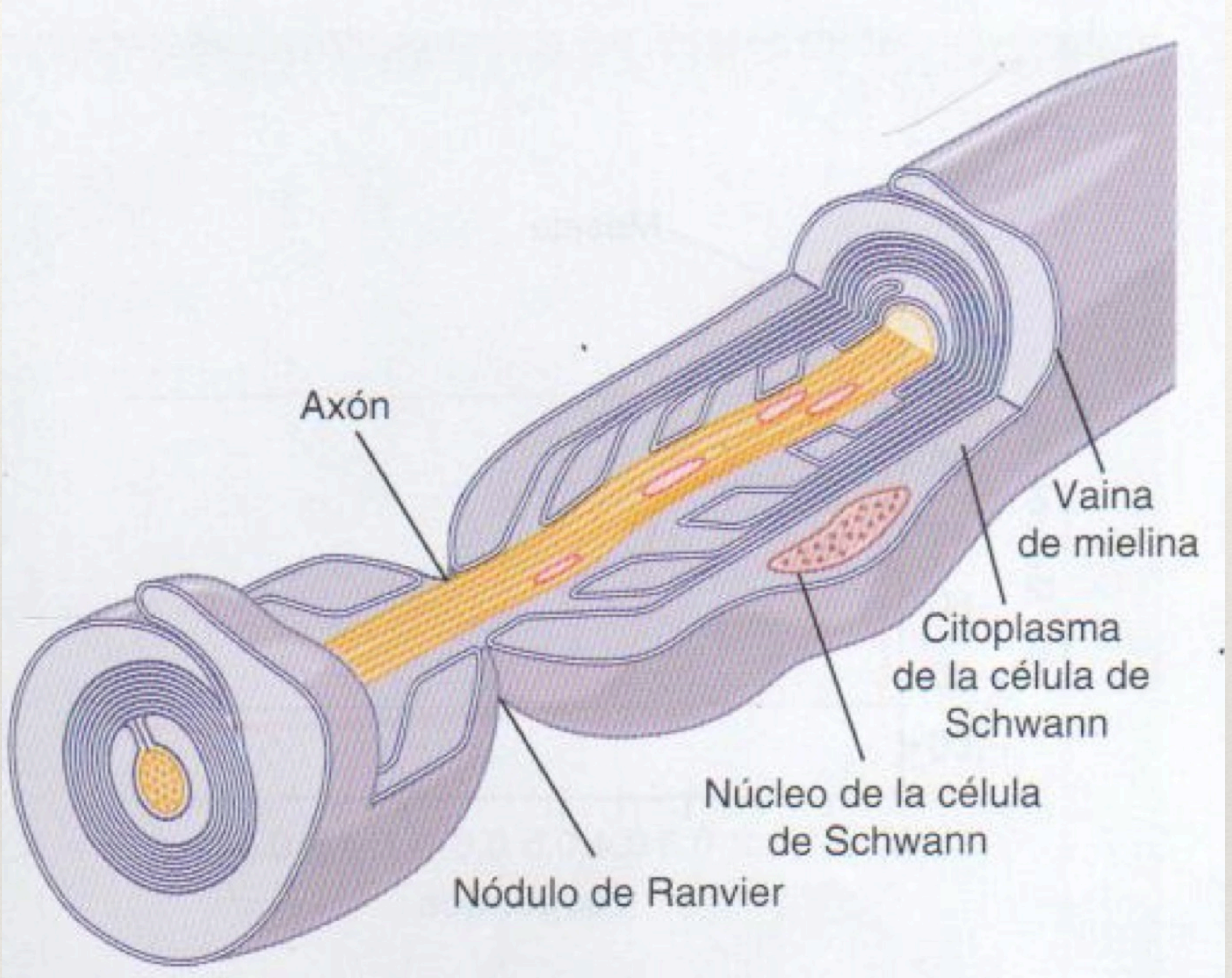
Bomba de Na-K



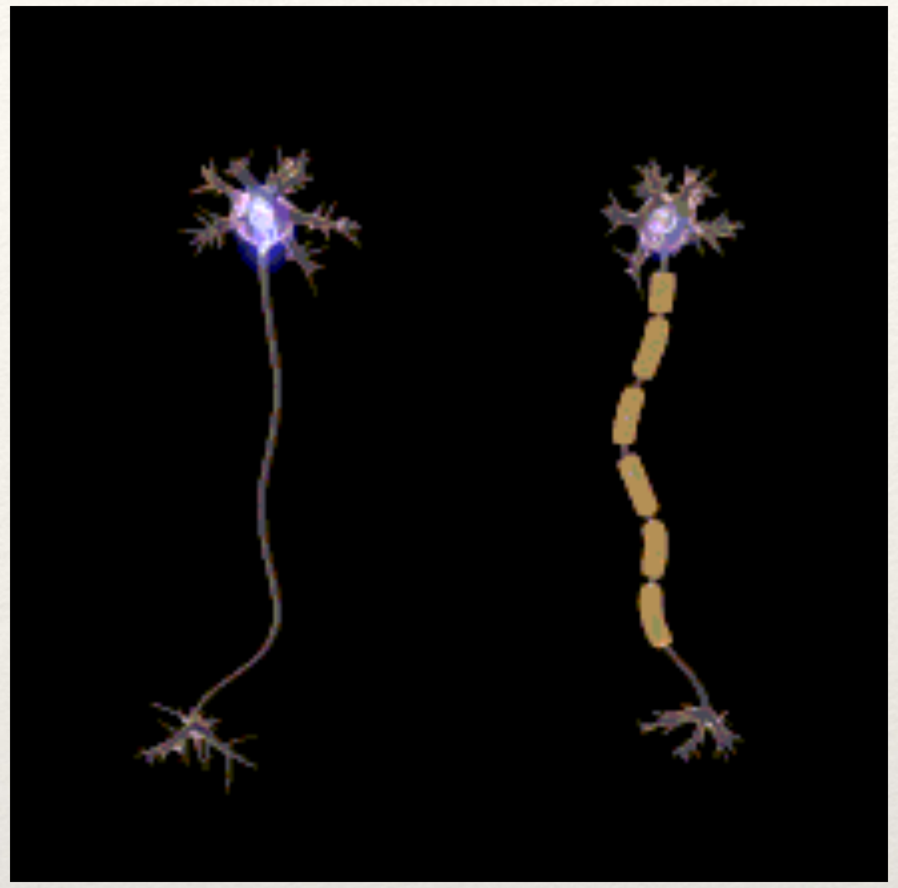
Canales de Fuga

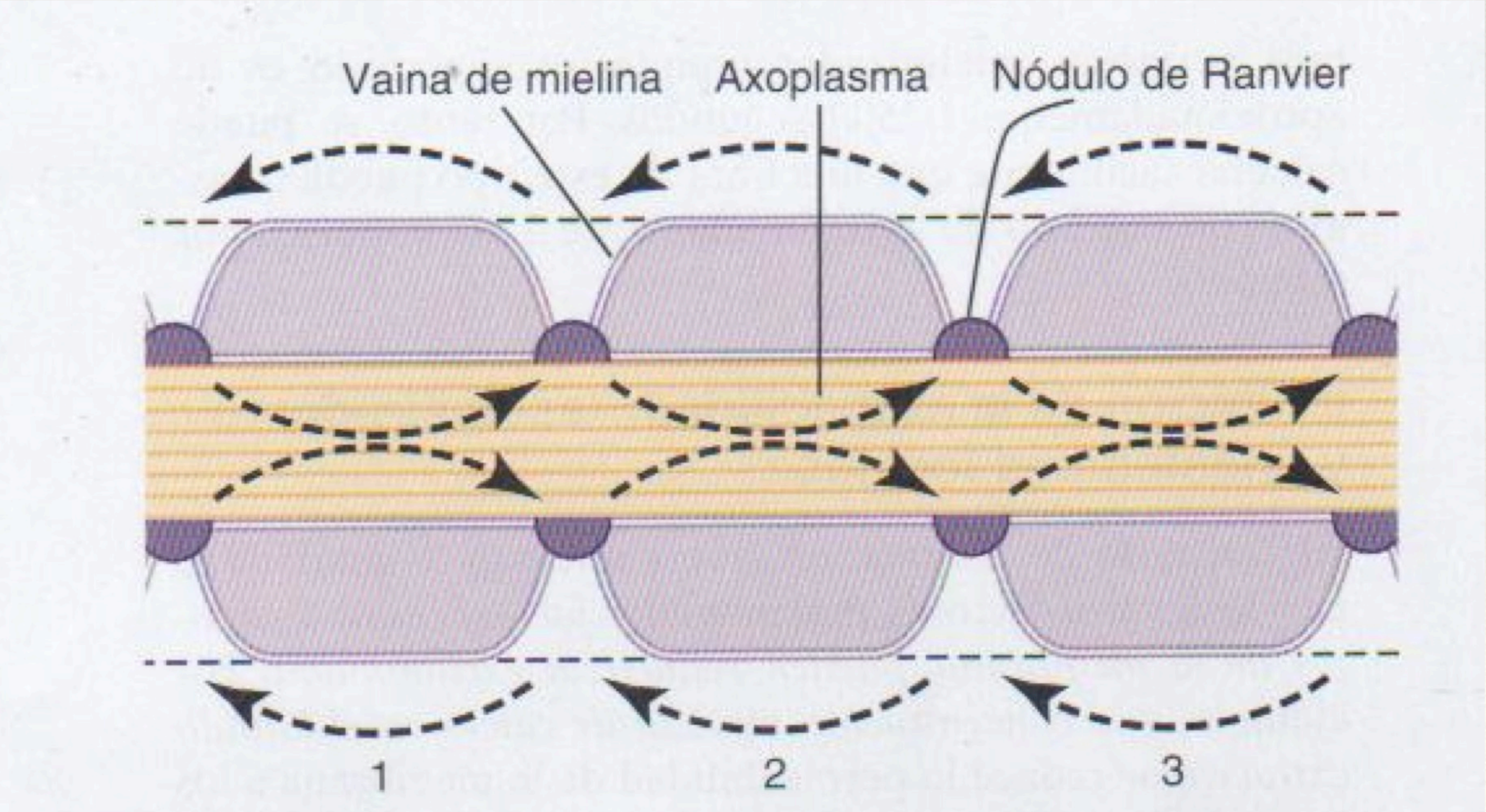


Saltatory Conduction



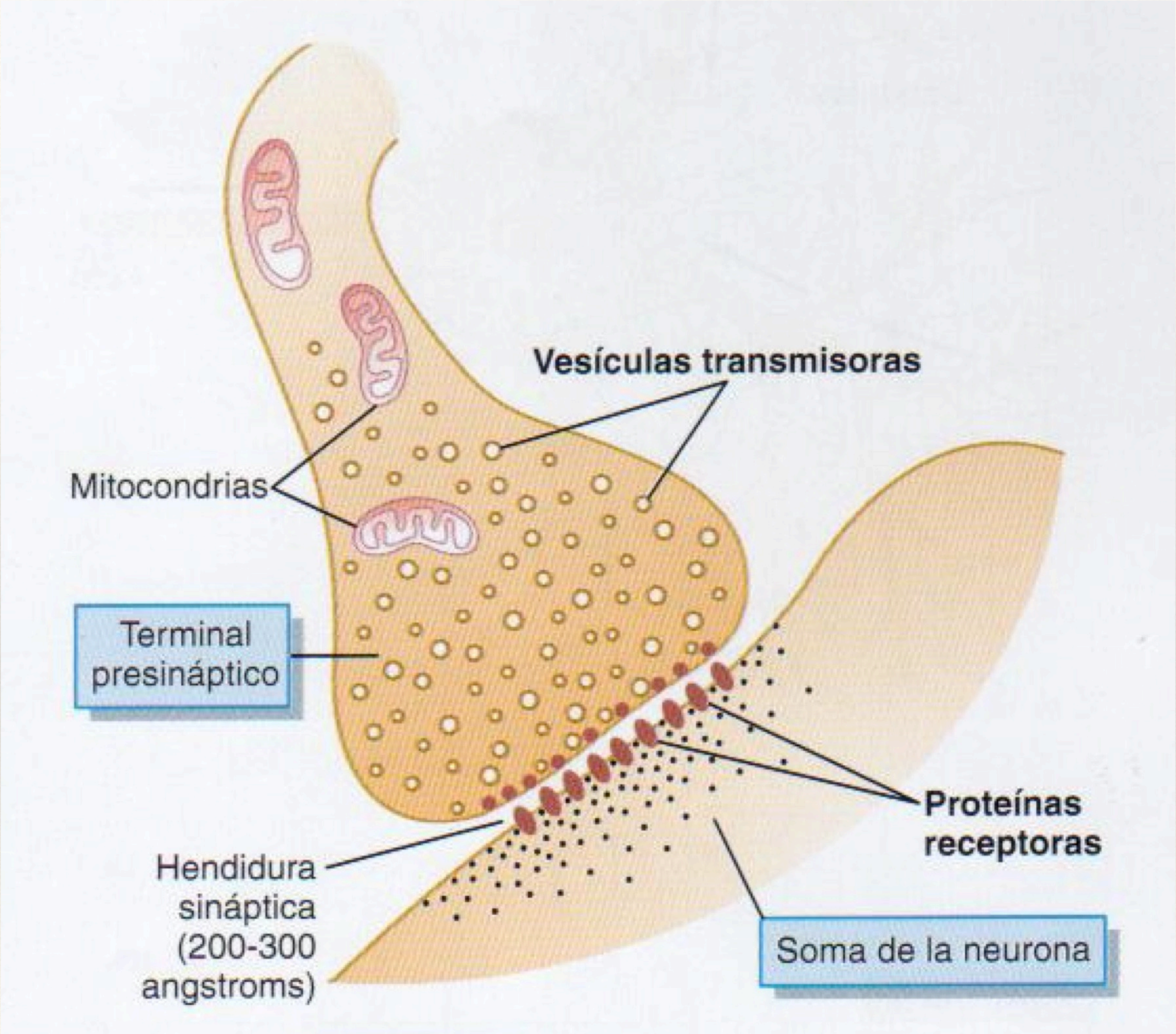
- Vainas de mielina (oligodendrocitos, células de Schwann)
- Hasta 100 m/s (sin mielina 0.25 m/s)





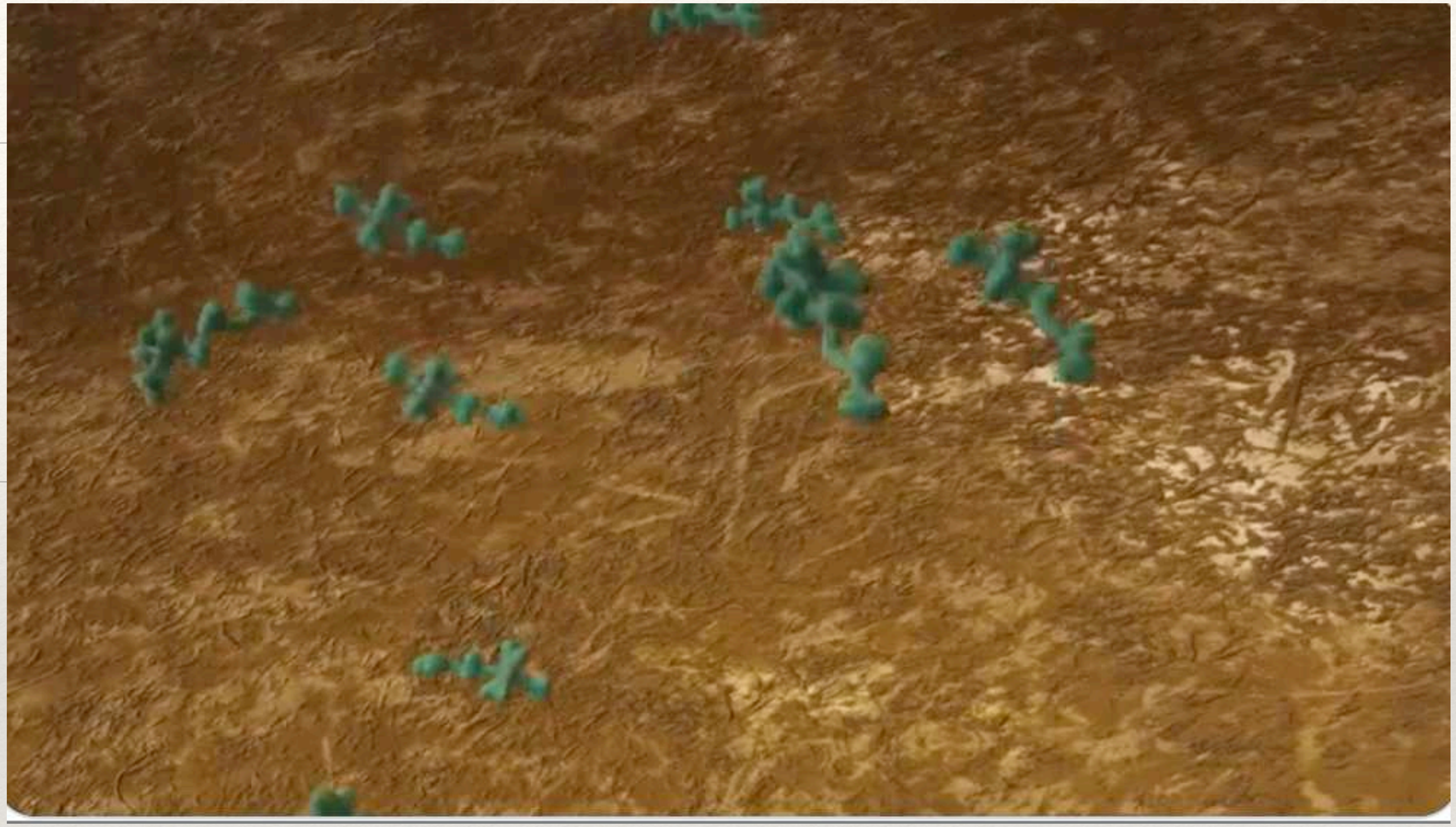
Synapse

- Canales por ligando (40 neurotransmisores)
- Iónico:
  - Excitatorio (Na^+)
  - Inhibitorio(Cl^-, K^+)
- Activación de Segundo Mensajero:
  - Proteinas G cAMP, cGMP
  - Ccambios enzimáticos, transcripción, activación de canales K^+
- One-Way



Synaptic Weights

Synapse



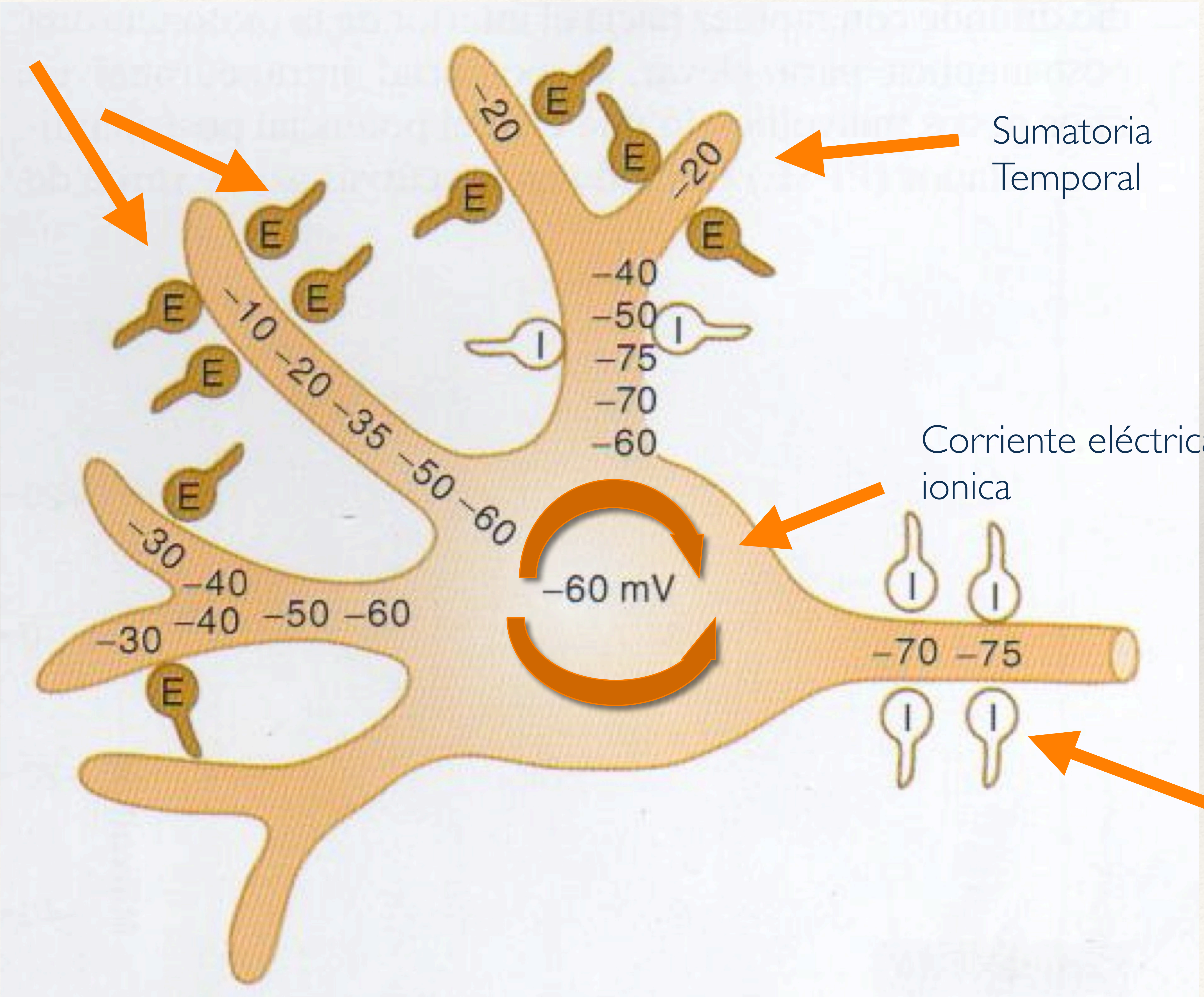
https://twitter.com/i/status/1502259198603808774

Synaptic Weights

Synapse

Sumation and Propagation

Sumatoria
Espacial



Sumatoria
Temporal

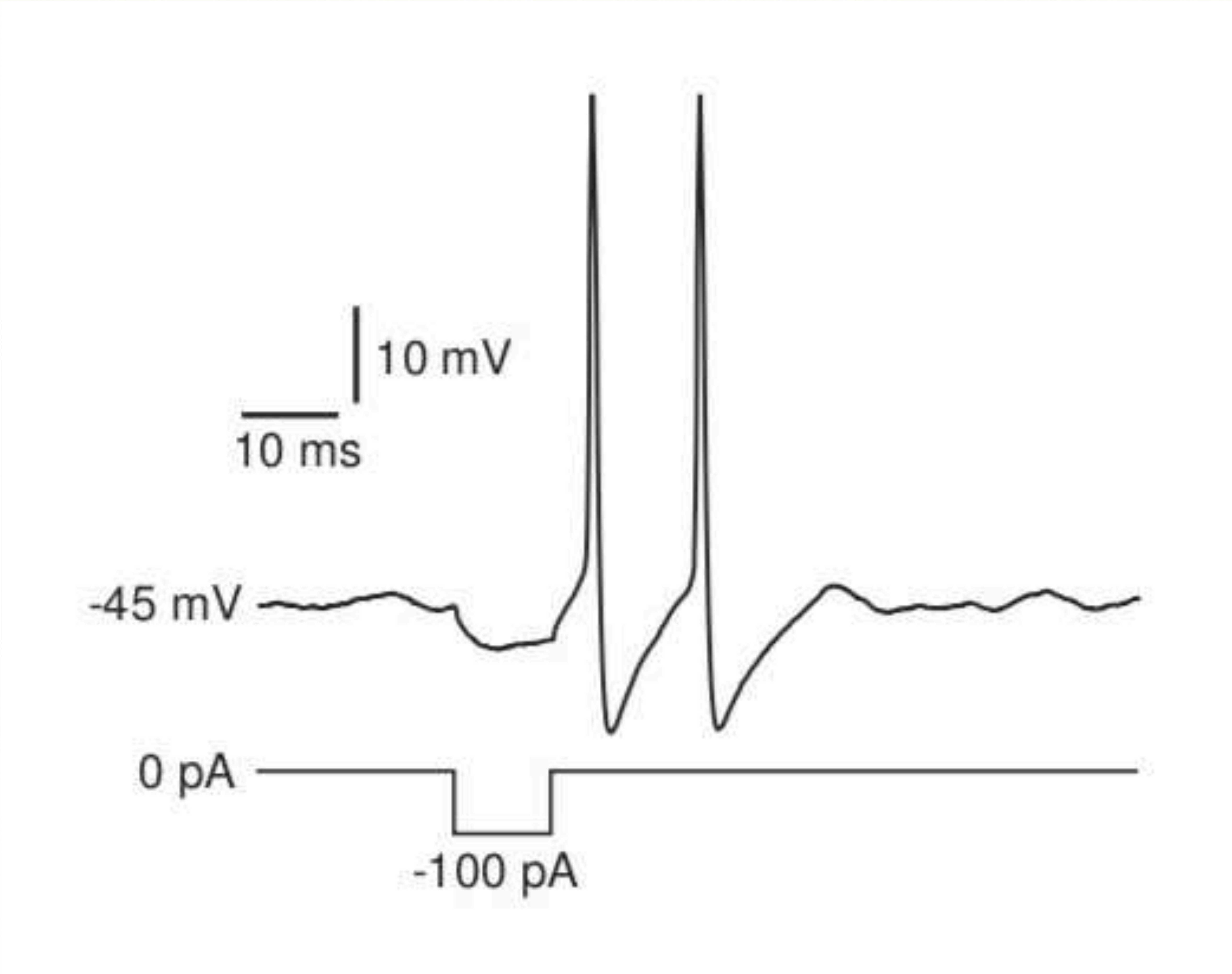
Corriente eléctrica
ionica

Segmento inicial del axon
(mayor presencia de
canales)

(+) Gap Junction
(+) Ephatic Coupling
(+) Mechanical

Neural Networks

Sumation and Propagation



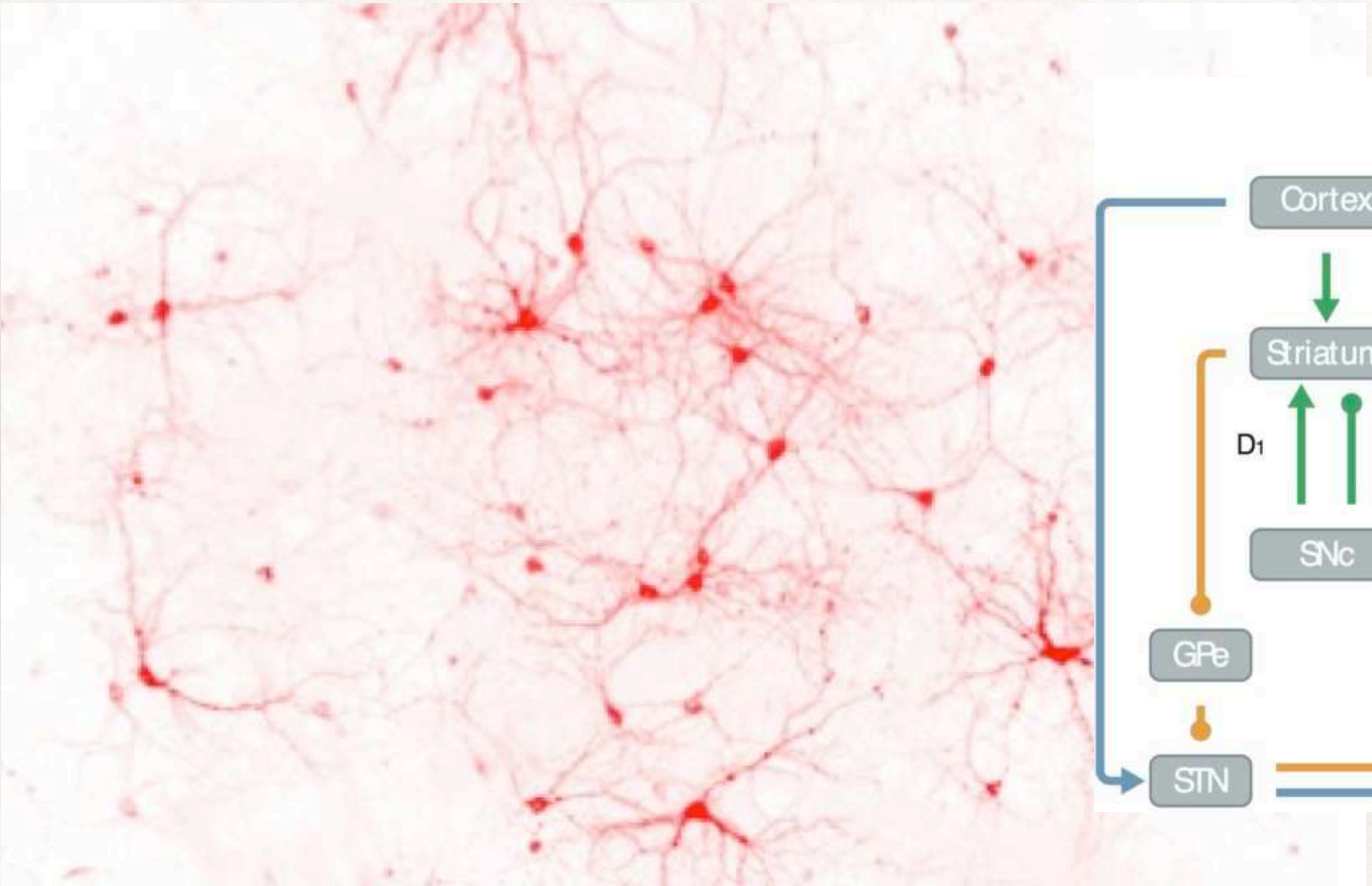
Hyperpolarized pulse of current

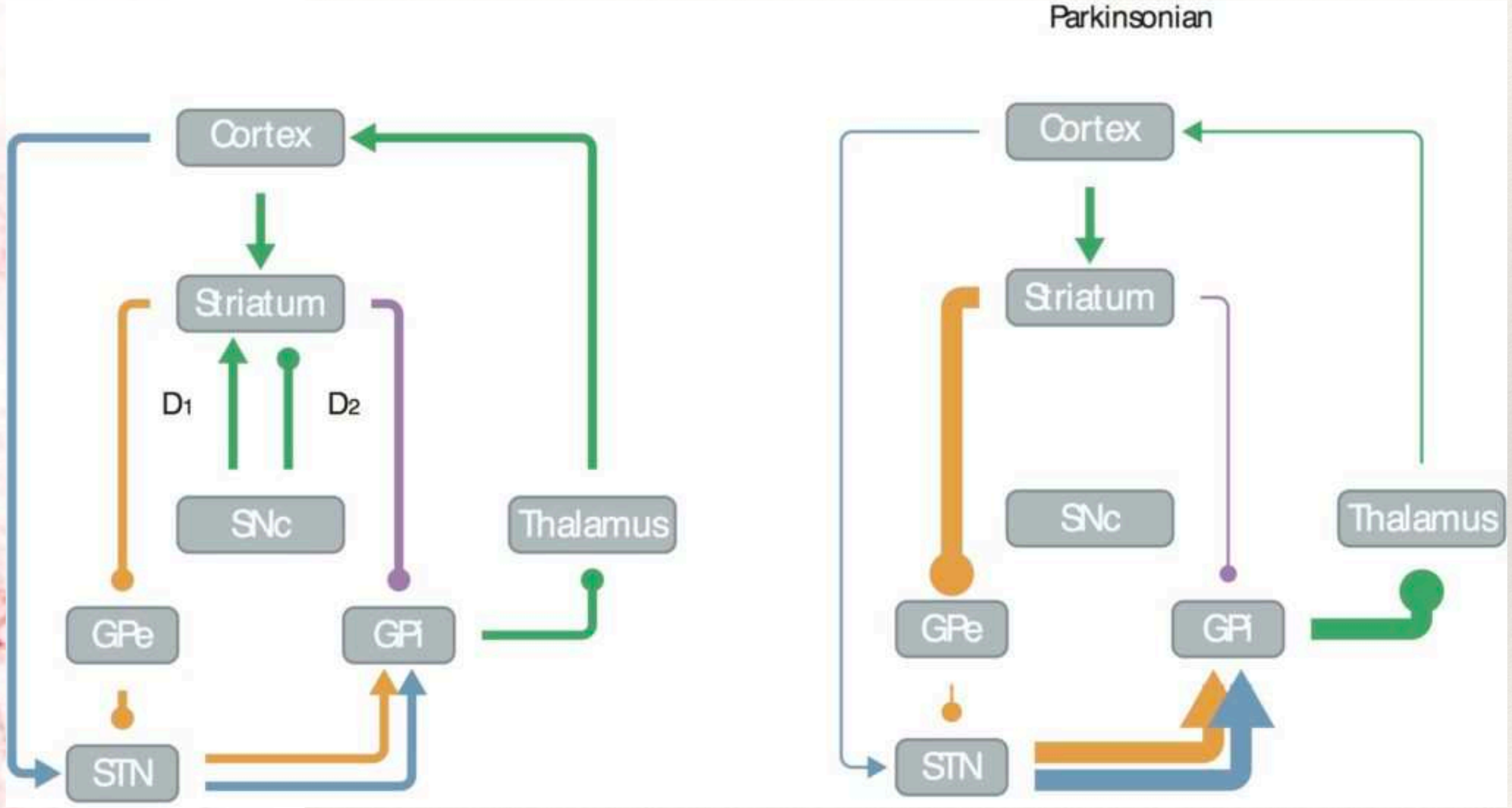
Refractory Period

Rebound Spike

Temporal Summation

Cortical Networks

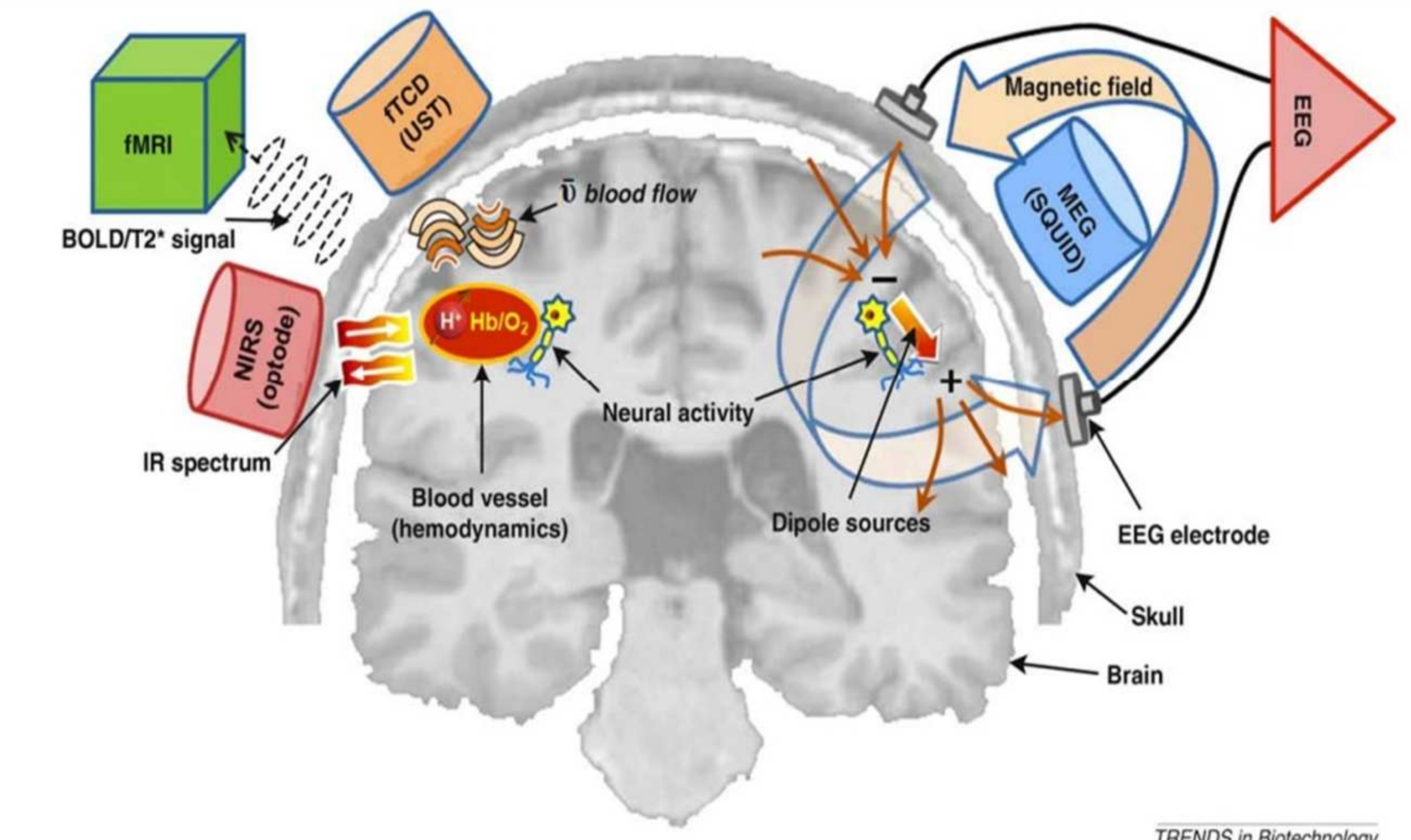




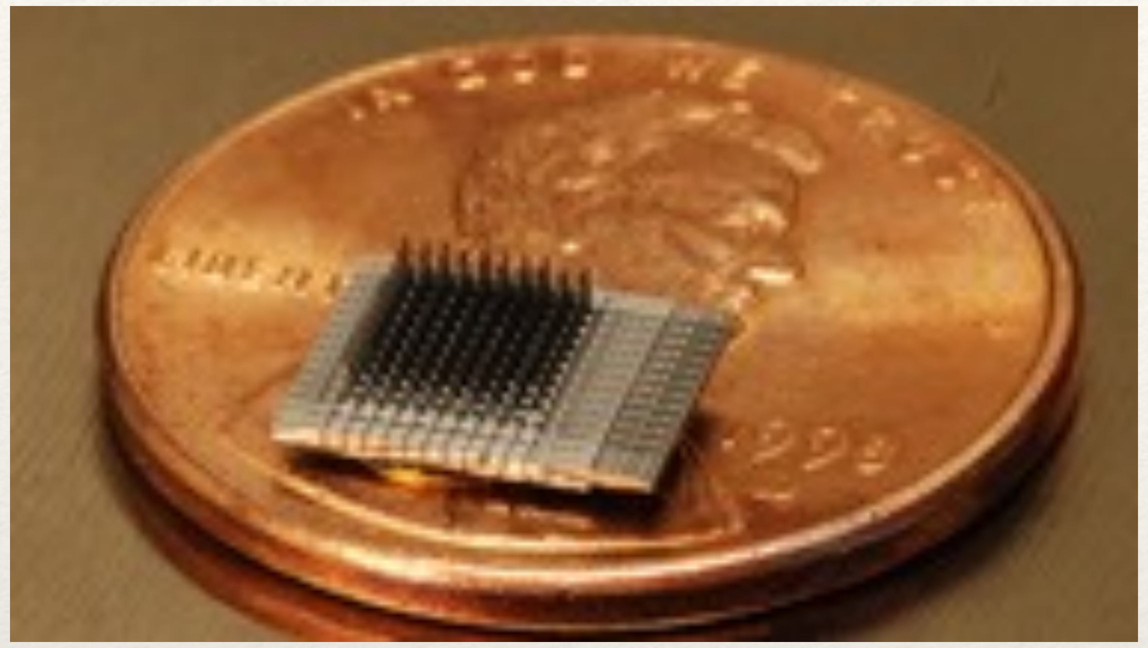
Barral J & Reyes AD (2016)

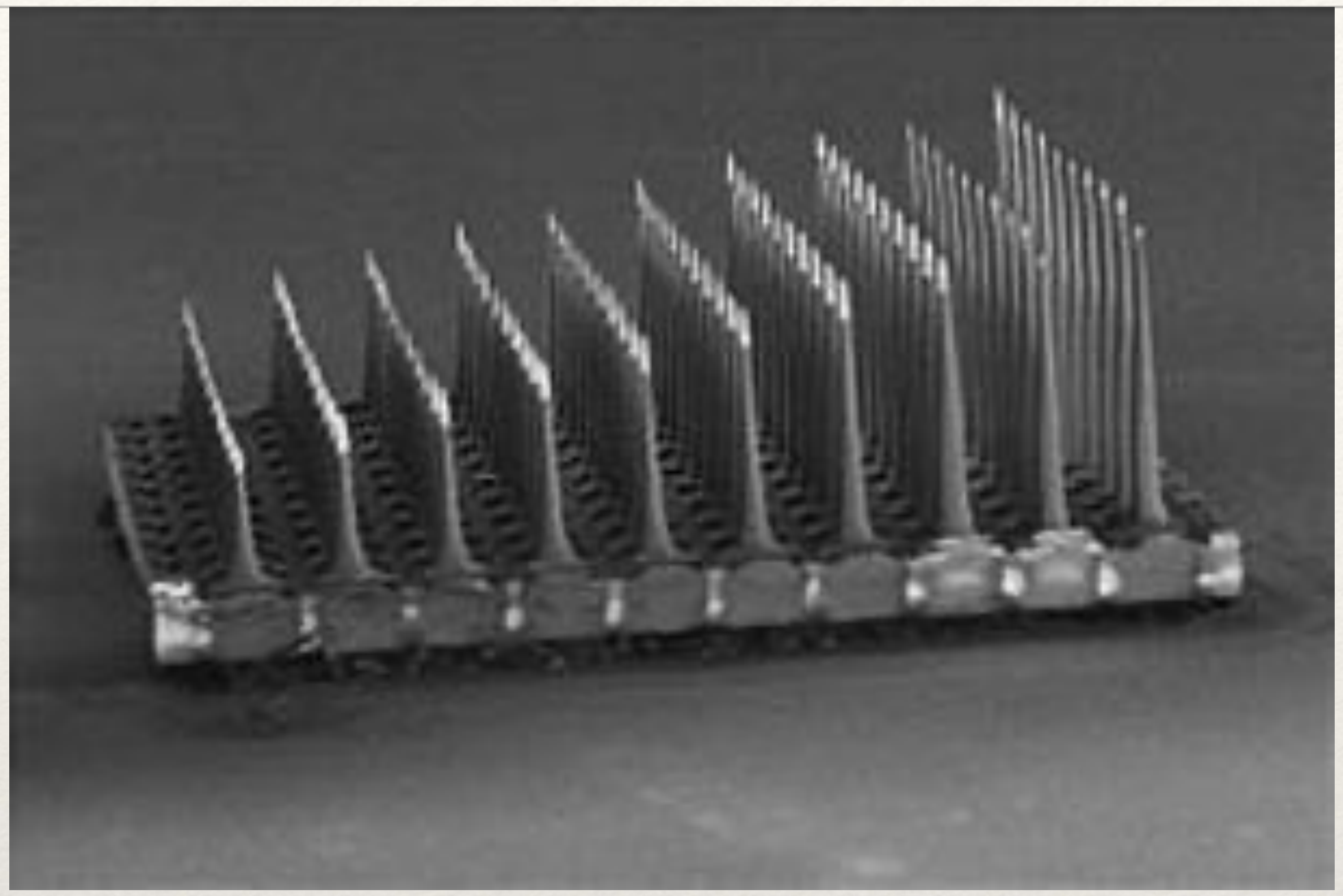
Brain Imaging

Medical Imaging



Invasive Scanning





Utah Grid Microarray

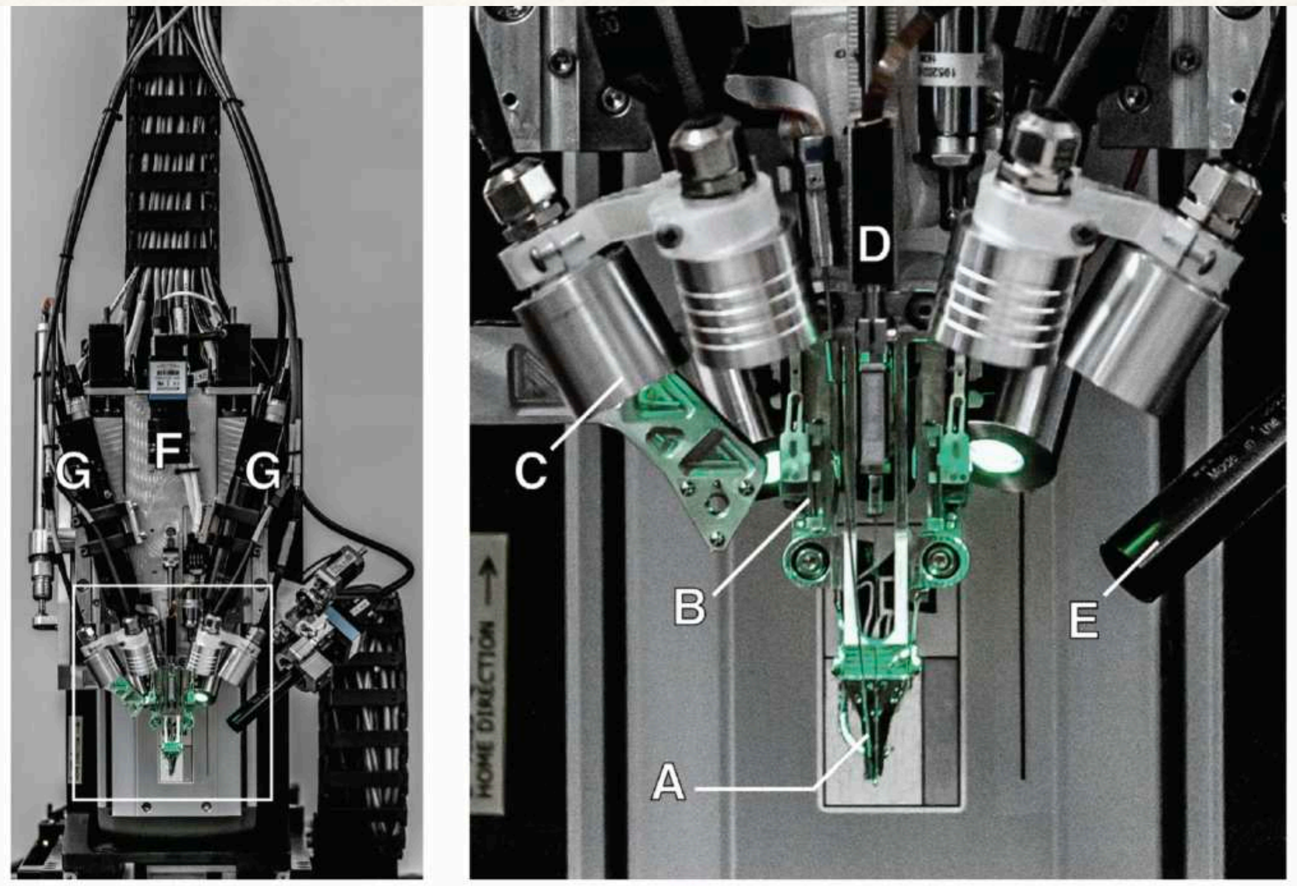
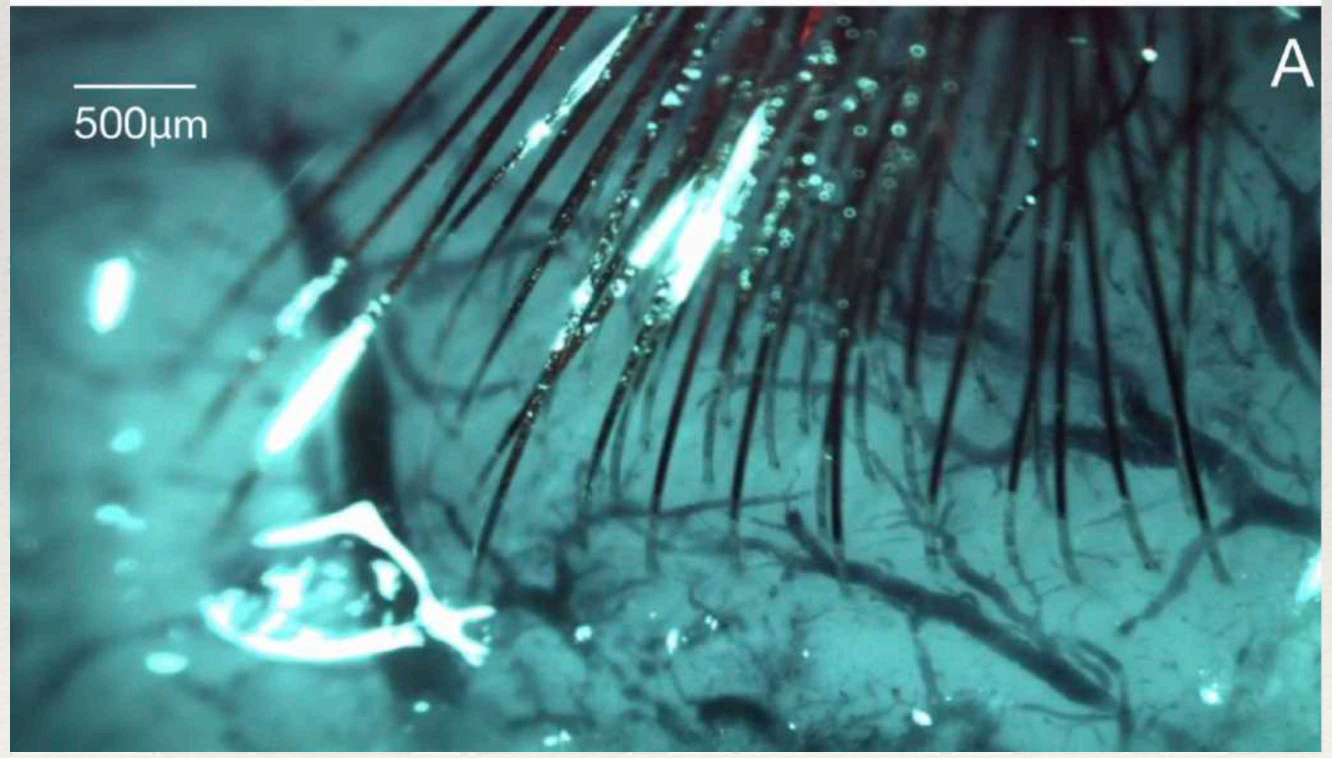
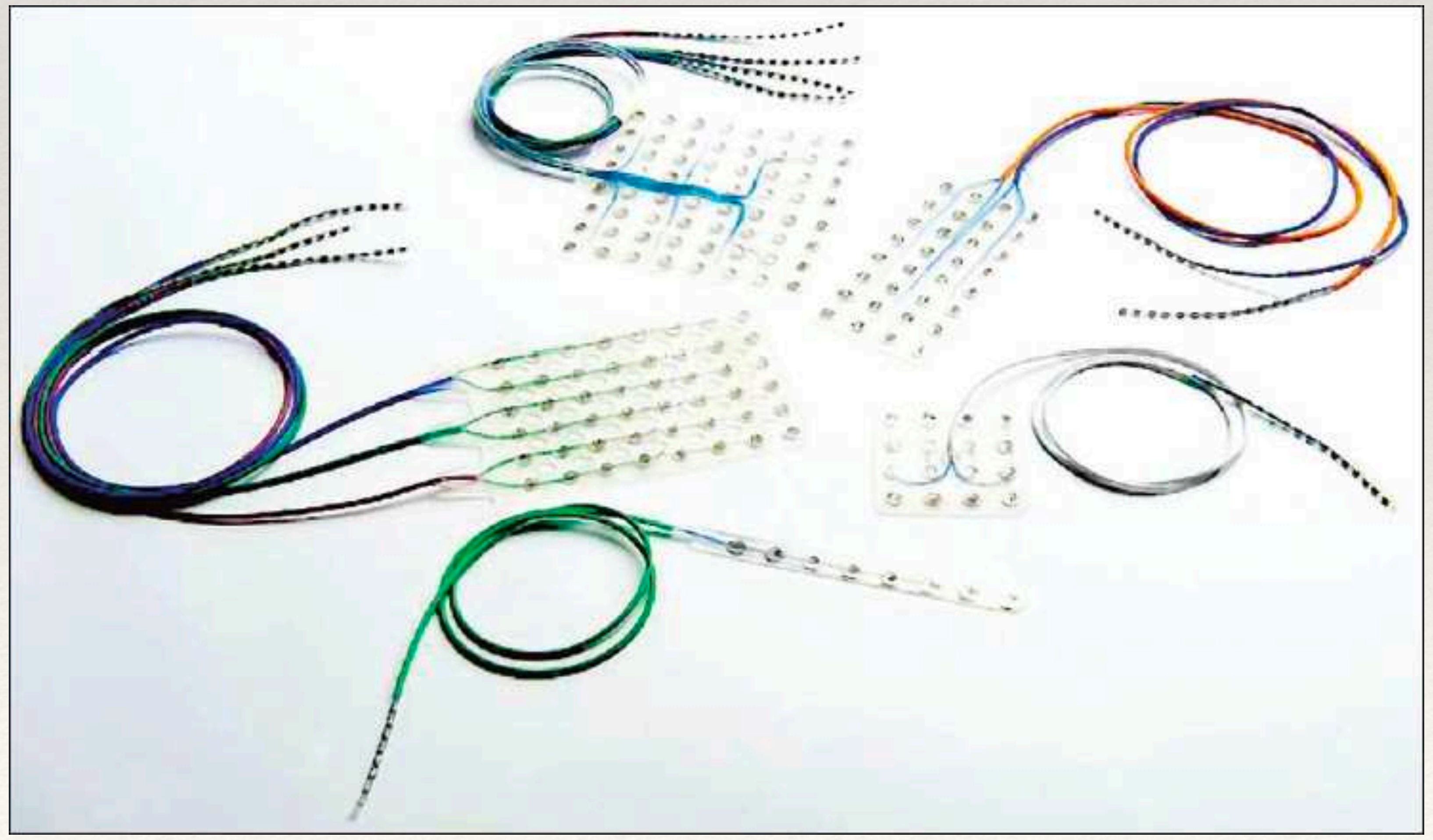


Figure 3: The robotic electrode inserter; enlarged view of the inserter-head shown in the inset. **A.** Loaded needle pincher cartridge. **B.** Low-force contact brain position sensor. **C.** Light modules with multiple independent wavelengths. **D.** Needle motor. **E.** One of four cameras focused on the needle during insertion. **F.** Camera with wide angle view of surgical field. **G.** Stereoscopic cameras.



Deep Brain Stimulation



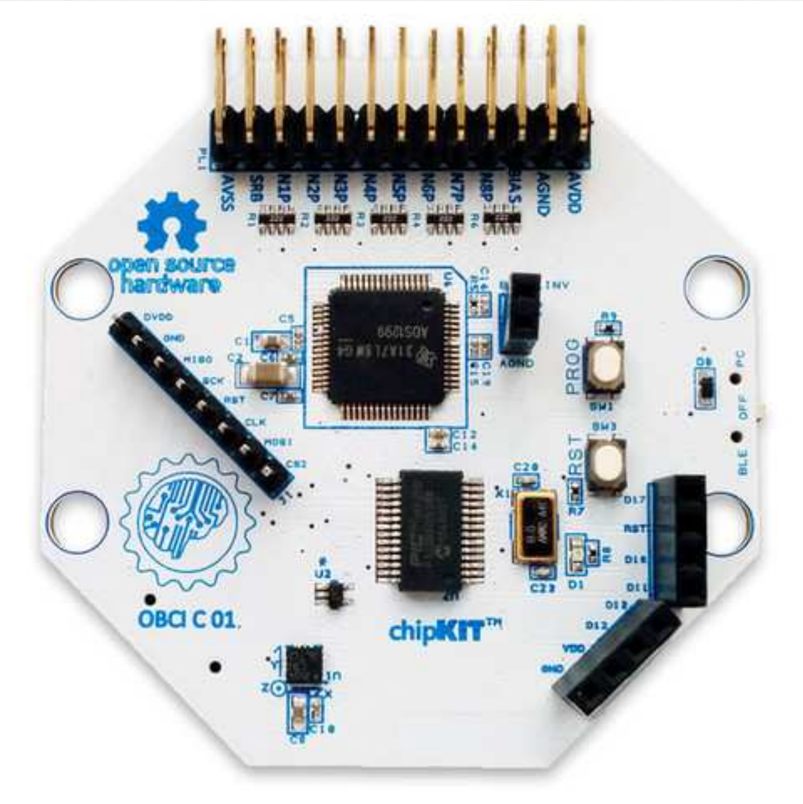
PMT Corporation, Chanhassen, MN, USA

Elon Musks’ Neuralink

Non-invasive Scanning

- EEG *Electroencefalográma*
- ECoG *Electrocorticography*
- MEG *Magnetoencefalografía*
- fMRI *Resonancia Magnética Funcional*
- PET *Tomografía por Emisión de Positrones*
- fTCD *Functional TransCranial Doppler*
- fNIR *functional Near InfraRed*

Non-invasive Scanning - EEG



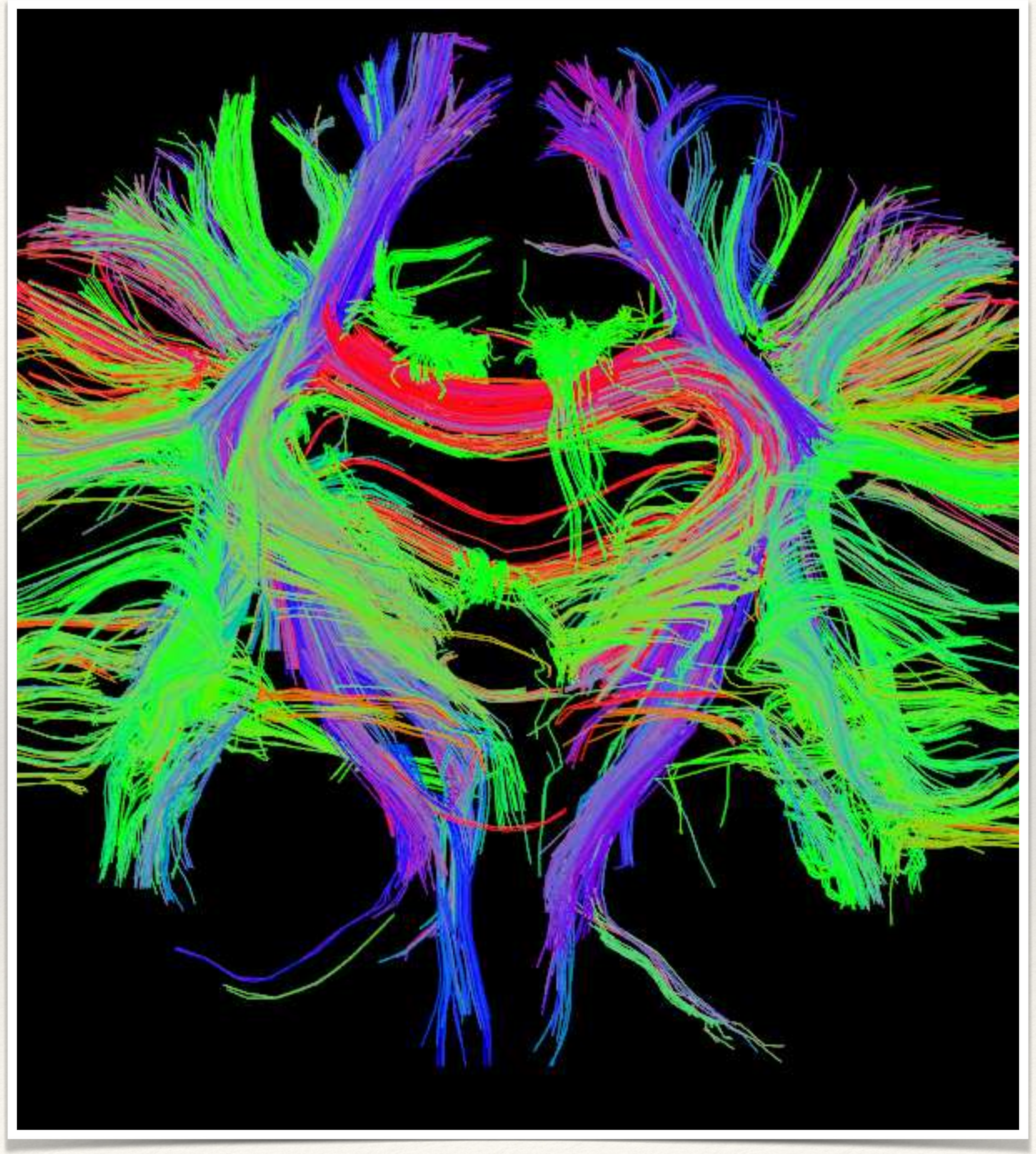
Non-invasive Scanning - fNIR MEG

University of Nottingham, 2018

Drexel University, 2016

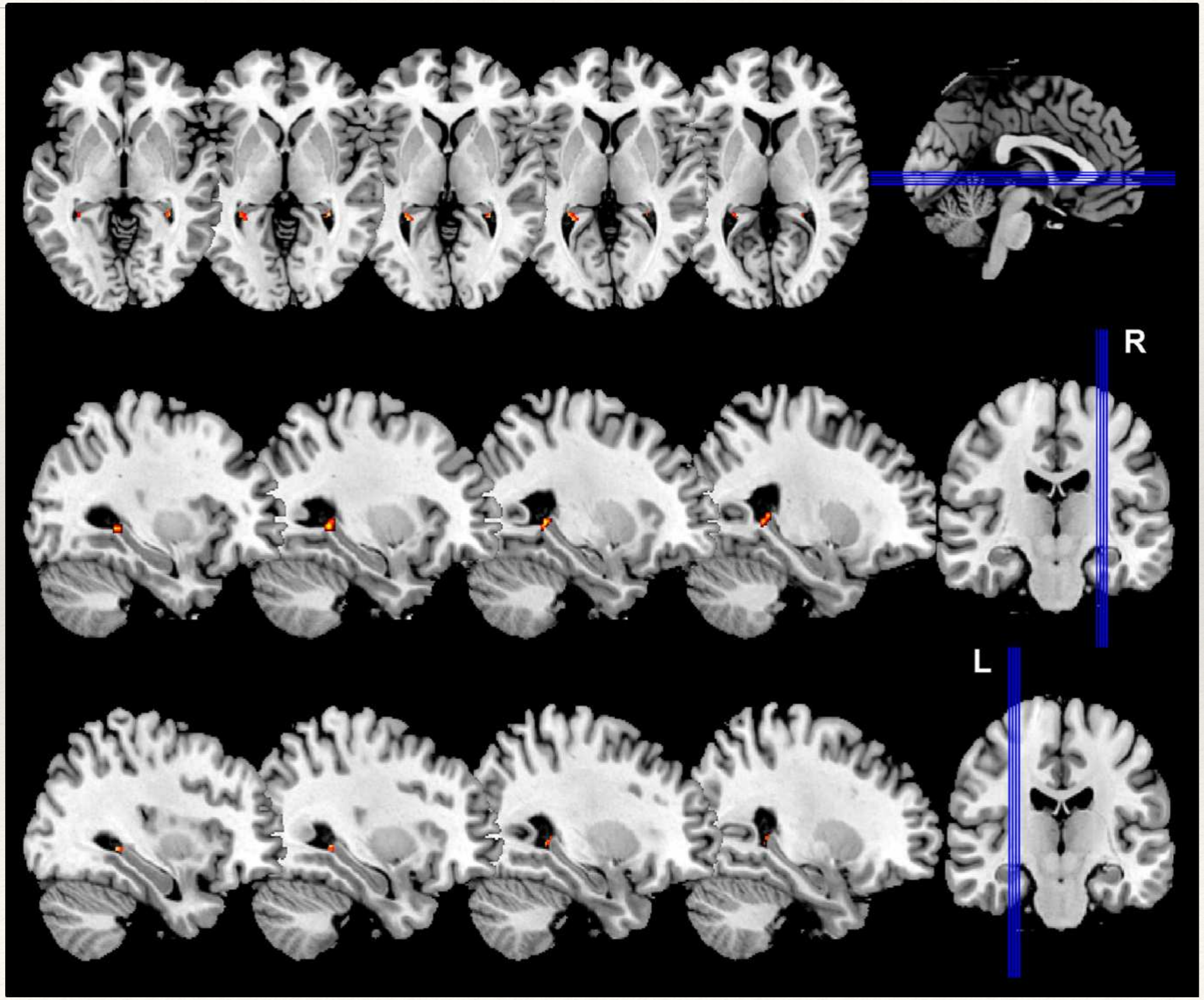
Diffusion Tensor Trachtography

The tools for the Connectome



Brain GPS

Moser, 2008

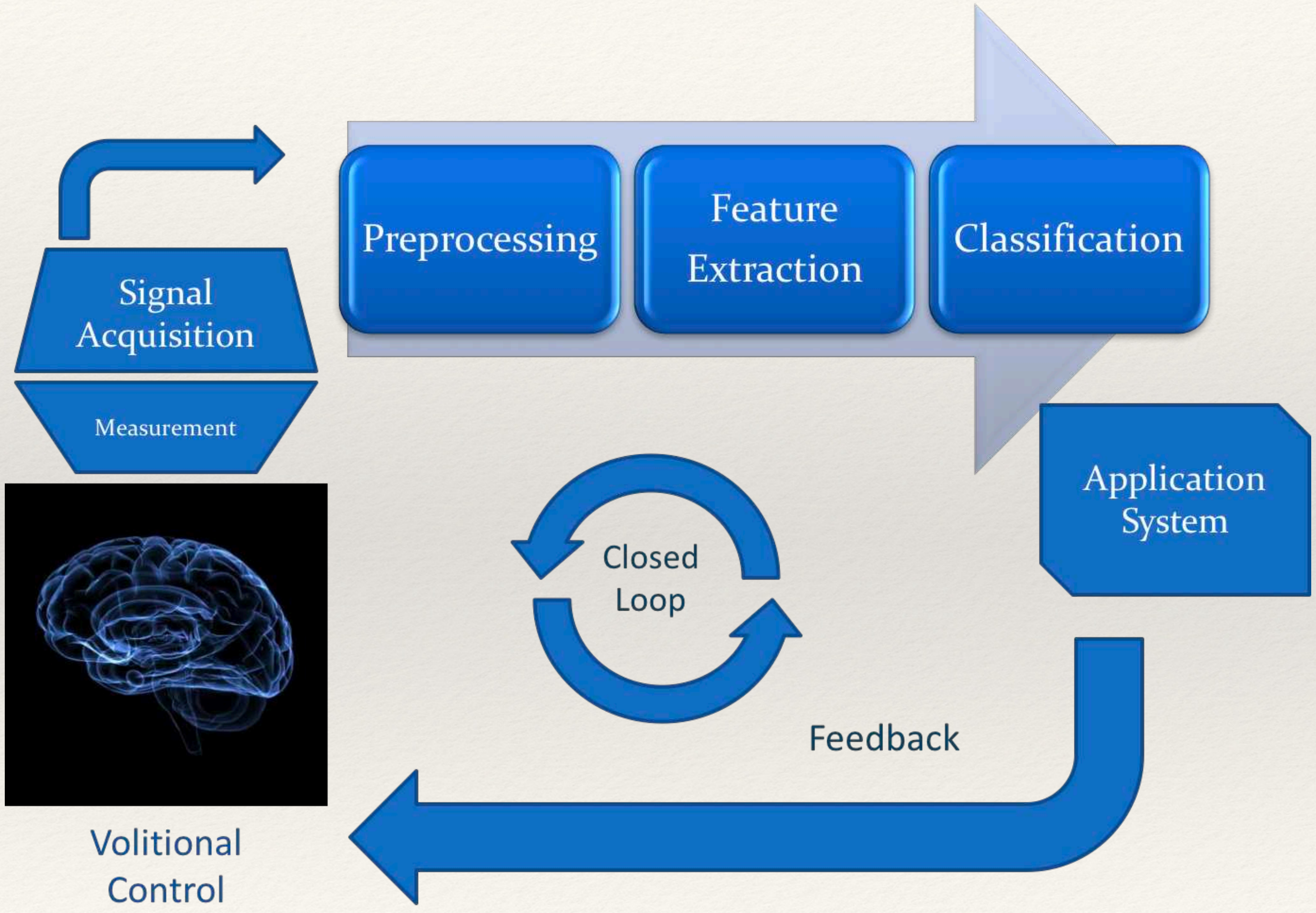


The materialization of Statistical Mathematics

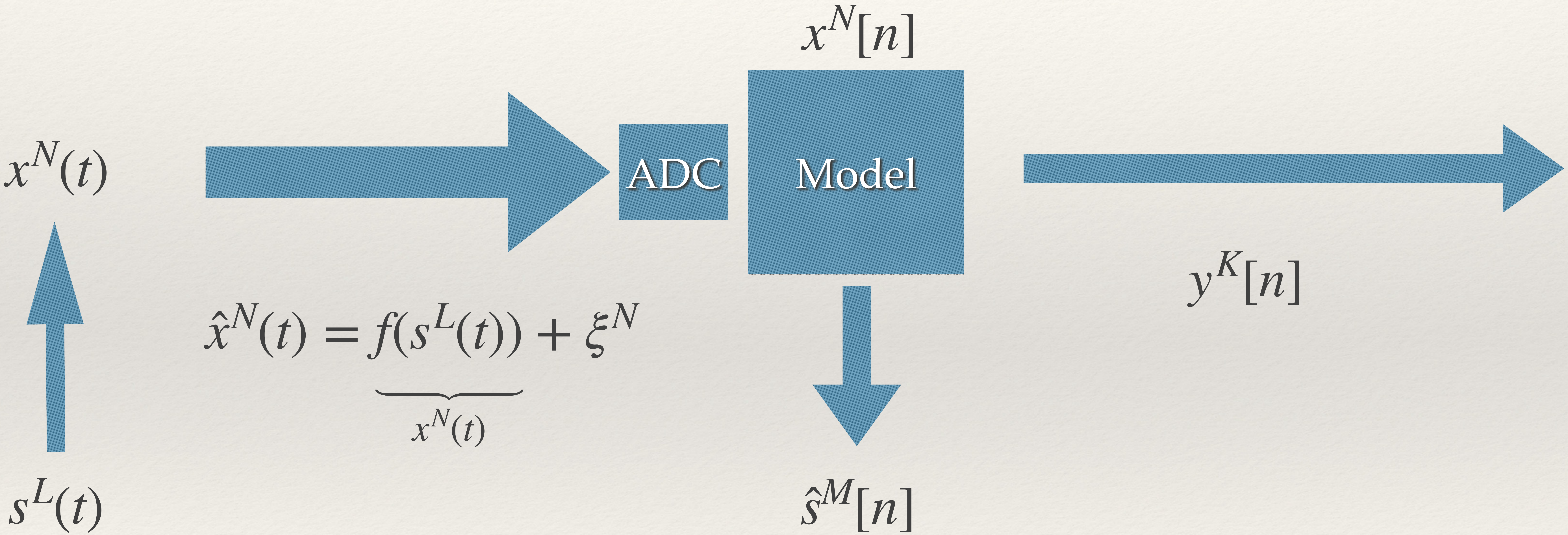
Machine Learning

"Learning" computers

Machine Learning Model



Engineering Inverse Problem



$$\hat{x}^N(t) = \underbrace{f(s^L(t))}_{x^N(t)} + \xi^N$$

What is “learning”?

Model Free Approach
To determine the mapping between some input X and some output Y by “adjusting” free parameters.

These free parameters are adjusted based on an Optimization Process

What is “learning”?

Optimization Problems with Constraints

(1) A company wants to manufacture cylindrical aluminum cans with a volume of $1000cm^3$ (1 liter). What should the radius and height of the can be to minimize the amount of aluminum used?

Volume of a Cylinder: $V = \pi r^2 h$

Surface Area of a Cylinder: $S = 2\pi r^2 + 2\pi rh$

We wish to optimize (minimize) the surface area $S = 2\pi r^2 + 2\pi rh$ with respect to the constraint $\pi r^2 h = 1000$. From this latter condition we have $h = \frac{1000}{\pi r^2}$, which gives

$$S = 2\pi r^2 + 2\pi r\left(\frac{1000}{\pi r^2}\right) = 2\pi r^2 + \frac{2000}{r}.$$

Differentiating,

$$S' = 4\pi r - \frac{2000}{r^2}.$$

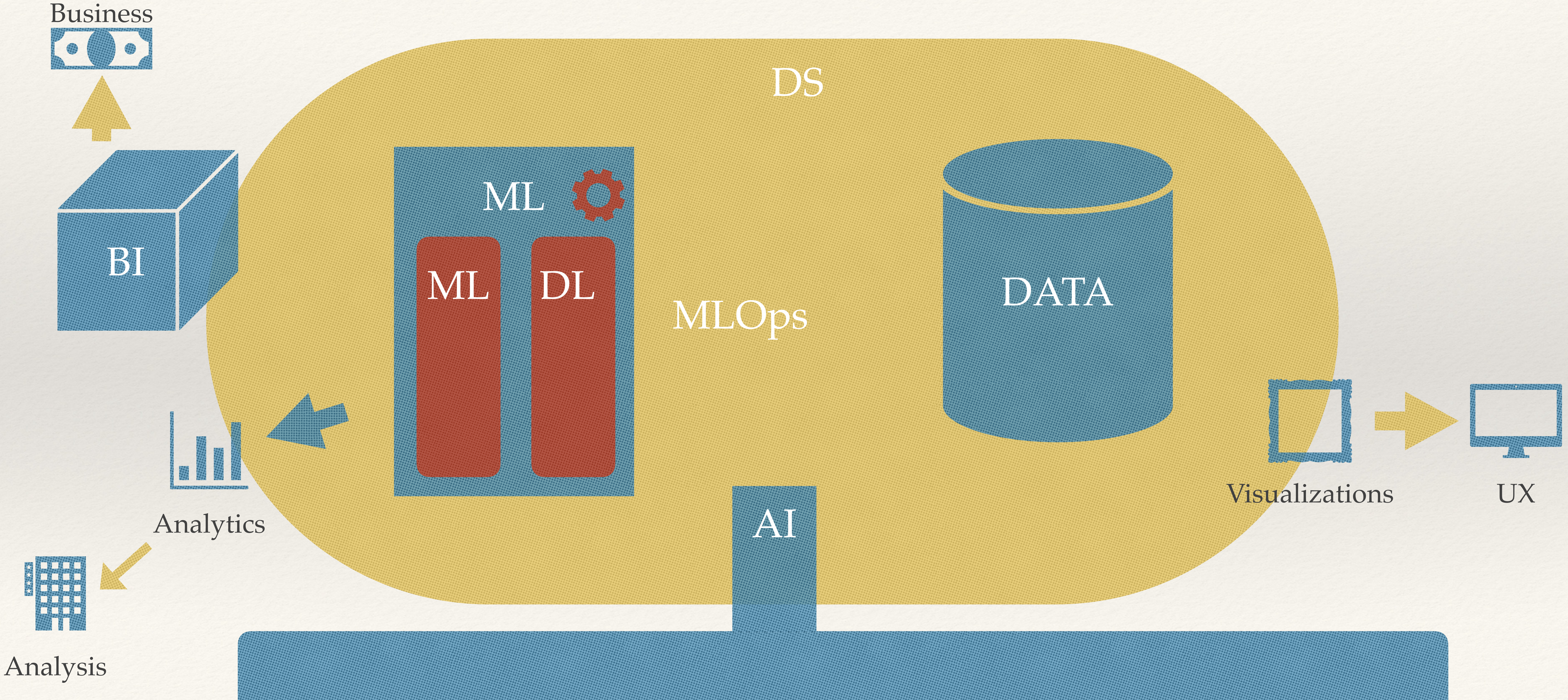
Observe that $r = 0$ is a critical point. We solve the equation $S' = 0$, giving

$$\begin{aligned} S' &= 0 \\ 4\pi r - \frac{2000}{r^2} &= 0 \\ 4\pi r &= \frac{2000}{r^2} \\ 4\pi r^3 &= 2000 \\ r^3 &= \frac{500}{\pi} \end{aligned}$$

$$r = \left(\frac{500}{\pi}\right)^{1/3}.$$

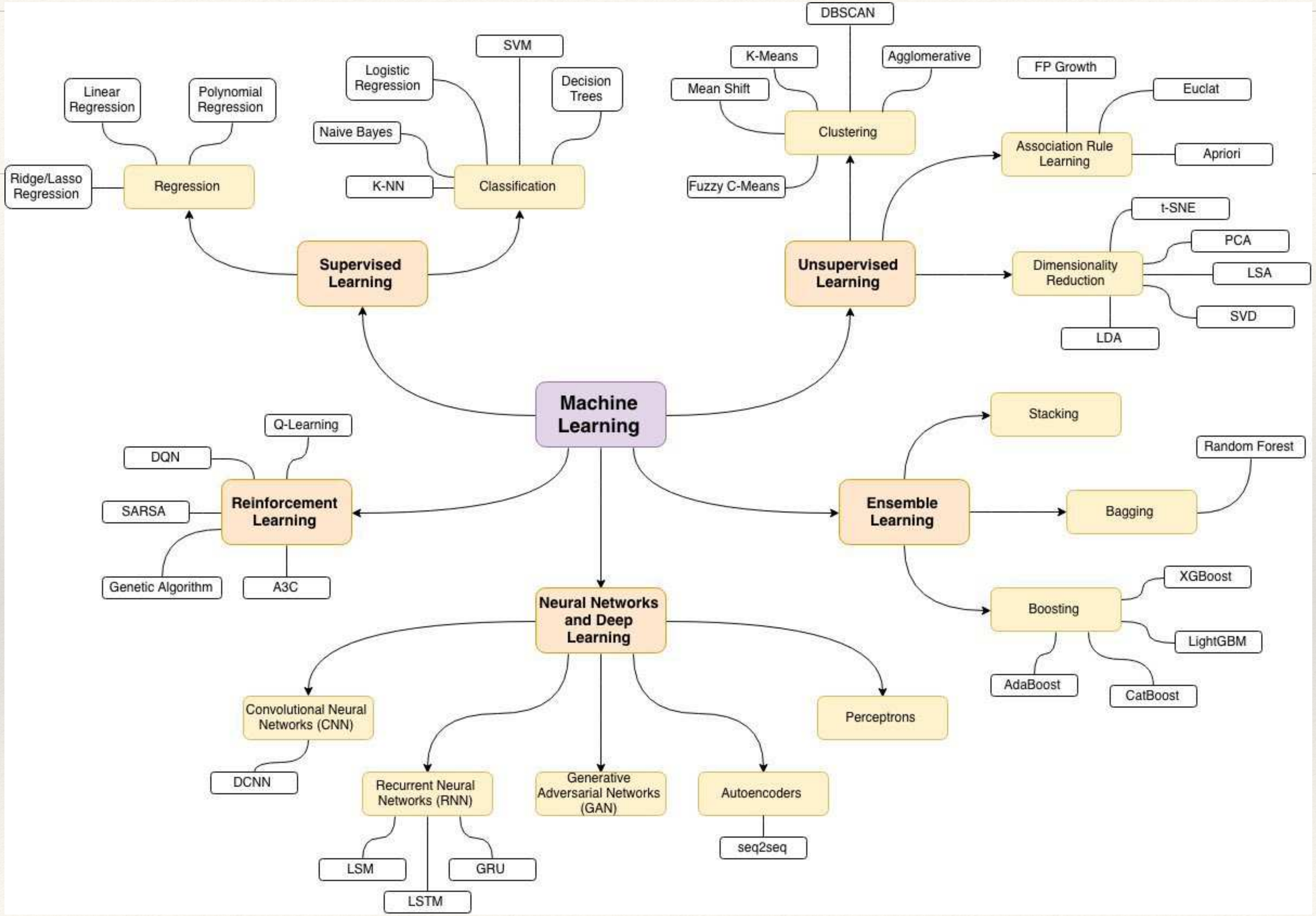
The derivative S' is negative for r in the interval $(0, \left(\frac{500}{\pi}\right)^{1/3})$ and positive for $r > \left(\frac{500}{\pi}\right)^{1/3}$, so we see that $r = \left(\frac{500}{\pi}\right)^{1/3}$ is a minimum. For this value of r the equation $h = \frac{1000}{\pi r^2}$ gives $h = \frac{1000}{\pi\left(\frac{500}{\pi}\right)^{1/3}}$.

Terminology

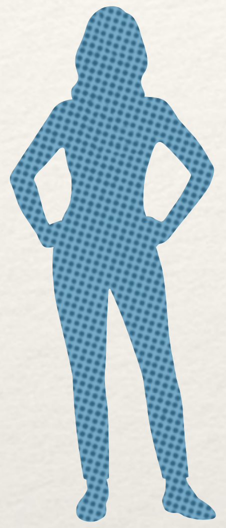


DL Map

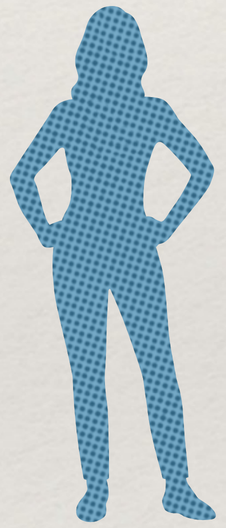

START HERE
Cove of Actuarial Science
Gulf of Descriptive Statistics
Data Visualization
Experimentation
Sea of Probability & Statistics
Business Intelligence
Biostatistics
Operations Research
Inferential
Hypothesis Testing
Causal Inference
Bayesian Inference
Statistical Learning
Econometrics
Financial Quantitative Analysis
Unsupervised Learning
Recurrent NN
Spark
Deep Learning
Convolutional NN
SVM
Hadoop
Kafka
Point
Transformers
Machine Learning
Big Data Technologies
Feed Forward NN
Reinforcement Learning
Ensembled Decision Trees
NoSQL
Pig
Computer Vision
CircleCI
MLops
Docker
Data Engineering
ETL
DVC
Computer Graphics
Database Admin/Mgmt
Data Structures
SQL
Data architecture
Computer Networks
Bay of Computer Science
Algorithms
mare mullius
Programming


Machine Learning
Supervised Learning
Regression
Linear Regression
Polynomial Regression
Ridge/Lasso Regression
Classification
Logistic Regression
SVM
Decision Trees
Naive Bayes
K-NN
Unsupervised Learning
Clustering
DBSCAN
K-Means
Agglomerative
Mean Shift
Fuzzy C-Means
Association Rule Learning
FP Growth
Euclat
Apriori
Dimensionality Reduction
t-SNE
PCA
LSA
SVD
LDA
Reinforcement Learning
Q-Learning
DQN
SARSA
Genetic Algorithm
A3C
Ensemble Learning
Stacking
Bagging
Random Forest
Boosting
XGBoost
LightGBM
AdaBoost
CatBoost
Neural Networks and Deep Learning
Convolutional Neural Networks (CNN)
DCNN
Recurrent Neural Networks (RNN)
LSM
LSTM
GRU
Generative Adversarial Networks (GAN)
Autoencoders
seq2seq
Perceptrons

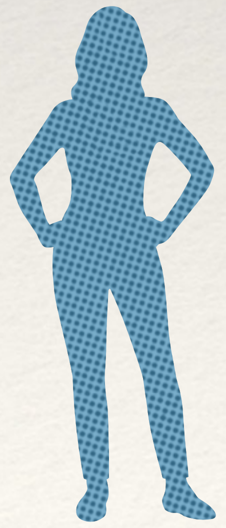
Roles



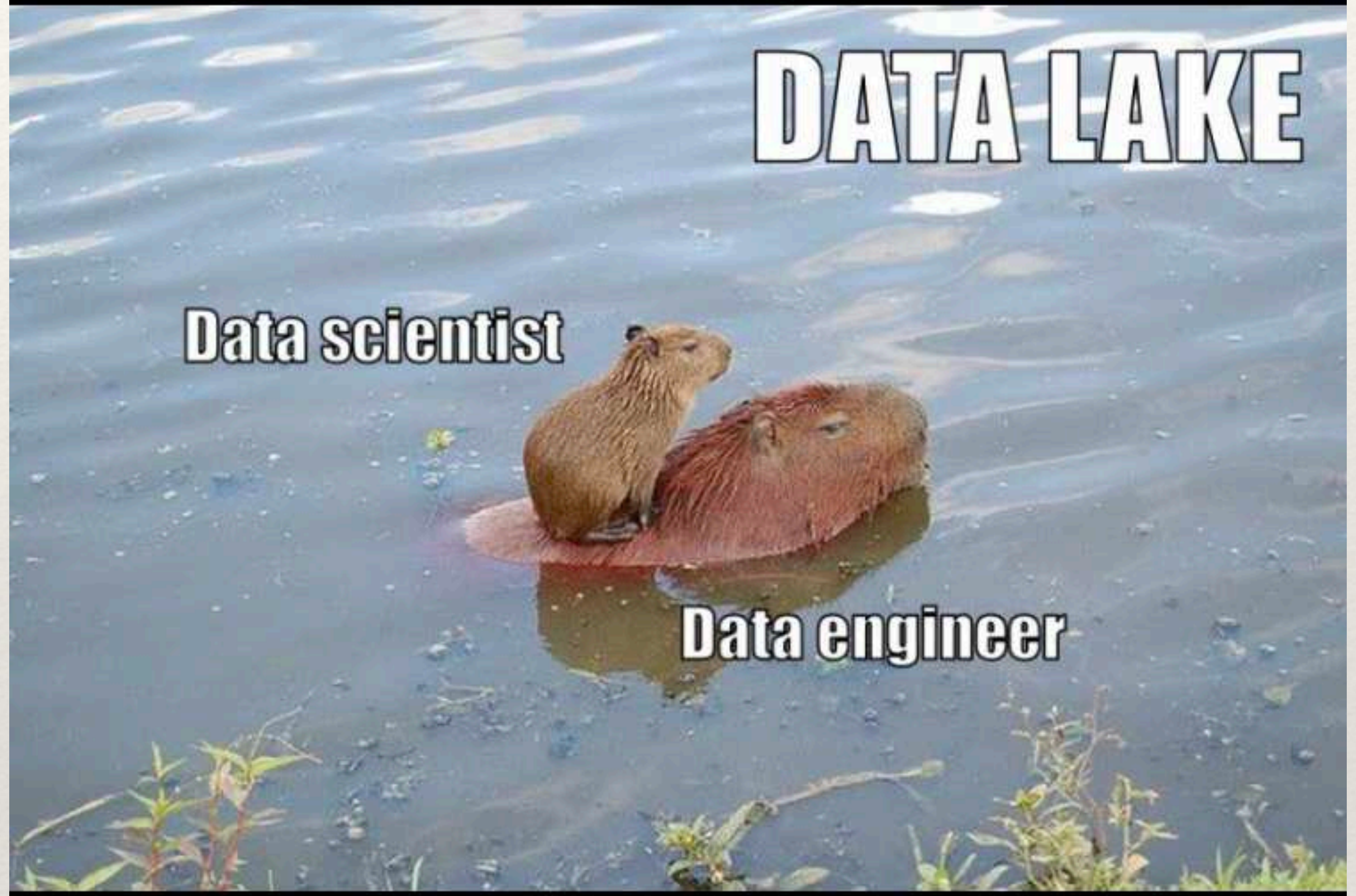
Data Engineer



Data Scientist



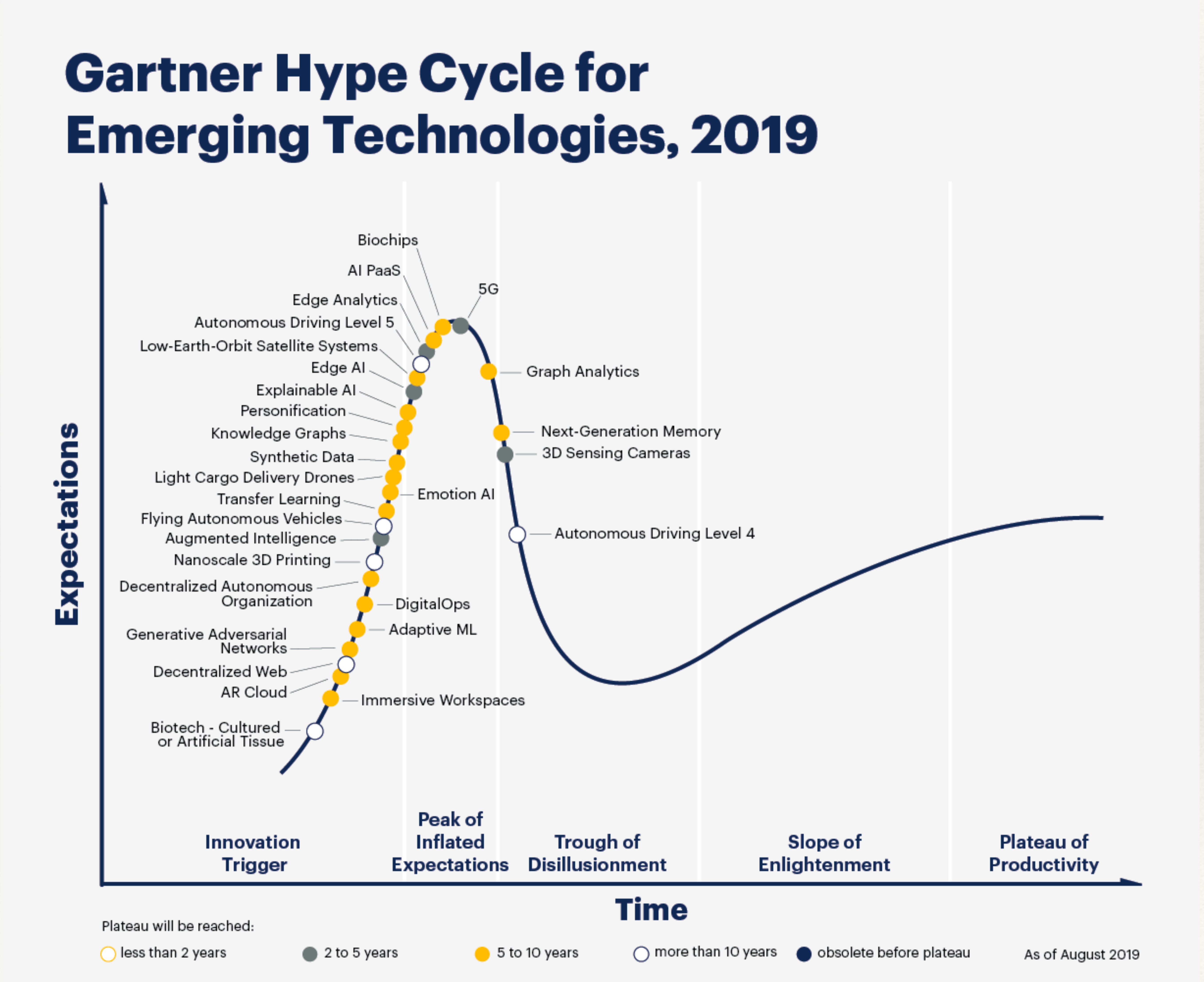
Data Analyst



Evolving Artificial Intelligence

Emerging Technologies

- GANs
- Explainable AI
- Edge AI
- AI PaaS
- Adaptive ML
- Emotion AI



Learning by Data

In ML there are three approaches to force a machine to "learn"

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

Semi Supervised Learning

Self Supervised Learning

Classification

Regression

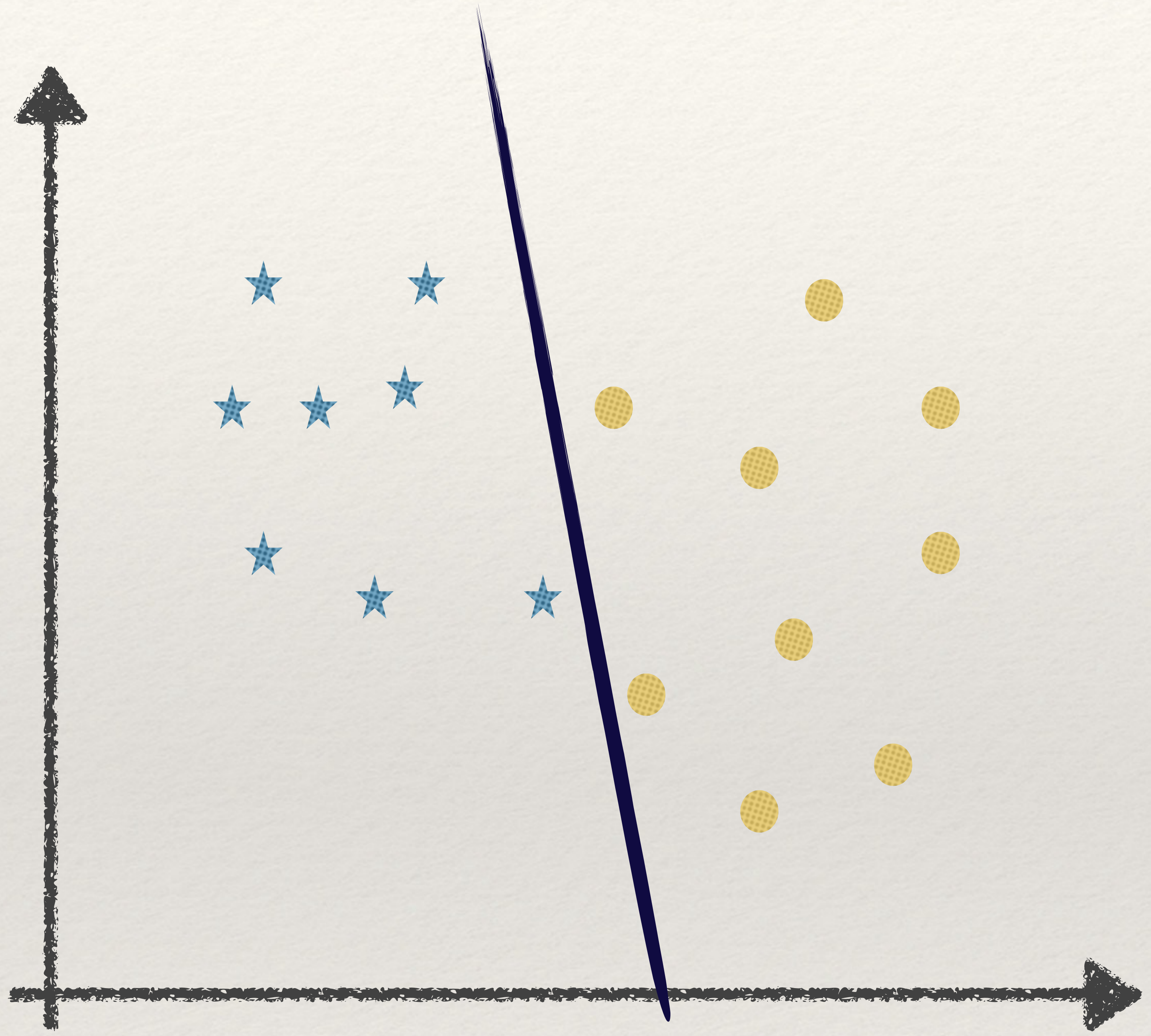
Generation

Optimize

Machine Learning Techniques

SVM

LDA



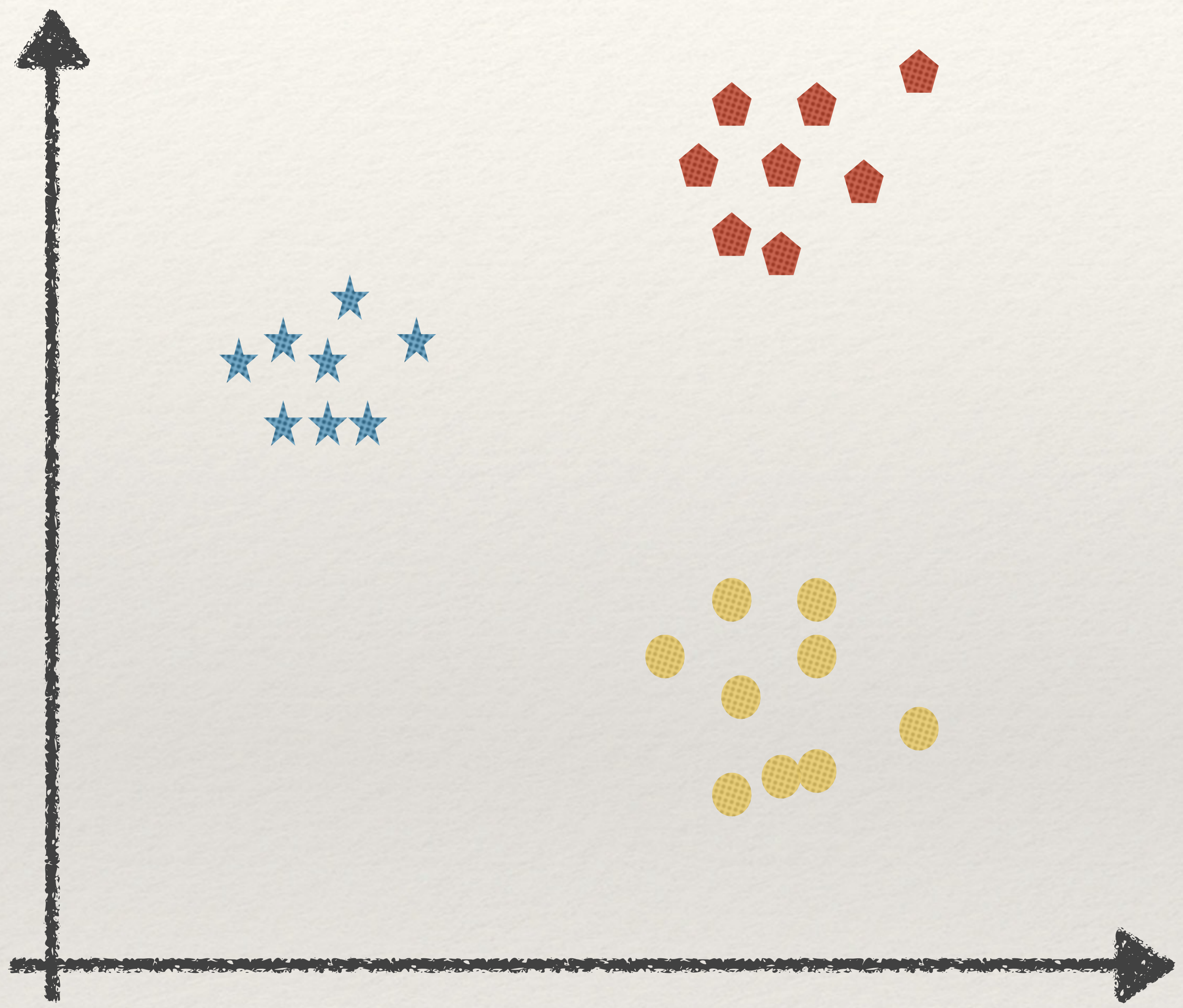
LogReg

kNN

Autoregressive

Machine Learning Techniques

DBSCAN



k-means

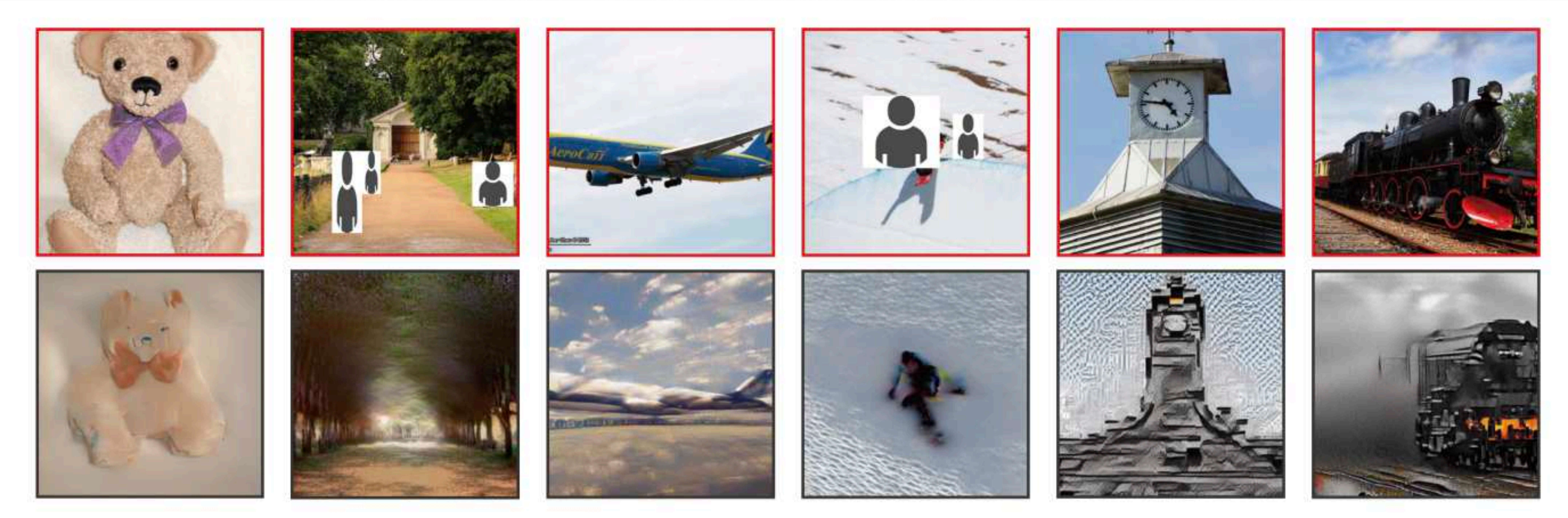
SOM

OPTICS

Watching Dreams?

Nishimoto,Cell 2011, Nytimes

Watching Thoughts?



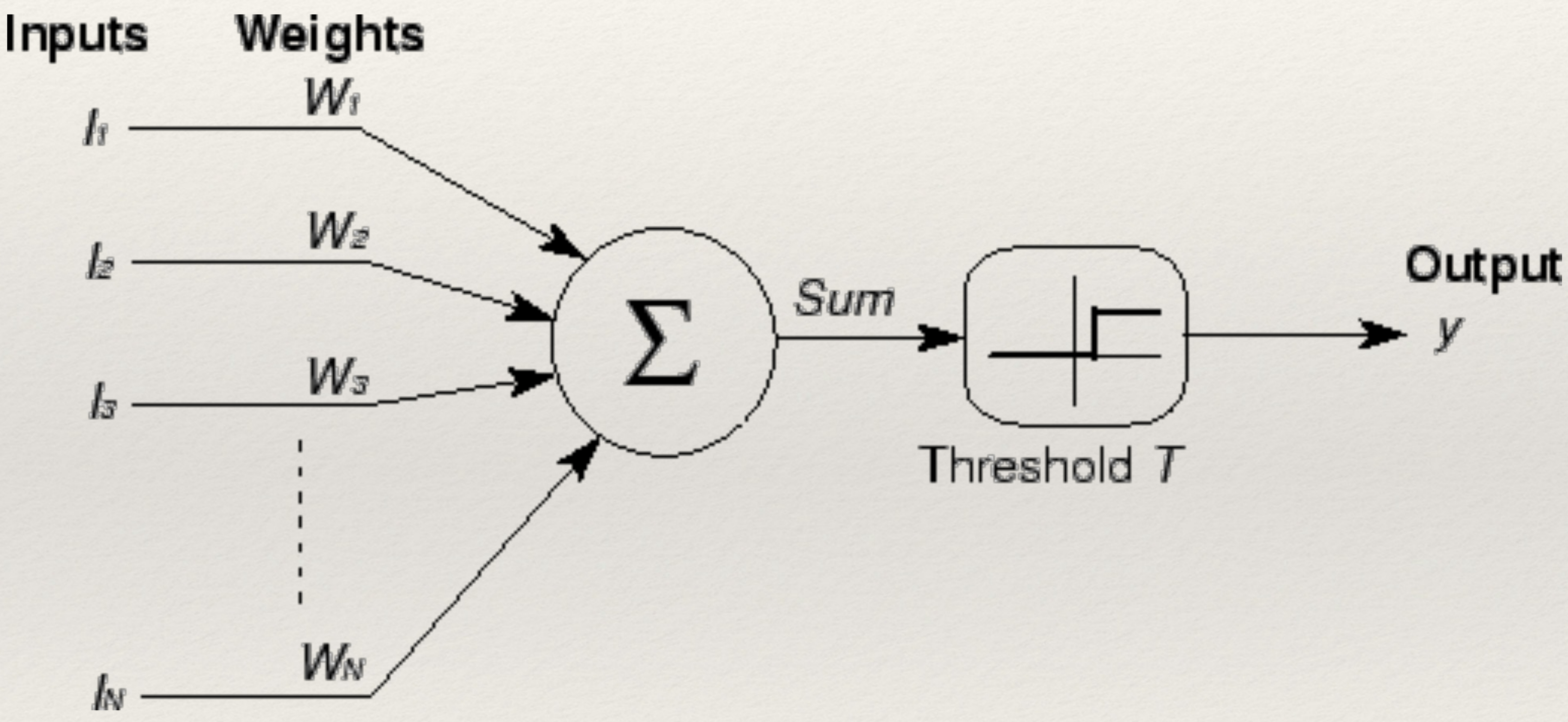
High-resolution image reconstruction with latent diffusion models from human brain activity, Takagi, Nishimoto 2022

Biomimetic neuroscience

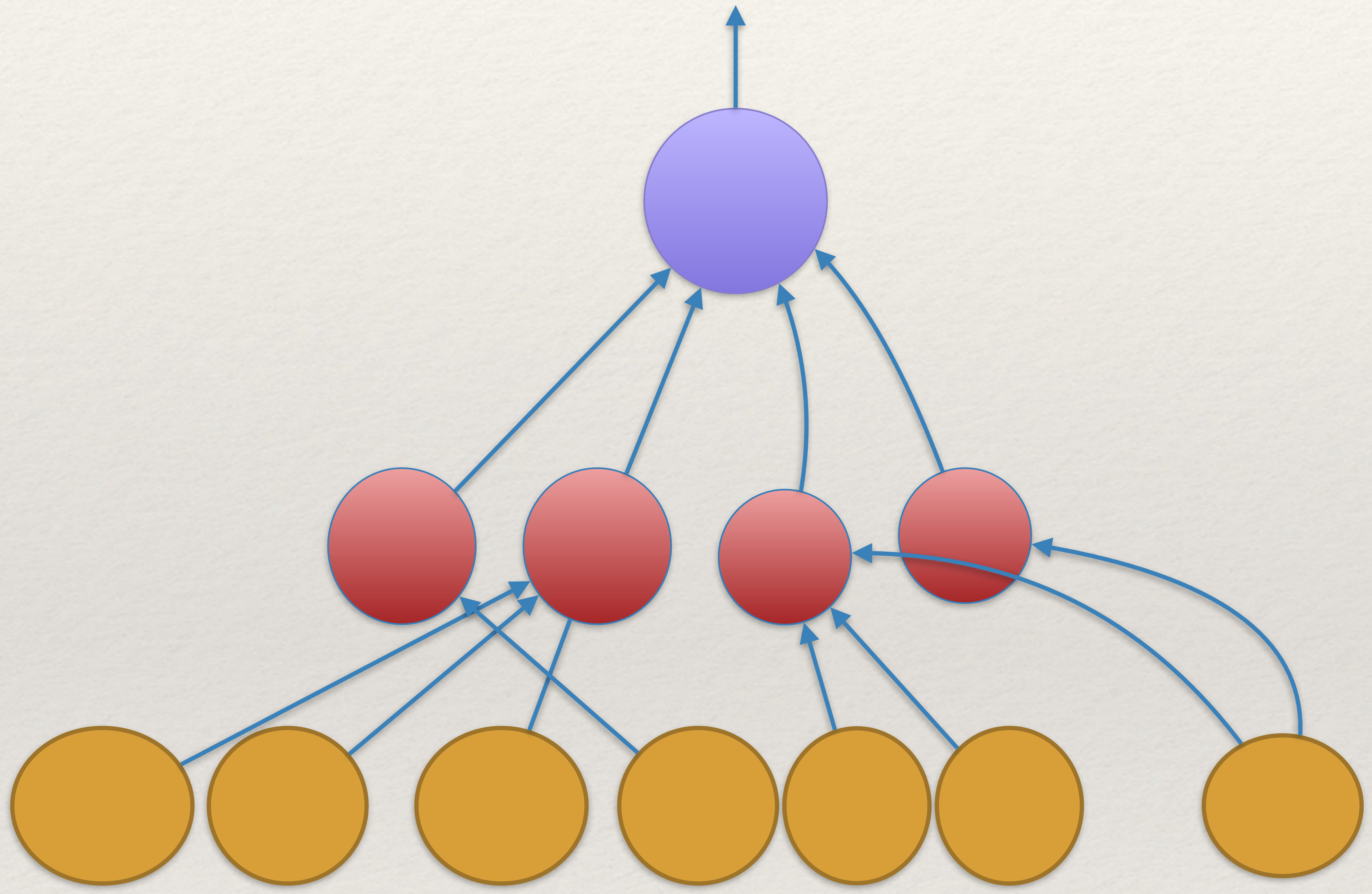
Deep Brain

The Revolution of Deep Learning

Neural Networks - McCulloch and Pitts



Neural Networks - MLP Multilayer Perceptron



$$o = \sigma(p + \sigma(b + \overline{x}W)V)$$

$$\overline{o} = \sigma(p + \overline{h}V)$$

$$\overline{h} = \sigma(b + \overline{x}W)$$

Deep Learning

- Cybernetics (McCulloch and Pits, Hebb 1949, Rosenblatt 1958)
- Connectionism (Backprop: Rumelhart and Hinton, Hopfield, 1986)
- Dechter 1986
- Aizenberg 2000
- Krizhevsky AlexNet 2012

Origins

- Deep Learning (Greedy layer-wise pretraining Hinton, Bengio and LeCun* 2006, Ranzato 2007)

− Minsky and Papert 1969

− Olshausen and Field 2005

+ Fukushima Neocognitron 1980

+ Distributed Representation 1986

+ Beating ImageNet Krizhevsky 2012

+ Playing Atari Games, Mnih 2015

(*) Turing Award 2019

What Deep Learning is NOT?

"Los algoritmos de IA no se programan."

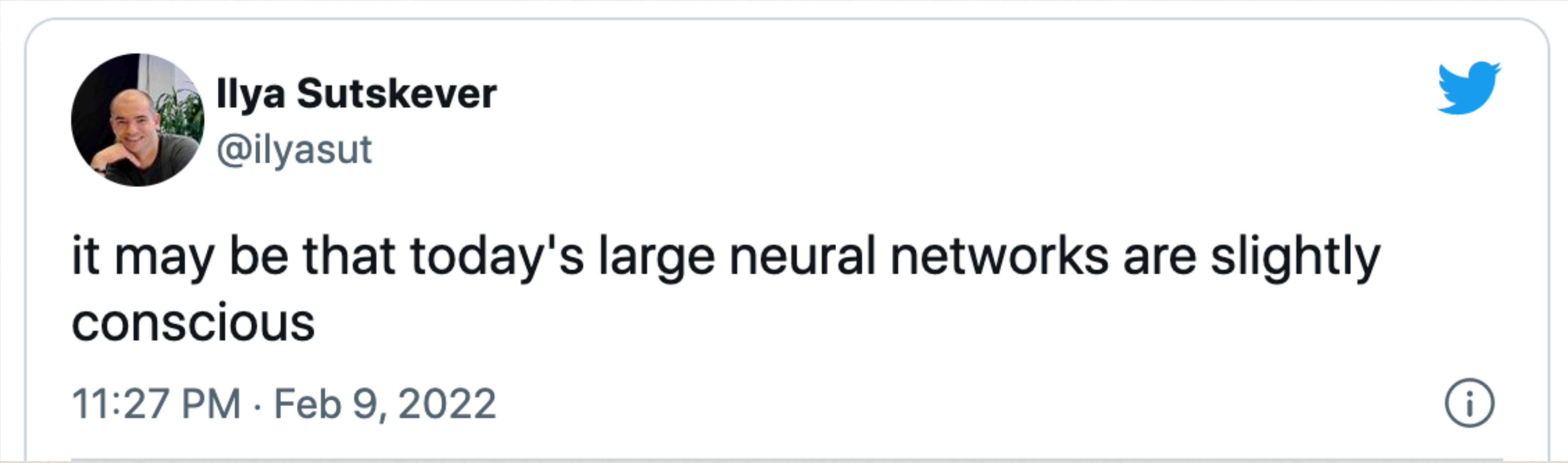
"Los robots vienen a reemplazarnos."

"Deep Learning permite reemplazar a los seres humanos."

"La tecnología empezó a reemplazarnos, y más nos vale sacarle provecho que quedarnos amargados por la herida que nos provoque."

"Los algoritmos tienen además la ventaja que no se cansan y pueden interactuar con el contexto a velocidades supersónicas en comparación con las personas"

"Nada será igual" M.Tetaz



"Trolley Train Problem."

"AGI Artificial General Intelligence."

"Make decisions based on data."

Laura Accion, et al , "Desmistificando la Inteligencia Artificial" en "Inteligencia artificial, una mirada interdisciplinaria" , 2020

Why the AI is so dumb, IEEE Spectrum Nov 2021

What Deep Learning is NOT?

MIT Technology Review

Featured Topics Newsletters Events Podcasts Sign in Subscribe

ARTIFICIAL INTELLIGENCE

Hundreds of AI tools have been built to catch covid. None of them helped.

Some have been used in hospitals, despite not being properly tested. But the pandemic could help make medical AI better.

What Deep Learning is?

Revolutionary Technology to move ahead the digital boundary

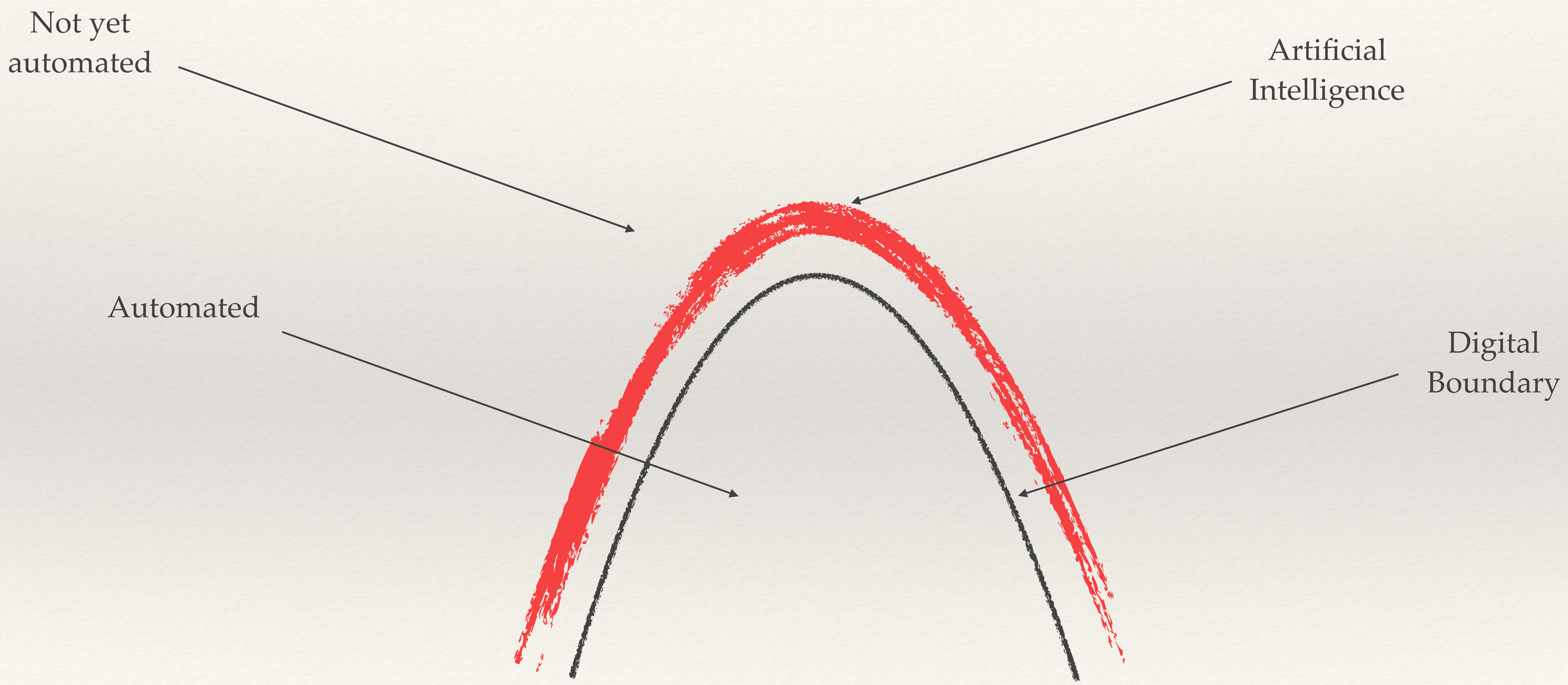
Computer Vision

Natural Language Processing

Speech Recognition

"Artificial Intelligence and Life in 2030", Sept 2016

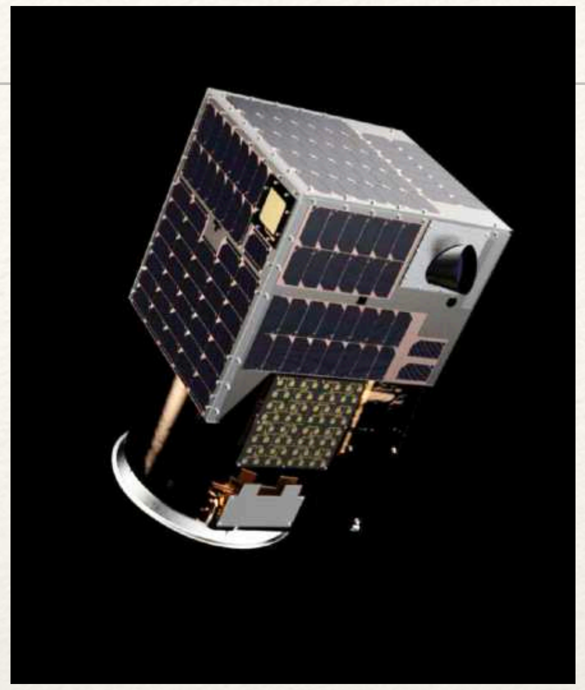
Digital Boundary



What Deep Learning is?

Magnetic control of tokamak plasmas through deep reinforcement learning

Jonas Degrave, Federico Felici, ... Martin Riedmiller + Show authors



How a Plucky Robot Found the Long-Lost *Endurance* Shipwreck

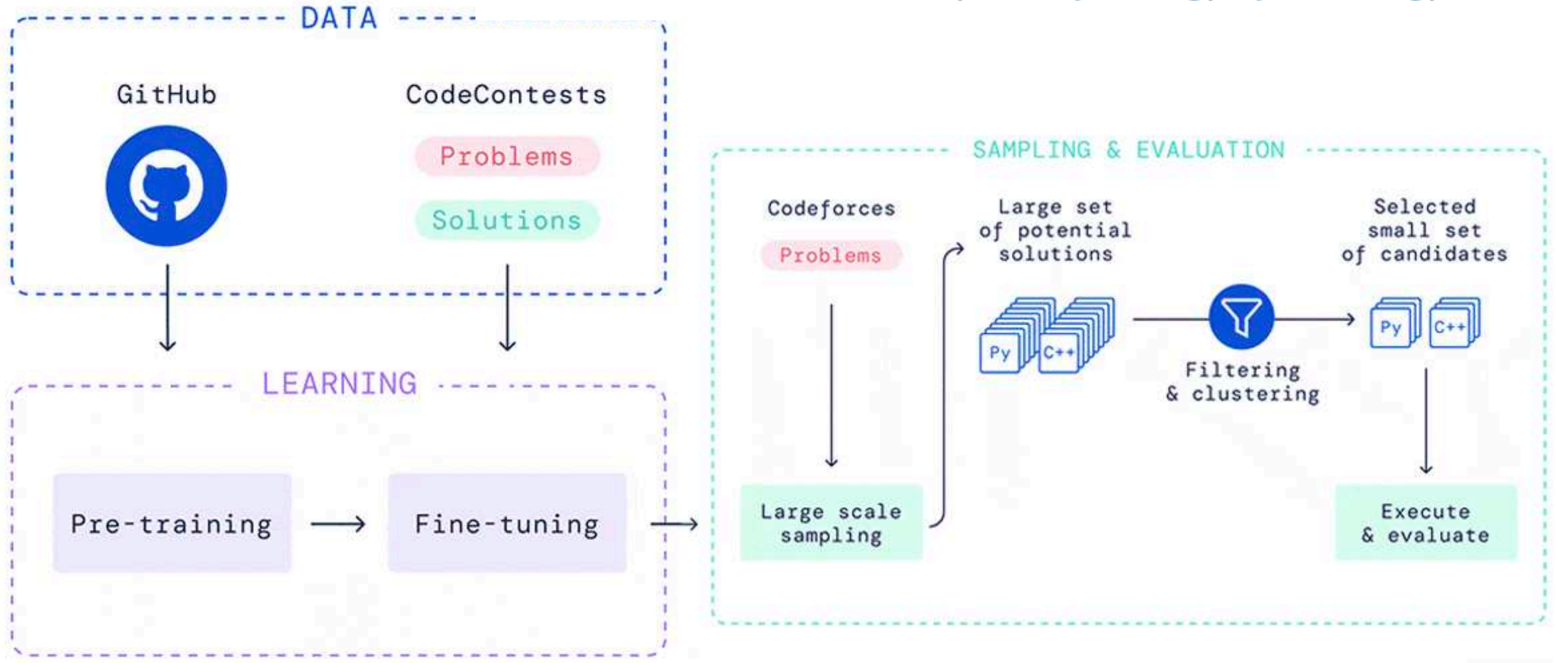
Over a century ago, two dozen men were stranded in Antarctica. Here's how a robot dove 10,000 feet to glimpse their lost ship for the first time.

AlphaFold AI predicts SARS-CoV-2 omicron variant might not evade antibody neutralization

AlphaFold: a solution to a 50-year-old grand challenge in biology

Discovering faster matrix multiplication algorithms with reinforcement learning

Alhussein Fawzi, Matej Balog, Aja Huang, Thomas Hubert, Bernardino Romera-Paredes,



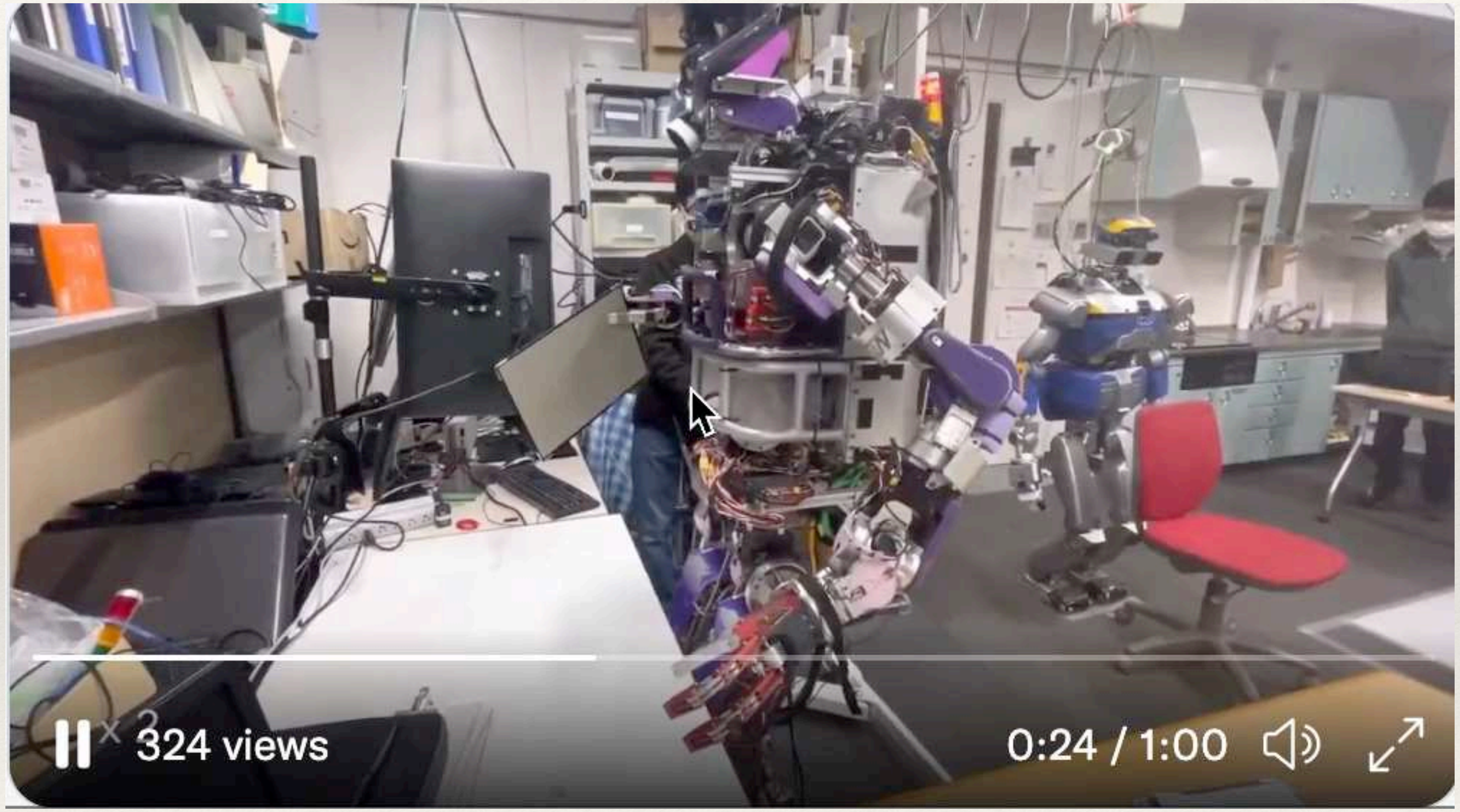
Competitive Coder

DeepMind's StarCraft-playing AI beats 99.8 per cent of human gamers

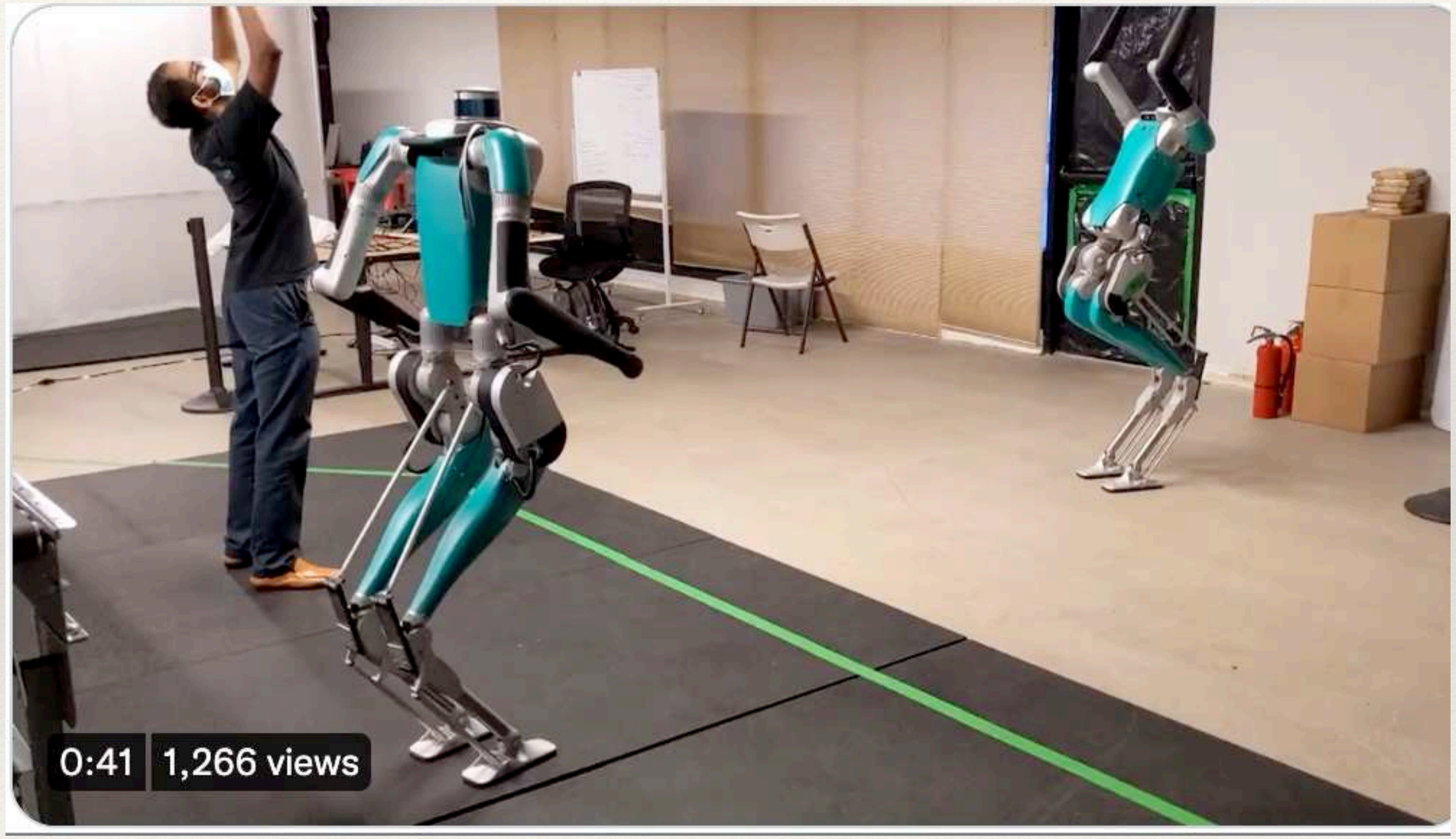
Nature, Wired, The Batch, Newscientist

What Deep Learning is?

Where Deep Learning may have something to do?

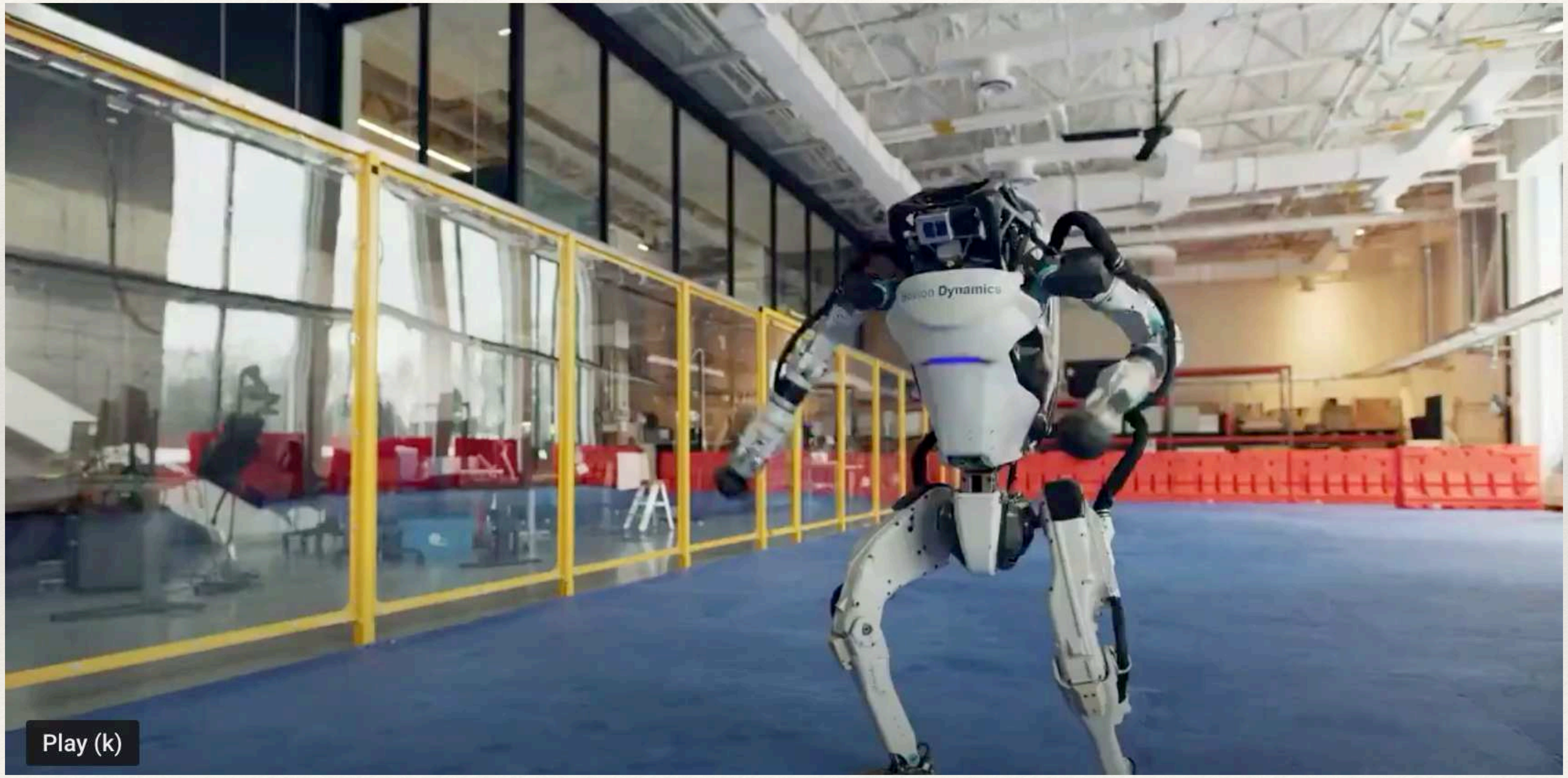


Where Deep Learning may have something to do?



Agility Robotics

Where Deep Learning may have something to do?



Boston Dynamics

Realistic Applications

きつい・汚い・危険

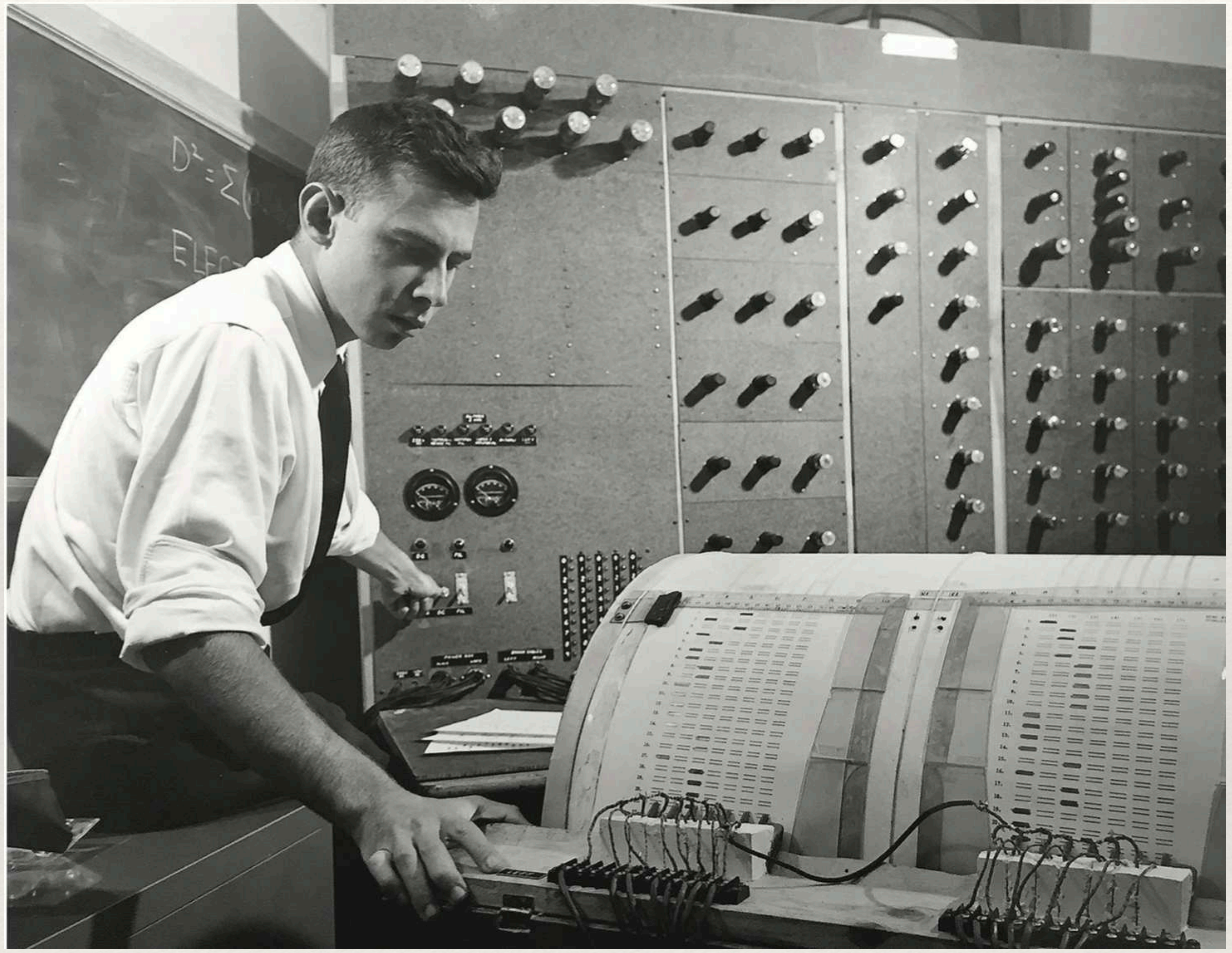
- Aburrido
- Descuidado
- Peligroso

Every successful deployment has either one of two expedients:

- It has a person somewhere in the loop, or
- The cost of failure, should the system blunder, is very low

Rodney Brooks, “An inconvenient truth About AI”, IEEE Spectrum, Sept 2021

Deep Learning



1958 New York Times...

NEW NAVY DEVICE LEARNS BY DOING

Psychologist Shows Embryo of Computer Designed to Read and Grow Wiser

WASHINGTON, July 7 (UPI) —The Navy revealed the embryo of an electronic computer today that it expects will be able to walk, talk, see, write, reproduce itself and be conscious of its existence.

The embryo—the Weather Bureau's $2,000,000 "704" computer—learned to differentiate between right and left after fifty attempts in the Navy's demonstration for newsmen.

The service said it would use this principle to build the first of its Perceptron thinking machines that will be able to read and write. It is expected to be finished in about a year at a cost of $100,000.

Dr. Frank Rosenblatt, designer of the Perceptron, conducted the demonstration. He said the machine would be the first device to think as the human brain. As do human beings, Perceptron will make mistakes at first, but will grow wiser as it gains experience, he said.

Dr. Rosenblatt, a research psychologist at the Cornell Aeronautical Laboratory, Buffalo, said Perceptrons might be fired to the planets as mechanical space explorers.

Without Human Controls

The Navy said the perceptron would be the first non-living mechanism "capable of receiving, recognizing and identifying its surroundings without any human training or control."

The "brain" is designed to remember images and information it has perceived itself. Ordinary computers remember only what is fed into them on punch cards or magnetic tape.

Later Perceptrons will be able to recognize people and call out their names and instantly translate speech in one language to speech or writing in another language, it was predicted.

Mr. Rosenblatt said in principle it would be possible to build brains that could reproduce themselves on an assembly line and which would be conscious of their existence.

In today's demonstration, the "704" was fed two cards, one with squares marked on the left side and the other with squares on the right side.

Learns by Doing

In the first fifty trials, the machine made no distinction between them. It then started registering a "Q" for the left squares and "O" for the right squares.

Dr. Rosenblatt said he could explain why the machine learned only in highly technical terms. But he said the computer had undergone a "self-induced change in the wiring diagram."

The first Perceptron will have about 1,000 electronic "association cells" receiving electrical impulses from an eye-like scanning device with 400 photo-cells. The human brain has 10,000,000,000 responsive cells, including 100,000,000 connections with the eyes.

IEEE Spectrum Nov 2021, Why AI is harder than we think?[49]

Deep Learning

1956 Dartmouth AI Conference

Why AI is harder than we think?[49]

The Four "Deep Learning" pillars

- Multiple Level of Composition
- Representational Learning
- Massive and standardized datasets
- GPU

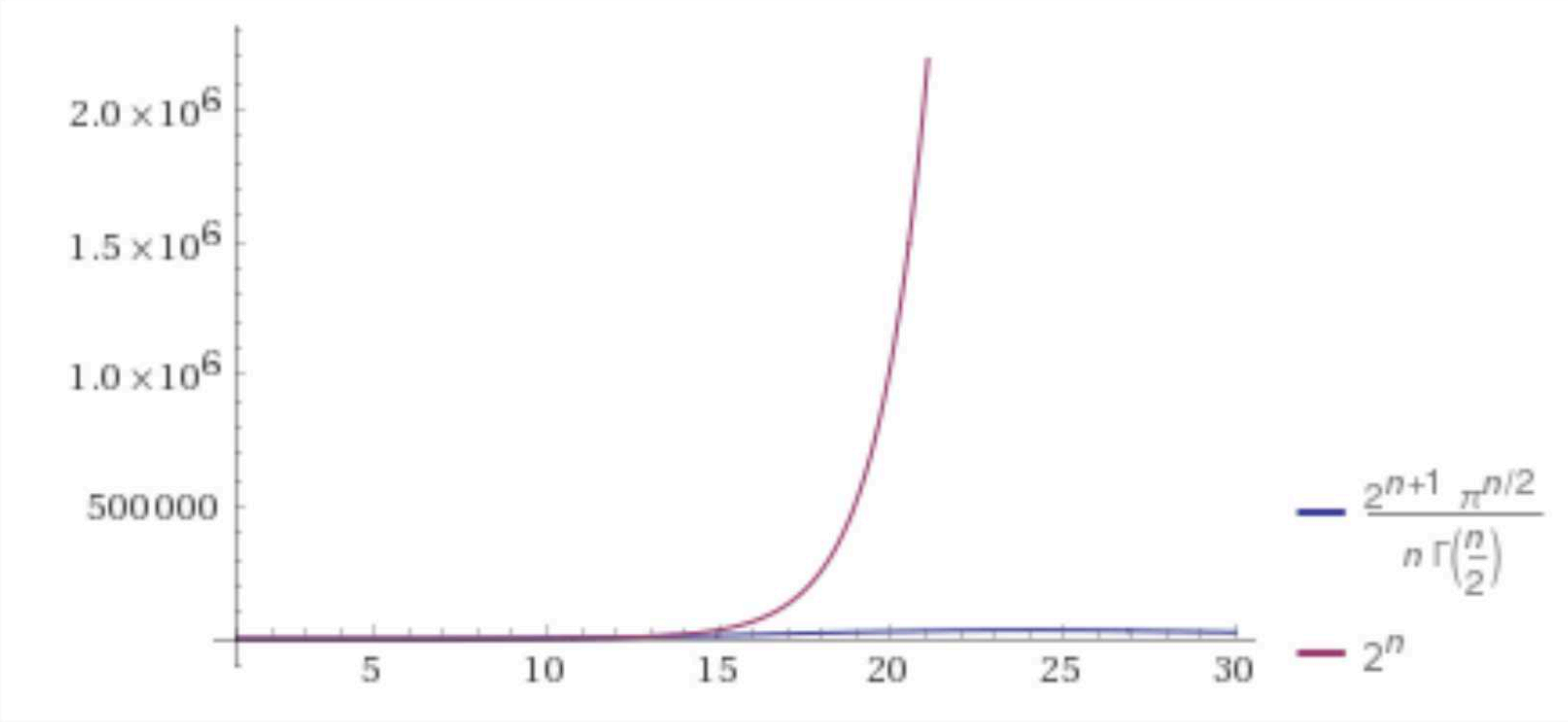
Why deep?

Limitations of Machine Learning on world problems

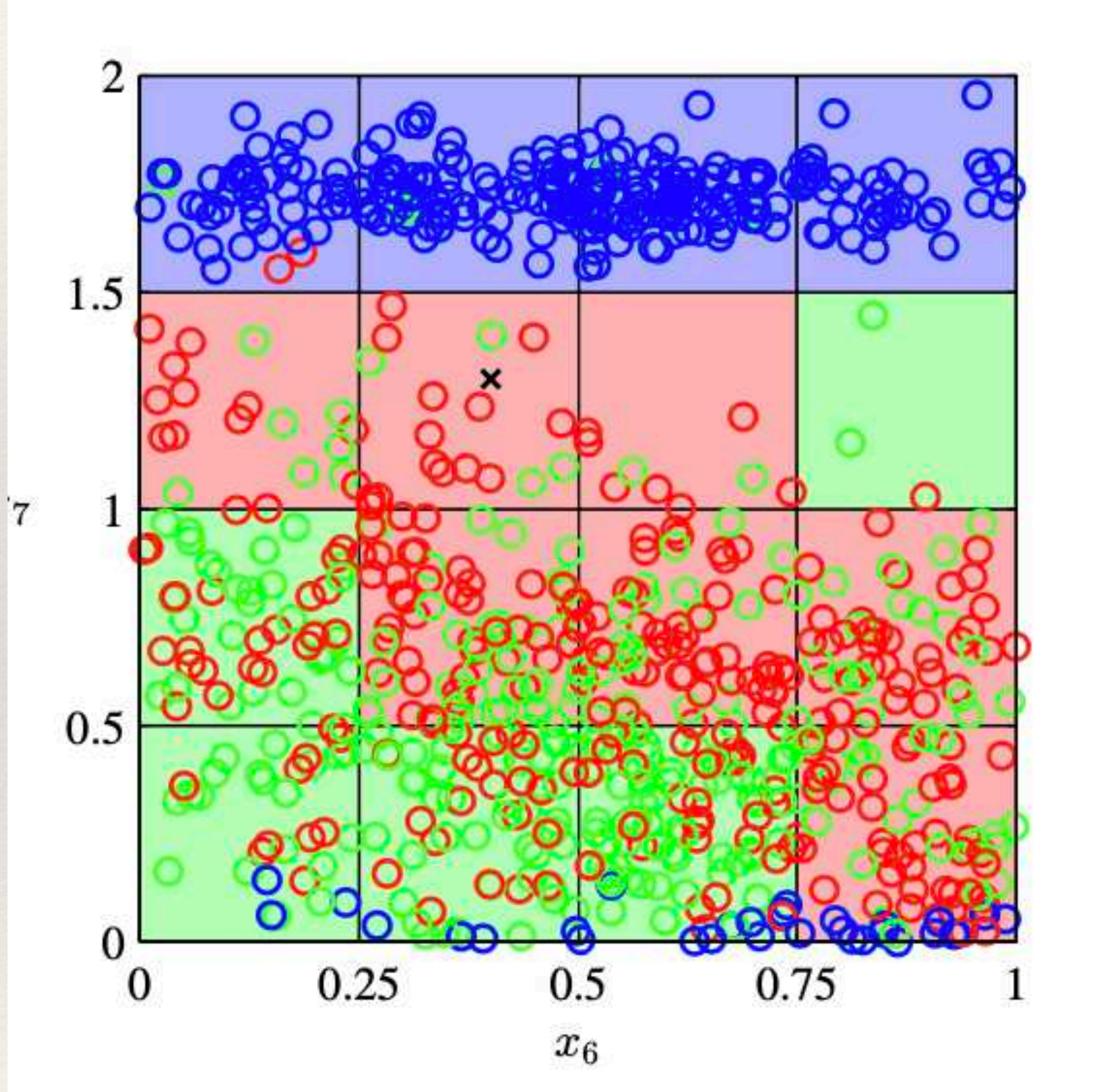
Schmidhuber 2014

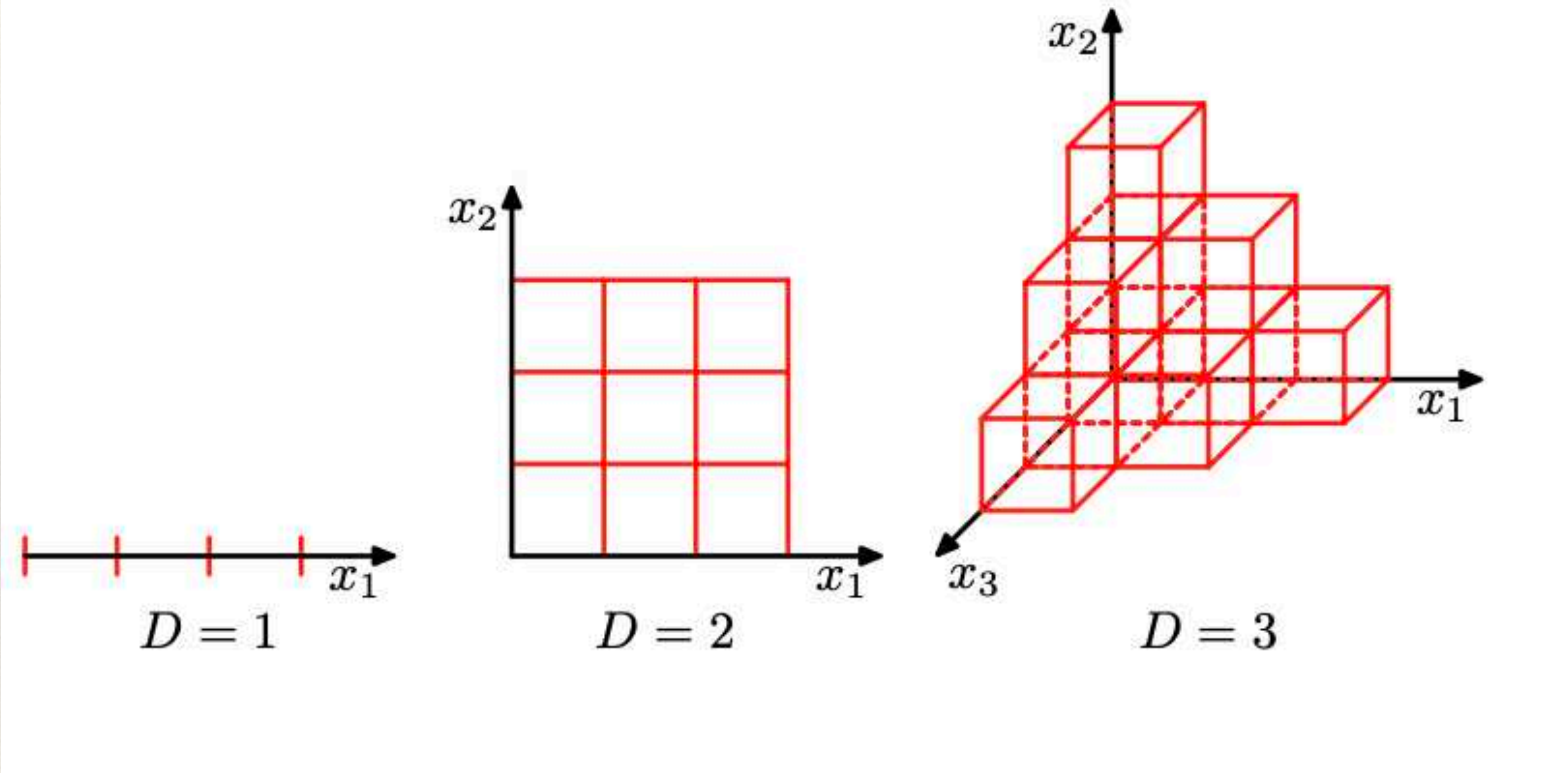
Hochreiter 1991, Hinton 2006

ML Problem - The Curse of Dimensionality



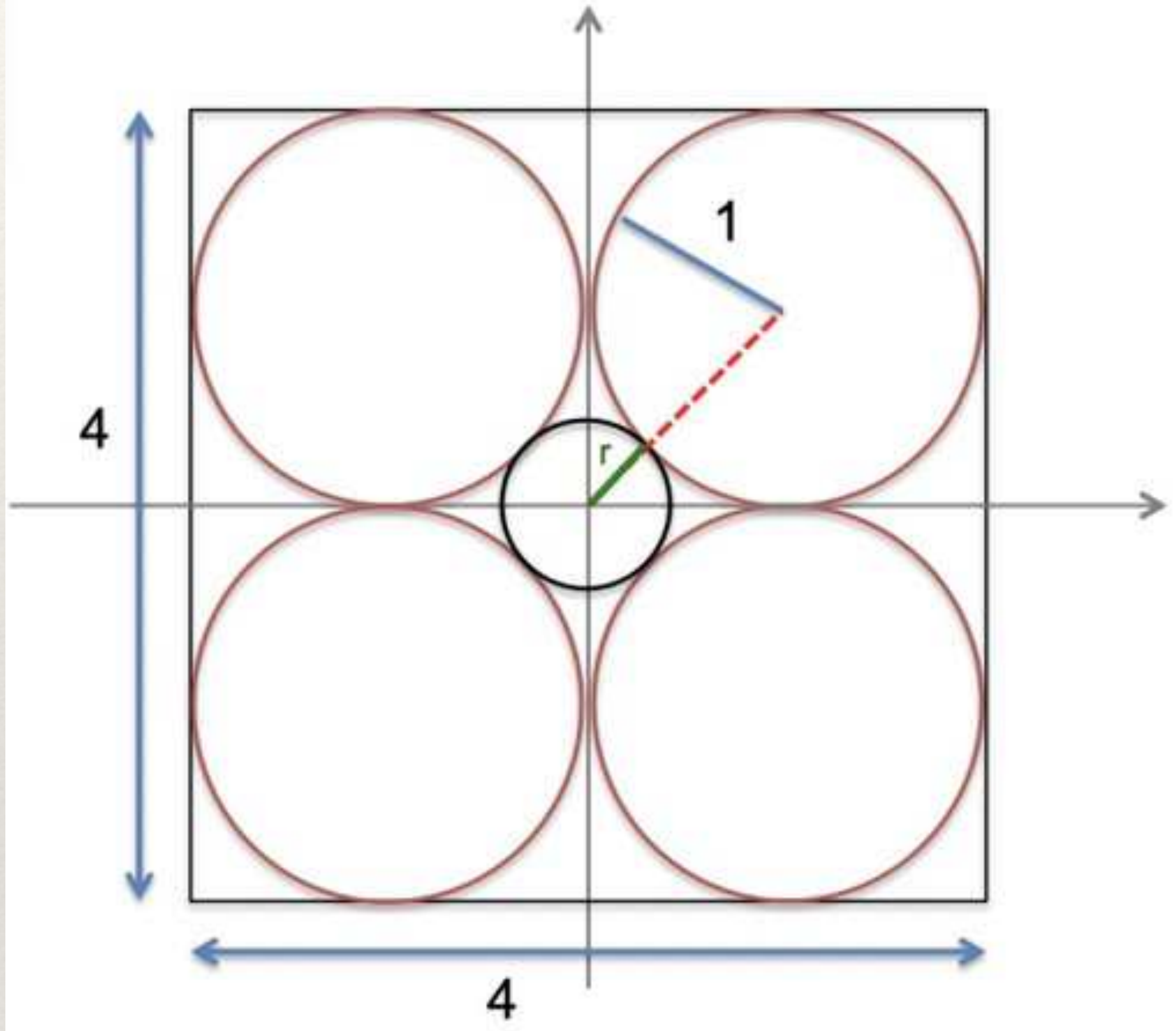
ML Problem - The Curse of Dimensionality





ML Problem - The Curse of Dimensionality

Dessert: High-Dimensional Intuition Failure



- J. Michael Steele, *The Cauchy-Schwarz Master Class*, Cambridge, 2004.

- In $\mathbb{R}^2$:
  - 4 unit-radius circles inside $[-2, 2]^2$
  - circle of radius r centred at the origin, confined by the unit-radius circles
  - distance from origin to centers: $\sqrt{2}$
  - $r = \sqrt{2} - 1$
- In $\mathbb{R}^d$:
  - 2^d unit-radius spheres inside $[-2, 2]^d$
  - radius-r sphere centred at the origin, confined by the unit-radius spheres
  - $r = \sqrt{d} - 1$
  - for $d \geq 10$, we have $r > 2$

If $d \geq 10$, the sphere confined by the spheres in the box, goes outside the box!

Why deep?

Universal Approximation Theorem

A feedforward neural network with a single hidden layer containing a finite number of neurons can approximate any continuous function on a compact subset of R, given appropriate activation functions.

“Approximation by superpositions of a sigmoidal function”, Cybenko 1989

Why deep?

Limitations of ANN on real world problems

Shallow Network → UAT ← **Deep Network**

Difficult to train
Difficult to find the optimal
Difficult to generalize well

Compositional Structure $\left\{ \begin{array}{l} \text{Shift Invariance} \\ \text{Locality at different scales} \end{array} \right.$

"Many natural questions on images corresponds to algorithms that are compositional"

"The unreasonable effectiveness of deep learning in artificial intelligence", Sejinowski 2019
"When and why..", Mhaskar 2017

Why deep?

Limitations of ANN on real world problems

The Curse of Dimensionality

Local Consistency Assumption

Interests points are Sparse

Limitations of Machine Learning on world problems

Deep

Fundamental Credit Assignment Problem

Credit Assignment Path

Vanishing or Exploding Gradients

Fundamental Deep Learning Problem

"When and why..", Mhaskar 2017

Schmidhuber 2014

Hochreiter 1991, Hinton 2006

Add more layers

Overcoming the FDLP

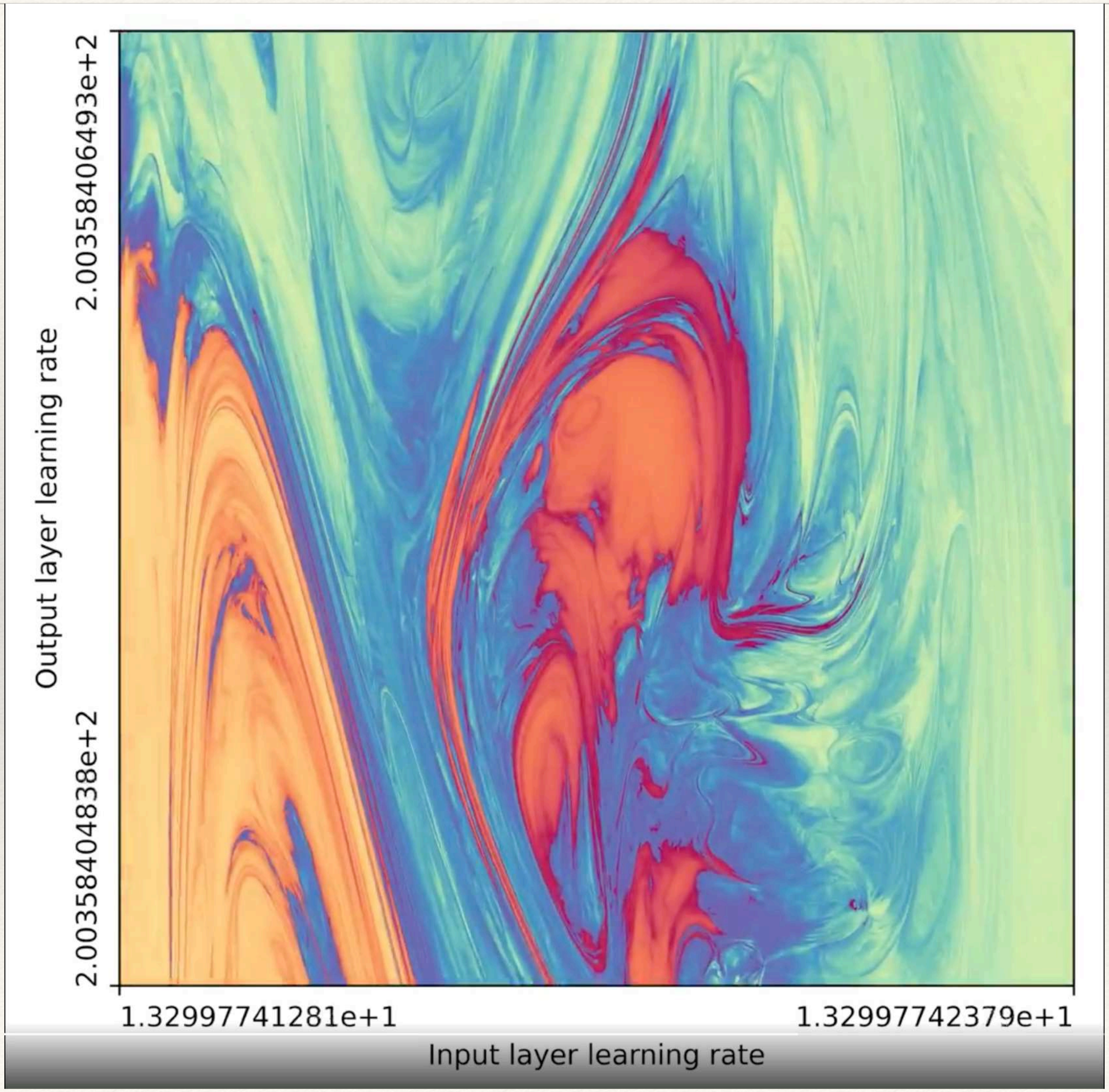
Limitations of ANN on real world problems

Fundamental Deep Learning Problem

- Unsupervised Pretraining
- Greedy Layer-Wise
- LSTM cells
- Hessian Free optimization, Dropout, Xavier initialization
- Random weight guessing

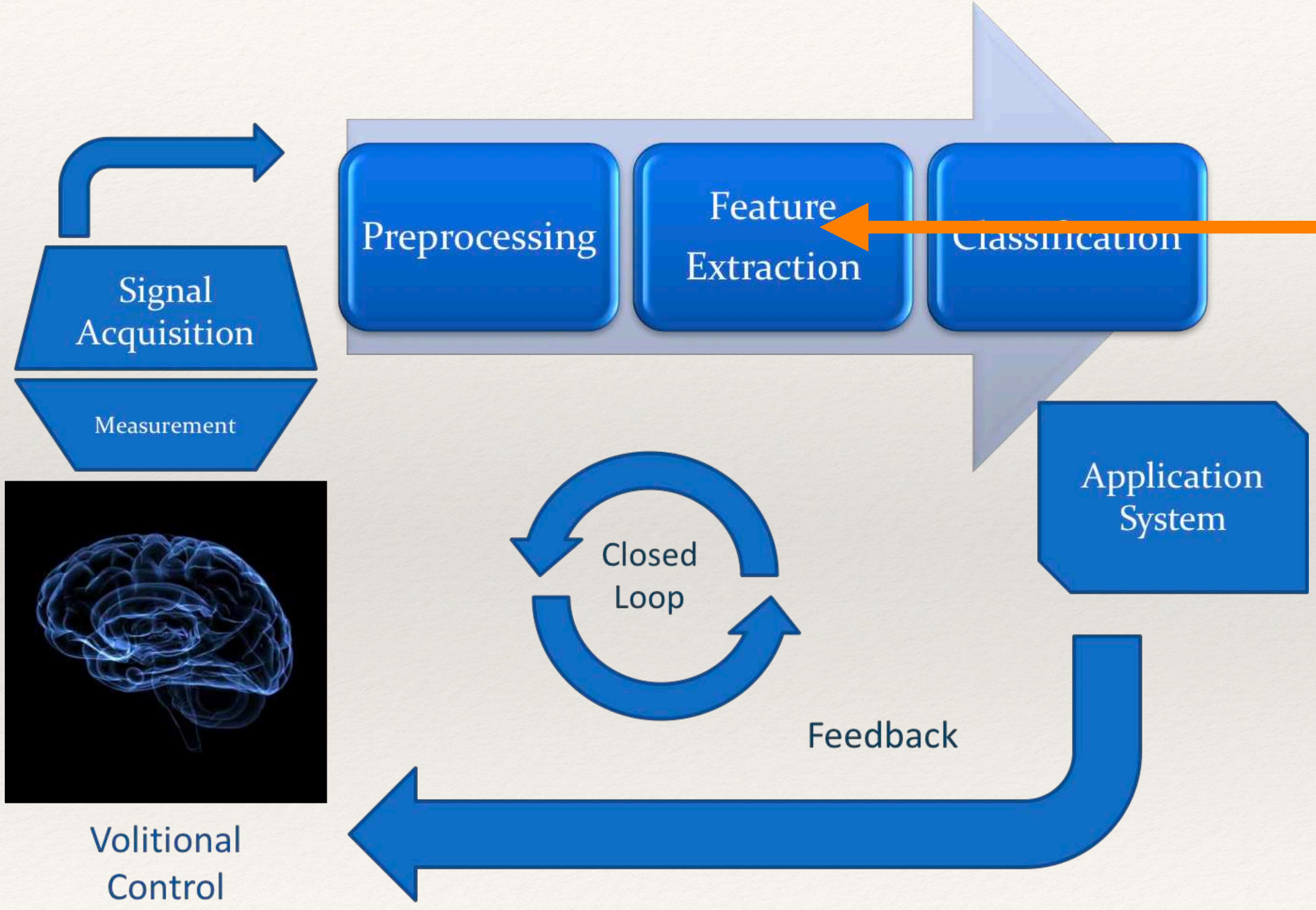
Schmidhuber 2014
Hochreiter 1991, Hinton 2006

Neural Networks - Convergence Paradox



"The unreasonable effectiveness of deep learning in artificial intelligence", Sejinowski 2019

Representational Learning



Handcrafted Feature

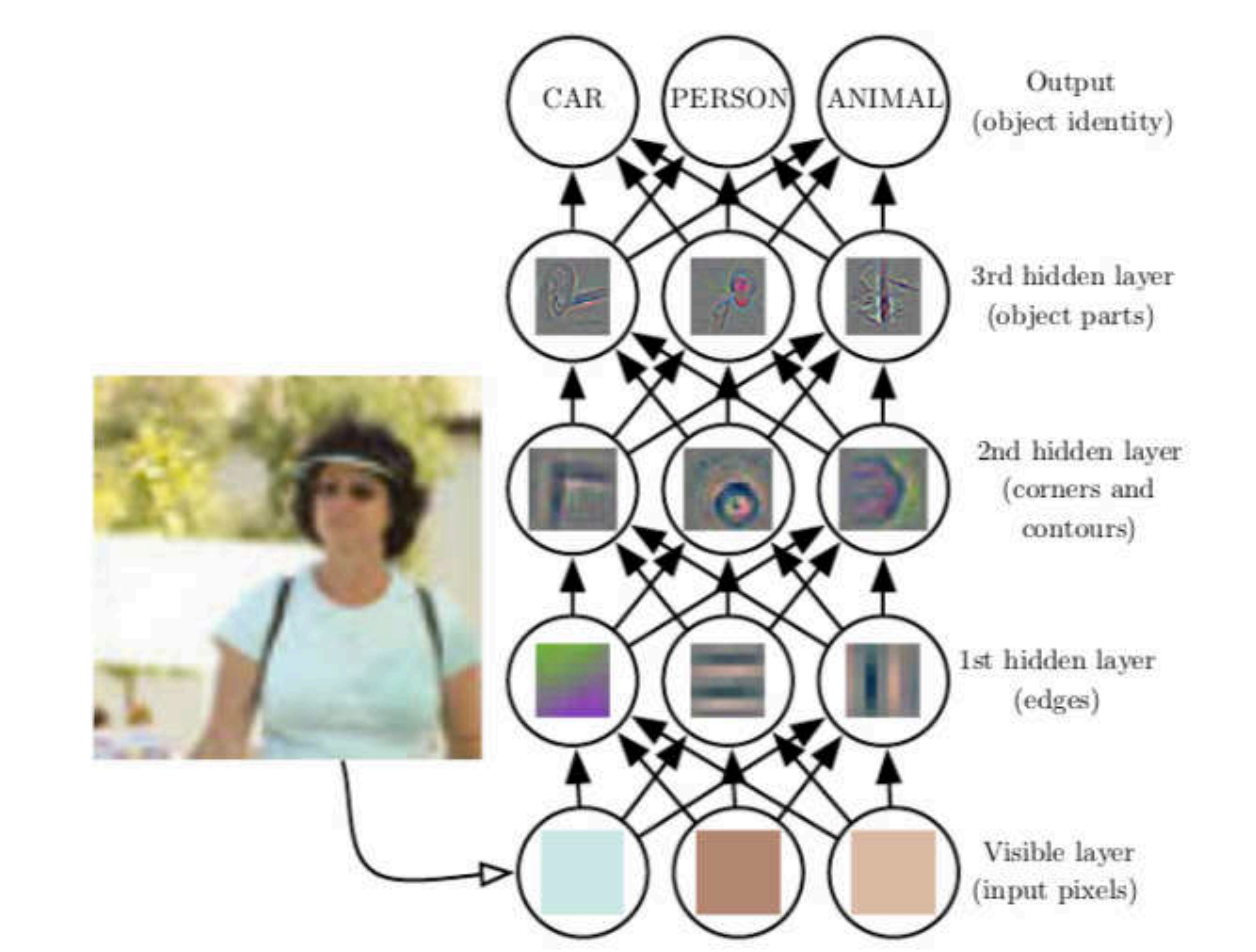
Representational Learning

In ML there are three approaches to force a machine to "learn"

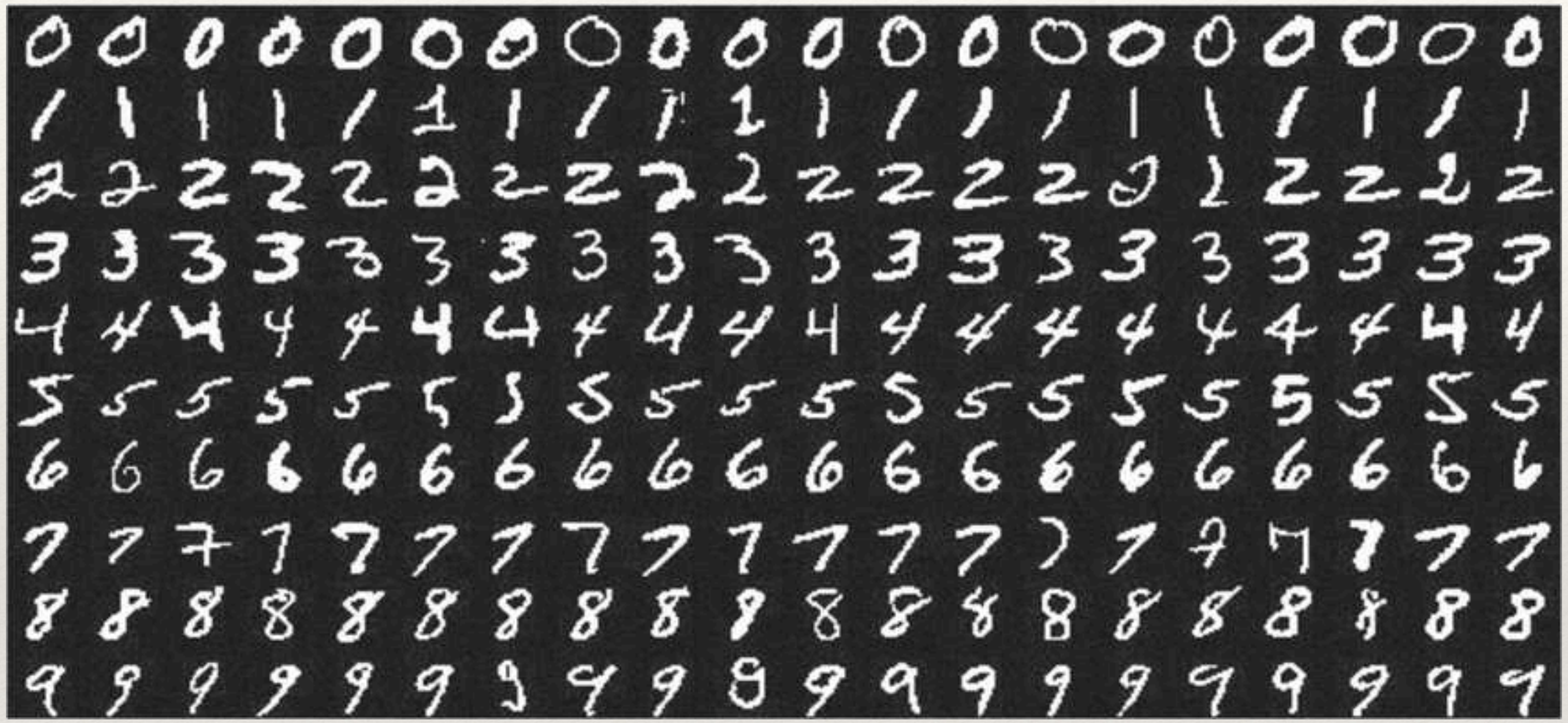
- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

Semi Supervised Learning

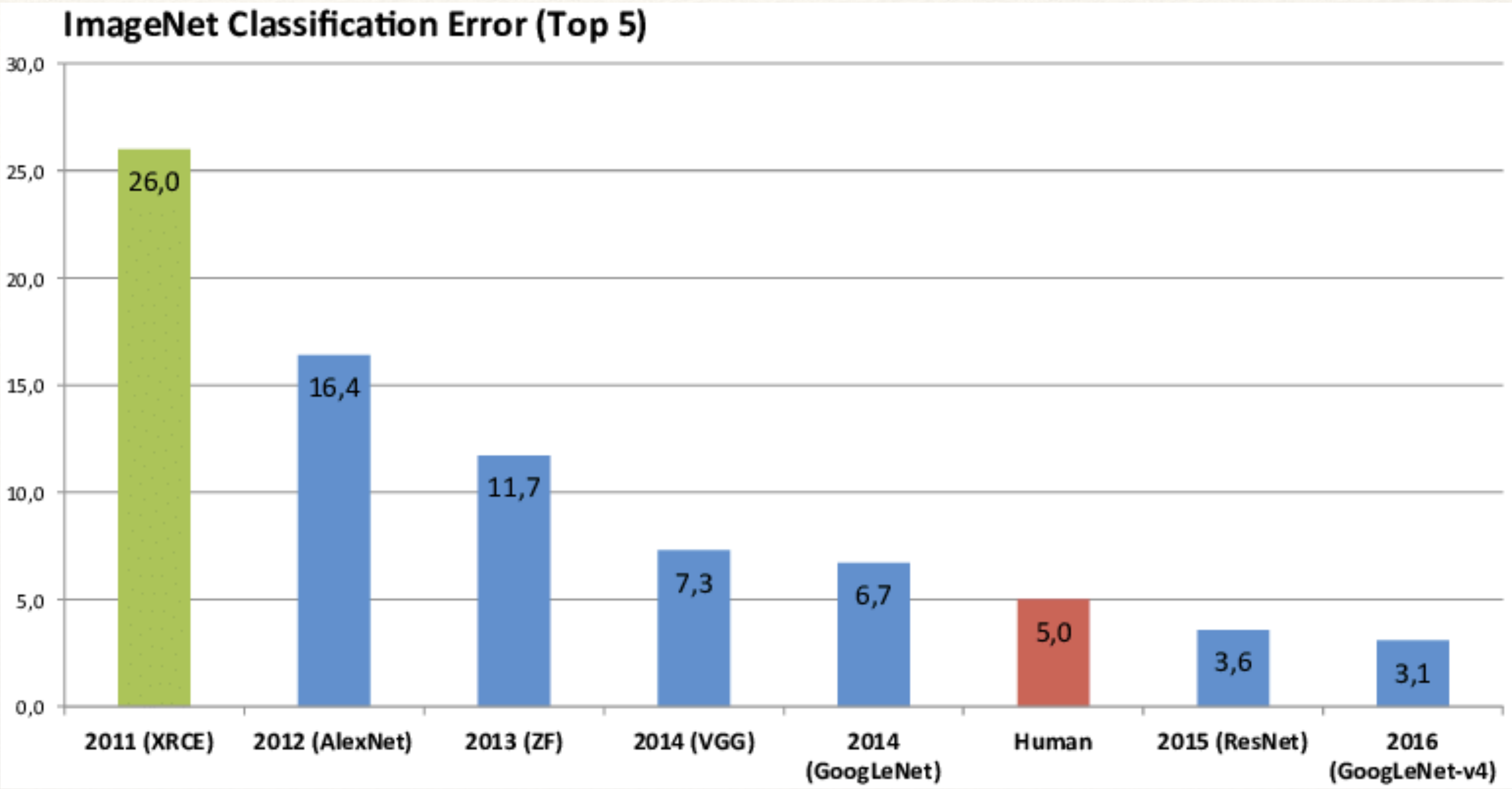
Self Supervised Learning



Massive Datasets



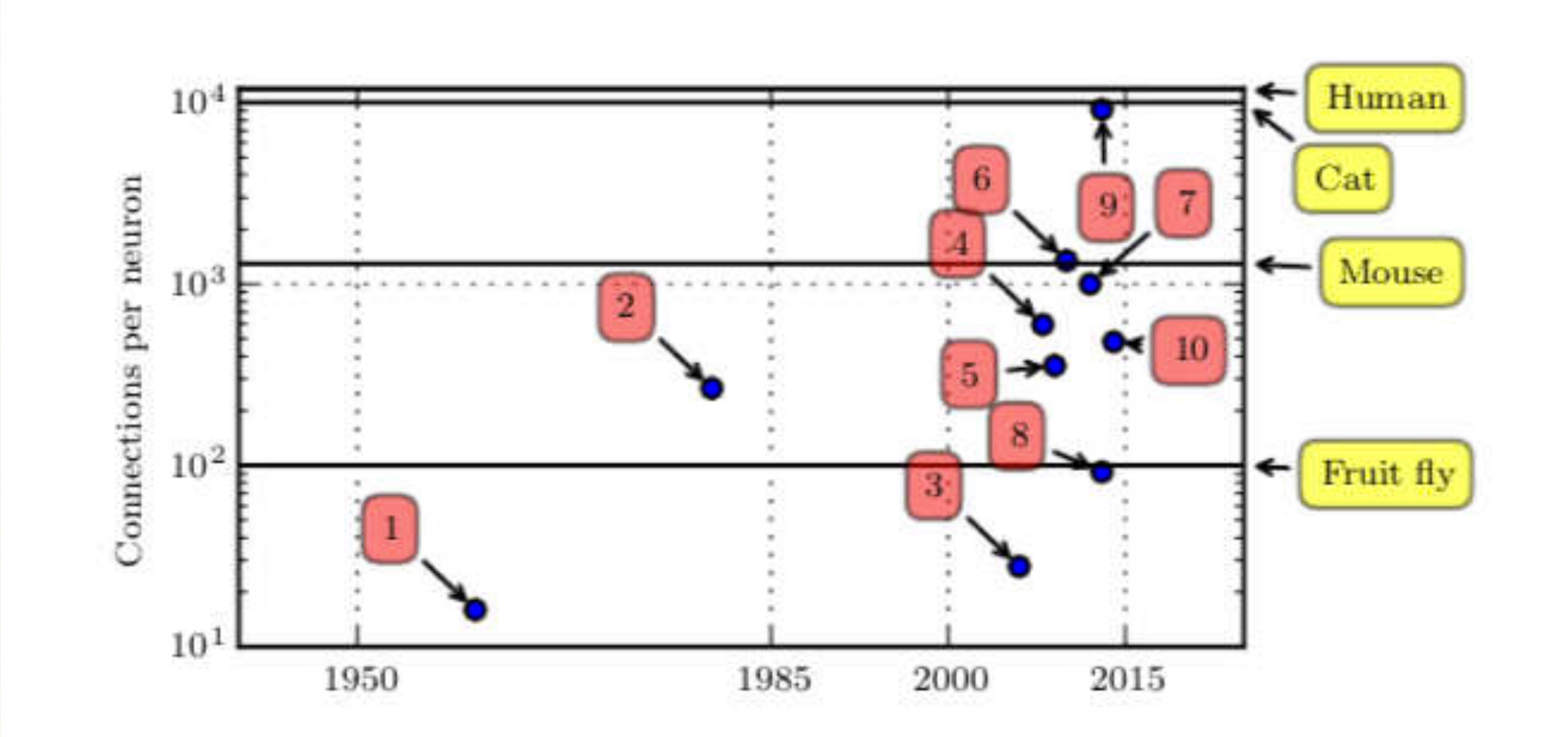
MNIST handwritten digit dataset



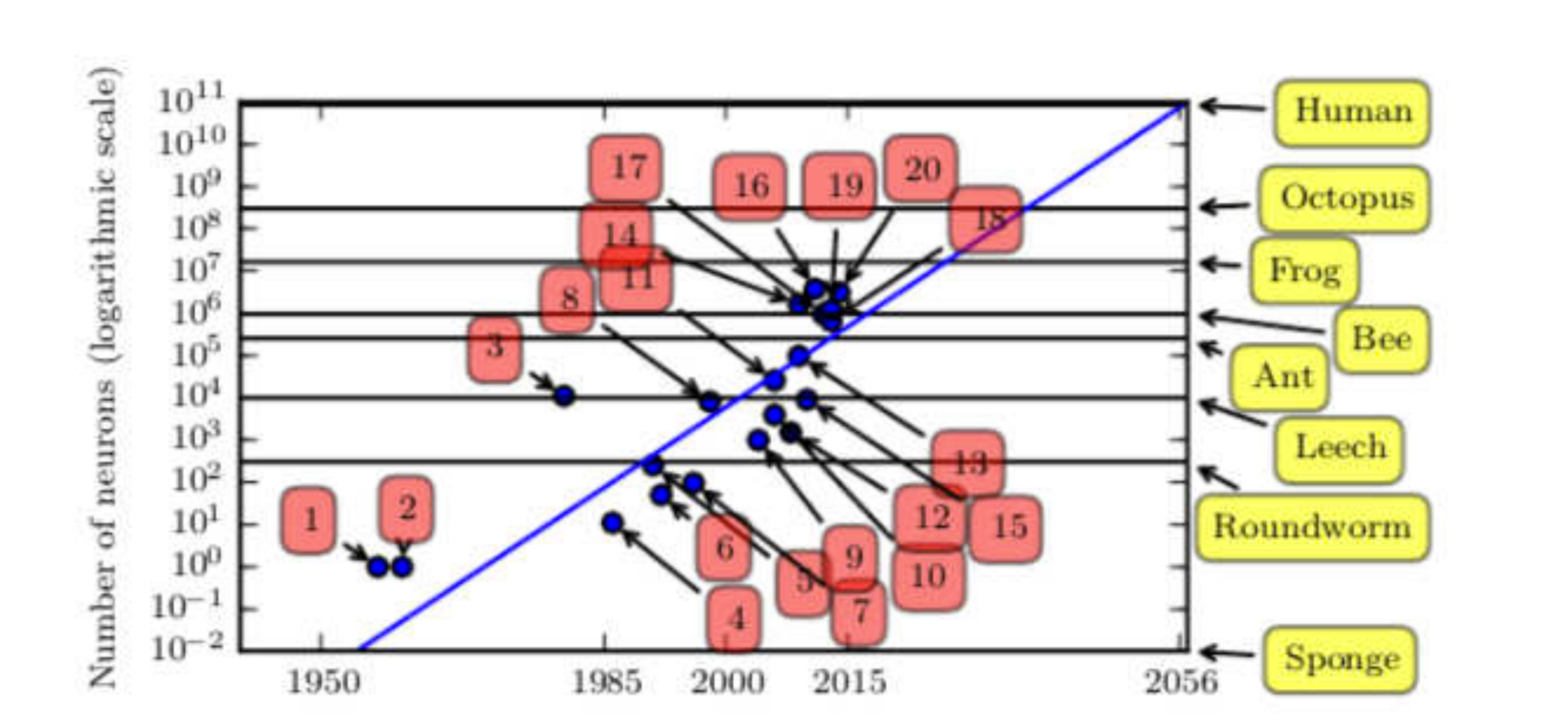
ImageNet Classification Error

Iris 10^2 , ImageNet 5000 images per category, WMT 10^9 total size,

Mammalian Brain Like?

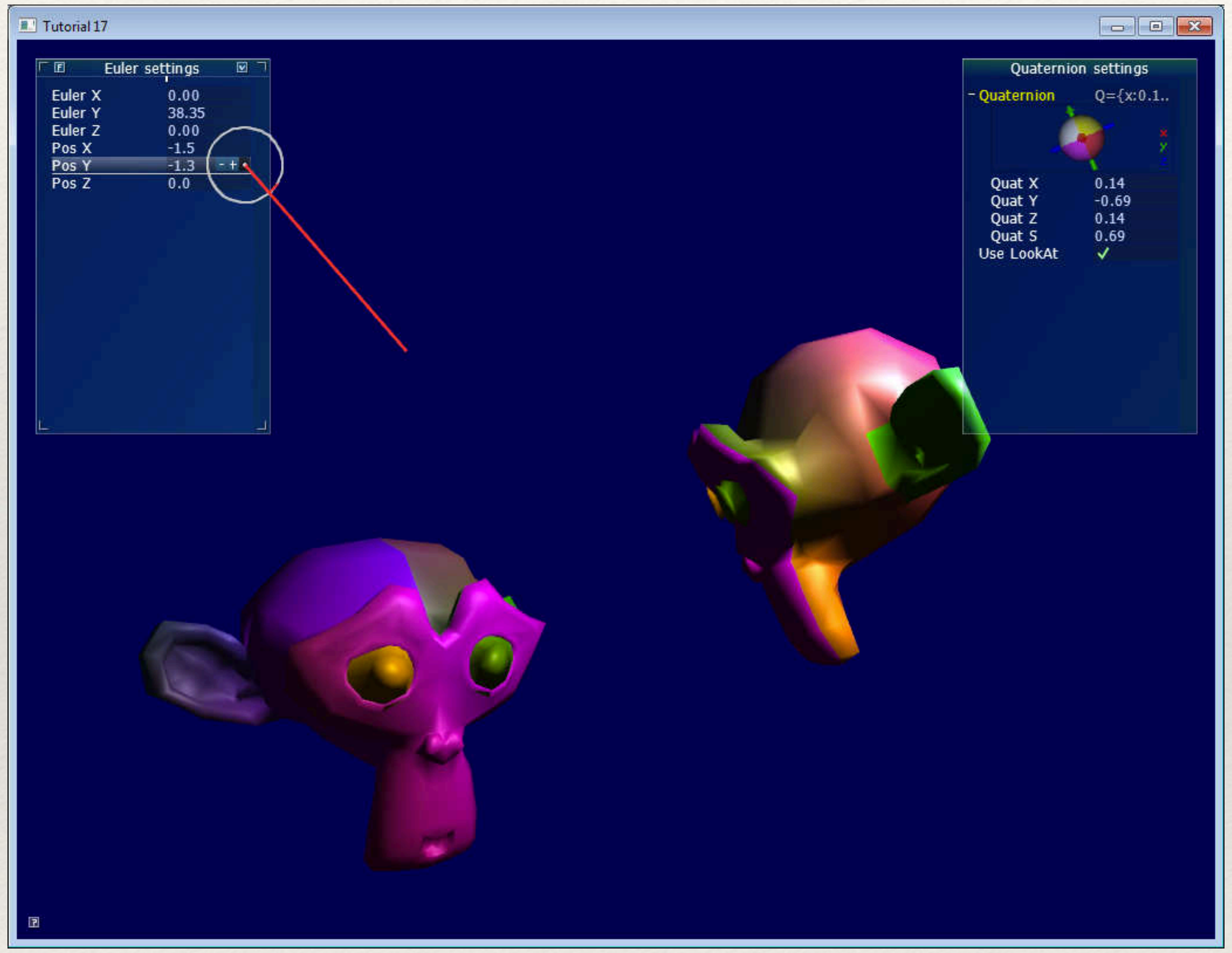


[9] GoogLeNet Szegedy 2014



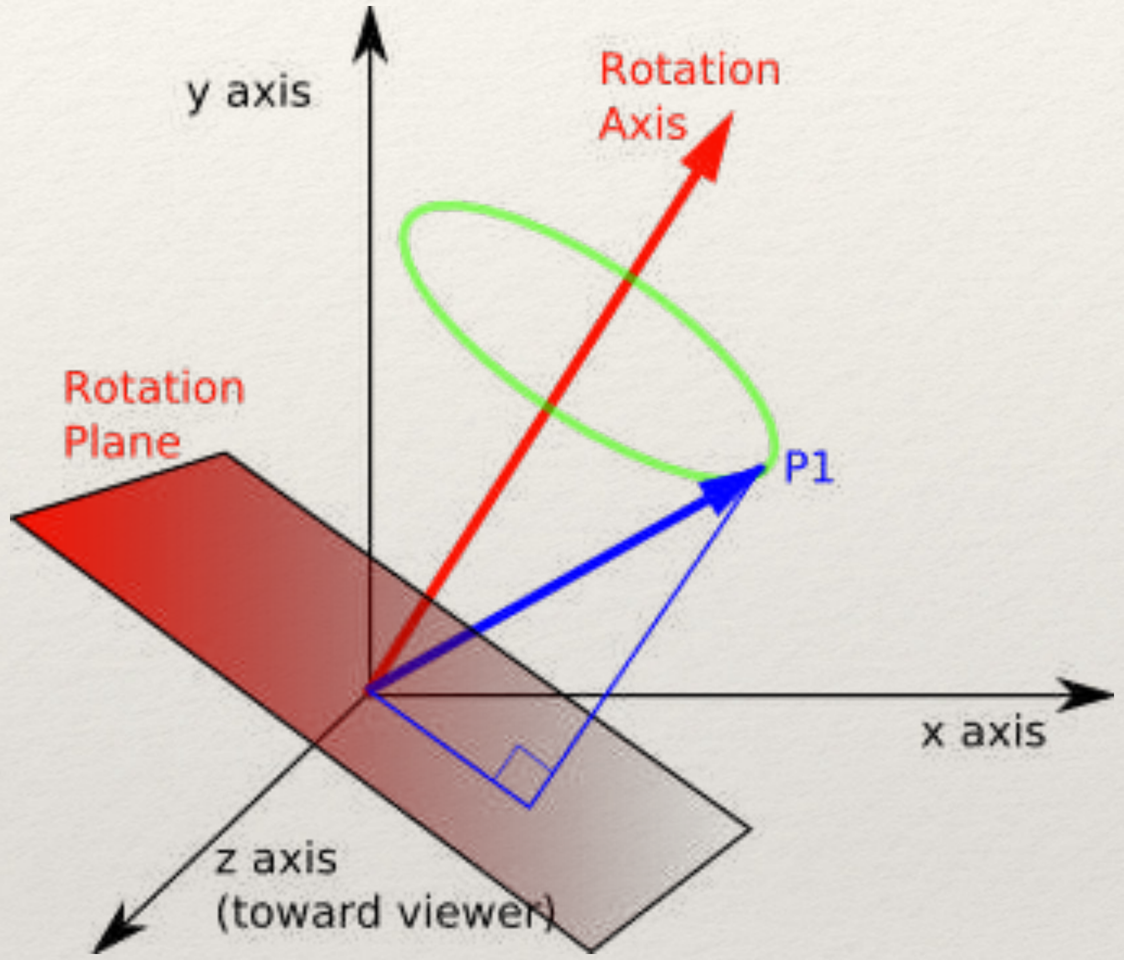
[18] MultiGPU ConvNet Krizhevsky 2012

GPU



CUDA | OpenCL

Shaders

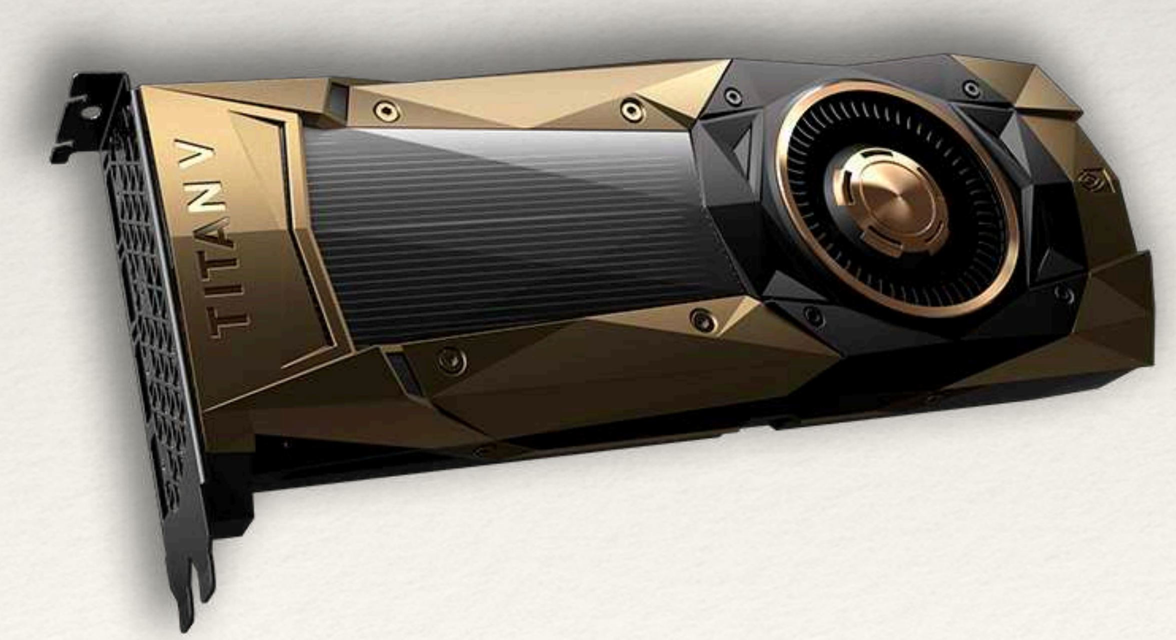


GPUs

Google Coral Dev Board

Nvidia Jetson Tk1

Intel Movidius Neural Stick



Nvidia Titan V

Nvidia Jetson Nano. RPi para GPU, Hardware Lottery [44]

Money is hardware

ImageNet Training in 24 Minutes

Yang You, Zhao Zhang, James Demmel, Kurt Keutzer, Cho-Jui Hsieh

(Submitted on 14 Sep 2017)

Finishing 90-epoch ImageNet-1k training with ResNet-50 on a NVIDIA M40 GPU takes 14 days. This training requires 10^18 single precision operations in total. On the other hand, the world's current fastest supercomputer can finish 2 * 10^17 single precision operations per second (Dongarra et al 2017). If we can make full use of the supercomputer for DNN training, we should be able to finish the 90-epoch ResNet-50 training in five seconds. However, the current bottleneck for fast DNN training is in the algorithm level. Specifically, the current batch size (e.g. 512) is too small to make efficient use of many processors

For large-scale DNN training, we focus on using large-batch data-parallelism synchronous SGD without losing accuracy in the fixed epochs. The LARS algorithm (You, Gitman, Ginsburg, 2017) enables us to scale the batch size to extremely large case (e.g. 32K). We finish the 100-epoch ImageNet training with AlexNet in 24 minutes, which is the world record. Same as Facebook's result (Goyal et al 2017), we finish the 90-epoch ImageNet training with ResNet-50 in one hour. However, our hardware budget is only 1.2 million USD, which is 3.4 times lower than Facebook's 4.1 million USD.

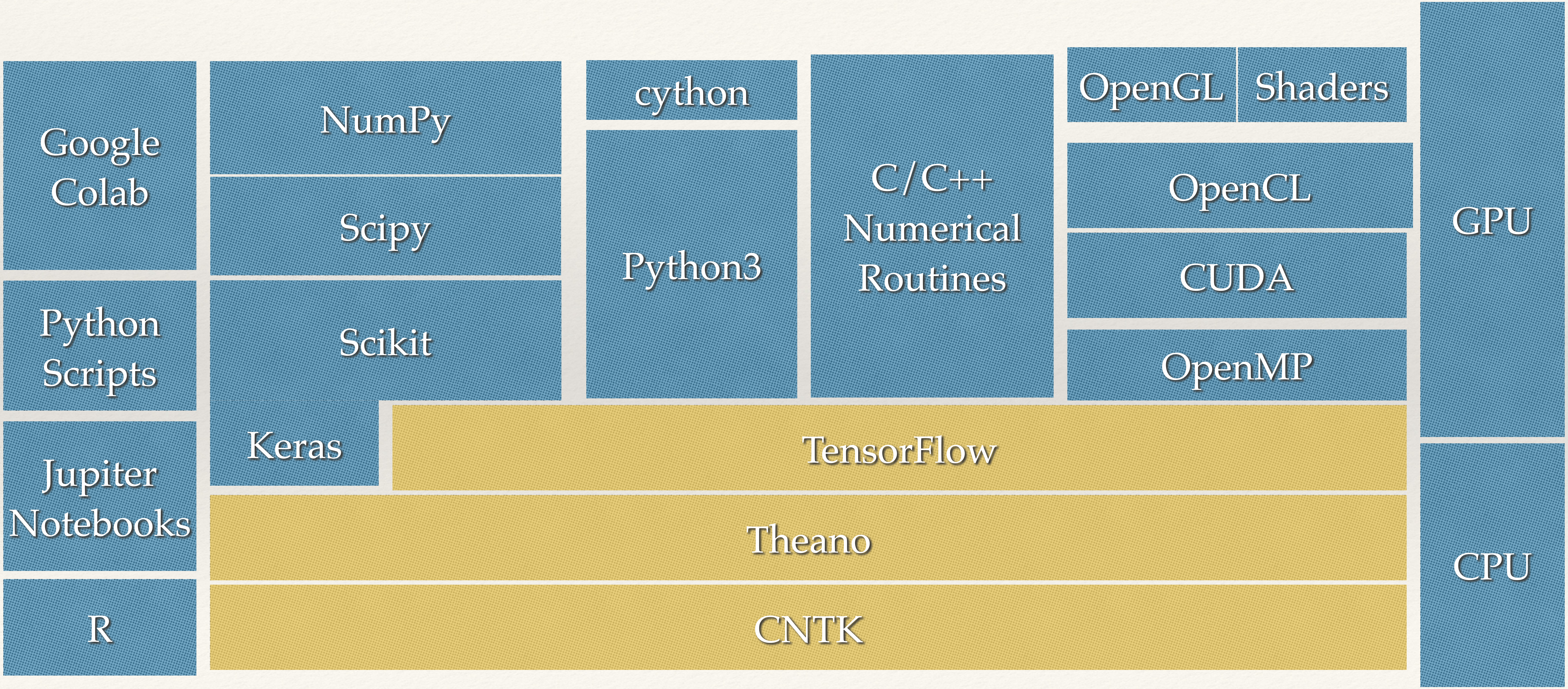
Software and Cloud Based Tools

- **Amazon**: Machine Learning, DSSTNE, Orbeus
- **Facebook**: TorchLibrary
- **Google**: Cloud Machine Learning, TensorFlow
- **IBM**: Watson Analytics, IBM SystemsML (AlchemyAPI)
- **Microsoft**: Azure Machine Learning, Computational Network Toolkit (SwiftKey)

- OpenAI
- Theano
- PyLearn2
- Torch
- DistBelief
- Caffe
- MXNet
- Tensorflow
- Keras

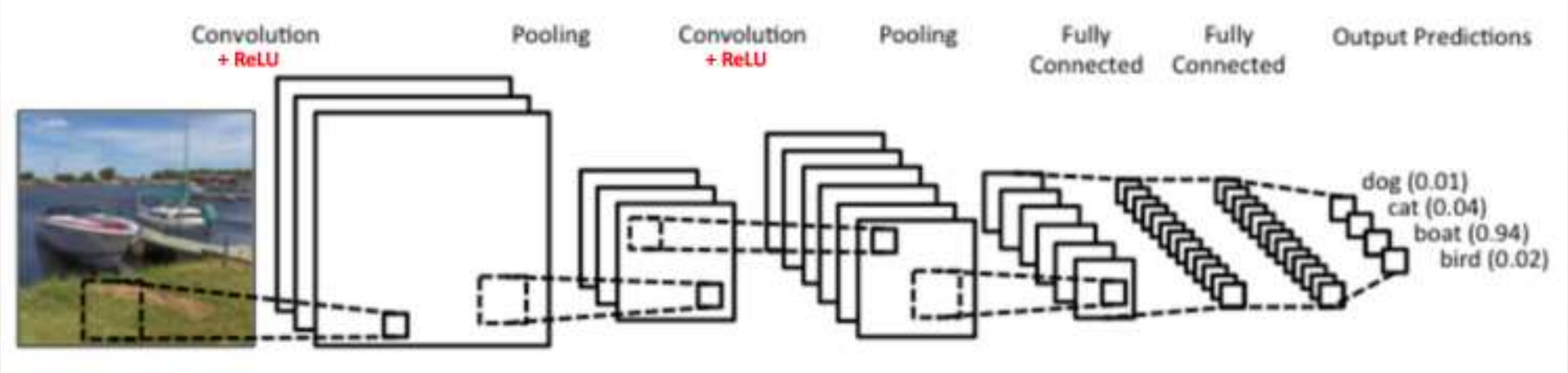
https://spectrum.ieee.org/computing/software/now-you-too-can-buy-cloudbased-deep-learning

Keras and TensorFlow

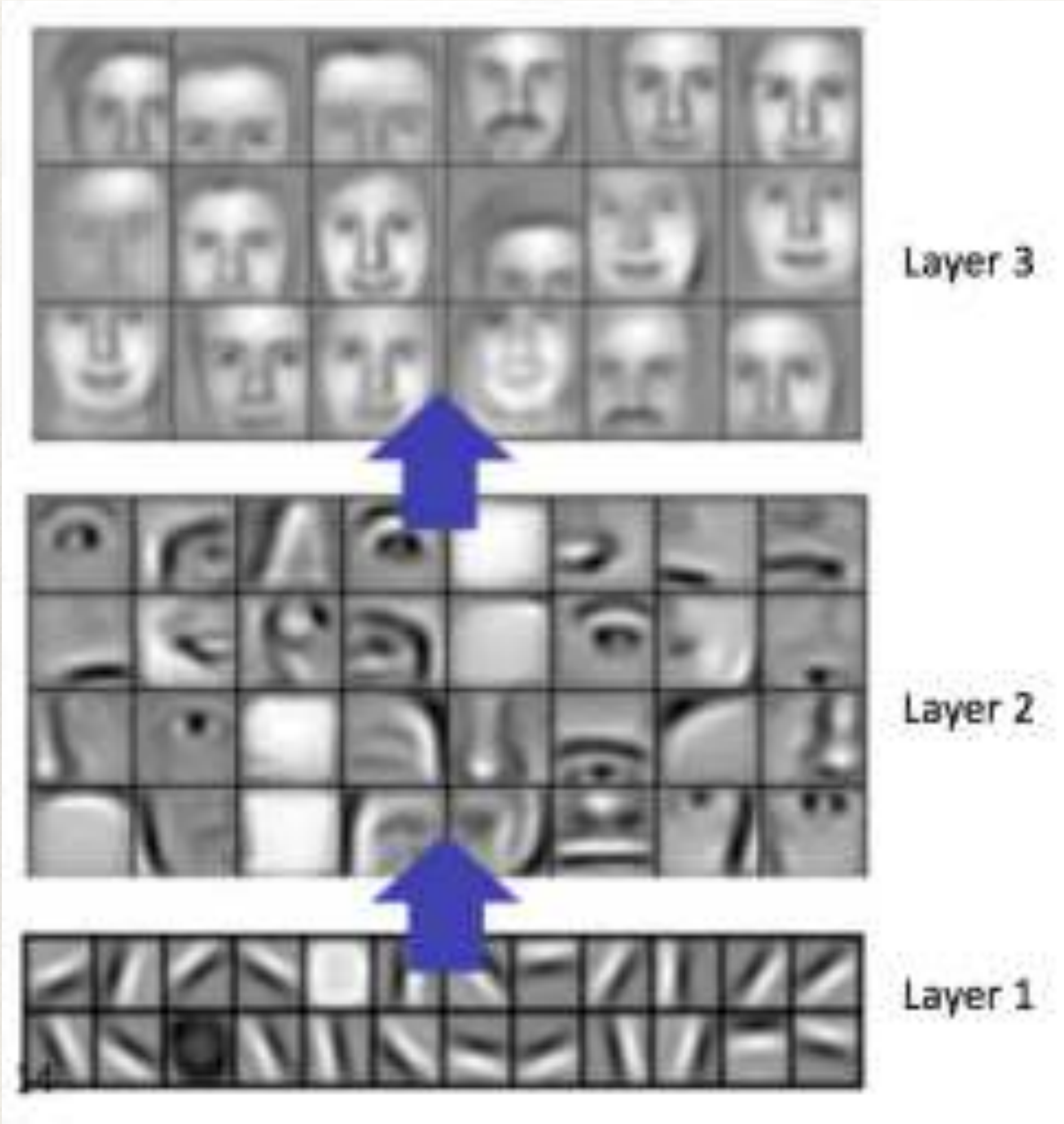


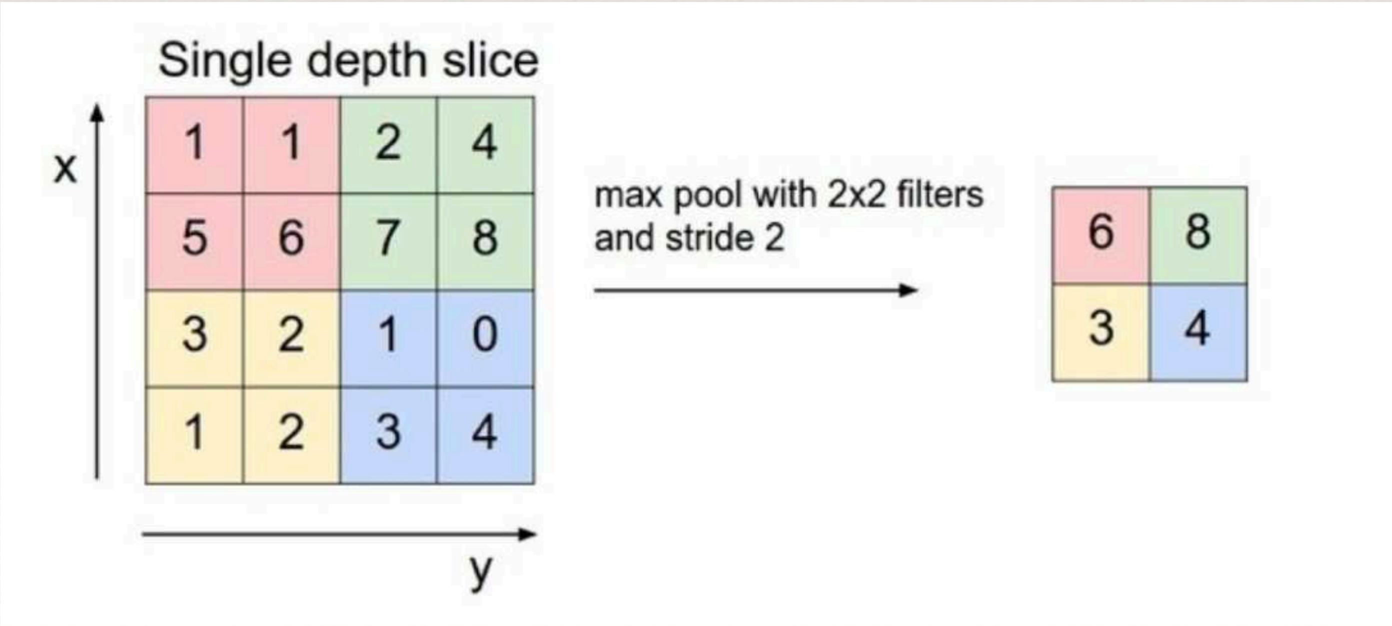
Deep Learning Networks

ConvNet - Convolutional Networks



Representational Learning



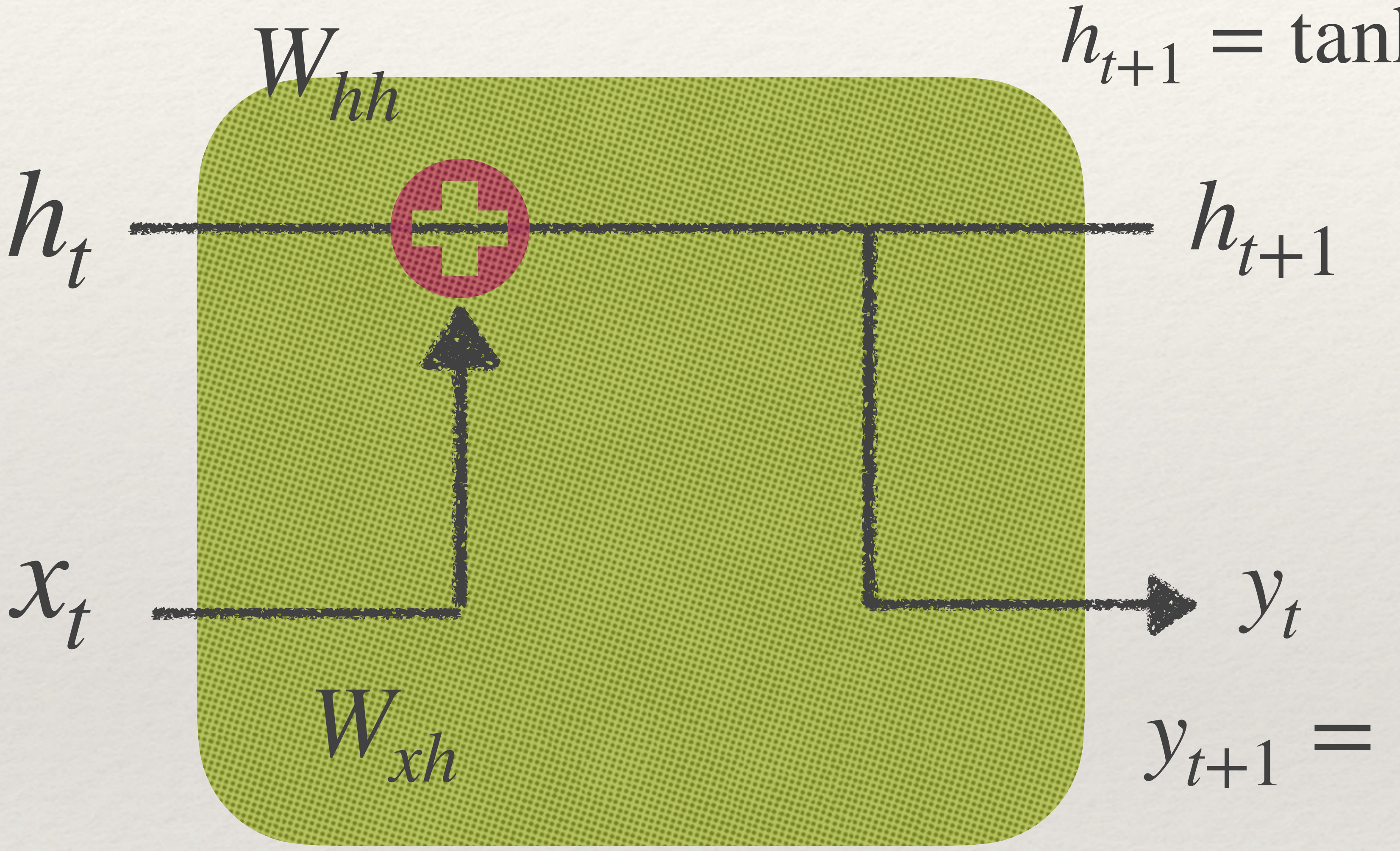


- Pooling = downsampling
- ReLU: Rectified Linear Neurons

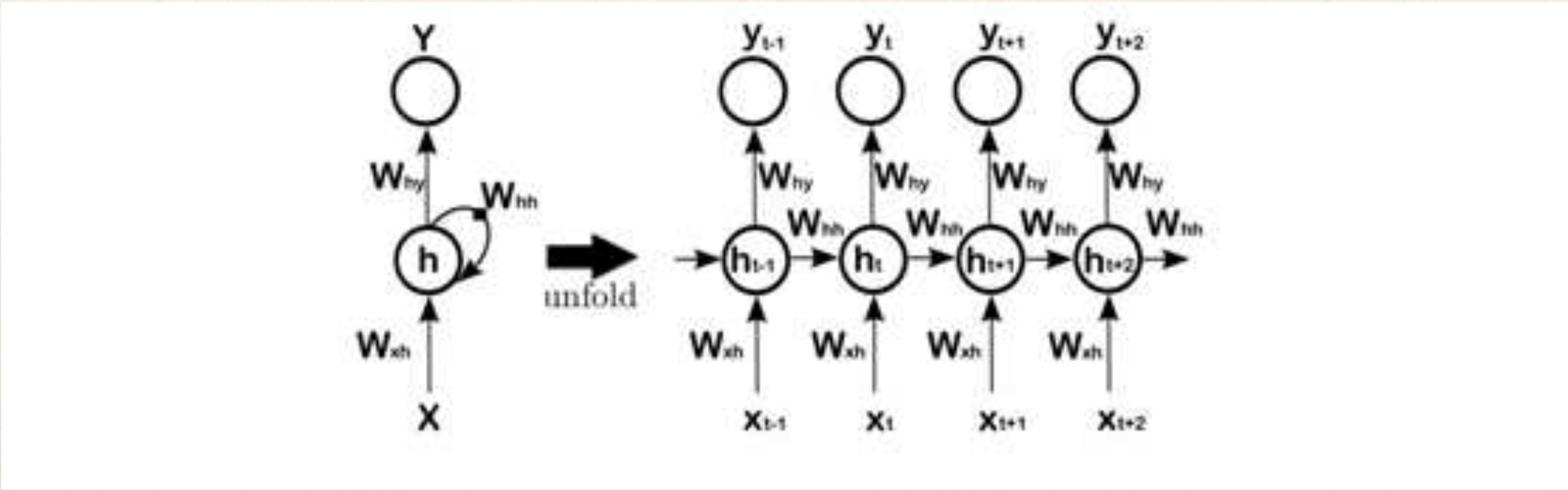
Fukushima 1980, Krizhevsky 2012

RNN - Recurrent Neural Network

$$h_{t+1} = \tanh(W_{xh}x_t + W_{hh}h_t)$$



$$y_{t+1} = g(h_t)$$



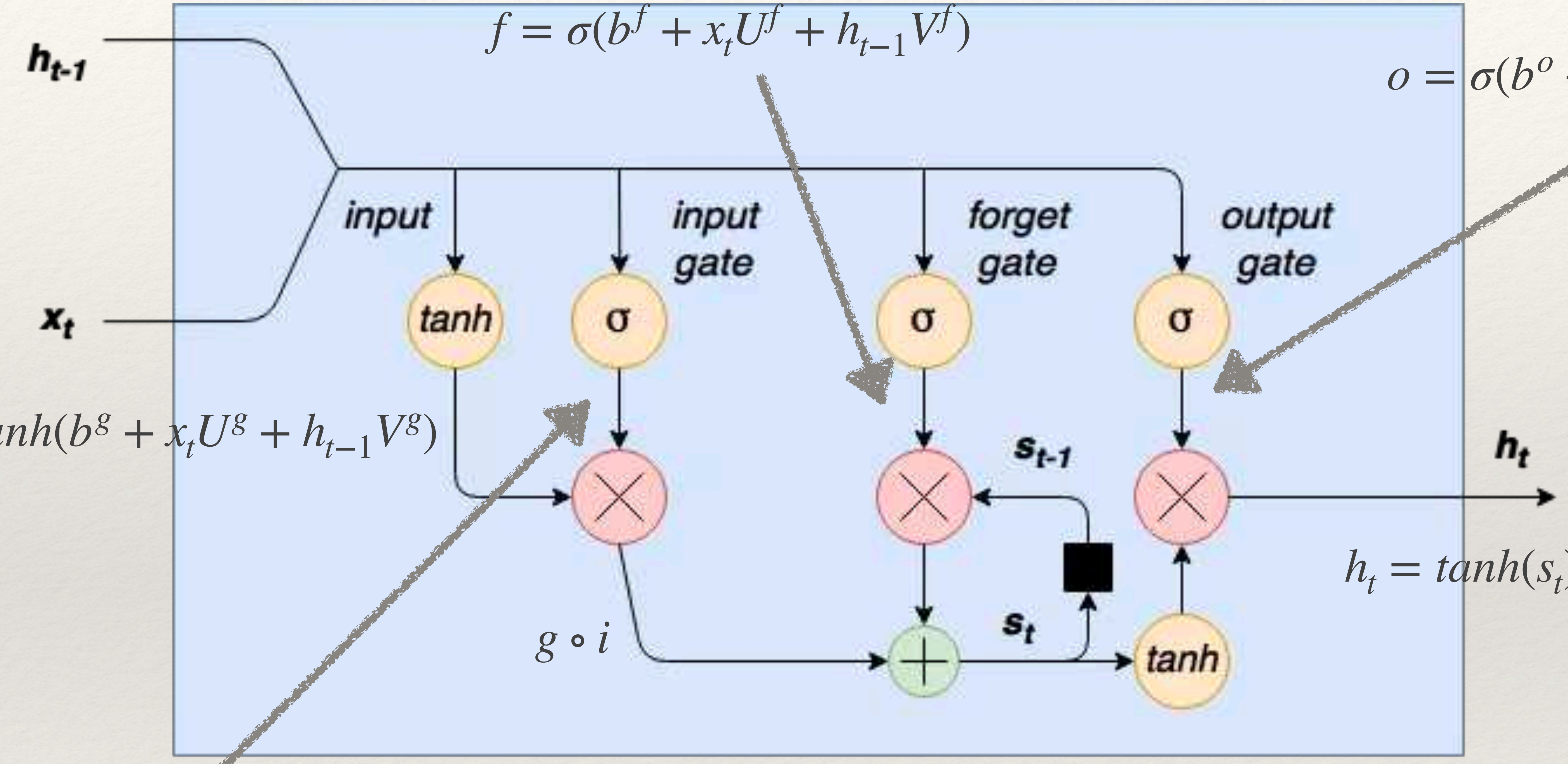
Time unfolding

"On the Computational Power of Neural Nets" Siegelmamn 1995, Chen 2016

LSTM - Long Short Term Memory

$f = \sigma(b^f + x_t U^f + h_{t-1} V^f)$

$o = \sigma(b^o + x_t U^o + h_{t-1} V^o)$



$g = tanh(b^g + x_t U^g + h_{t-1} V^g)$

$h_t = tanh(s_t) \circ o$

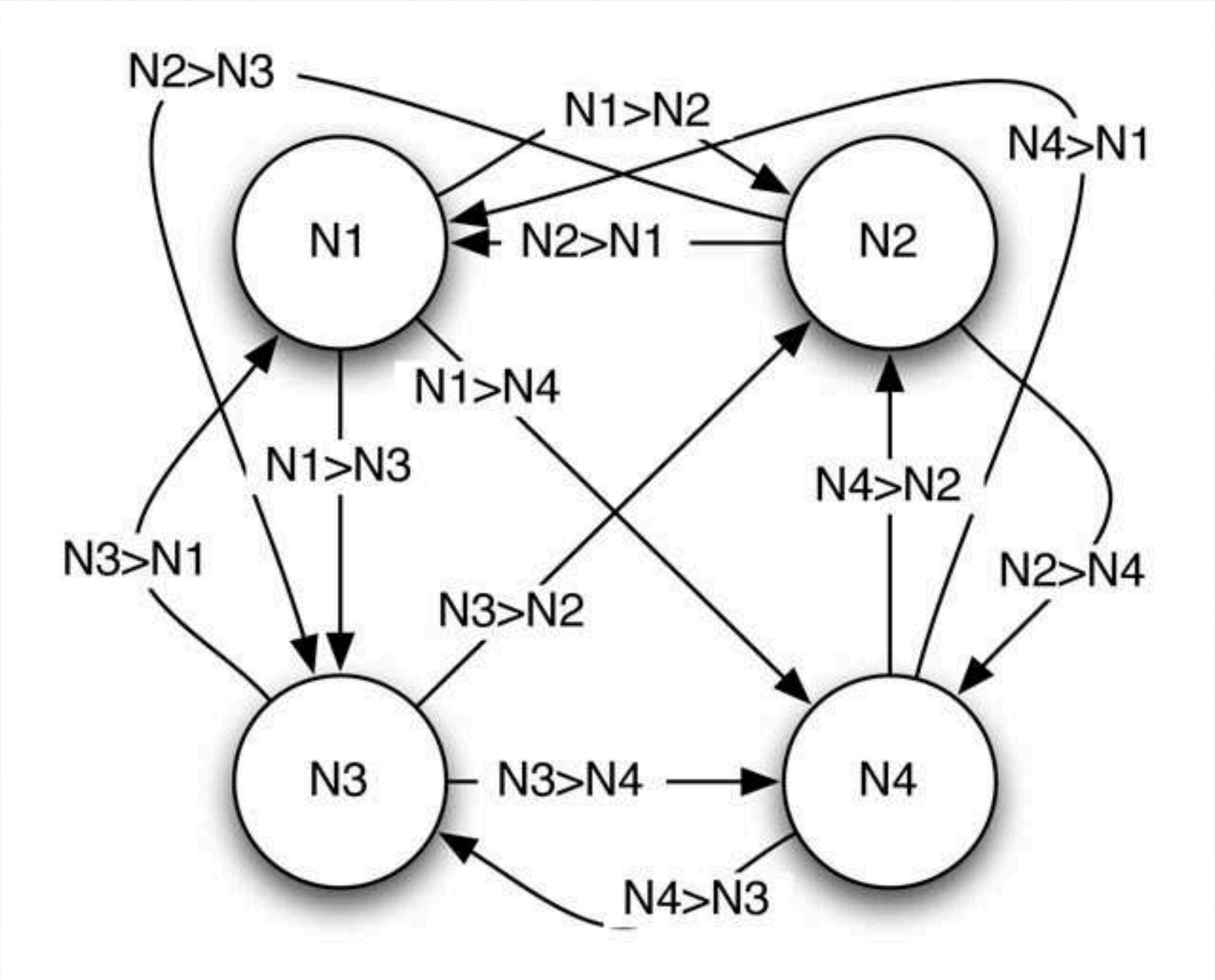
$g \circ i$

$s_t = s_{t-1} \circ f + g \circ i$

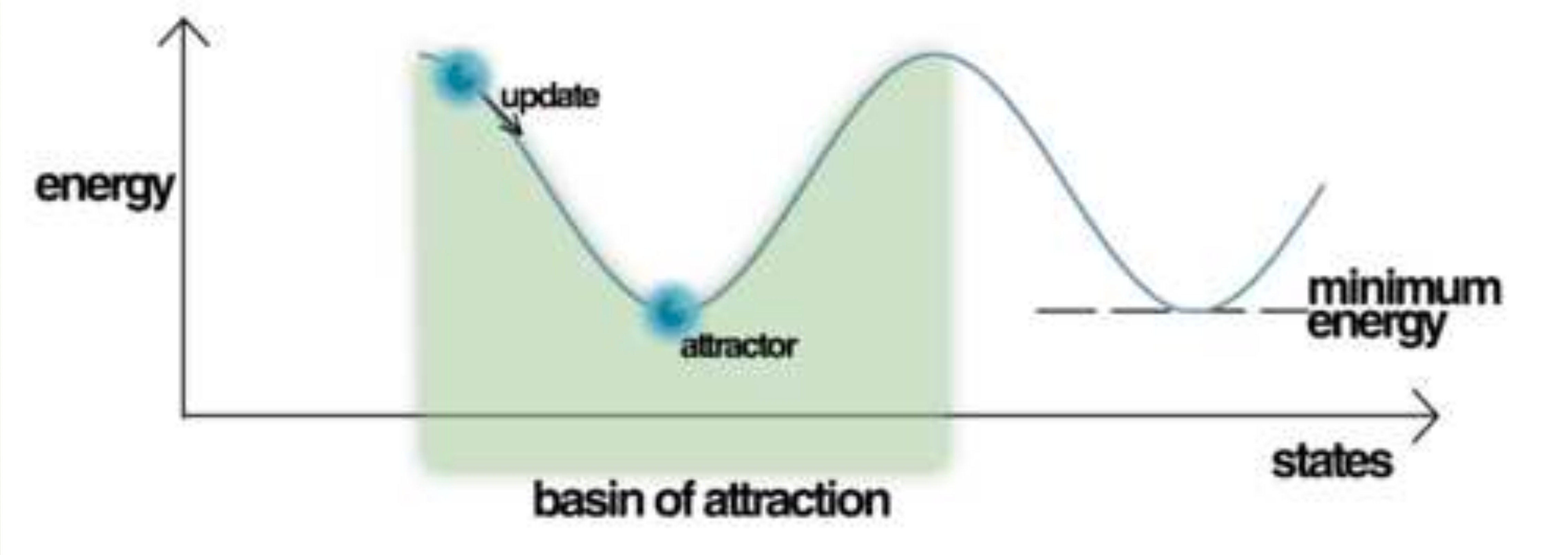
$i = \sigma(b^i + x_t U^i + h_{t-1} V^i)$

Hochreiter & Schmidhuber 1997, Chen 2016

DBN - Deep Belief Networks

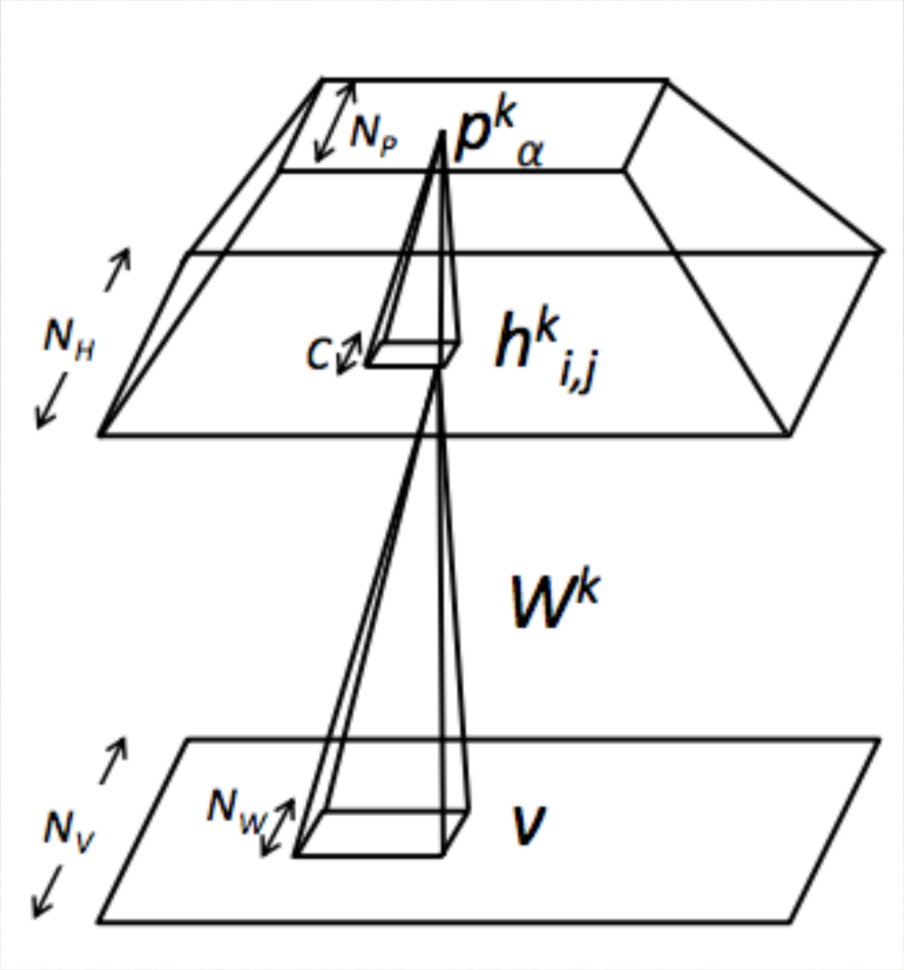


Hopfield Neural Network



$$E = \sum_i W_{ij} XY$$

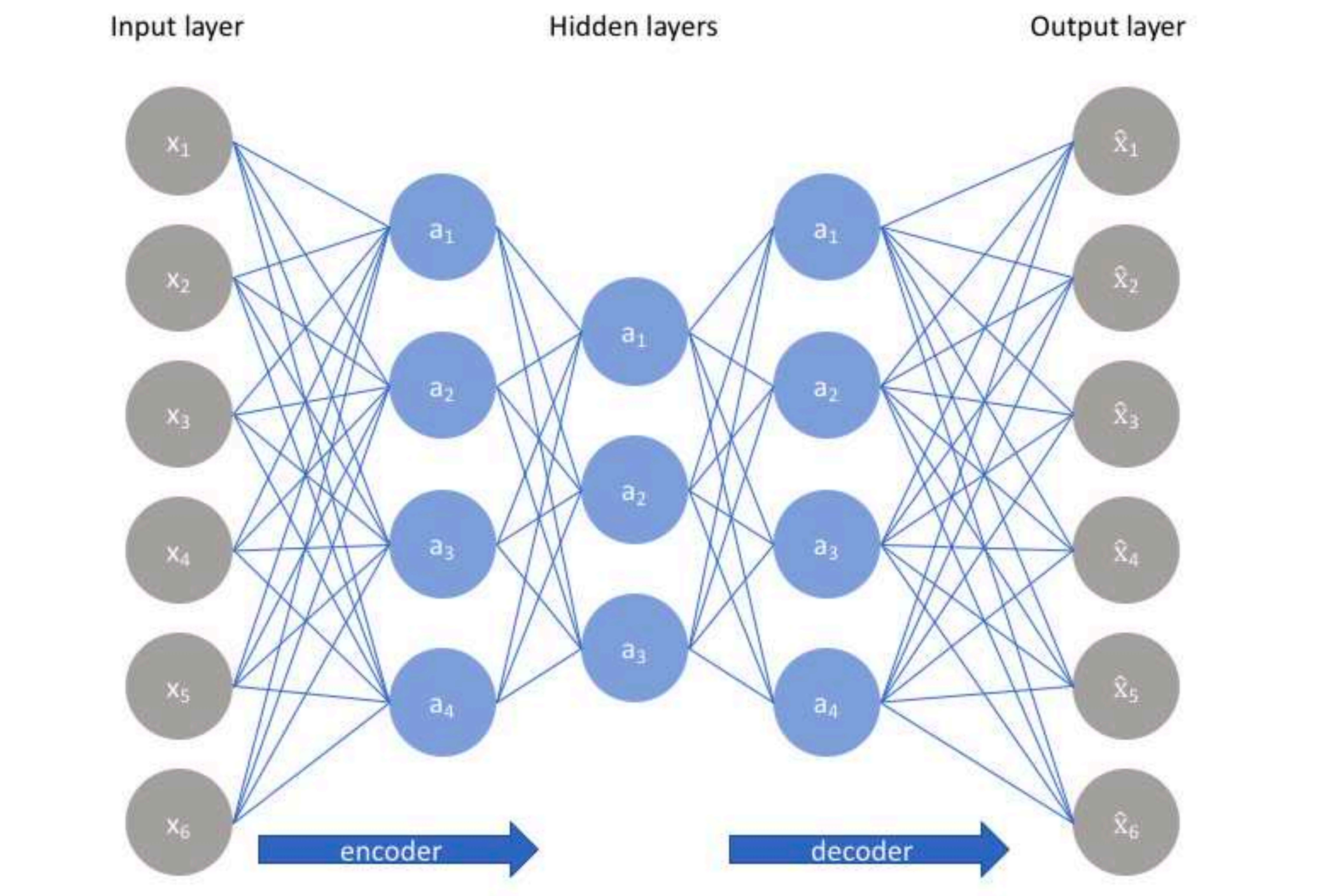
Hebian Energy Function



Stacked Boltzmann Machine

“Deep Belief Nets” Hinton 1995

Autoencoders



Dimensionality Reduction
PCA

Sparse Autoencoder
(Kohonnen)

Denoising Autoencoder

Contracting Autoencoder

Variational Autoencoders Generators

https://colab.research.google.com/drive/1Hobi6plfrrUQ9DCiFPZ6vaztOu5Q05ND

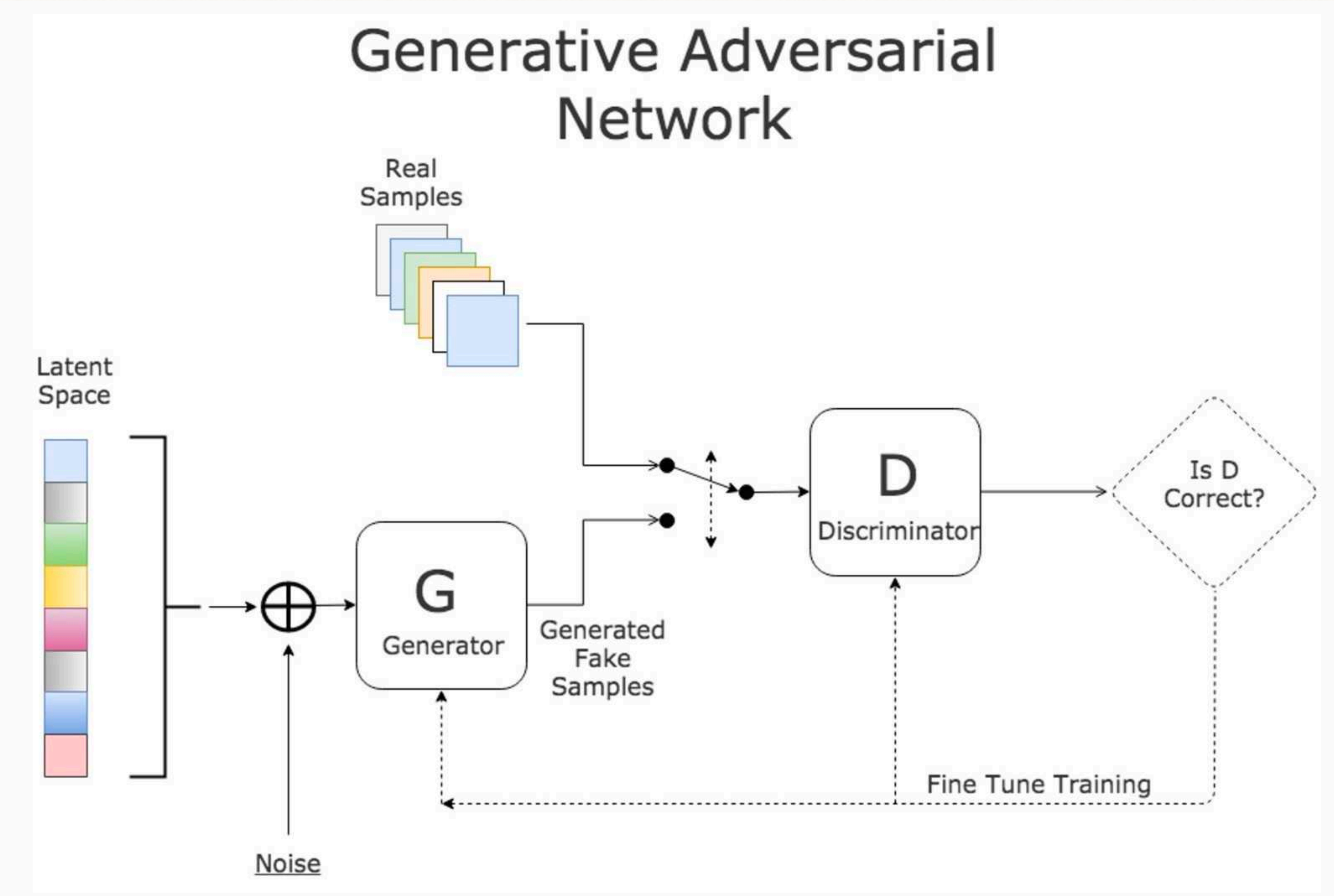
GAN - Generative Adversarial Networks

Playing a counterfeit game.

G network is trained to increase the chance of D assuming and output of G being valid.

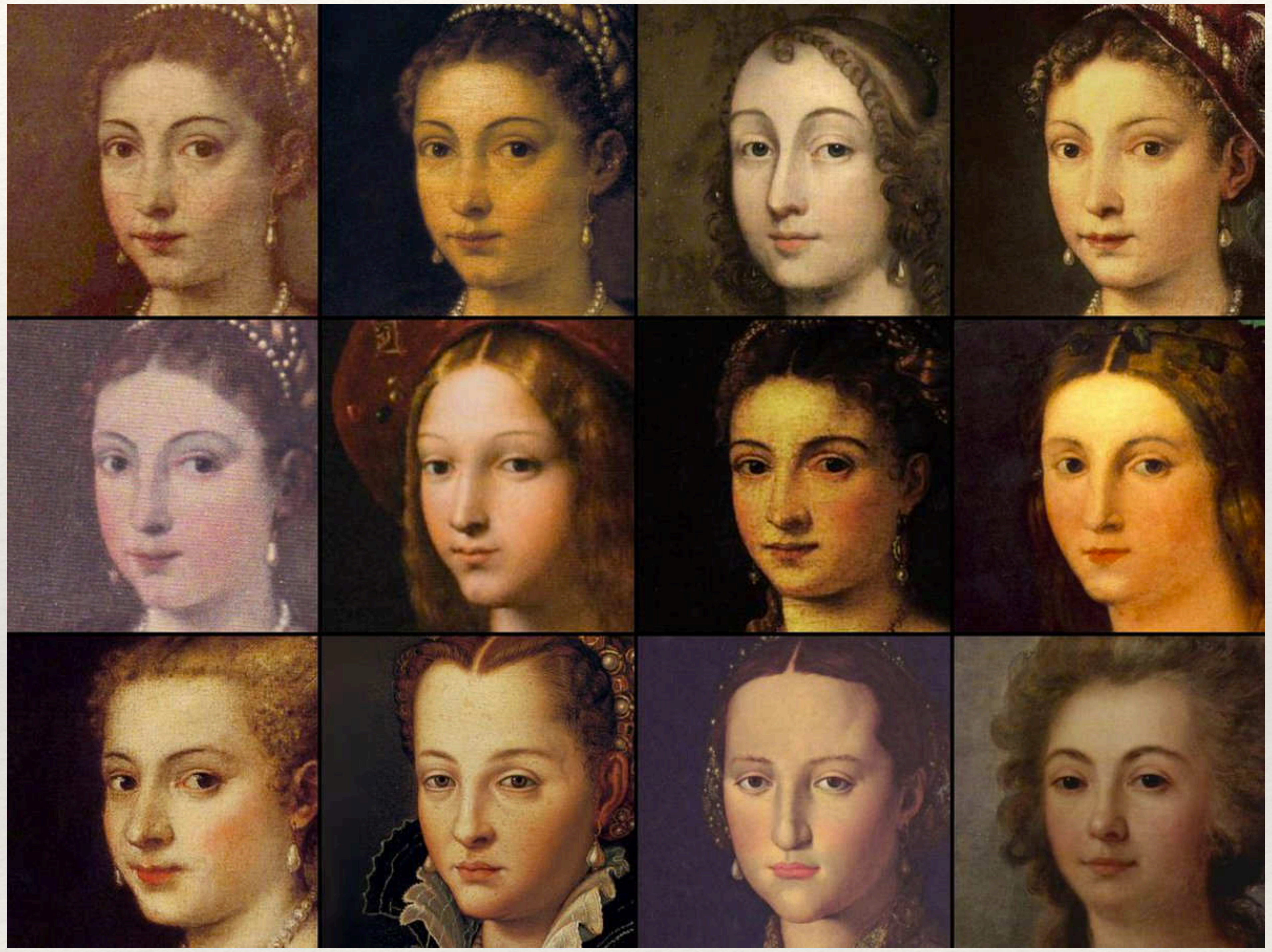
D network is trained to decrease the chance of assuming G output to be valid.

$$\min_G \max_D V(D, G) = E[\log D(x)] + E[\log(1 - D(G(z))]$$



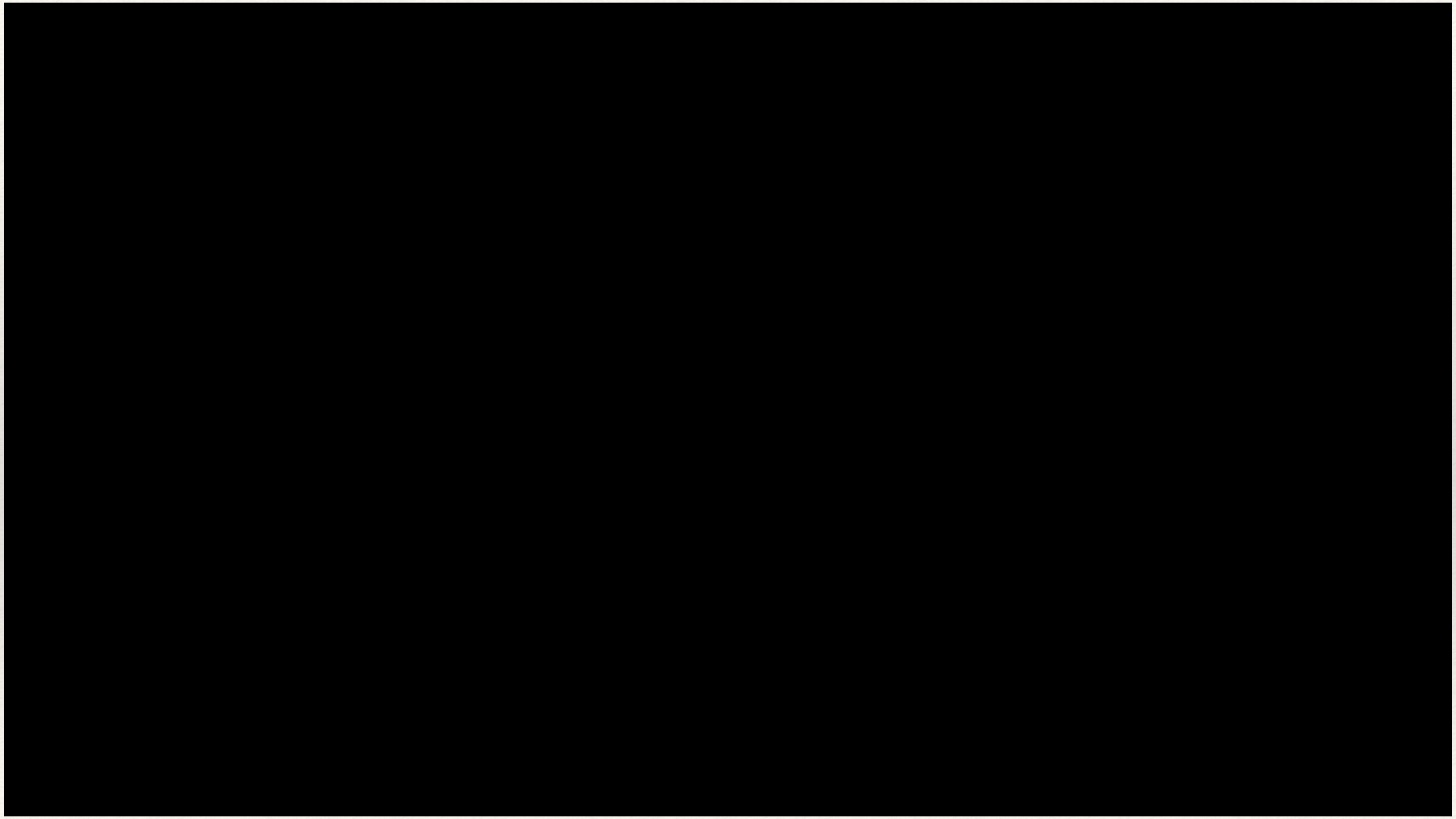
Goodfellow 2014

GAN - Generative Adversarial Networks



Mario Klingermann

GAN - Generative Adversarial Networks



Ming-Yu Liu et al https://arxiv.org/pdf/1905.01723.pdf

Computer Art

COMPUTERWORLD September 11, 1968

Computer Art Is Called Genuine New Art Form

Computer art, drawn by a plotter under the direction of a computer, has become something of a fad this year, and at least one expert thinks such art represents a genuine new art form.

The computer has definite potential as an art medium and should be thoroughly explored, according to Dr. Thomas S. Fern, chairman of the Department of Art, University of Notre Dame.

Fern said the computer meets the first essential condition of art: it can serve as a medium for human expression. If computer plotting were merely a mechanical matter in which creativity did not play a role, then it would not be a valid art form, he explained.

Similar Art Forms

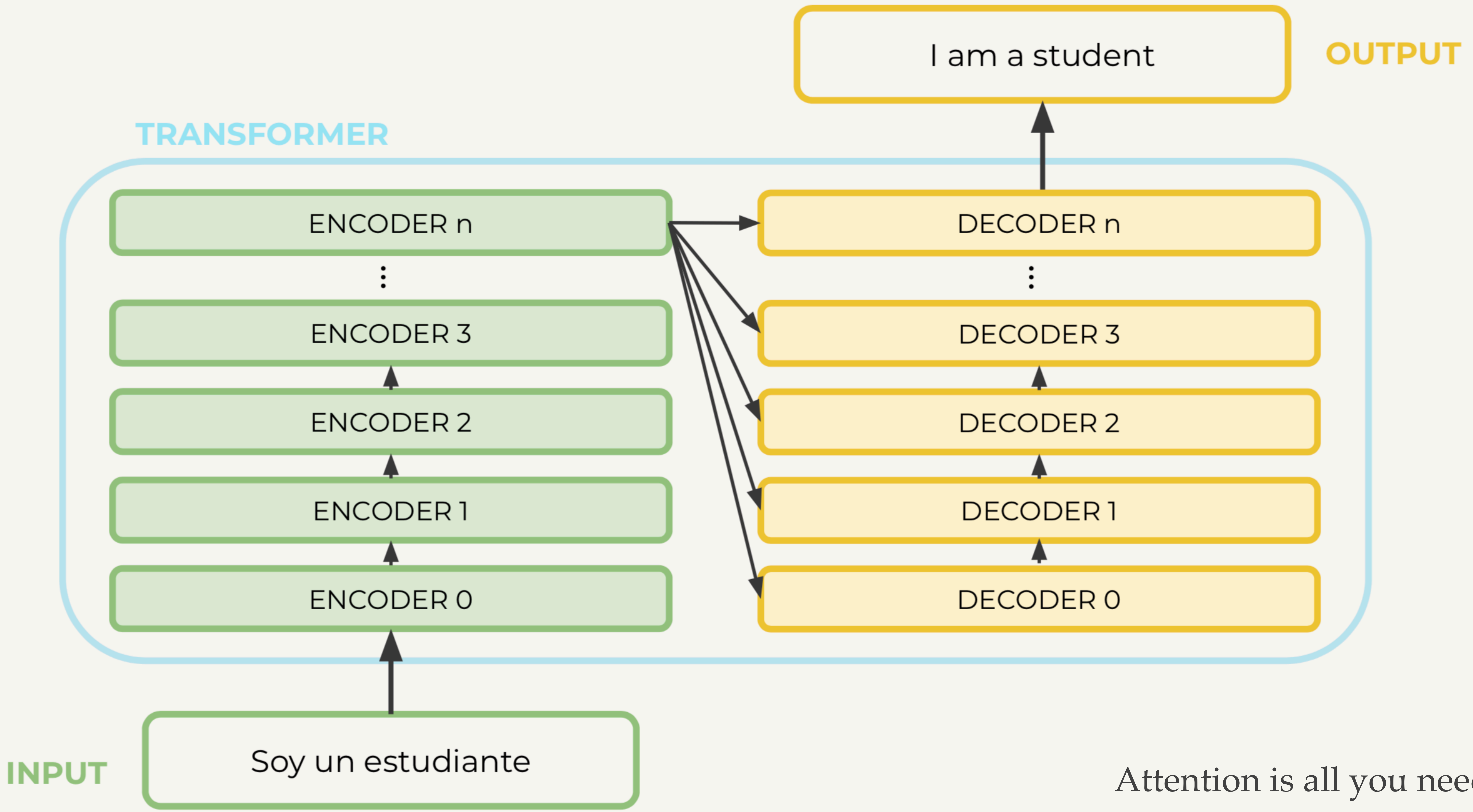
There are other forms of art in which the artist doesn't necessarily create a finished product with his own hands, Fern pointed out. For example, an architect visualizes the form and substance of a building but leaves the construction to someone else. The fact that the architect produces blueprints, or instructions to the builder, rather than the building itself, does not diminish his creative role. Logically then, Fern added, a human who instructs a computer also is creative and can be considered an artist.

In Fern's opinion, computer produced abstract drawings are more exciting than representational work.

"Orderliness and precision are the outstanding qualities of computer drawings," he observed. "I personally think it's more interesting to see such qualities used in an abstract way than for a literal representation of a scene."

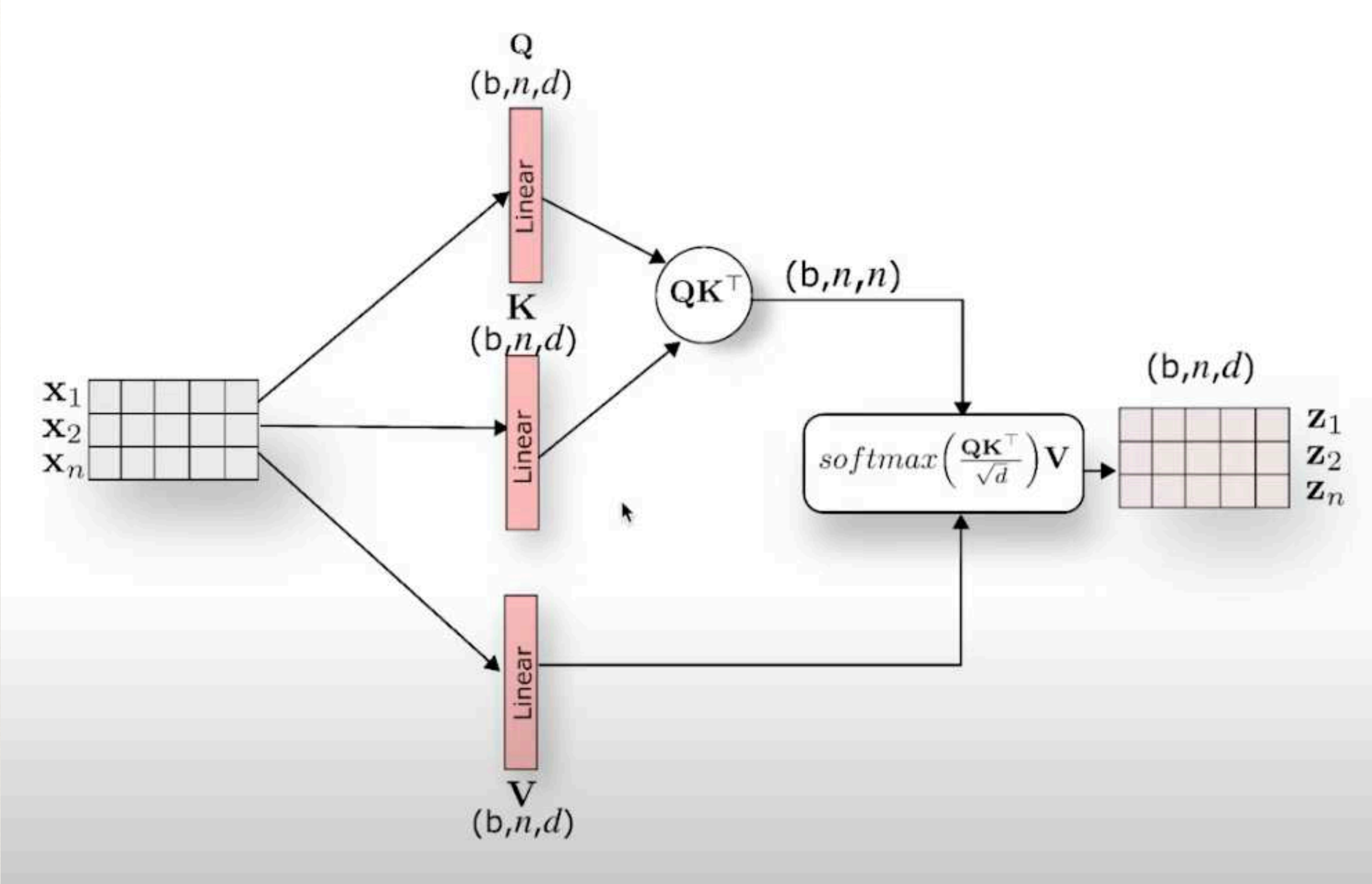
https://x.com/quasimondo/status/1706947023037927690?s=20

Transformers



Attention is all you need, Vaswani 2017

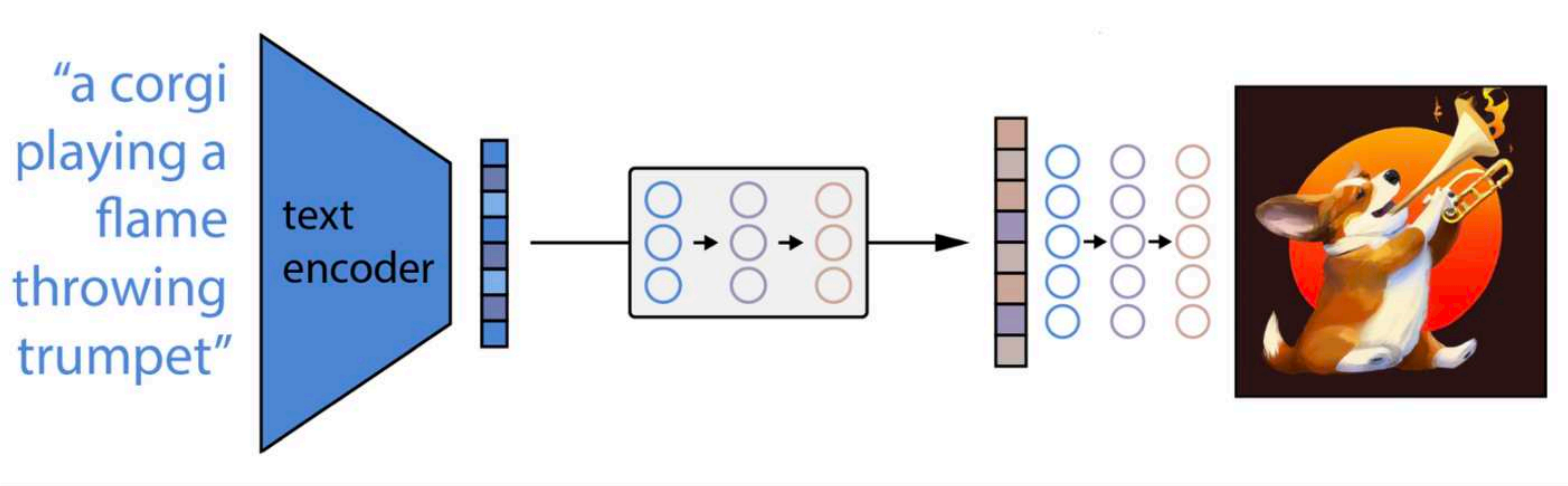
Transformers



Attention is all you need, Vaswani 2017

Diffusion Models

"a bowl of soup that is a portal to another dimension as digital art"

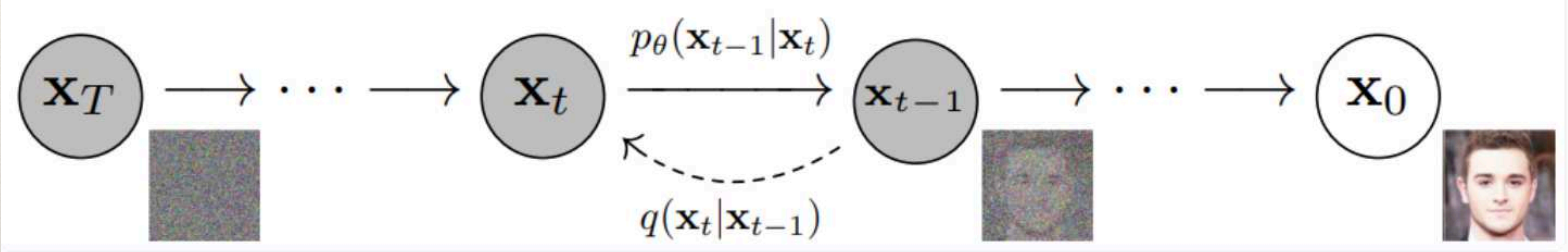


CLIP Contrastive Learning

GLIDE Diffusion Model

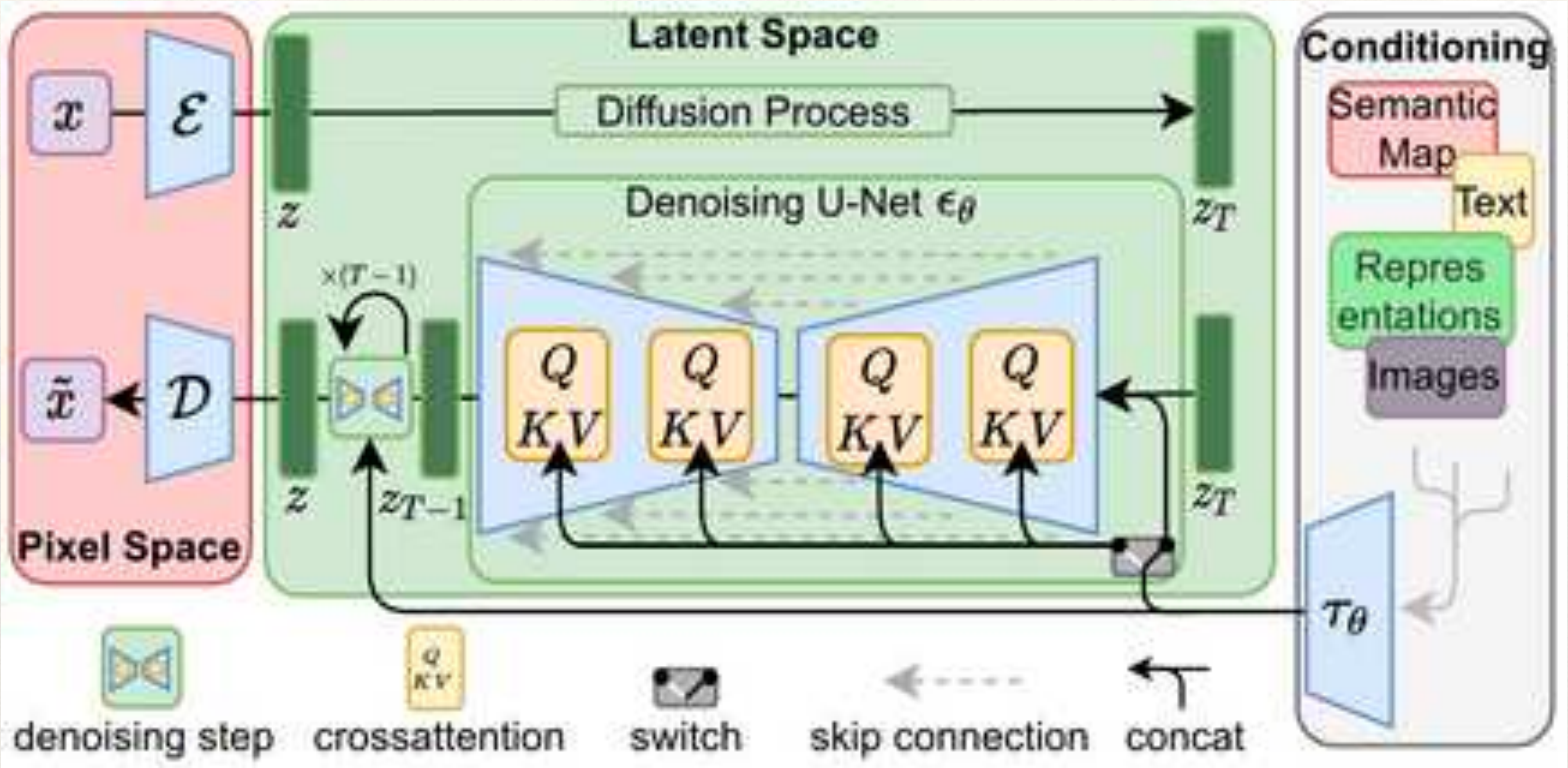
Transformer Model

Diffusion Models

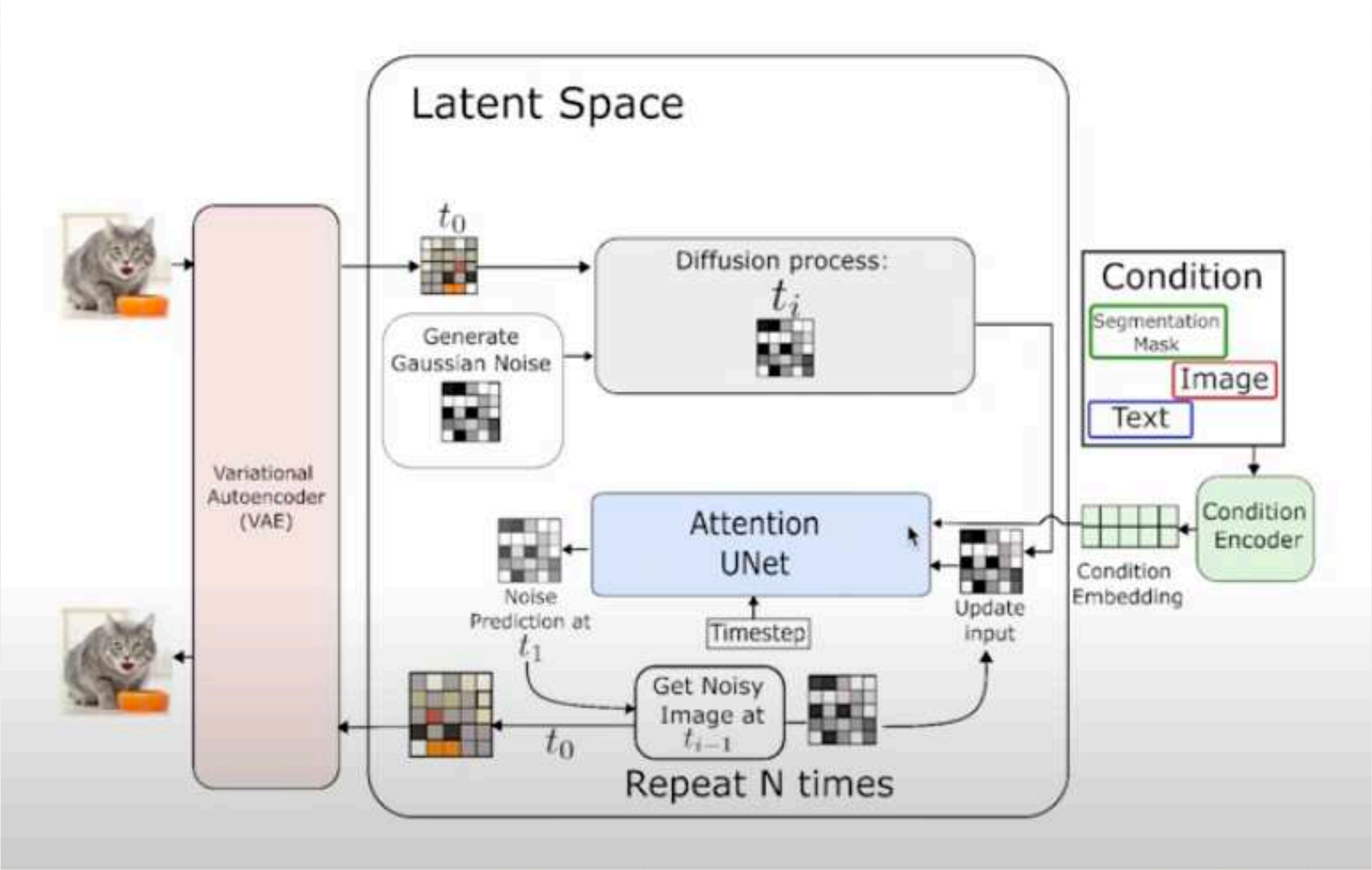


High-Resolution Image Synthesis With Latent Diffusion Models, Rombach 2022

Stable Diffusion



Latent Stable Diffusion

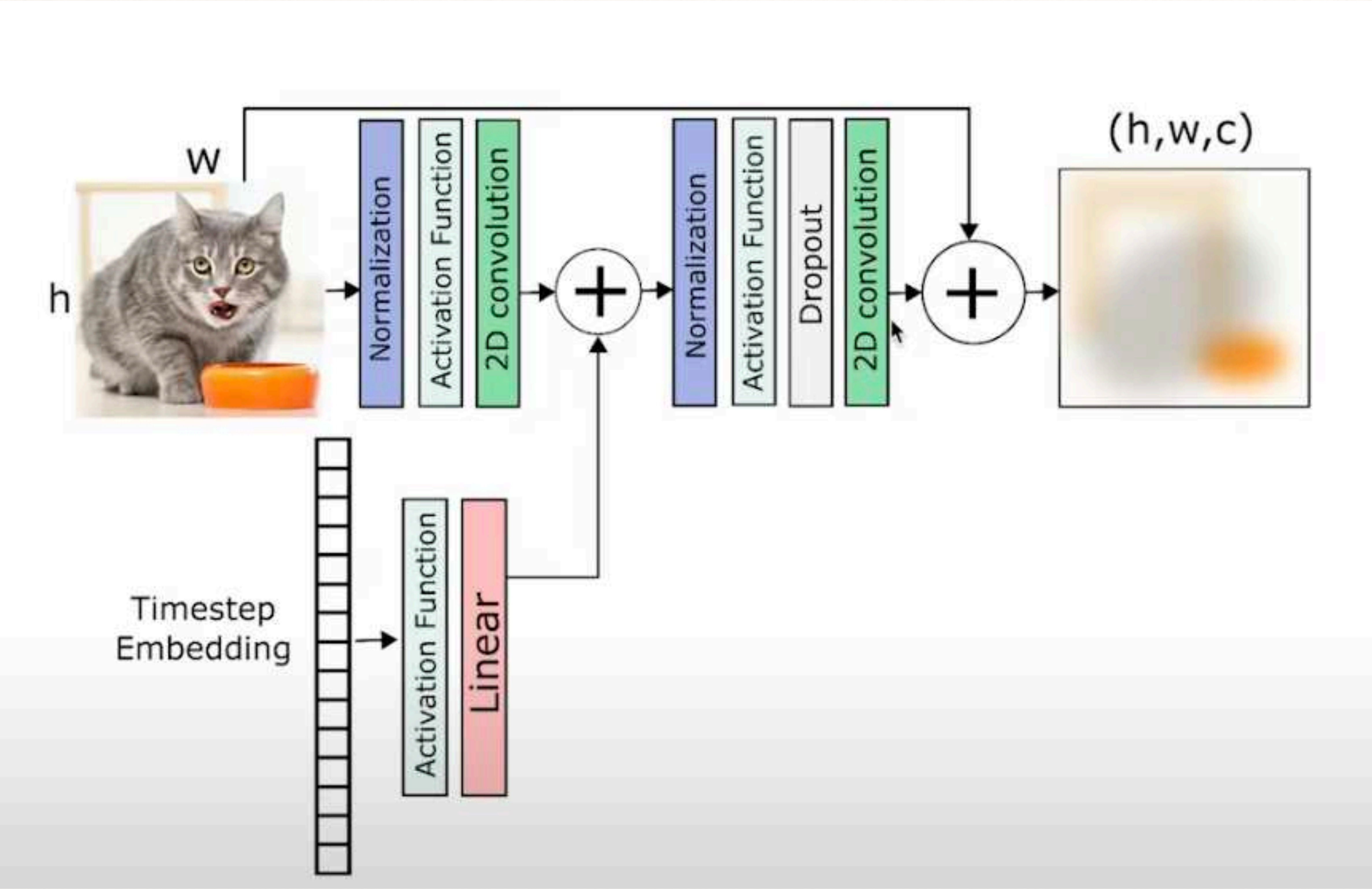


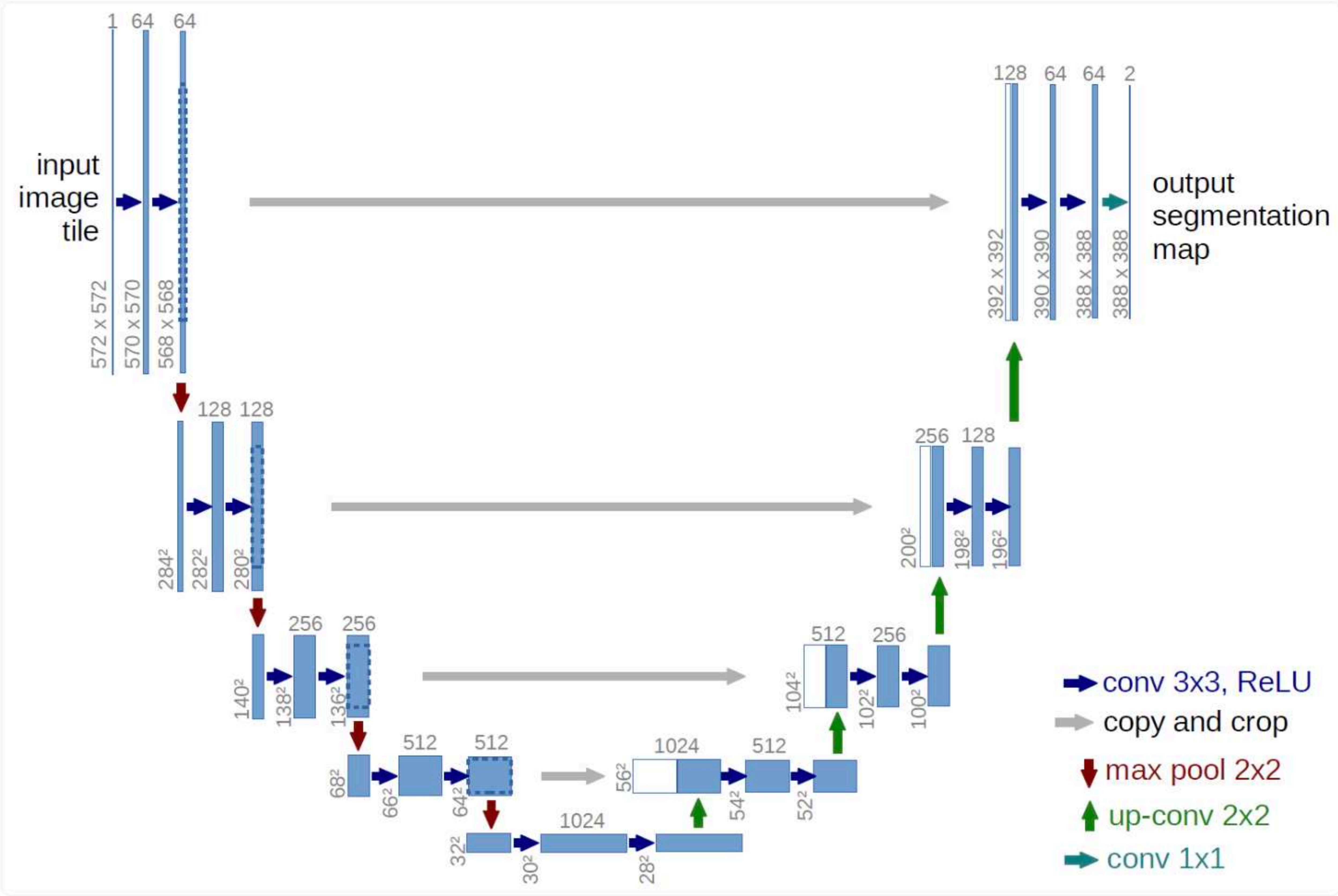
U-Net Network conditioned to a Condition Embedding. It reverses the diffusion process.

VAE Variational Autoencoder to convert to and from the Latent Space.

Condition Encoder CLIP Generate Condition Embedding for the UNET

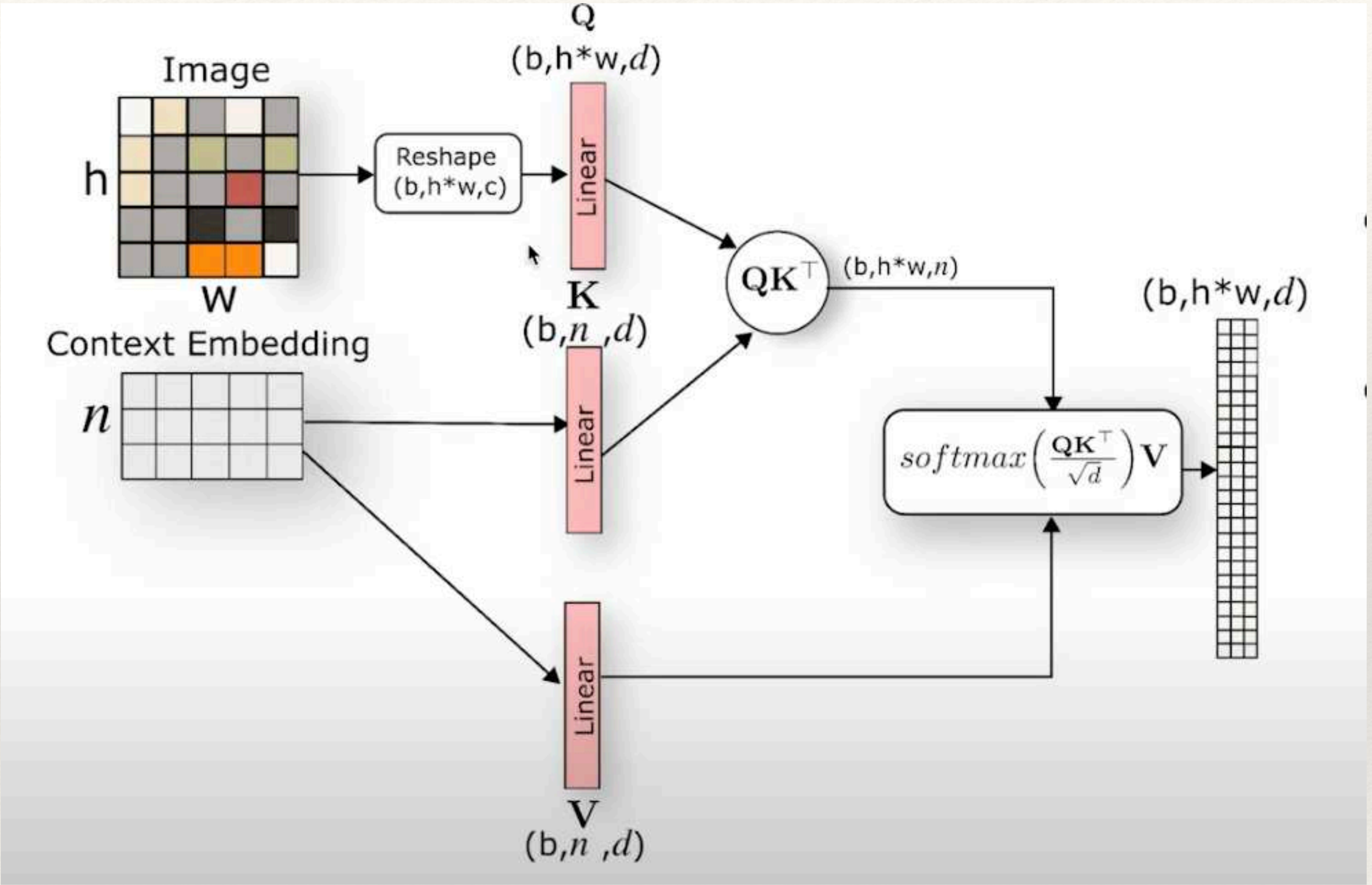
Residual Neural Networks and U-Nets





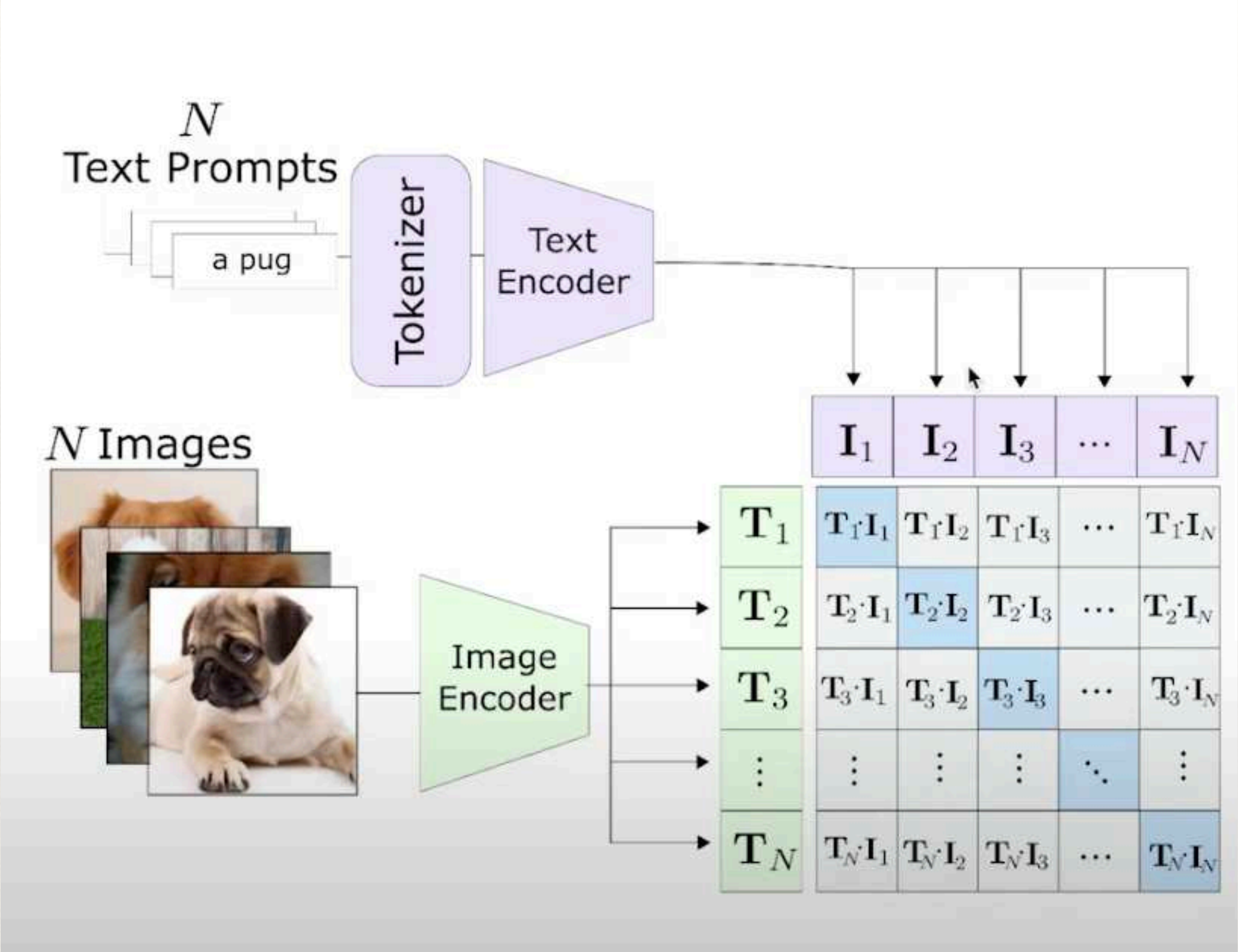
U-Net: Convolutional Networks for Biomedical Image Segmentation, Ronneberger 2015

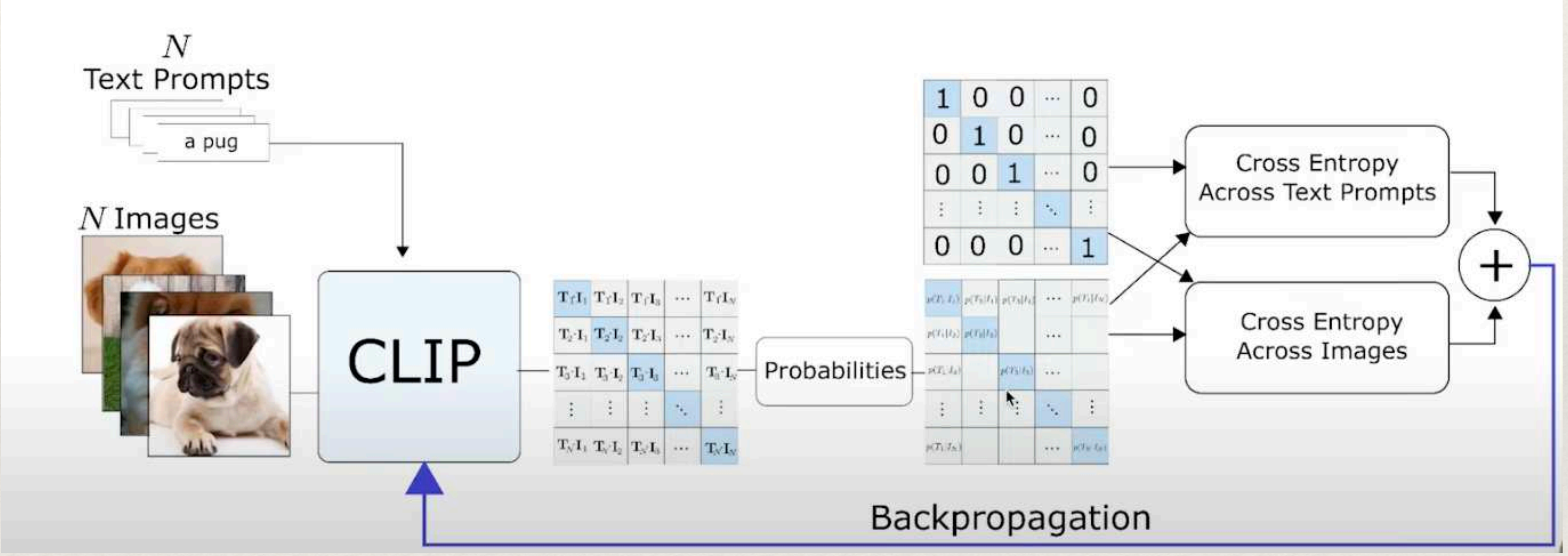
Cross Attention



U-Net: Convolutional Networks for Biomedical Image Segmentation, Ronneberger 2015

Conditional Encoder





U-Net: Convolutional Networks for Biomedical Image Segmentation, Ronneberger 2015

Reinforcement Learning

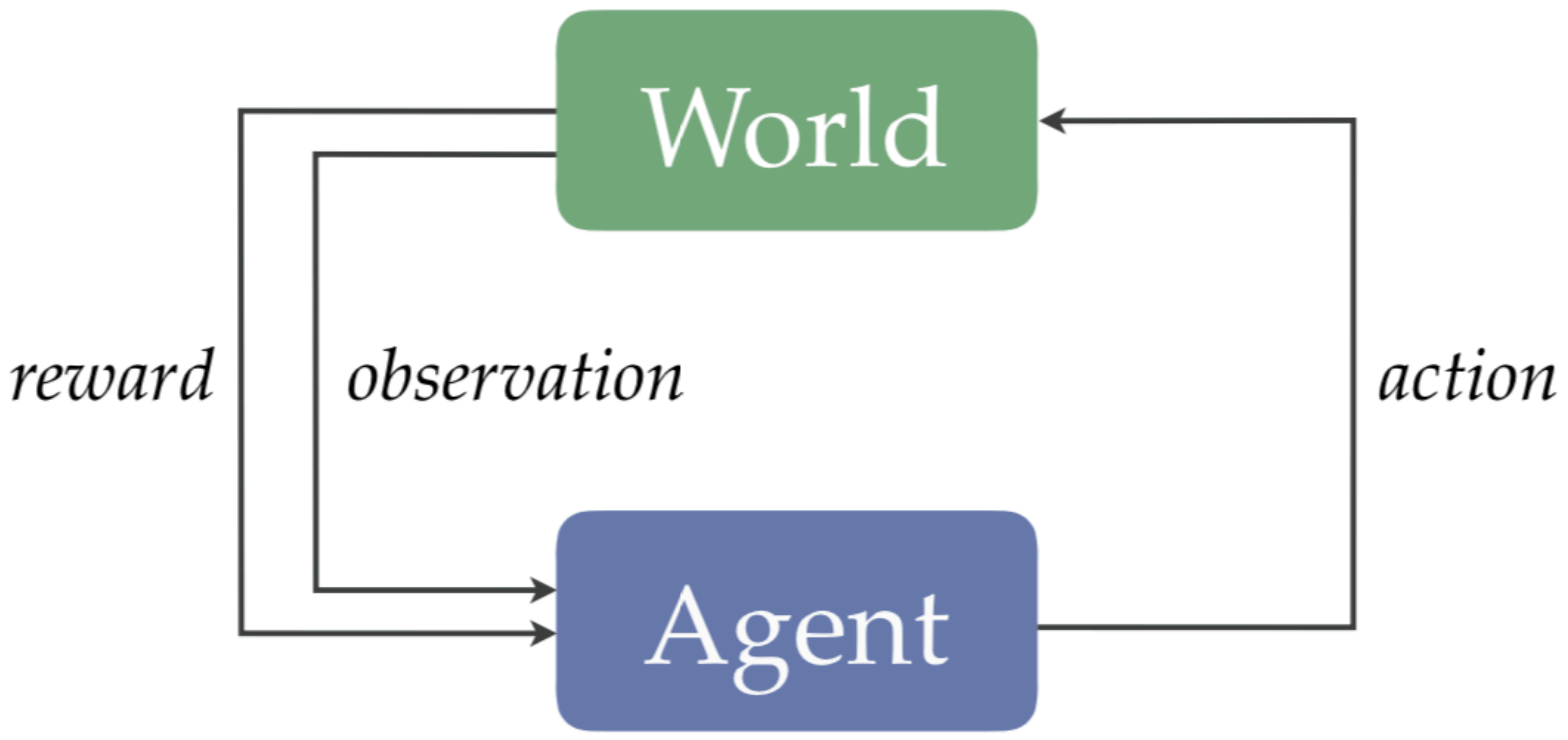
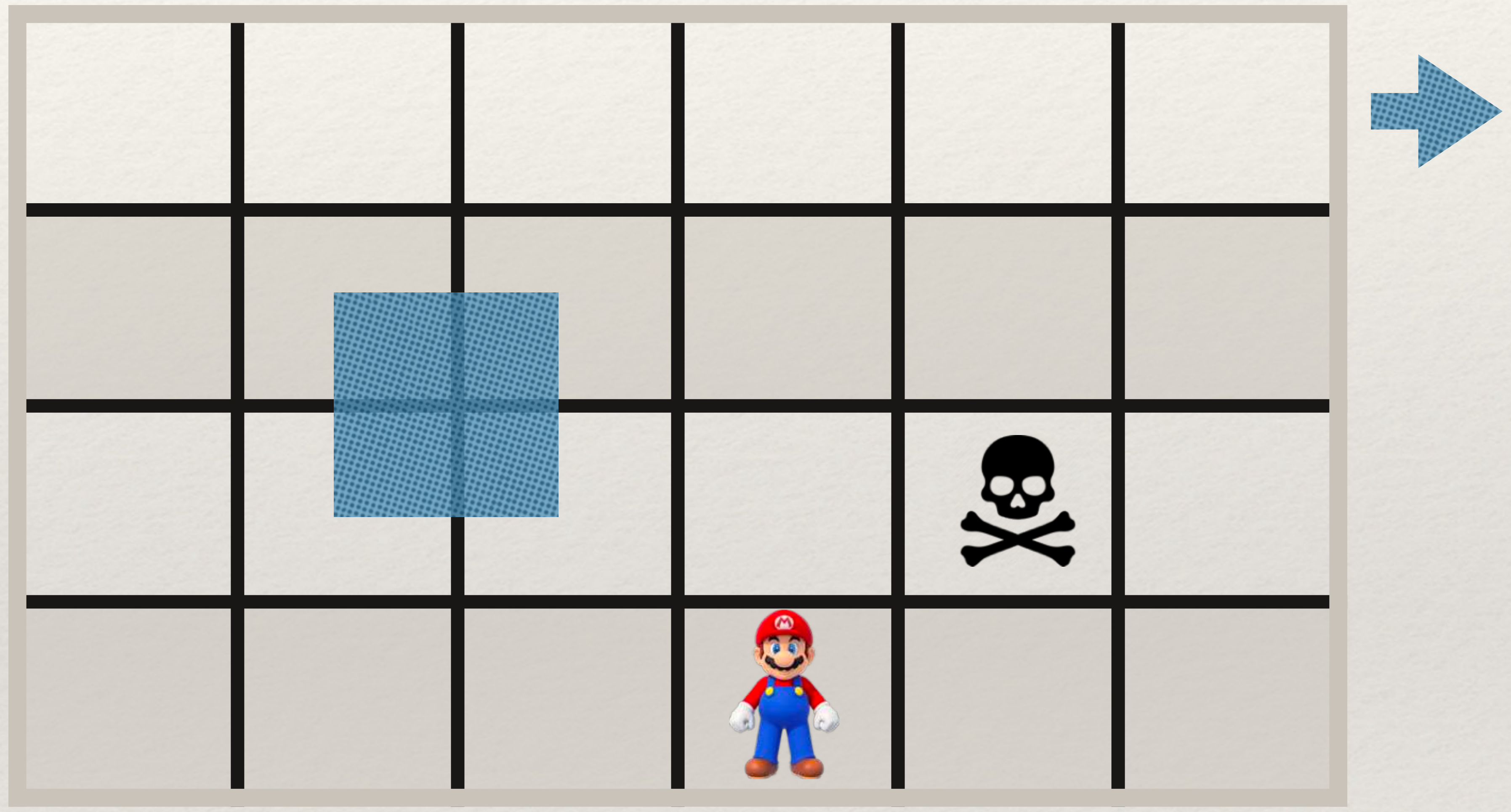


Figure 1.2: The RL problem.

Mnih 2015,Lavet 2018,Sutton 2018

Deep Reinforcement Learning

- Start at s:state, a:action
- Q(s,a) based on greediest a
- Iterate
  - Apply a
  - Get s′
  - Get r(s′)
  - $Q(s,a) = \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$

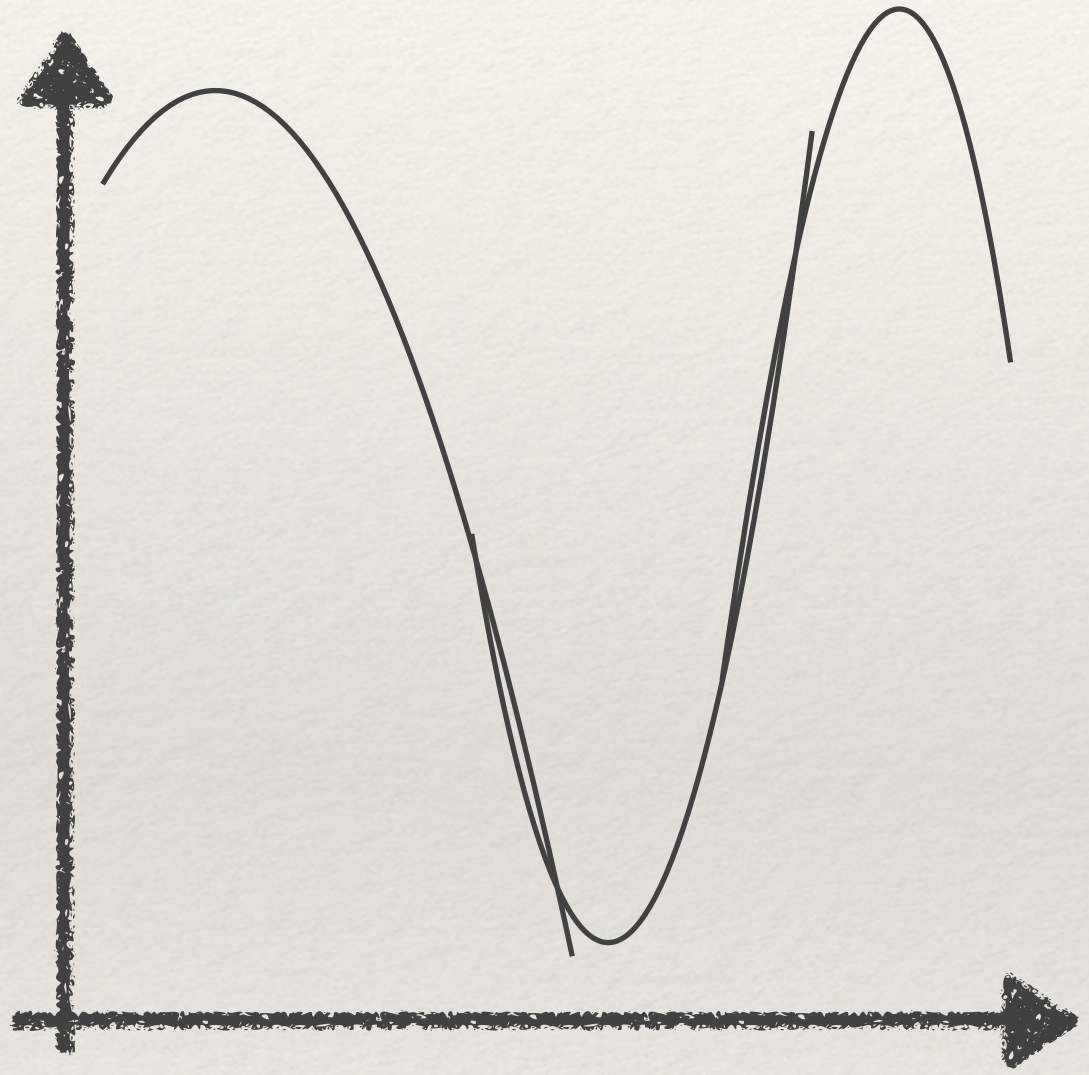


Q: Deep Neural Network

Mnih 2015,Lavet 2018,Sutton 2018

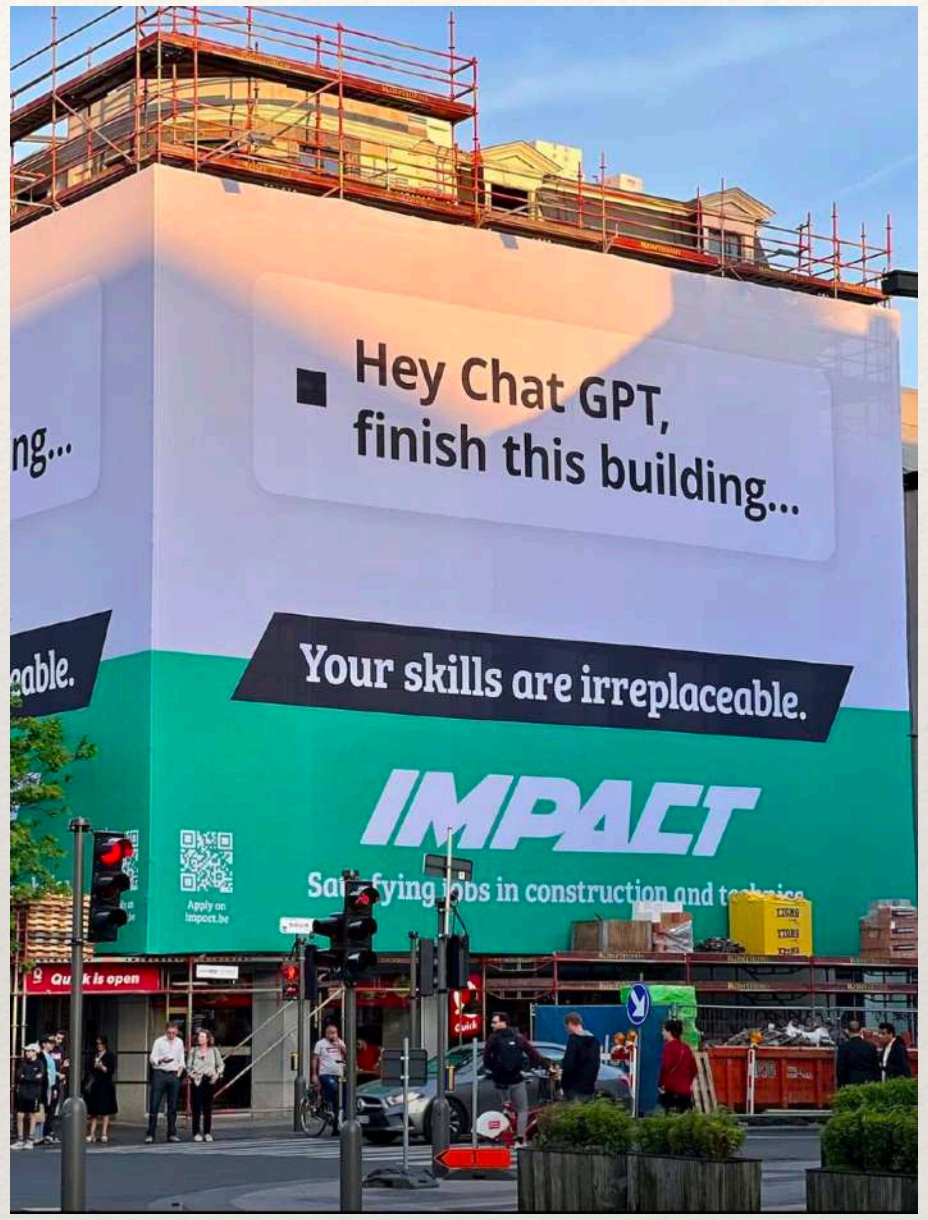
Are our jobs been displaced by robotics?

Income



Dextrous

Cognitive



Plenty of work to be done

Let the robot do the more dangerous and boring stuff

Noam Chomsky https://www.youtube.com/watch?v=ojzNM6tRiyc

Privacidad

HIPAA

GDPR

Deep Fake

Neuroderechos

What are current real risks of DL?

Is there anything unbiased?

Biased

Minority Report

Lost Agency

Cambridge Analytical

Privacy

DeepFake

Digital Boundary

Hacking Autonomous Car

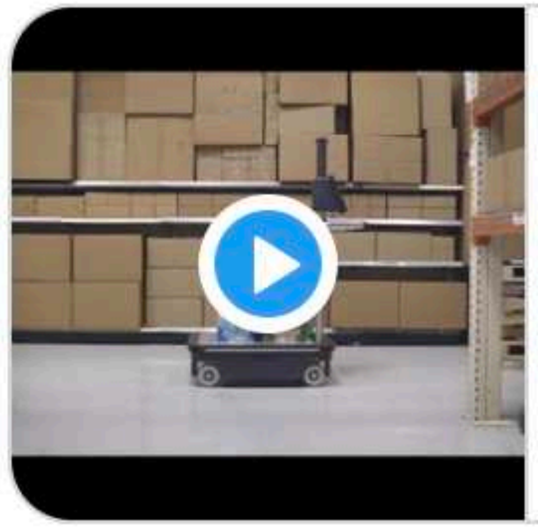
Accountability

https://digital-strategy.ec.europa.eu/en/library/ethics-guidelines-trustworthy-ai

Unsettled discussion

Rodney Brooks @rodneyabrooks · 12h

Anthony Jules and I explain @RobustAI's new software/hardware combo Grace and Carter, and how they change the way people can work in warehouses. Workers can control and interact with the robots whenever they choose to do so. Collaborative Productivity.



youtube.com

Robust.AI on Collaborative Productivity

At Robust.AI, we make robots work for people.More info: https://www.robust.ai/

3 16 58

Yann LeCun @ylecun · 2h

And your perception system doesn't use deep learning, right?

2 5

Rodney Brooks @rodneyabrooks · 16m

Are you trolling me @ylecun?? As always, I send you respect and sincere thanks for all that you have done. To answer: one component of our perception system is multiple model DL inference at frame rates.

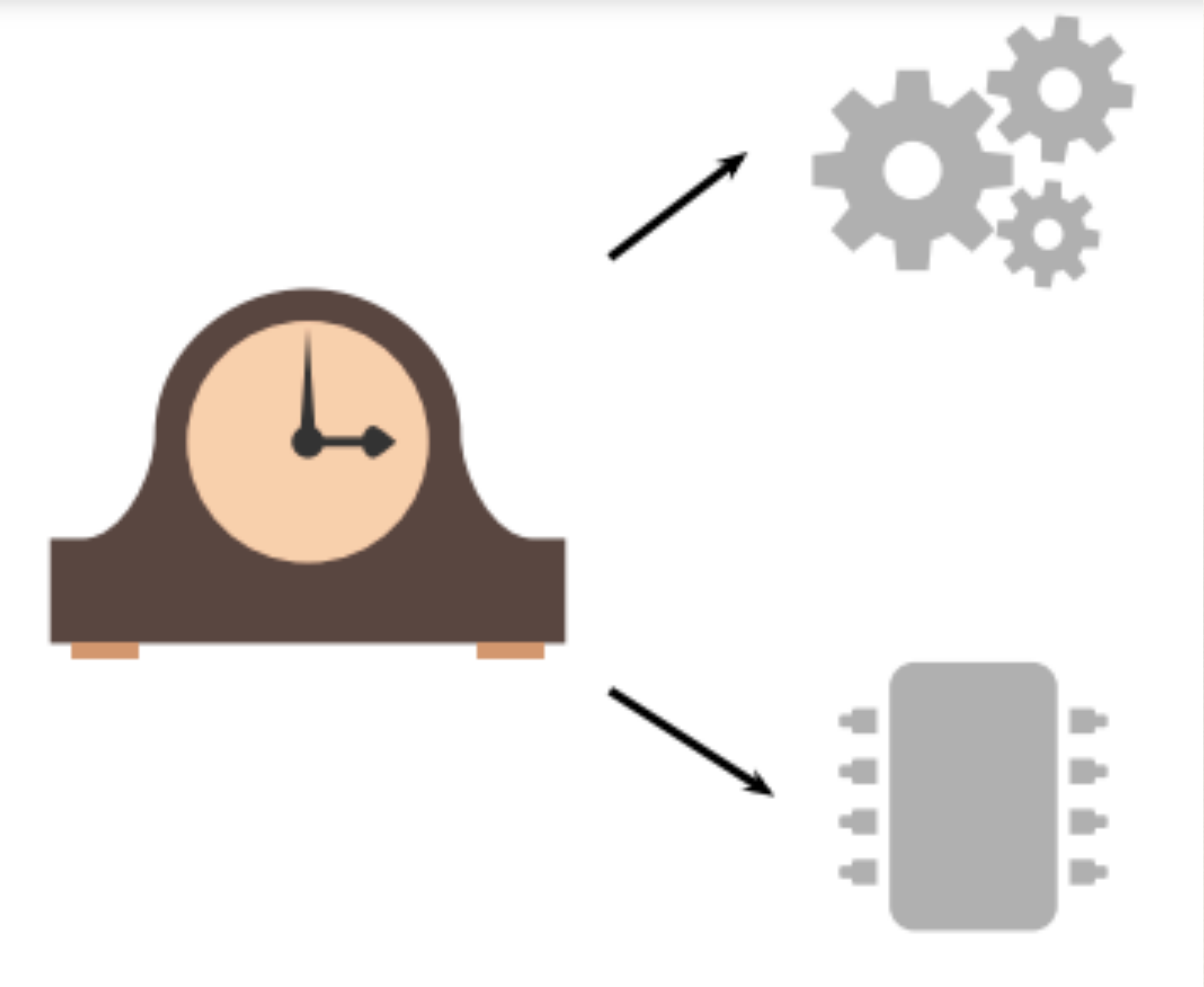
El pato de Vaucanson

Planes don’t flap their wings

Principle of multiple realizability:
“different substrates can nonetheless perform the same input-output mapping but they are not the same”.

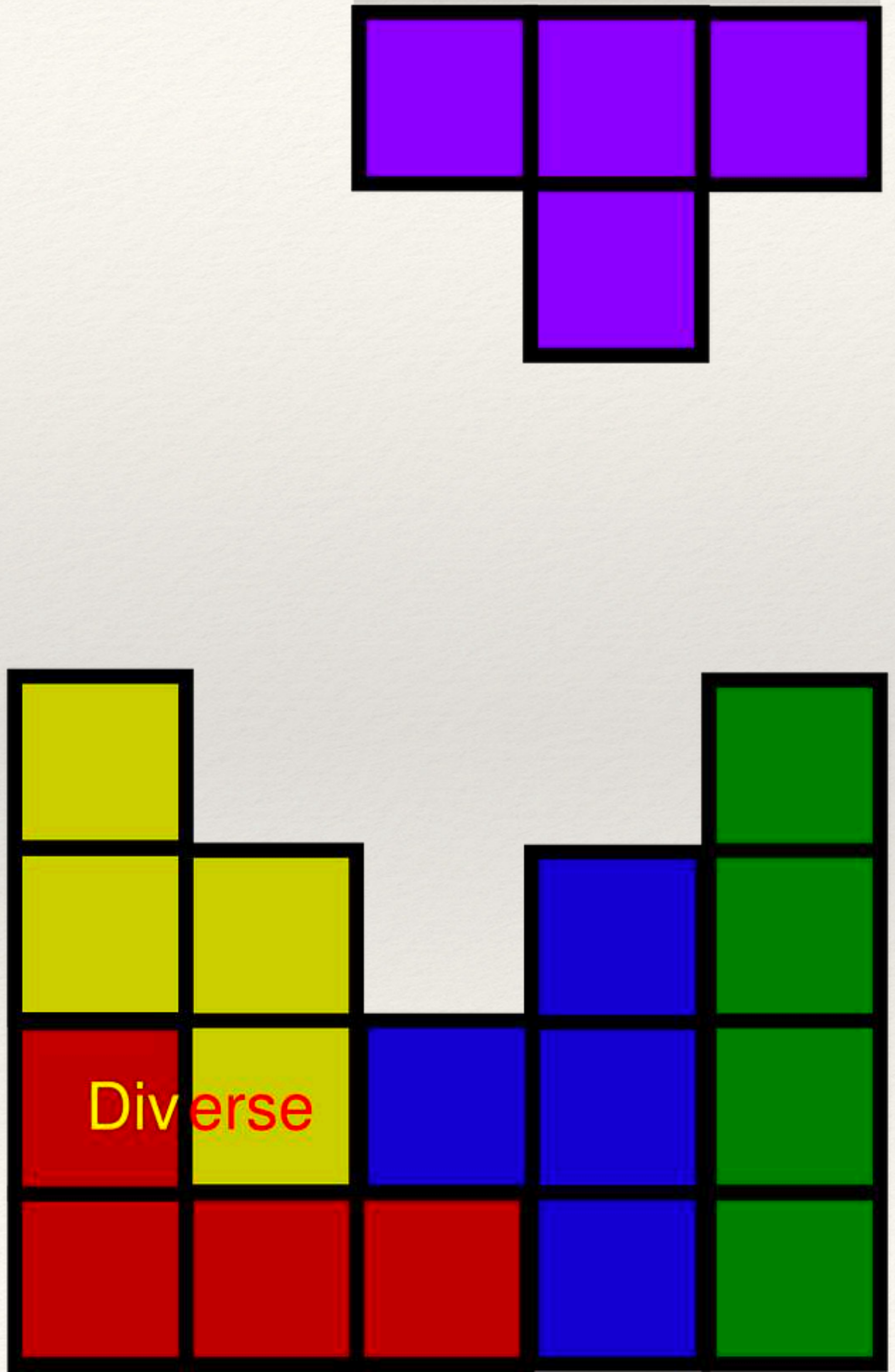
Grace Lindsay[47], Guest 2021 [52]

Functionalism vs Mechanism

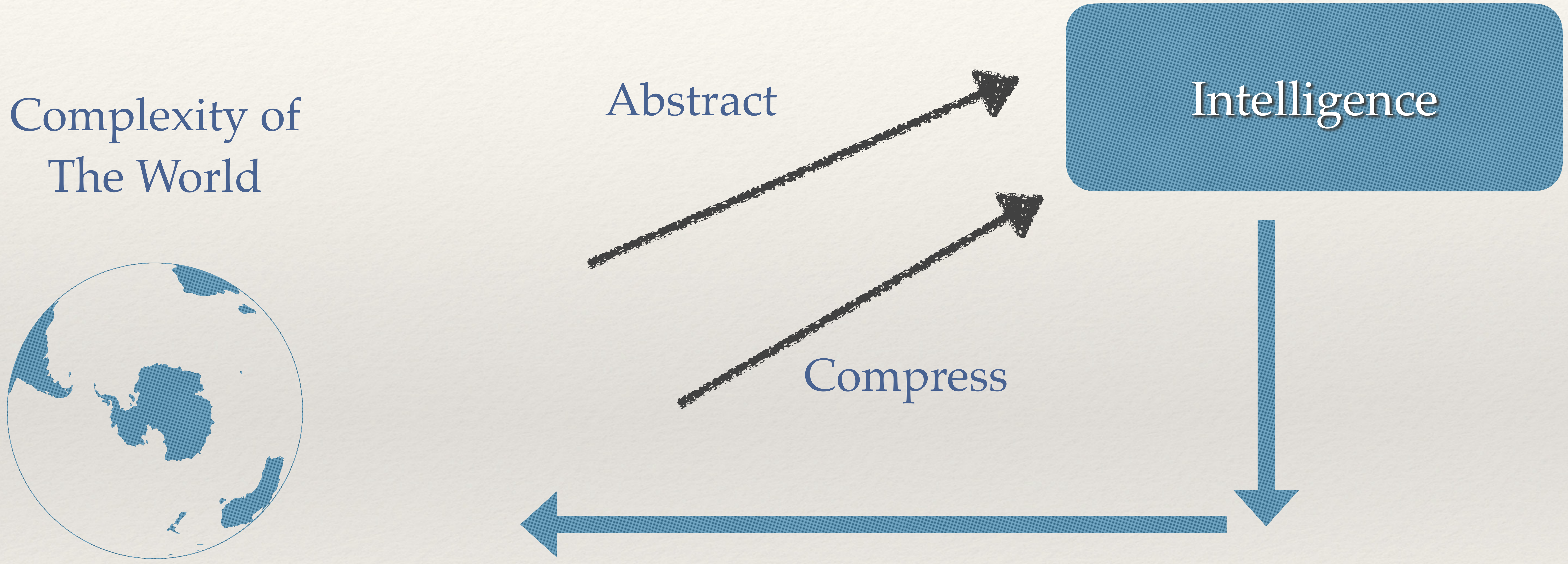


if the clockwork clock behaves like the digital clock, then the clockwork clock is a digital clock

The ability to fit

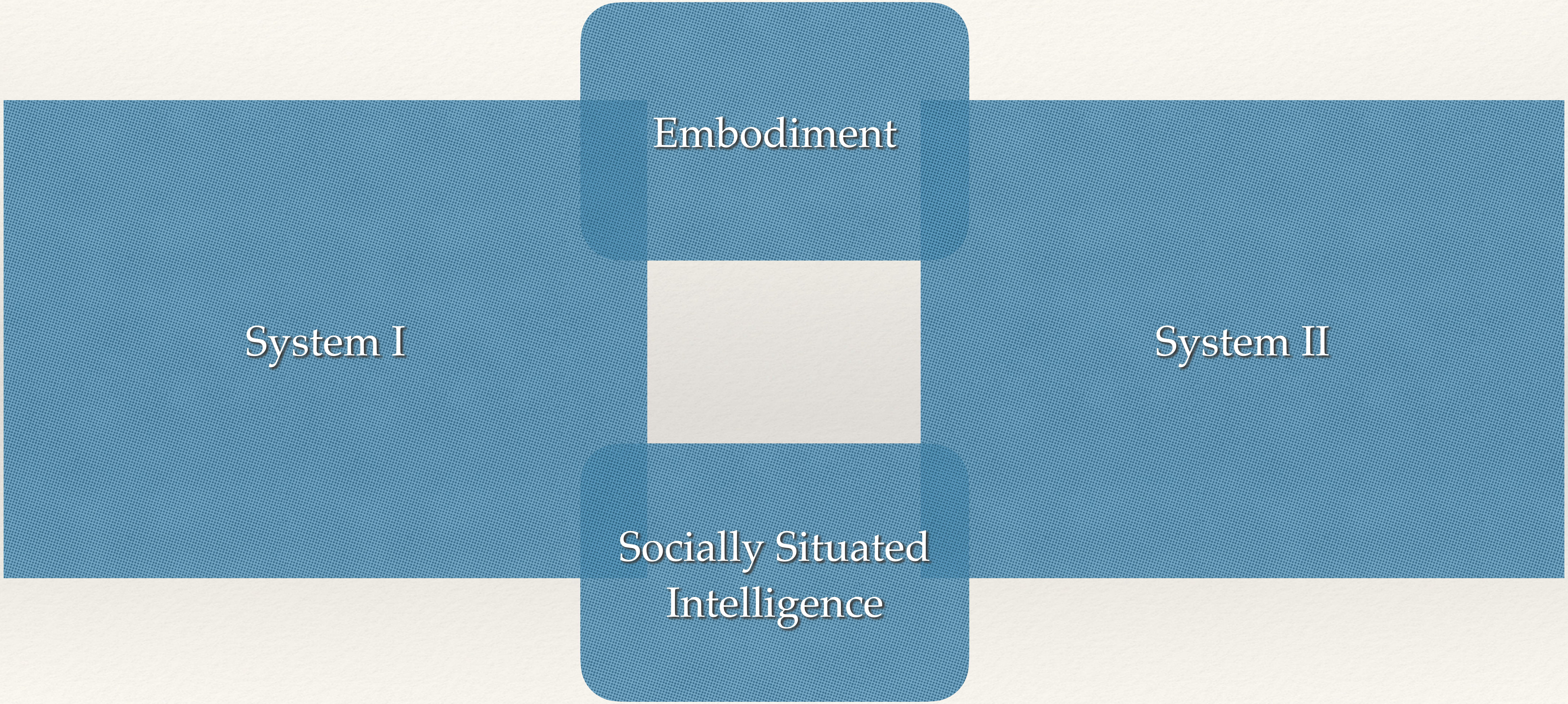


The ability to summarize



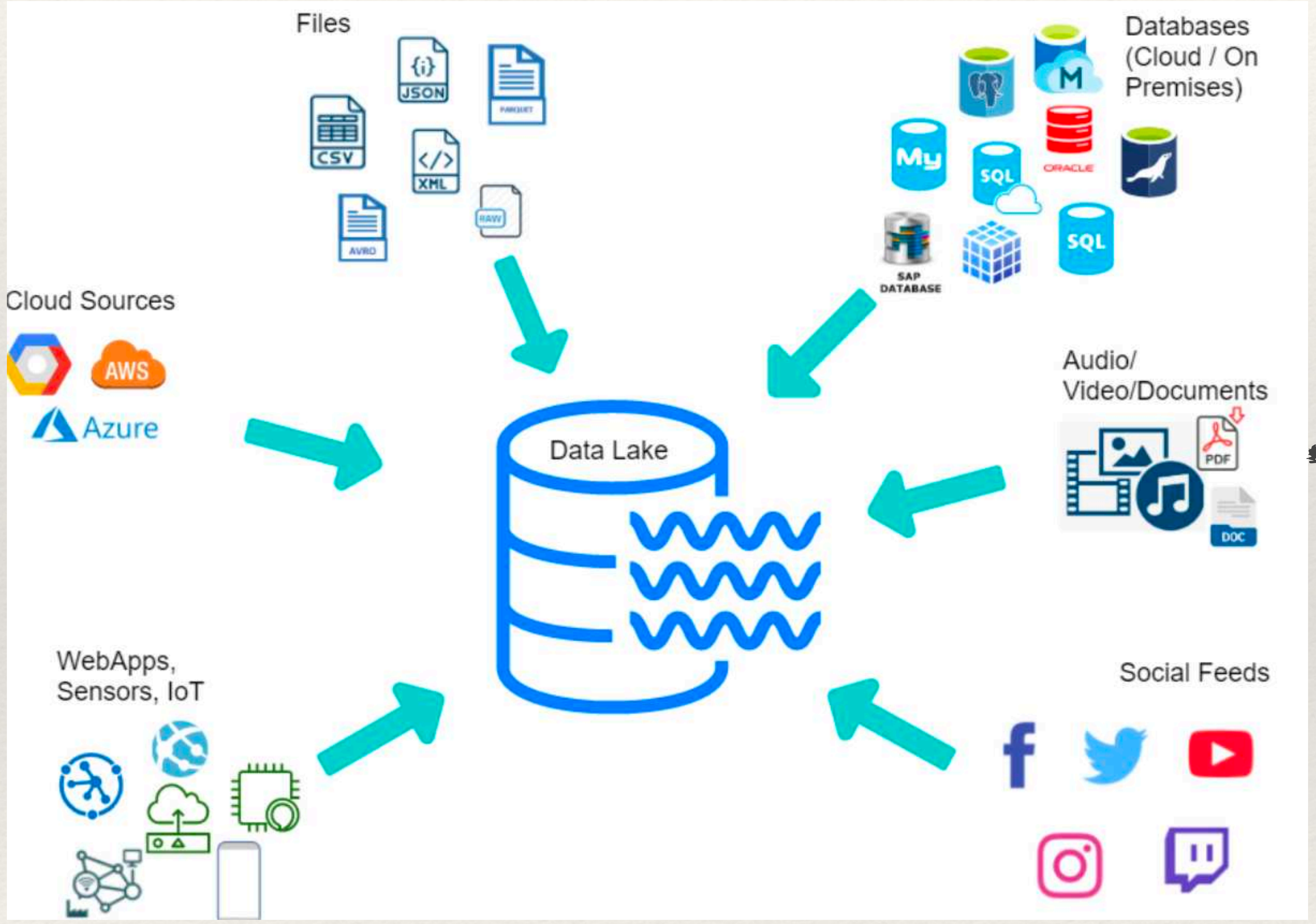
A Theory of Abstraction in RL, David Abel
Del Rigor de la Ciencia, Borges

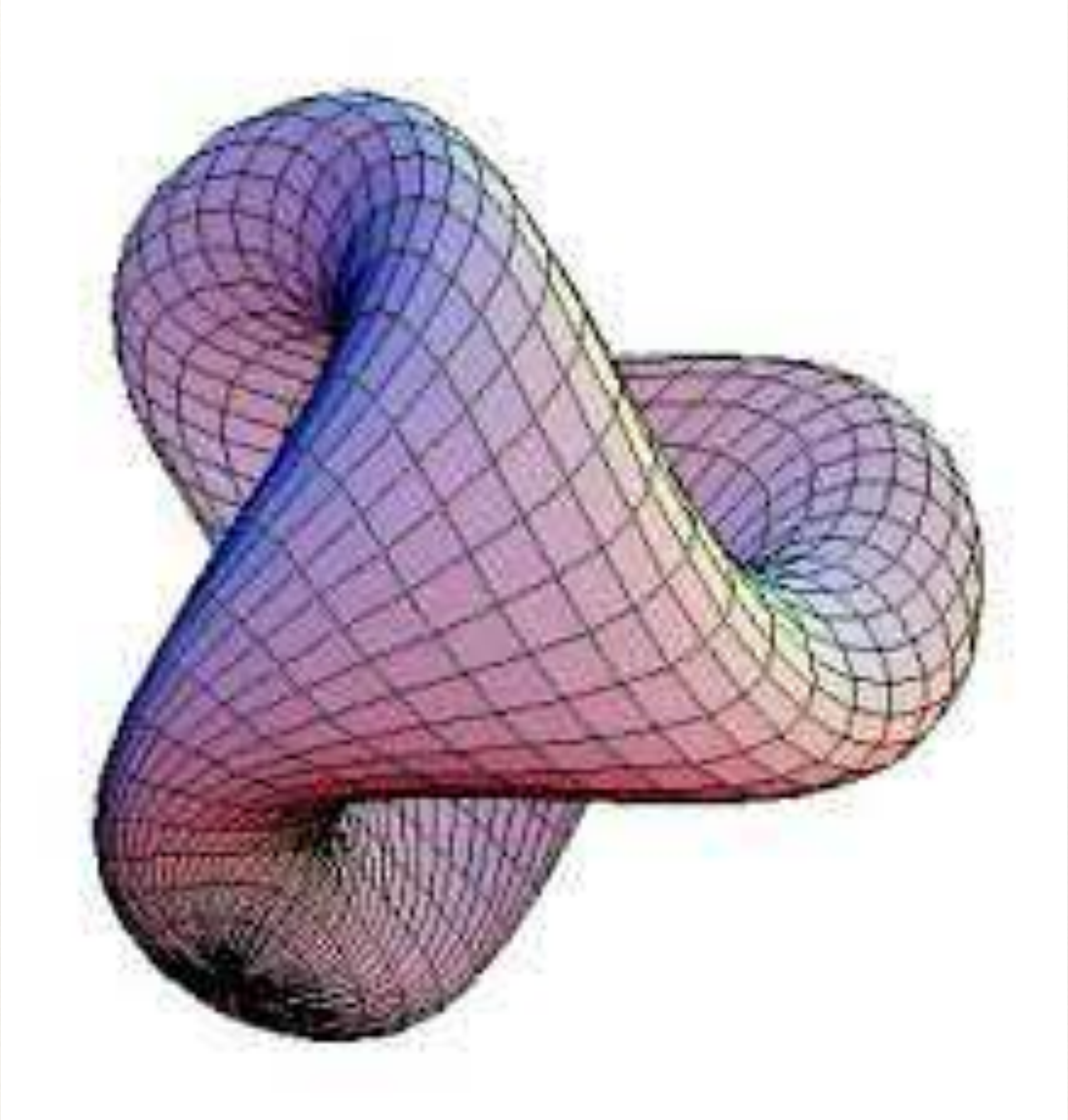
Connectionism vs Symbolism

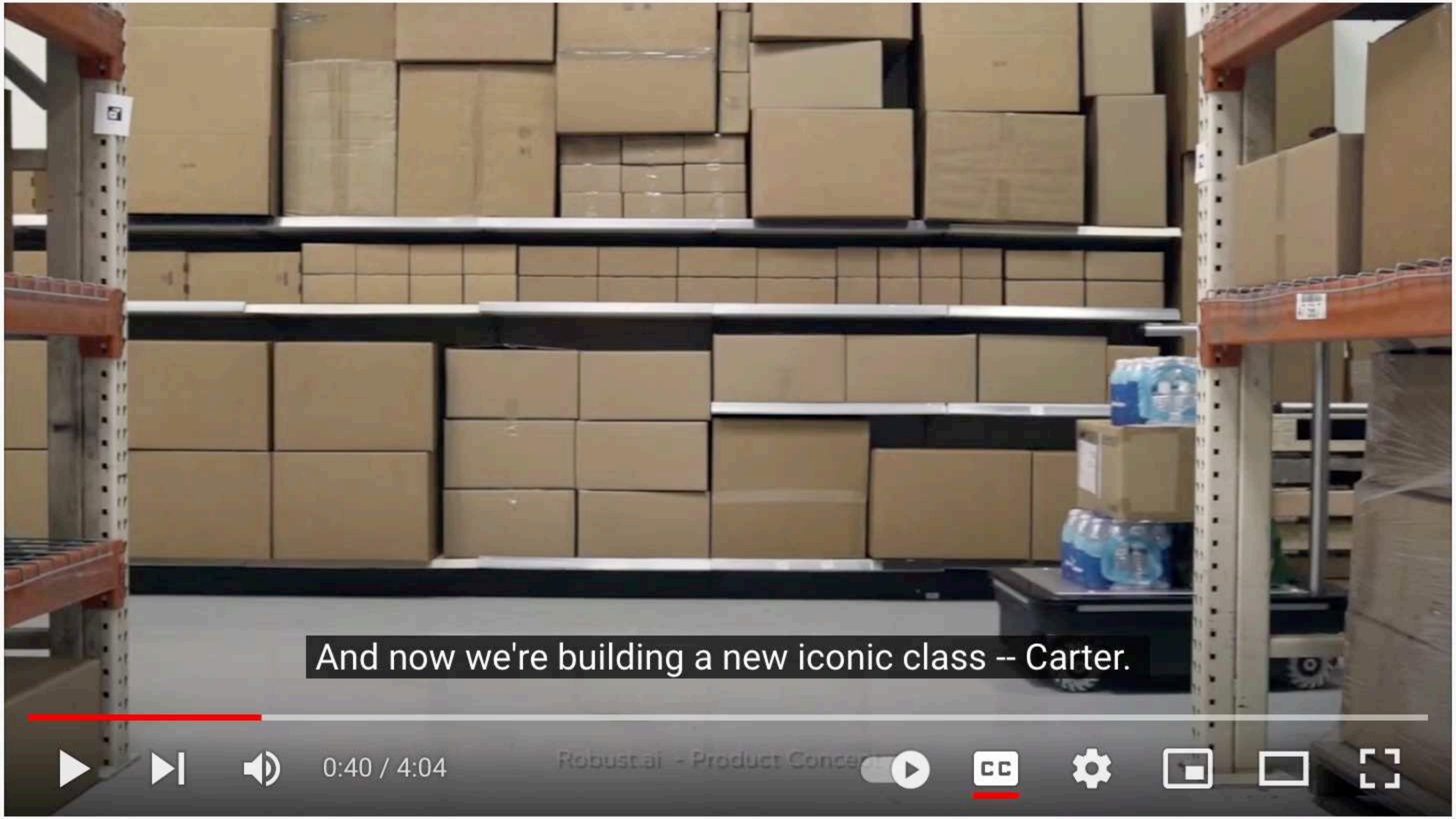


Looking back, looking ahead: Symbolism vs connections AI, Goel 2021

The Age of Data Manifolds







Robust.AI on Collaborative Productivity

https://www.robust.ai/

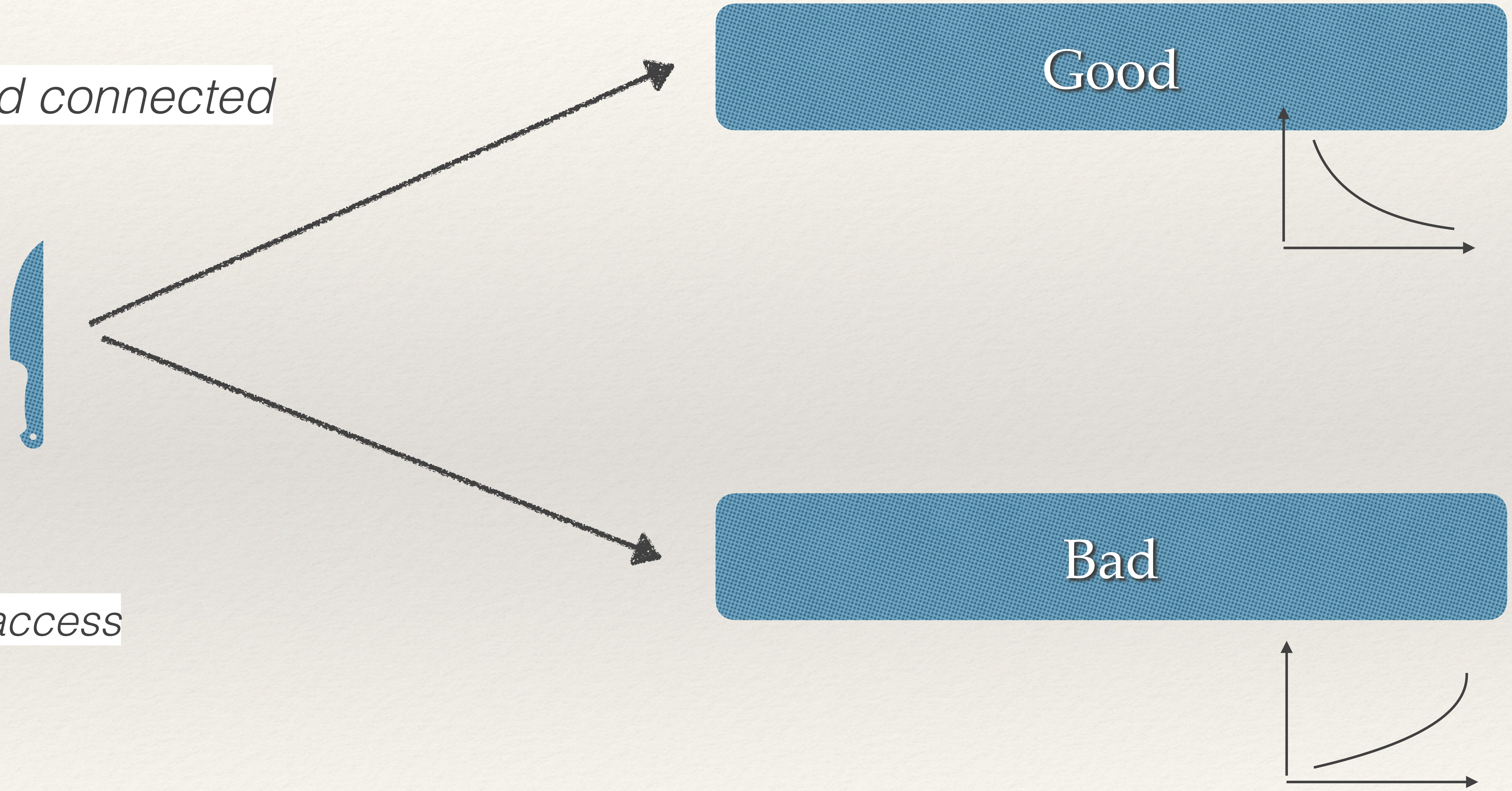
IEEE Spectrum DARPA Challenge

Is DL++ good or bad?

make the world more open and connected

Don't be evil

Internet will provide to everyone, access
To humanity's knowledge



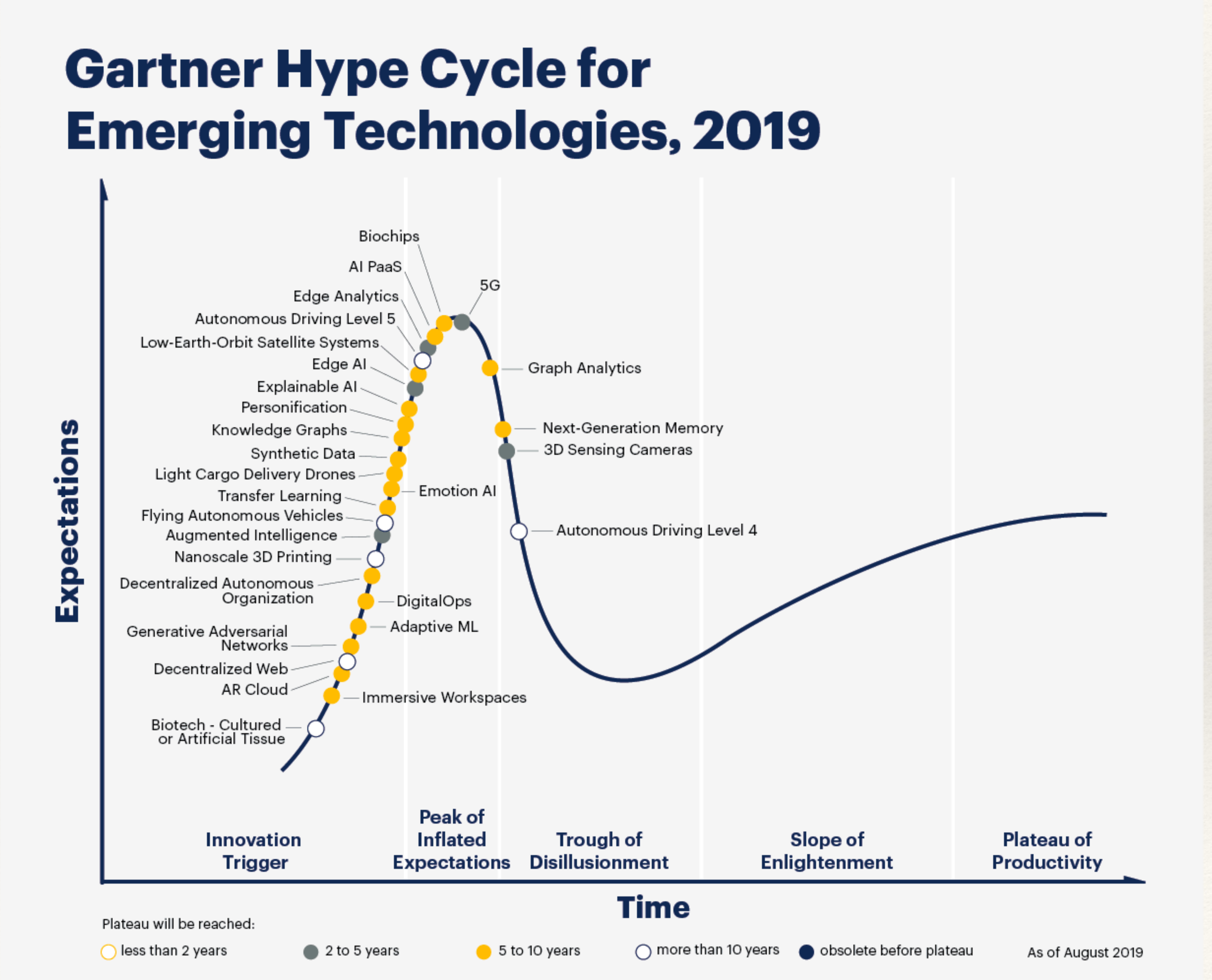
Is DL++ good or bad?

Technology
Is
Inherently
Good

Allows more DOF

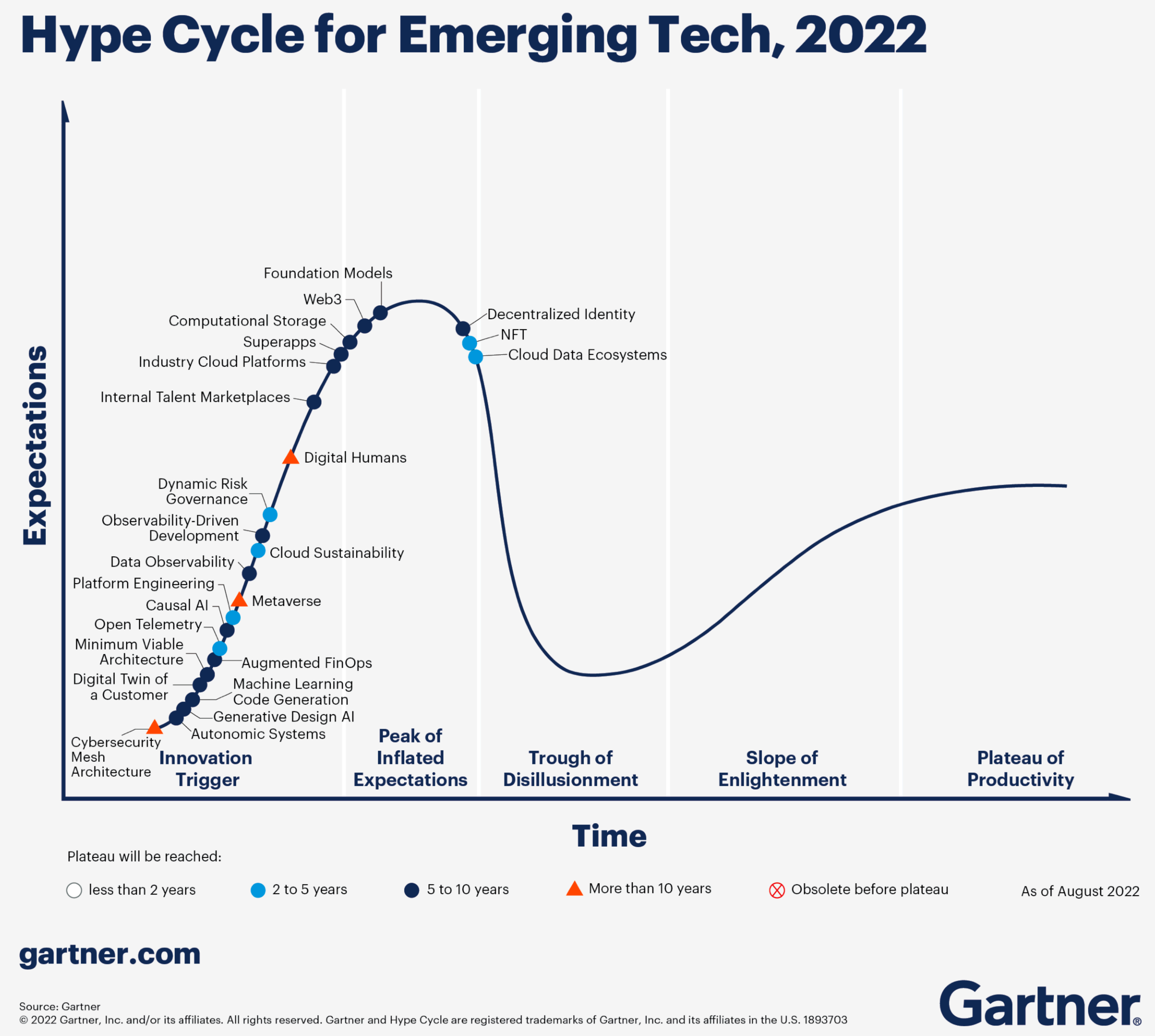
Emerging Technologies

- GANs
- Explainable AI
- Edge AI
- AI PaaS
- Adaptive ML
- Emotion AI



Emerging Technologies

- Autonomic Systems
- Generative Design AI
- Causal IA
- Metaverse
- Web3 / Decentralized Identity
- Data Observability



Applied AI

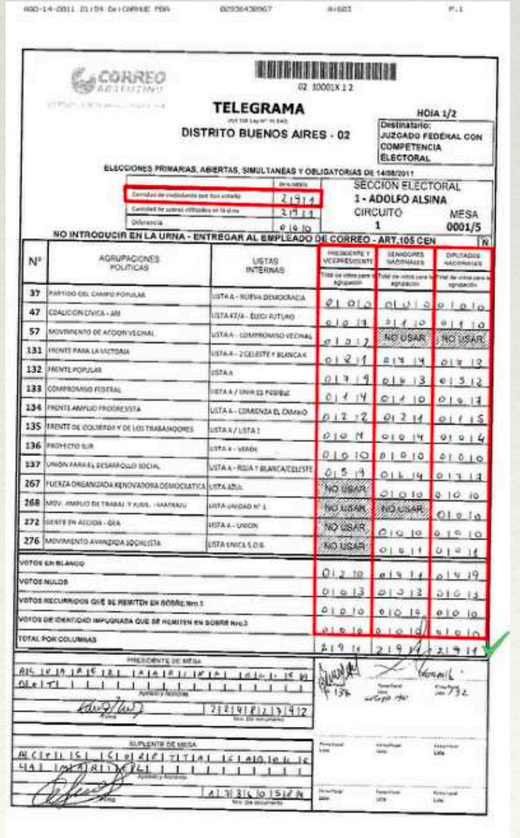
DS

Liveness Detection

Estado de la fruta
Seguros de Exportaciones

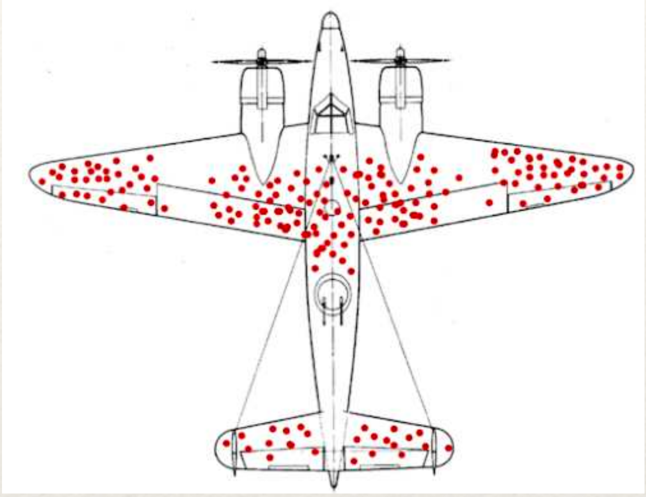
Retail

Customer Satisfaction
NLP



OCR de los Telegrama Electorales

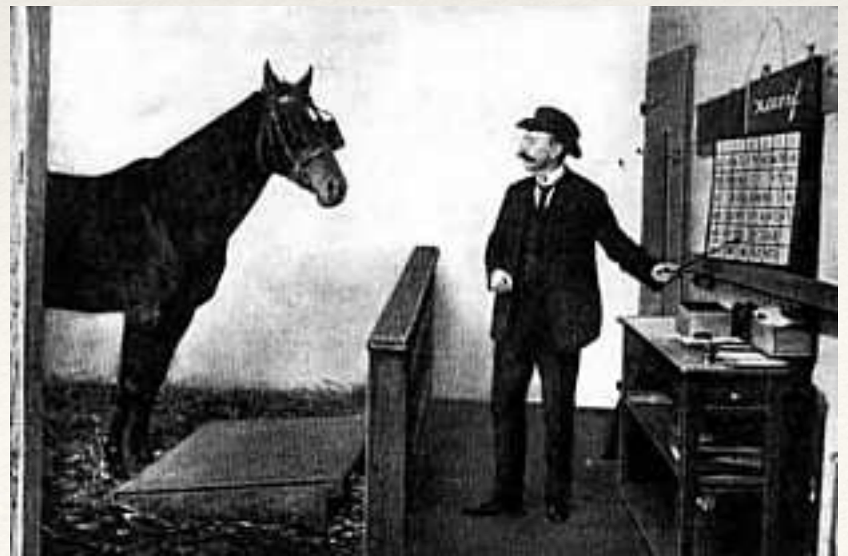
Fallacies of Data Science



- Biased Cherry Picking: Selecting results that fit your claim.
- Data Dredging: Results of correlation was due to chance.
- Survivorship bias: Drawing conclusions from an incomplete set of data that survived.
- Perverse incentive: the incentive produces the opposite result (wrong metrics, KPIs).
- False causality: two things occur together, they are related (falsely)
- Garbage in, garbage out.

💩 ➡️💻➡️💩

- Sampling bias (cross validation, bootstrapping)
- Gamblers fallacy: believing that when something happen the chance of happening again is reduced.
- Unbalanced Datasets: medical images
- Simpsons paradox: a trend is present in a group but disappears when data is combined.
- Clever Hans Effect: https://arxiv.org/pdf/2006.10609.pdf



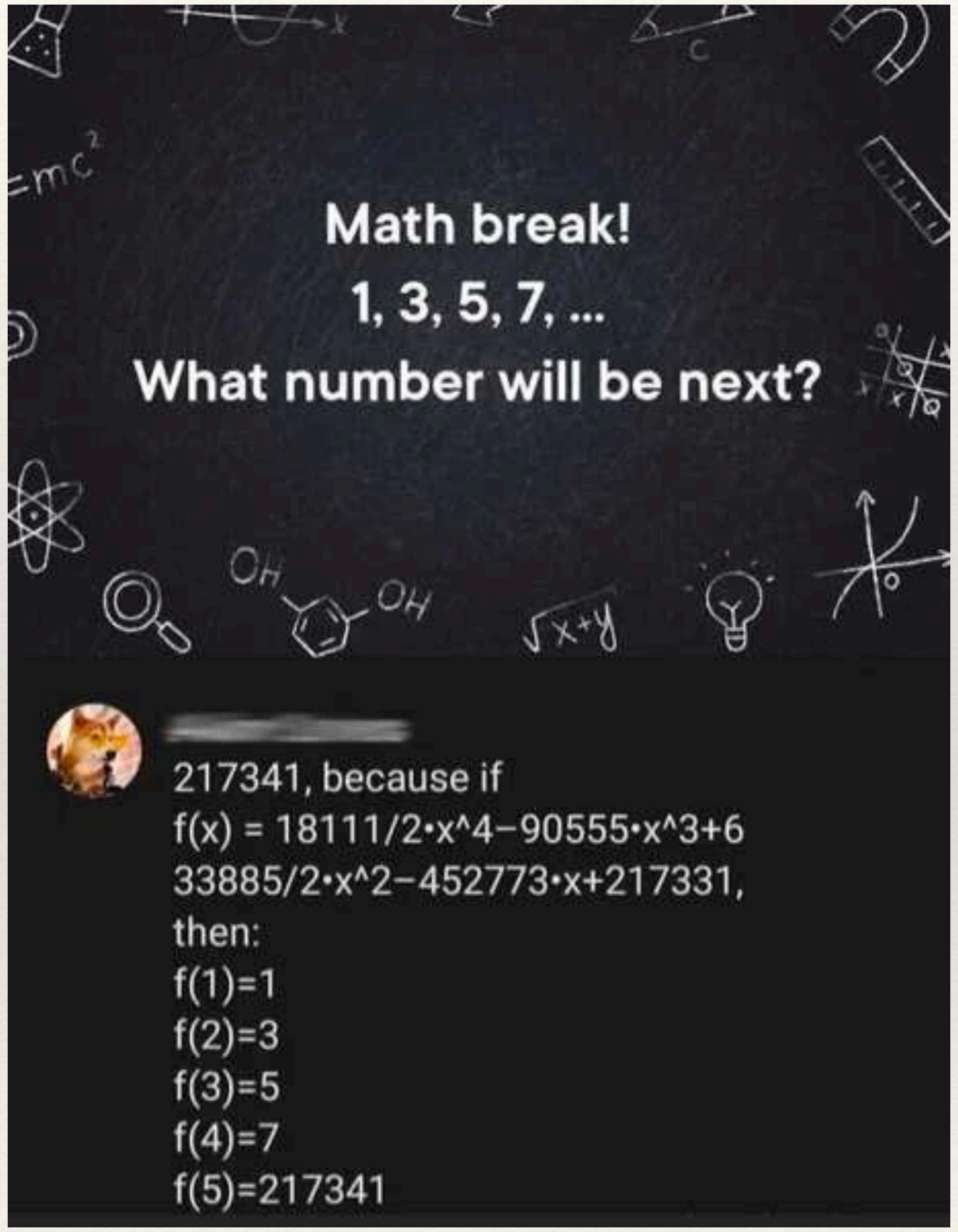
Fallacies of Data Science

It also muddies the origin of certain data sets. This can mean that researchers miss important features that skew the training of their models. Many unwittingly used a data set that contained chest scans of children who did not have covid as their examples of what non-covid cases looked like. But as a result, the AIs learned to identify kids, not covid.

Driggs's group trained its own model using a data set that contained a mix of scans taken when patients were lying down and standing up. Because patients scanned while lying down were more likely to be seriously ill, the AI learned wrongly to predict serious covid risk from a person's position.

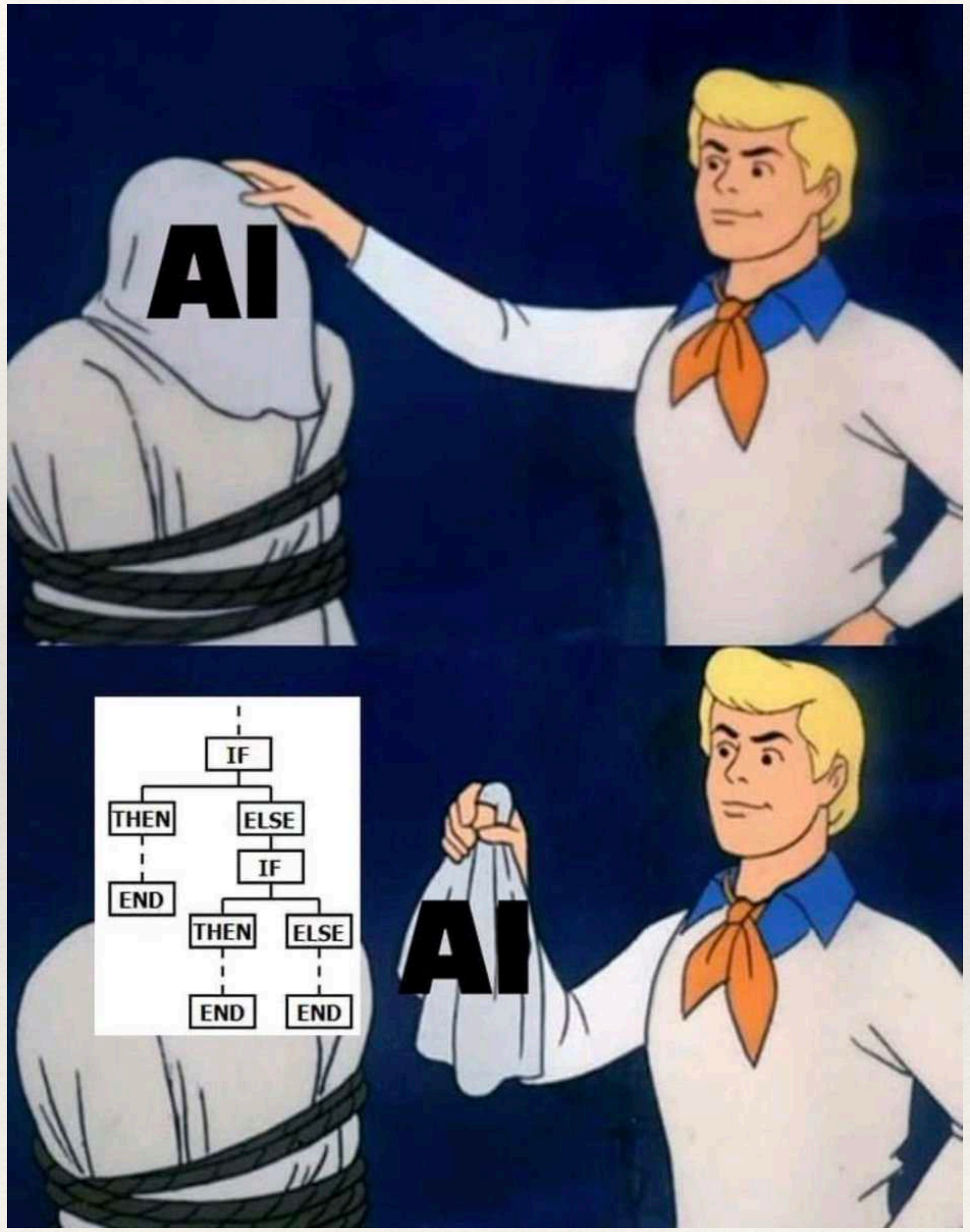
In yet other cases, some AIs were found to be picking up on the text font that certain hospitals used to label the scans. As a result, fonts from hospitals with more serious caseloads became predictors of covid risk.

Machine Learning



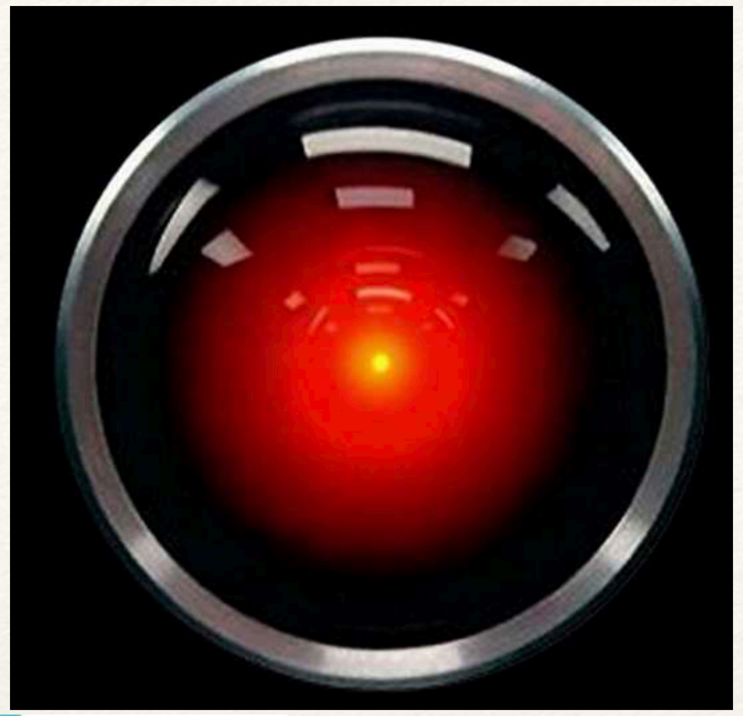
https://twitter.com/vardi/status/1566070939586027520?s=20&t=MHwWjzTDfRfm2d4emglImg

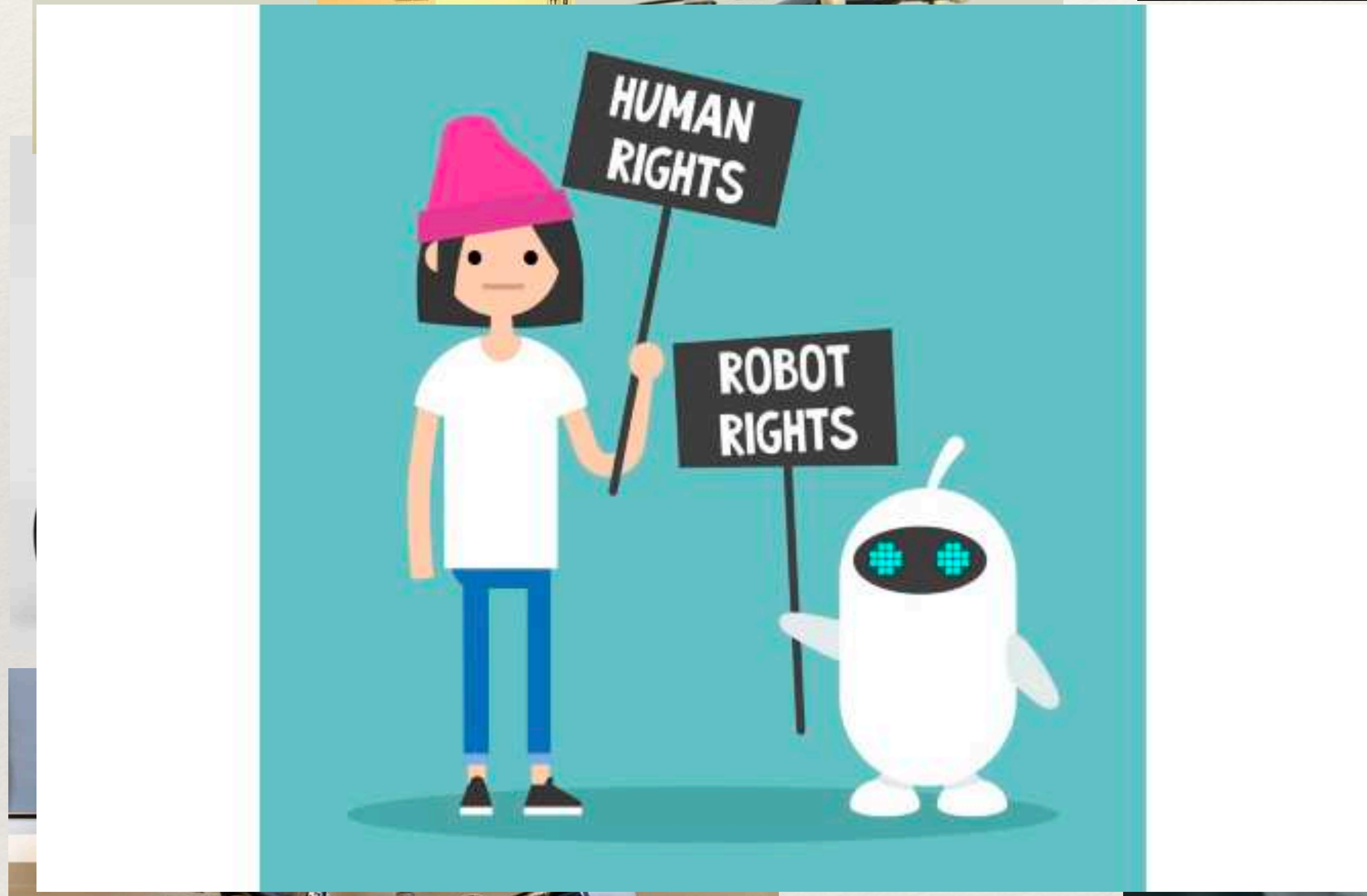
Artificial Intelligence

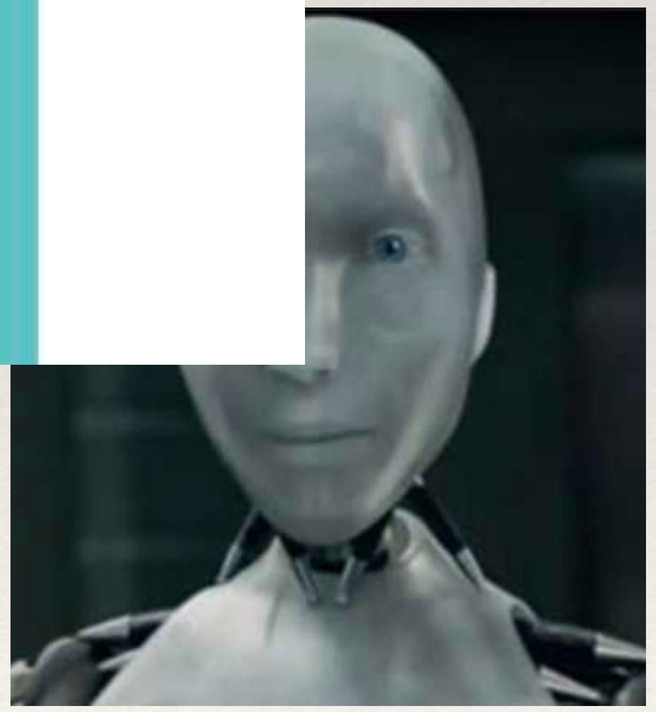


https://twitter.com/ExcelHumor/status/1589098712177217538?s=20&t=w97fO1jJWtTOw8R5kLwJmQ

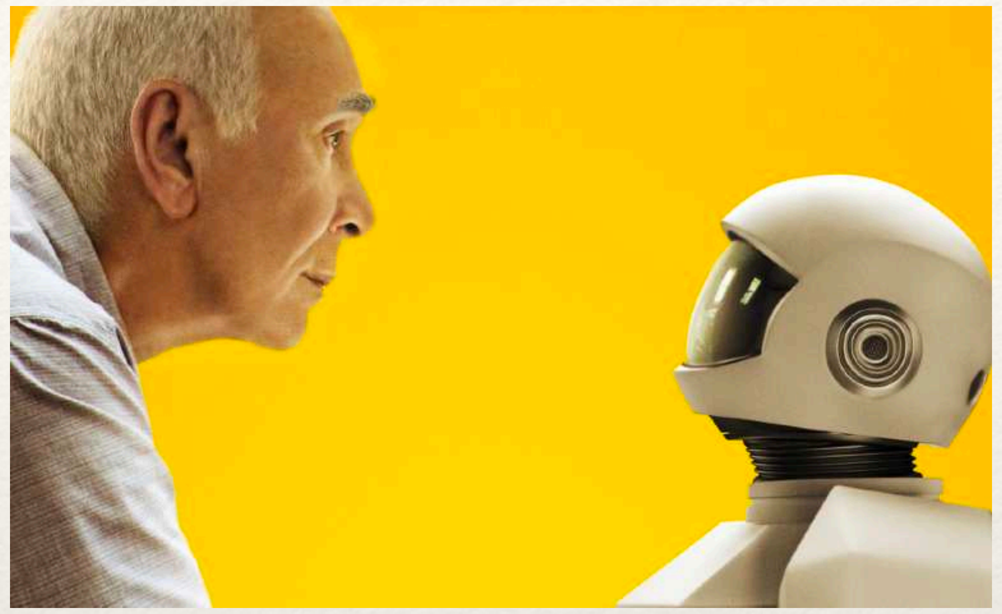
2025







2050



2075

En resumen

Sutton

http://www.incompleteideas.net/IncIdeas/BitterLesson.html

Rodney Brooks

https://rodneybrooks.com/a-better-lesson/

The bitter lesson

En resumen

IEEE Spectrum March 2020

En resumen

- Focalizado para representar datos que intrínsecamente están estructurados jerárquicamente.
- Las capas intermedias ofrecen contenido que tienen valor semántico (es como una salida intermedia parcial de la red). E.g.: la primer capa me separa arriba y abajo y la segunda derecha e izquierda.
- El "deep" implica que la cantidad de saltos en la dependencia causal es mayor.
- Estos layers adicionales permiten que los features los genere y encuentre la misma red (en vez de ponérselos a mano). La red hace la reducción de la dimensionalidad.
- Algunos ajustes a los algoritmos de arquitectura y aprendizaje que aprovechan estas arquitecturas y el paralelismo.
- No estaríamos hablando de esto sin GPUs y sin Moore´s Law.
- Extraordinarios resultados en tres áreas particulares: image processing, speech recognition y natural language processing.
- Excelentes para implementaciones Cheery-Picking en escenarios donde no existe actualmente Digitalización.
- El componente estocástico que tienen las redes hace muy complejo su parametrización.
- Limitaciones cuando la cantidad de muestras es baja[26].
- Inteligible Property[25]
- Hyperhype: Through of Disillusionment (Gartner Hype Cycle)

どうもありがとう

Gracias !

Centro Inteligencia Computacional
ITBA

Follow up

- Tutorials at Conferences (e.g. ICML, NeurIPS)
- Two Minute Papers - Youtube Channel
- The Batch - Andrew NG newsletter
- Yannic Kilcher - Youtube Channel
- Arxiv Insight - Youtube Channel
- http://monostuff.logdown.com/posts/7835544-aprendizaje-modo-mquina

[1] Alberts, Bruce, et al. *Essential cell biology*. Garland Science, 2013.

[2] Guyton and Hall, “Textbook of Medical Physiology”, 12th edition

[3] Izhikevich, “Dynamical systems in neuroscience··, 1st edition

[4] Abott & Dayan, Theoretical Neuroscience, 2012

[5] Ramele, R., A. J. Villar, and J. M. Santos. "A Brain Computer Interface Classification Method Based on Signal Plots." 4th Winter Conference on Brain Computer Interfaces, Yongpyong, Korea, February 2015. IEEE Signal and Processing, 2016.

[6] Chia-Chu et al, “Slow periodic activity in the longitudinal hippocampal slice can self-propagate non-synaptically by a mechanism consistent with ephaptic coupling”, The Physiological Society Journal, 2018

[7] Belousov et al, “Neuronal gap junctions: making and breaking connections during development and injury”, Trends in Neuroscience, 2012

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